

Question Bank

2400 questions across 8 topics

Computer Networks

1. In the following pairs of OSI protocol layer/sub-layer and its functionality, the INCORRECT pair is
2. An IP machine Q has a path to another IP machine H via three IP routers R1, R2, and R3. Q---R1---R2---R3---H H acts as an HTTP server, and Q connects to H via HTTP and downloads a file. Session layer encryption is used, with DES as the shared key encryption protocol. Consider the following four pieces of information: [I1] The URL of the file downloaded by Q [I2] The TCP port numbers at Q and H [I3] The IP addresses of Q and H [I4] The link layer addresses of Q and H. Which of I1, I2, I3, and I4 can an intruder learn through sniffing at R2 alone?
3. To send same bit sequence, NRZ encoding require
4. If there are n devices (nodes) in a network, what is the number of cable links required for a fully connected mesh and a star topology respectively
5. In networking terminology UTP means
6. Physical topology of FDDI is?
7. Bit stuffing refers to
8. How many characters per sec (7 bits + 1 parity) can be transmitted over a 2400 bps line if the transfer is synchronous (1 start and 1 stop bit)?
9. What frequency range is used for microwave communications, satellite and radar?
10. A T-switch is used to
11. Consider a 50 kbps satellite channel with a 500 milliseconds round trip propagation delay. If the sender wants to transmit 1000 bit frames, how much time will it take for the receiver to receive the frame?
12. In the Ethernet, which field is actually added at the physical layer and is not part of the frame.
13. The voltage ranges for a logic high and a logic low in RS-232 C standard is
14. IEEE 1394 is related to
15. What is the maximum number of characters (7 bits + parity) that can be transmitted in a second on a 19.2 Kbps line. This asynchronous transmission requires 1 start bit and 1 stop bit.
16. One SAN switch has 24 ports. All 24 supports 8 Gbps Fiber Channel technology. What is the aggregate bandwidth of that SAN switch?
17. Data is transmitted continuously at 2.048 Mbps rate for 10 hours and received 512 bits errors. What is the bit error rate?
18. What is the bit rate of a video terminal unit with 80 characters/line, 8 bits/character and horizontal sweep time of 100 μ s (including 20 μ s of retrace time)?

- 19.** The encoding technique used to transmit the signal in giga ethernet technology over fiber optic medium is
- 20.** Assume that each character code consists of 8 bits. The number of characters that can be transmitted per second through an asynchronous serial line at 2400 baud rate, and with two stop bits is
- 21.** What is the bandwidth of the signal that ranges from 40 kHz to 4 MHz?
- 22.** How many bytes of data can be sent in 15 seconds over a serial link with baud rate of 9600 in asynchronous mode with odd parity and two stop bits in the frame?
- 23.** By using an eight bit optical encoder the degree of resolution that can be obtained is (approximately)
- 24.** Assume that each character code consists of 8 bits. The number of characters that can be transmitted per second through a synchronous serial line at 2400 baud rate, and with two stop bits is
- 25.** If the bandwidth of a signal is 5 kHz and the lowest frequency is 52 kHz, what is the highest frequency
- 26.** Phase transition for each bit are used in
- 27.** Bit stuffing refers to
- 28.** How many 8-bit characters can be transmitted per second over a 9600 baud serial communication link using asynchronous mode of transmission with one start bit, eight data bits, two stop bits, and one parity bit?
- 29.** A serial transmission T1 uses 8 information bits, 2 start bits, 1 stop bit and 1 parity bit for each character. A synchronous transmission T2 uses 3 eight-bit sync characters followed by 30 eight-bit information characters. If the bit rate is 1200 bits/second in both cases, what are the transfer rates of T1 and T2?
- 30.** In Ethernet when Manchester encoding is used, the bit rate is:
- 31.** In the waveform (a) given below, a bit stream is encoded by Manchester encoding scheme. The same bit stream is encoded in a different coding scheme in wave form (b). The bit stream and the coding scheme are
- 32.** Let us consider a statistical time division multiplexing of packets. The number of sources is 10. In a time unit, a source transmits a packet of 1000 bits. The number of sources sending data for the first 20 time units is 6, 9, 3, 7, 2, 2, 2, 3, 4, 6, 1, 10, 7, 5, 8, 3, 6, 2, 9, 5 respectively. The output capacity of multiplexer is 5000 bits per time unit. Then the average number of backlogged of packets per time unit during the given period is
- 33.** Assume that each character code consists of 8 bits. The number of characters that can be transmitted per second through an asynchronous serial line at 2400 baud rate, and with two stop bits is
- 34.** Purpose of a start bit in RS-232 serial communication protocol is:
- 35.** In serial communication employing 8 data bits, a parity bit and 2 stop bits, the minimum band rate required to sustain a transfer rate of 300 characters per second is

- 36.** In serial data transmission, every byte of data is padded with a '0' in the beginning and one or two '1's at the end of byte because
- 37.** Consider an Ethernet segment with a transmission speed of 108 bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is 2×10^8 meters/sec, then the minimum frame size (in bits) required for collision detection is _____
- 38.** Suppose two hosts are connected by a point-to-point link and they are configured to use Stop-and-Wait protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest.
- 39.** Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 3×10^8 m/s. The time taken (in milliseconds, rounded off to two decimal places) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender is
- 40.** Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 1 Mbps (=106 bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is _____
- 41.** Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial $X^3 + X + 1$. Suppose the message $m_4 \parallel m_3 \parallel m_2 \parallel m_1 \parallel m_0 = 11000$ is to be transmitted. Check bits $c_2 \parallel c_1 \parallel c_0$ are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4 \parallel m_3 \parallel m_2 \parallel m_1 \parallel m_0 \parallel c_2 \parallel c_1 \parallel c_0$. The value of the checkbit sequence $c_2 \parallel c_1 \parallel c_0$ is
- 42.** Consider the following two statements. S1: Destination MAC address of an ARP reply is a broadcast address. S2: Destination MAC address of an ARP request is a broadcast address. Which one of the following choices is correct?
- 43.** Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate up link ports. The minimum number of switches needed is _____.
- 44.** Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?
- 45.** Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 ms and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use?
- 46.** _____ can detect burst error of length less than or equal to degree of the polynomial and detects burst errors that affect odd number of bits.

47. Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like Ethernet or wireless). The nodes in the system use the following carrier-sense based medium access protocol. A node that receives a packet to transmit will carrier-sense the medium for 5 units of time. If the node does not detect any other transmission in this duration, it starts transmitting its packet in the next time unit. If the node detects another transmission, it waits until this other transmission finishes, and then begins to carrier-sense for 5 time units again. Once they start to transmit, nodes do not perform any collision detection and continue transmission even if a collision occurs. All transmissions last for 20 units of time. Assume that the transmission signal travels at the speed of 10 meters per unit time in the medium. Assume that the system has two nodes P and Q, located at a distance d meters from each other. P starts transmitting a packet at time $t=0$ after successfully completing its carrier-sense phase. Node Q has a packet to transmit at time $t=0$ and begins to carrier-sense the medium. The maximum distance d (in meters, rounded to the closest integer) that allows Q to successfully avoid a collision between its proposed transmission and P's ongoing transmission is _____.

48. An Ethernet frame that is less than the IEEE 802.3 minimum length of 64 octets is called

49. Which media access control protocol is used by IEEE 802.11 wireless LAN?

50. Consider a binary code that consists of only four valid code words as given below:
00000, 01011, 10101, 11110 Let the minimum Hamming distance of the code be p and the maximum number of erroneous bits that can be corrected by the code be q . Then the values of p and q are

51. The values of parameters for the Stop-and-Wait ARQ protocol are as given below: Bit rate of the transmission channel = 1Mbps Propagation delay from sender to receiver = 0.75 ms Time to process a frame = 0.25ms Number of bytes in the information frame = 1980 Number of bytes in the acknowledge frame = 20 Number of overhead bytes in the information frame = 20. Assume that there are no transmission errors. Then the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters is _____ (correct to 2 decimal places)

52. A computer network uses polynomials over $GF(2)$ for error checking with 8 bits as information bits and uses x^3+x+1 as the generator polynomial to generate the check bits. In this network, the message 01011011 is transmitted as

53. Frames of 1000 bits are sent over a 106 bps duplex link between two hosts. The propagation time is 25 ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

54. What is the minimum number of bits (I) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.

55. The message 11001001 is to be transmitted using the CRC polynomial x^3+1 to protect it from errors. The message that should be transmitted is:

56. In a token ring network the transmission speed is 107 bps and the propagation speed is 200 meters/ μ s. The 1-bit delay in this network is equivalent to:

57. In Ethernet CSMA/CD, the special bit sequence transmitted by media access management to handle collision is called

- 58.** Consider a 128×10^3 bits/second satellite communication link with one way propagation delay of 150 milliseconds. Selective retransmission(repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____.
- 59.** For the IEEE802.11 MAC protocol for wireless communication, which of the following statements is/are TRUE?
- 60.** For the IEEE802.11 MAC protocol for wireless communication, which of the following statements is/are TRUE? I. At least three non-overlapping channels are available for transmissions. II. The RTS-CTS mechanism is misused for collision detection. III. Unicast frames are ACKed.
- 61.** A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _____ bytes.
- 62.** In an Ethernet local area network, which one of the following statements is TRUE?
- 63.** A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps ($1 \text{ Kbps} = 1000 \text{ bits/second}$). Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds. Assuming no frame is lost, the sender throughput is _____ bytes/second.
- 64.** A certain population of ALOHA users manages to generate 70 request/sec. If the time is slotted in units of 50 msec, then channel load would be
- 65.** In CRC if the data unit is 100111001 and the divisor is 1011 then what is dividend at the receiver?
- 66.** Two hosts are connected via a packet switch with 107 bits per second links. Each link has a propagation delay of 20 microseconds. The switch begins forwarding a packet 35 microseconds after it receives the same. If 10000 bits of data are to be transmitted between the two hosts using a packet size of 5000 bits, the time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds is _____.
- 67.** Consider a network connecting two systems located 8000 kilometers apart. The bandwidth of the network is 500×10^6 bits per second. The propagation speed of the media is 4×10^6 meters per second. It is needed to design a Go-Back-N sliding window protocol for this network. The average packet size is 107 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. Then, the minimum size in bits of the sequence number field has to be
- 68.** Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km (kilometer) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?
- 69.** A link has a transmission speed of 10^6 bits/sec. It uses data packets of size 1000 bytes each. Assume that the acknowledgment has negligible transmission delay, and that its propagation delay is the same as the data propagation delay. Also assume that the processing delays at nodes are negligible. The

efficiency of the stop-and-wait protocol in this setup is exactly 25%. The value of the one-way propagation delay (in milliseconds) is _____.

70. Consider a LAN with four nodes S1■, S2■, S3■ and S4■. Time is divided into fixed-size slots, and a node can begin its transmission only at the beginning of a slot. A collision is said to have occurred if more than one node transmits in the same slot. The probabilities of generation of a frame in a time slot by S1■, S2■, S3■ and S4■ are 0.1, 0.2, 0.3 and 0.4, respectively. The probability of sending a frame in the first slot without any collision by any of these four stations is _____.

71. Suppose that the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Assume that the transmission time for the acknowledgement and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of at least 50% is _____.

72. Which of the following is not a valid multicast MAC address?

73. A mechanism or technology used in Ethernet by which two connected devices choose common transmission parameters such as speed, duplex mode and flow control is called

74. A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 01111100101, then the input bit-string is

75. Consider the store and forward packet switched network given below. Assume that the bandwidth of each link is 106 bytes / sec. A user on host A sends a file of size 103 bytes to host B through routers R1 and R2 in three different ways. In the first case a single packet containing the complete file is transmitted from A to B. In the second case, the file is split into 10 equal parts, and these packets are transmitted from A to B. In the third case, the file is split into 20 equal parts and these packets are sent from A to B. Each packet contains 100 bytes of header information along with the user data. Consider only transmission time and ignore processing, queuing and propagation delays. Also assume that there are no errors during transmission. Let T1, T2 and T3 be the times taken to transmit the file in the first, second and third case respectively. Which one of the following is CORRECT?

76. Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is _____.

77. Consider a token ring network with a length of 2km having 10 stations including a monitoring station. The propagation speed of the signal is 2×10^8 m/s and the token transmission time is ignored. If each station is allowed to hold the token for 2 μ sec, the minimum time for which the monitoring station should wait (in μ sec) before assuming that the token is lost is _____.

78. What will be the efficiency of a Stop and Wait protocol, if the transmission time for a frame is 20ns and the propagation time is 30ns?

79. If the frame to be transmitted is 1101011011 and the CRC polynomial to be used for generating checksum is $x^4 + x + 1$, then what is the transmitted frame?

80. Ethernet layer-2 switch is a network element type which gives.

- 81.** How many check bits are required for 16 bit data word to detect 2 bit errors and single bit correction using hamming code?
- 82.** Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.
- 83.** Consider a source computer (S) transmitting a file of size 106 bits to a destination computer (D) over a network of two routers (R1 and R2) and three links (L1, L2, and L3). L1 connects S to R1; L2 connects R1 to R2; and L3 connects R2 to D. Let each link be of length 100 km. Assume signals travel over each link at a speed of 108 meters per second. Assume that the link bandwidth on each link is 1Mbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D?
- 84.** The hamming distance between the octets of 0xAA and 0x55 is
- 85.** The network protocol which is used to get MAC address of a node by providing IP address is
- 86.** Which of the following is a MAC address?
- 87.** Frames of 1000 bits are sent over a 106 bps duplex link between two hosts. The propagation time is 25ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link). What is the minimum number of bits (I) that will be required to represent the sequence numbers distinctly? Assume that no time gap needs to be given between transmission of two frames.
- 88.** Frames of 1000 bits are sent over a 106 bps duplex link between two hosts. The propagation time is 25ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link). Suppose that the sliding window protocol is used with the sender window size of 2I, where I is the number of bits identified in the earlier part and acknowledgements are always piggy backed. After sending 2I frames, what is the minimum time the sender will have to wait before starting transmission of the next frame? (Identify the closest choice ignoring the frame processing time.)
- 89.** Let $G(x)$ be the generator polynomial used for CRC checking. What is the condition that should be satisfied by $G(x)$ to detect odd number of bits in error?
- 90.** Which Project 802 standard provides for a collision-free protocol?
- 91.** Which of the following transmission media is not readily suitable to CSMA operation?
- 92.** In Ethernet, the source address field in the MAC frame is the _____ address.
- 93.** Data transmitted on a link uses the following 2D parity scheme for error detection: Each sequence of 28 bits is arranged in a 4×7 matrix (rows r_0 through r_3 , and columns d_7 through d_1) and is padded with a column d_0 and row r_4 of parity bits computed using the Even parity scheme. Each bit of column d_0 (respectively, row r_4) gives the parity of the corresponding row (respectively, column). These 40 bits are transmitted over the data link. The table shows data received by a receiver and has n corrupted bits. What is the minimum possible value of n ?
- 94.** The minimum frame size required for a CSMA/CD based computer network running at 1Gbps on a 200m cable with a link speed of 2×10^8 m/sec is:

- 95.** A 1 Mbps satellite link connects two ground stations. The altitude of the satellite is 36,504 km and speed of the signal is 3×10^8 m/s. What should be the packet size for a channel utilization of 25% for a satellite link using go-back-127 sliding window protocol? Assume that the acknowledgment packets are negligible in size and that there are no errors during communication.
- 96.** A computer on a 10Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2Mbps. It is initially filled to capacity with 16Megabits. What is the maximum duration for which the computer can transmit at the full 10Mbps?
- 97.** An Ethernet hub
- 98.** Consider a token ring topology with N stations (numbered 1 to N) running token ring protocol where the stations are equally spaced. When a station gets the token it is allowed to send one frame of fixed size. Ring latency is t_p , while the transmission time of a frame is t_t . All other latencies can be neglected. The maximum utilization of the token ring when $t_t = 5$ ms, $t_p = 3$ ms, $N = 15$ is:
- 99.** Consider a token ring topology with N stations (numbered 1 to N) running token ring protocol where the stations are equally spaced. When a station gets the token it is allowed to send one frame of fixed size. Ring latency is t_p , while the transmission time of a frame is t_t . All other latencies can be neglected. The maximum utilization of the token ring when $t_t = 3$ ms, $t_p = 5$ ms, $N = 10$ is:
- 100.** A broadcast channel has 10 nodes and total capacity of 10 Mbps. It uses polling for medium access. Once a node finishes transmission, there is a polling delay of $80 \mu s$ to poll the next node. Whenever a node is polled, it is allowed to transmit a maximum of 1000 bytes. The maximum throughput of the broadcast channel is:
- 101.** An error correcting code has the following code words: 00000000, 00001111, 01010101, 10101010, 11110000. What is the maximum number of bit errors that can be corrected?
- 102.** The distance between two stations M and N is L kilometers. All frames are K bits long. The propagation delay per kilometer is t seconds. Let R bits/second be the channel capacity. Assuming that processing delay is negligible, the minimum number of bits for the sequence number field in a frame for maximum utilization, when the sliding window protocol is used, is:
- 103.** The message 11001001 is to be transmitted using the CRC polynomial x^3+1 to protect it from errors. The message that should be transmitted is:
- 104.** In a token ring network the transmission speed is 107 bps and the propagation speed is 200 metres/ μs . The 1-bit delay in this network is equivalent to:
- 105.** There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that ONLY one station transmits in a given time slot?
- 106.** On a wireless link, the probability of packet error is 0.2. A stop-and-wait protocol is used to transfer data across the link. The channel condition is assumed to be independent of transmission to transmission. What is the average number of transmission attempts required to transfer 100 packets?

107. A link of capacity 100 Mbps is carrying traffic from a number of sources. Each source generates an on-off traffic stream; when the source is on, the rate of traffic is 10 Mbps, and when the source is off, the rate of traffic is zero. The duty cycle, which is the ratio of on-time to off-time, is 1:2. When there is no buffer at the link, the minimum number of sources that can be multiplexed on the link so that link capacity is not wasted and no data loss occurs is S1. Assuming that all sources are synchronized and that the link is provided with a large buffer, the maximum number of sources that can be multiplexed so that no data loss occurs is S2. The values of S1 and S2 are, respectively

108. A router has two full-duplex Ethernet interfaces each operating at 100 Mb/s. Ethernet frames are at least 84 bytes long (including the Preamble and the Inter-Packet-Gap). The maximum packet processing time at the router for wirespeed forwarding to be possible is (in micro-seconds)

109. Suppose that it takes 1 unit of time to transmit a packet (of fixed size) on a communication link. The link layer uses a window flow control protocol with a window size of N packets. Each packet causes an ack or a nak to be generated by the receiver, and ack/nak transmission times are negligible. Further, the round trip time on the link is equal to N units. Consider time $i > N$. If only acks have been received till time i (no naks), then the goodput evaluated at the transmitter at time i (in packets per unit time) is

110. Consider the diagram shown below where a number of LANs are connected by (transparent) bridges. In order to avoid packets looping through circuits in the graph, the bridges organize themselves in a spanning tree. First, the root bridge is identified as the bridge with the least serial number. Next, the root sends out (one or more) data units to enable the setting up of the spanning tree of shortest paths from the root bridge to each bridge. Each bridge identifies a port (the root port) through which it will forward frames to the root bridge. Port conflicts are always resolved in favour of the port with the lower index value. When there is a possibility of multiple bridges forwarding to the same LAN (but not through the root port), ties are broken as follows: bridges closest to the root get preference and between such bridges, the one with the lowest serial number is preferred. Host H1 sends out a broadcast ping packet. Which of the following options represents the correct forwarding table on B3?

111. Consider the diagram shown below where a number of LANs are connected by (transparent) bridges. In order to avoid packets looping through circuits in the graph, the bridges organize themselves in a spanning tree. First, the root bridge is identified as the bridge with the least serial number. Next, the root sends out (one or more) data units to enable the setting up of the spanning tree of shortest paths from the root bridge to each bridge. Each bridge identifies a port (the root port) through which it will forward frames to the root bridge. Port conflicts are always resolved in favour of the port with the lower index value. When there is a possibility of multiple bridges forwarding to the same LAN (but not through the root port), ties are broken as follows: bridges closest to the root get preference and between such bridges, the one with the lowest serial number is preferred. For the given connection of LANs by bridges, which one of the following choices represents the depth first traversal of the spanning tree of bridges?

112. Station A needs to send a message consisting of 9 packets to Station B using a sliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no acks from B ever get lost), then what is the number of packets that A will transmit for sending the message to B ?

113. Station A uses 32 byte packets to transmit messages to Station B using a sliding window protocol. The round trip delay between A and B is 80 milliseconds and the bottleneck bandwidth on the path between A and B is 128 kbps. What is the optimal window size that A should use ?

114. Consider the following message $M = 1010001101$. The cyclic redundancy check (CRC) for this message using the divisor polynomial $x^5+x^4+x^2+1$ is :

115. In a TDM medium access control bus LAN, each station is assigned one time slot per cycle for transmission. Assume that the length of each time slot is the time to transmit 100bits plus the end-to-end propagation delay. Assume a propagation speed of 2×10^8 m/sec. The length of the LAN is 1 km with a bandwidth of 10Mbps. The maximum number of stations that can be allowed in the LAN so that the throughput of each station can be 2/3Mbps is

116. A channel has a bit rate of 4 kbps and one-way propagation delay of 20 ms. The channel uses stop and wait protocol. The transmission time of the acknowledgment frame is negligible. To get a channel efficiency of at least 50%, the minimum frame size should be

117. A network with CSMA/CD protocol in the MAC layer is running at 1Gbps over a 1km cable with no repeaters. The signal speed in the cable is 2×10^8 m/sec. The minimum frame size for this network should be:

118. Which of the following statements is TRUE about CSMA/CD:

119. Suppose the round trip propagation delay for a 10 Mbps Ethernet having 48-bit jamming signal is $46.4 \mu\text{s}$. The minimum frame size is

120. In a packet switching network, packets are routed from source to destination along a single path having two intermediate nodes. If the message size is 24 bytes and each packet contains a header of 3 bytes, then the optimum packet size is:

121. The maximum window size for data transmission using the selective reject protocol with n-bit frame sequence numbers is:

122. Consider a simplified time slotted MAC protocol, where each host always has data to send and transmits with probability $p = 0.2$ in every slot. There is no backoff and one frame can be transmitted in one slot. If more than one host transmits in the same slot, then the transmissions are unsuccessful due to collision. What is the maximum number of hosts which this protocol can support if each host has to be provided a minimum throughput of 0.16 frames per time slot?

123. Consider a parity check code with three data bits and four parity check bits. Three of the Code Words are 0101011, 1001101 and 1110001. Which of the following are also code words? I. 0010111 II. 0110110 III. 1011010 IV. 0111010

124. A 20 Kbps satellite link has a propagation delay of 400 ms. The transmitter employs the "go back n ARQ" scheme with n set to 10. Assuming that each frame is 100 byte long, what is the maximum data rate possible?

125. Consider a 10 Mbps token ring LAN with a ring latency of $400\mu\text{s}$. A host that needs to transmit seizes the token. Then it sends a frame of 1000 bytes, removes the frame after it has circulated all around the ring, and finally releases the token. This process is repeated for every frame. Assuming that only a single host wishes to transmit, the effective data rate is

126. In a sliding window ARQ scheme, the transmitter's window size is N and the receiver's window size is M . The minimum number of distinct sequence numbers required to ensure correct operation of the ARQ scheme is

127. In a data link protocol, the frame delimiter flag is given by 0111. Assuming that bit stuffing is employed, the transmitter sends the data sequence 01110110 as

128. Consider three IP networks A, B and C. Host HA in network A sends messages each containing 180 bytes of application data to a host HC in network C. The TCP layer prefixes a 20 byte header to the message. This passes through an intermediate network B. The maximum packet size, including 20 byte IP header, in each network is A : 1000 bytes B : 100 bytes C : 1000 bytes. The network A and B are connected through a 1 Mbps link, while B and C are connected by a 512 Kbps link (bps = bits per second). What is the rate at which application data is transferred to host Hc? Ignore errors, acknowledgements, and other overheads.

129. A and B are the only two stations on an Ethernet. Each has a steady queue of frames to send. Both A and B attempt to transmit a frame, collide, and A wins the first backoff race. At the end of this successful transmission by A, both A and B attempt to transmit and collide. The probability that A wins the second backoff race is

130. Host A is sending data to host B over a full duplex link. A and B are using the sliding window protocol for flow control. The send and receive window sizes are 5 packets each. Data packets (sent only from A to B) are all 1000 bytes long and the transmission time for such a packet is $50\mu\text{s}$. Acknowledgment packets (sent only from B to A), are very small and require negligible transmission time. The propagation delay over the link is $200\mu\text{s}$. What is the maximum achievable throughput in this communication ?

131. A 2 km long broadcast LAN has 107 bps bandwidth and uses CSMA/CD. The signal travels along the wire at $2 \times 10^8 \text{ m/s}$. What is the minimum packet size that can be used on this network ?

132. Start and stop bits do not contain any "information" but are used in serial communication for

133. Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

134. Which of the following fields of an IP header is/are always modified by any router before it forwards the IP packet?

135. Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection. Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

136. Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is _____

137. Consider the entries shown below in the forwarding table of an P router. Each entry consists of an IP prefix and the corresponding next hop router for packets whose destination IP address matches the prefix. The notation "/N" in a prefix indicates a subnet mask with the most significant N bits set to 1. This router forwards 20 packets each to 5 hosts. The IP addresses of the hosts are 10.1.1.16, 10.1.1.72, 10.1.1.132, 10.1.1.191, and 10.1.1.205. The number of packets forwarded via the next hop router R2 is

138. Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

139. The forwarding table of a router is shown below. A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID _____.

140. Which of the following statements is/are INCORRECT about the OSPF (Open Shortest Path First) routing protocol used in the Internet?

141. Consider a network with three routers P, Q, R shown in the figure below. All the links have cost of unity. The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q-R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation (rounded off to one decimal place) between P and Q, leading to count-to-infinity problem, is _____

142. Consider routing table of an organization's router shown below: Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?

143. Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via three routers as shown below. What is the number of subnets inside the enterprise network?

144. Consider a computer network using the distance vector routing algorithm in its network layer. The partial topology of the network is shown below. The objective is to find the shortest-cost path from the router R to routers P and Q. Assume that R does not initially know the shortest routes to P and Q. Assume that R has three neighboring routers denoted as X, Y and Z. During one iteration, R measures its distance to its neighbors X, Y, and Z as 3, 2 and 5, respectively. Router R gets routing vectors from its neighbors that indicate that the distance to router P from routers X, Y and Z are 7, 6 and 5, respectively. The routing vector also indicates that the distance to router Q from routers X, Y and Z are 4, 6 and 8 respectively. Which of the following statement(s) is/are correct with respect to the new routing table of R, after updation during this iteration?

145. Which of the following classes of languages can validate an IPv4 address in dotted decimal format? It is to be ensured that the decimal values lie between 0 and 255.

146. An organization requires a range of IP address to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space 202.61.0.0/17. The ISP wants to assign an

address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one of the organization?

147. Consider the following statements about the functionality of an IP based router. I. A router does not modify the IP packets during forwarding. II. It is not necessary for a router to implement any routing protocol. III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet. Which of the above statements is/are TRUE?

148. Consider three machines M, N and P with IP addresses 100.10.5.2, 100.10.5.5 and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all the three machines. Which one of the following is true?

149. Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and a 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0. The fragmentation offset value stored in the third fragment is _____.

150. The default subnet mask for a class B network can be

151. The maximum number of IPv4 router addresses that can be listed in the record route (RR) option field of an IPv4 header is _____.

152. The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

153. Consider the following statements about the routing protocols, Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network. I. RIP uses distance vector routing. II. RIP packets are sent using UDP. III. OSPF packets are sent using TCP. IV. OSPF operation is based on link-state routing. Which of the statements above are CORRECT?

154. The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

155. Which network protocol allows hosts to dynamically get a unique IP number on each bootup

156. Dynamic routing protocol enable routers to

157. An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes. The number of fragments that the IP datagram will be divided into for transmission is _____.

158. Which layers of the OSI reference model are host-to-host layers?

159. In a class B subnet, we know the IP address of one host and the mask as given below: IP address: 125.134.112.66 Mask: 255.255.224.0. What is the first address (Network address)?

- 160.** In the network 200.10.11.144/27, the fourth octet (in decimal) of the last IP address of the network which can be assigned to a host is _____.
- 161.** How many bits internet address is assigned to each host on a TCP/IP internet which is used in all communication with the host?
- 162.** Which one of the following fields of an IP header is NOT modified by a typical IP router?
- 163.** An organization is granted the block 130.34.12.64/26. It needs to have 4 subnets. Which of the following is not an address of this organization?
- 164.** Assume the following information. Original timestamp value = 46 Receive timestamp value = 59 Transmit timestamp value = 60 Timestamp at arrival of packet = 69 Which of the following statements is correct?
- 165.** A supernet has a first address of 205.16.32.0 and a supernet mask of 255.255.248.0. A router receives 4 packets with the following destination addresses. which packet belongs to this supernet?
- 166.** An IP packet has arrived with the first 8 bits as 0100 0010. Which of the following is correct?
- 167.** A IP packet has arrived in which the fragmentation offset value is 100, the value of HLEN is 5 and the value of total length field is 200. What is the number of the last byte?
- 168.** What is routing algorithm used by OSPF routing protocol?
- 169.** The process of modifying IP address information in IP packet headers while in transit across a traffic routing device is called
- 170.** An IP router with a Maximum Transmission Unit (MTU) of 1500 bytes has received an IP packet of size 4404 bytes with an IP header of length 20 bytes. The values of the relevant fields in the header of the third IP fragment generated by the router for this packet are
- 171.** Every host in an IPv4 network has a 1-second resolution real-time clock with battery backup. Each host needs to generate up to 1000 unique identifiers per second. Assume that each host has a globally unique IPv4 address. Design a 50-bit globally unique ID for this purpose. After what period (in seconds) will the identifiers generated by a host wrap around?
- 172.** An IP router implementing Classless Inter-domain routing (CIDR) receives a packet with address 131.23.151.76. The router's routing table has the following entries:
- 173.** Host A (on TCP/IP v4 network A) sends an IP datagram D to host B (also on TCP/IP V4 network B). Assume that no error occurred during the transmission of D. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D? (i) TTL (ii) Checksum (iii) Fragment Offset
- 174.** In the diagram shown below, L1 is an Ethernet LAN and L2 is a Token-Ring LAN. An IP packet originates from sender S and traverses to R, as shown. The links within each ISP and across the two ISPs, are all point-to-point optical links. The initial value of the TTL field is 32. The maximum possible value of the TTL field when R receives the datagram is _____.

175. Which one of the following is TRUE about the interior gateway routing protocols - Routing Information Protocol (RIP) and Open Shortest Path First (OSPF)?

176. Consider the following three statements about link state and distance vector routing protocols, for a large network with 500 network nodes and 4000 links [S1] The computational overhead in link state protocols is higher than in distance vector protocols. [S2] A distance vector protocol (with split horizon) avoids persistent routing loops, but not a link state protocol. [S3] After a topology change, a link state protocol will converge faster than a distance vector protocol. Which one of the following is correct about S1, S2, and S3?

177. What is IP class and number of sub-networks if the subnet mask is 255.224.0.0?

178. IPv6 does not support which of the following addressing modes?

179. In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are

180. Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.

181. An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B?

182. In the IPv4 addressing format, the number of networks allowed under Class C addresses is

183. Lightweight Directory Access protocol is used for

184. The broadcast address for IP network 172.16.0.0 with subnet mask 255.255.0.0 is

185. Suppose computers A and B have IP addresses 10.105.1.113 and 10.105.1.91 respectively and they both use the same net mask N. Which of the values of N given below should not be used if A and B should belong to the same network?

186. One of the header fields in an IP datagram is the Time to Live (TTL) field. Which of the following statements best explains the need for this field?

187. What is the primary purpose of a VLAN?

188. The address resolution protocol (ARP) is used for

189. The subnet mask for a particular network is 255.255.31.0. Which of the following pairs of IP addresses could belong to this network?

190. The network 198.78.41.0 is a

191. On a LAN ,where are IP datagrams transported?

192. The subnet mask 255.255.255.192

193. Host X has IP address 192.168.1.97 and is connected through two routers R1 and R2 to another host Y with IP address 192.168.1.80. Router R1 has IP addresses 192.168.1.135 and 192.168.1.110. R2 has IP addresses 192.168.1.67 and 192.168.1.155. The netmask used in the network is 255.255.255.224. Which IP address should X configure its gateway as?

194. Host X has IP address 192.168.1.97 and is connected through two routers R1 and R2 to another host Y with IP address 192.168.1.80. Router R1 has IP addresses 192.168.1.135 and 192.168.1.110. R2 has IP addresses 192.168.1.67 and 192.168.1.155. The netmask used in the network is 255.255.255.224. Given the information above, how many distinct subnets are guaranteed to already exist in the network?

195. Two popular routing algorithms are Distance Vector(DV) and Link State (LS) routing. Which of the following are true? (S1): Count to infinity is a problem only with DV and not LS routing (S2): In LS, the shortest path algorithm is run only at one node (S3): In DV, the shortest path algorithm is run only at one node (S4): DV requires lesser number of network messages than LS

196. If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

197. Range of IP Address from 224.0.0.0 to 239.255.255.255 are

198. When a host on network A sends a message to a host on network B, which address does the router look at?

199. If there are five routers and six networks in intranet using link state routing, how many routing tables are there?

200. A group of 15 routers is interconnected in a centralized complete binary tree with a router at each tree node. Router i communicates with router j by sending a message to the root of the tree. The root then sends the message back down to router j. The mean number of hops per message, assuming all possible router pairs are equally likely is

201. For the network given in the figure below, the routing tables of the four nodes A, E, D and G are shown. Suppose that F has estimated its delay to its neighbors, A, E, D and G as 8, 10, 12 and 6 msec respectively and updates its routing table using distance vector routing technique.

202. The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

203. A subnetted Class B network has the following broadcast address: 144.16.95.255 Its subnet mask

204. Two computers C1 and C2 are configured as follows. C1 has IP address 203. 197.2.53 and netmask 255.255. 128.0. C2 has IP address 203.197.75.201 and netmask 255.255.192.0. Which one of the following statements is true?

205. For which one of the following reason does Internet Protocol (IP) use the timeto-live (TTL) field in the IP datagram header?

206. Consider a simple graph with unit edge costs. Each node in the graph represents a router. Each node maintains a routing table indicating the next hop router to be used to relay a packet to its destination and the cost of the path to the destination through that router. Initially, the routing table is empty. The routing table is synchronously updated as follows. In each updated interval, three tasks are performed. i. A node determines whether its neighbors in the graph are accessible. If so, it sets the tentative cost to each accessible neighbor as 1. Otherwise, the cost is set to ∞ . ii. From each accessible neighbor, it gets the costs to relay to other nodes via that neighbor (as the next hop). iii. Each node updates its routing table based on the information received in the previous two steps by choosing the minimum cost. Continuing from the earlier problem, suppose at some time t , when the costs have stabilized, node A goes down. The cost from node F to node A at time $(t+100)$ is :

207. Assume that "host1.mydomain.dom" has an IP address of 145.128.16.8. Which of the following options would be most appropriate as a subsequence of steps in performing the reverse lookup of 145.128.16.8 ? In the following options "NS" is an abbreviation of "nameserver".

208. A company has a class C network address of 204.204.204.0. It wishes to have three subnets, one with 100 hosts and two with 50 hosts each. Which one of the following options represents a feasible set of subnet address/subnet mask pairs?

209. Count to infinity is a problem associated with:

210. An organization has a class B network and wishes to form subnets for 64 departments. The subnet mask would be:

211. In a network of LANs connected by bridges, packets are sent from one LAN to another through intermediate bridges. Since more than one path may exist between two LANs, packets may have to be routed through multiple bridges. Why is the spanning tree algorithm used for bridge-routing?

212. The address resolution protocol (ARP) is used for:

213. In the TCP/IP protocol suite, which one of the following is NOT part of the IP header?

214. A host is connected to a Department network which is part of a University network. The University network, in turn, is part of the Internet. The largest network in which the Ethernet address of the host is unique is

215. A subnet has been assigned a subnet mask of 255.255.255.192. What is the maximum number of hosts that can belong to this subnet?

216. Which of the following is NOT true with respect to a transparent bridge and a router?

217. The subnet mask for a particular network is 255.255.31.0 Which of the following pairs of IP addresses could belong to this network ?

218. Which of the following assertions is false about the internet Protocol (IP) ?

219. Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window

size (in MSS) during the third RTT will be _____

220. Which of the following statements about IPv4 fragmentation is/are TRUE?

221. TCP client P successfully establishes a connection to TCP server Q. Let NP denote the sequence number in the SYN sent from P to Q. Let NQ denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

222. Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named myTCP, runs over a 100 Mbps network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes. Which of the following is/are valid lengths of the Sequence Number field in the myTCP header?

223. Consider the data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL is

224. Consider the three-way handshake mechanism followed during TCP connection establishment between hosts P and Q. Let X and Y be two random 32-bit starting sequence numbers chosen by P and Q respectively. Suppose P sends a TCP connection request message to Q with a TCP segment having SYN bit =1, SEQ number =X, and ACK bit =0. Suppose Q accepts the connection request. Which one of the following choices represents the information present in the TCP segment header that is sent by Q to P?

225. A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible?

226. Checksum field in TCP header is

227. The persist timer is used in TCP to

228. Consider a TCP connection between a client and a server with the following specifications; the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time $t=0$. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time $t+60$ ms after all acknowledgements are processed is _____

229. Assuming that for a given network layer implementation, connection establishment overhead is 100 bytes and disconnection overhead is 28 bytes. What would be the minimum size of the packet the transport layer needs to keep up, if it wishes to implement a datagram service above the network layer and needs to keep its overhead to a minimum of 12.5%. (ignore transport layer overhead)

230. Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps (= 10^9 bits-per-second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is _____

- 231.** Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which connect function has already been called. Which of the following statements is/are CORRECT ? I. A connected UDP socket can be used to communicate with multiple peers simultaneously. II. A process can successfully call connect function again for an already connected UDP socket.
- 232.** Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls close to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK which is received by the client-side TCP. As per the TCP connections state diagram (RFC 793), in which state does the client-side TCP connection wait for the FIN from the server-side TCP?
- 233.** What is the maximum size of data that the application layer can pass on to the TCP layer below?
- 234.** For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is _____ seconds.
- 235.** An ACK number of 1000 in TCP always means that
- 236.** Consider the following statements. I. TCP connections are full duplex. II. TCP has no option for selective acknowledgement. III. TCP connections are message streams.
- 237.** Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are
- 238.** Host A sends a UDP datagram containing 8880 bytes of user data to host B over an Ethernet LAN. Ethernet frames may carry data up to 1500 bytes (i.e. MTU=1500 bytes). Size of UDP header is 8 bytes and size of IP header is 20 bytes. There is no option field in IP header. How many total number of IP fragments will be transmitted and what will be the contents of offset field in the last fragment?
- 239.** Which one of the following socket API functions converts an unconnected active TCP socket into a passive socket?
- 240.** Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2KB. The time taken (in msec) by the TCP connection to get back to 32KB congestion window is _____
- 241.** The protocol data unit for the transport layer in the internet stack is
- 242.** Which algorithm is used to shape the bursty traffic into a fixed rate traffic by averaging the data rate?
- 243.** The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are

244. Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

245. Which of the following transport layer protocols is used to support electronic mail?

246. A layer-4 firewall (a device that can look at all protocol headers up to the transport layer) CANNOT

247. In networking, UTP stands for

248. The TCP sliding window

249. Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

250. What is WPA?

251. Which protocol suite designed by IETF to provide security for a packet at the Internet layer?

252. Which of the following protocol is used for transferring electronic mail messages from one machine to another?

253. When a DNS server accepts and uses incorrect information from a host that has no authority giving that information, then it is called

254. Which of the following is/are example(s) of stateful application layer protocols? (i) HTTP (ii) FTP (iii) TCP (iv) POP3

255. Consider three IP networks A, B and C. Host HA in networks A sends messages each containing 180 bytes of application data to a host HC in network C. The TCP layer prefixes a 20 byte header to the message. This passes through an intermediate network B. the maximum packet size, including 20 byte IP header, in each network is:

256. A TCP message consisting of 2100 bytes is passed to IP for delivery across two networks. The first network can carry a maximum payload of 1200 bytes per frame and the second network can carry a maximum payload of 400 bytes per frame, excluding network overhead. Assume that IP overhead per packet is 20 bytes. What is the total IP overhead in the second network for this transmission?

257. Suppose that the maximum transmit window size for a TCP connection is 12000 bytes. Each packet consists of 2000 bytes. At some point of time, the connection is slow start phase with a current transmit window of 4000 bytes. Subsequently, the transmitter receives two acknowledgments. Assume that no packet are lost and there are no time-outs. What is the maximum possible value of the current transmit window?

258. A client process P needs to make a TCP connection to a server process S. Consider the following situation: the server process S executes a `socket()`, a `bind()` and a `listen()` system call in that order, following which it is preempted. Subsequently, the client process P executes a `socket()` system call followed by `connect()` system call to connect to the server process S. The server process has not executed any `accept()` system call. Which one of the following events could take place?

259. While opening a TCP connection, the initial sequence number is to be derived using a time-of-day (ToD) clock that keeps running even when the host is down. The low order 32 bits of the counter of the ToD clock is to be used for the initial sequence numbers. The clock counter increments once per millisecond. The maximum packet lifetime is given to be 64s. Which one of the choices given below is closest to the minimum permissible rate at which sequence numbers used for packets of a connection can increase?

260. Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

261. Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is _____.

262. Assume that the bandwidth for a TCP connection is 1048560 bits/sec. Let be the value of RTT in milliseconds(rounded off to the nearest integer) after which the TCP window scale option is needed. Let be the maximum possible window size the window scale option. Then the values of and are

263. For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is _____ seconds.

264. Consider an packet with a length of bytes that includes a -byte header and a -byte header. The packet is forwarded to an router that supports a Maximum Transmission Unit of bytes. Assume that the length of the header in all the outgoing fragments of this packet is bytes. Assume that the fragmentation offset value stored in the first fragment is

265. Consider a TCP connection between a client and a server with the following specifications: the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time $t = 0$. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time $t + 60$ ms after all acknowledgements are processed is _____.

266. A TCP server application is programmed to listen on port number P on host S. A TCP client connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and is rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviours is/are possible?

267. Consider two hosts P and Q connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes, and between R and Q is 820 bytes. A TCP segment of size 1400 bytes was transferred from P to Q through R, with IP identification value as 0×1234 . Assume that the IP header size is 20 bytes. Further, the packet is allowed to be fragmented i. e., Don't Fragment (DF) flag in the IP header is not set by P. Which of the following statements is/are correct?

268. Consider the data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL is _____.

269. Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 3×10^8 m/s. The time taken (in milliseconds, rounded off to two decimal places) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender is _____.

270. Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named myTCP, runs over a 100 Mbps network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes. Which of the following is/are valid lengths of the Sequence Number field in the myTCP header?

271. Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (in MSS) during the third RTT will be _____.

272. Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is _____.

273. Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like Ethernet or wireless). The nodes in the system use the following carrier-sense based medium access protocol. A node that receives a packet to transmit will carrier-sense the medium for units of time. If the node does not detect any other transmission in this duration, it starts transmitting its packet in the next time unit. If the node detects another transmission, it waits until this other transmission finishes, and then begins to carrier-sense for time units again. Once they start to transmit, nodes do not perform any collision detection and continue transmission even if a collision occurs. All transmissions last for units of time. Assume that the transmission signal travels at the speed of meters per unit time in the medium. Assume that the system has two nodes and located at a distance meters from each other. starts transmitting a packet at time after successfully completing its carrier-sense phase. Node has a packet to transmit at time and begins to carrier-sense the medium. The maximum distance (in meters, rounded to the closest integer) that allows to successfully avoid a collision between its proposed transmission and ongoing transmission is _____.

274. In an RSA cryptosystem, the value of the public modulus parameter n is 3007. If it is also known that $(n) = 2880$, where $(\cdot)$ denotes Euler's Totient Function, then the prime factor of n which is greater than 50 is _____.

275. Consider that B wants to send a message m that is digitally signed to A. Let the pair of private and public keys for A and B be denoted by and for $x = A, B$, respectively. Let $K_x(m)$ represent the operation of encrypting m with a key K_x and $H(m)$ represent the message digest. Which one of the following indicates the CORRECT way of sending the message m along with the digital signature to A?

276. The total number of keys required for a set of n individuals to be able to communicate with each other using secret key and public key crypto-systems, respectively are:

277. Suppose that two parties A and B wish to setup a common secret key (D - H key) between themselves using the Diffie-Hellman key exchange technique. They agree on 7 as the modulus and 3 as the primitive root. Party A chooses 2 and party B chooses 5 as their respective secrets. Their D - H key is

278. Consider a source computer (S) transmitting a file of size 106 bits to a destination computer (D) over a network of two routers (R1 and R2) and three links (L1, L2, and L3). L1 connects S to R1; L2 connects R1 to R2; and L3 connects R2 to D. Let each link be of length 100 km. Assume signals travel over each line at a speed of 108 meters per second. Assume that the link bandwidth on each link is 1 Mbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D?

279. Suppose in a web browser, you click on the `www.gate-2023.in` URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers. Let RTT denote the round trip time between your local host and DNS servers in the DNS hierarchy. The group trip time between the local host and the web server hosting `www.gate-2023.in` is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server. Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?

280. A graphical HTML browser resident at a network client machine Q accesses a static HTML webpage from a HTTP server S. The static HTML page has exactly one static embedded image which is also at S. Assuming no caching, which one of the following is correct about the HTML webpage loading (including the embedded image)?

281. Consider a network path $P \rightarrow Q \rightarrow R$ between nodes P and R via router Q. Node P sends a file of size bytes to R via this path by splitting the file into chunks of bytes each. Node P sends these chunks one after the other without any wait time between the successive chunk transmissions. Assume that the size of extra headers added to these chunks is negligible, and that the chunk size is less than the MTU. Each of the links $P \rightarrow Q$ and $Q \rightarrow R$ has a bandwidth of bits/sec, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R, with negligible processing and queueing delays. Router Q can simultaneously receive on link $P \rightarrow Q$ and transmit on link $Q \rightarrow R$. Assume P starts transmitting the chunks at time. Which one of the following options gives the time (in seconds, rounded off to 3 decimal places) at which R receives all the chunks of the file?

282. Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is

- 283.** Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits/ second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgement packet is negligible. Which ONE of the following most accurately gives the channel utilization for the above scenario in percentage?
- 284.** Consider an Ethernet segment with a transmission speed of bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is meters/sec, then the minimum frame size (in bits) required for collision detection is _____
- 285.** Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial $X^3 + X + 1$. Suppose the message $m_4m_3m_2m_1m_0 = 11000$ is to be transmitted. Check bits $c_2c_1c_0$ are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$. The value of the check bit sequence $c_2c_1c_0$ is
- 286.** A link has a transmission speed of 106 bits/sec. It uses data packets of size 1000 bytes each. Assume that the acknowledgement has negligible transmission delay, and that its propagation delay is the same as the data propagation delay. Also assume that the processing delays at the nodes are negligible. The efficiency of the stop-and-wait protocol in this setup is exactly 25%. The value of the one-way propagation delay (in milliseconds) is _____.
- 287.** A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 01111100101, then the input bit-string is
- 288.** The maximum window size for data transmission using the selective reject protocol with n-bit frame sequence numbers is:
- 289.** A machine receives an IPv4 datagram. The protocol field of the IPv4 header has the protocol number of a protocol X . Which ONE of the following is NOT a possible candidate for X ?
- 290.** Consider a network that uses Ethernet and IPv4. Assume that IPv4 headers do not use any options field. Each Ethernet frame can carry a maximum of 1500 bytes in its data field. A UDP segment is transmitted. The payload (data) in the UDP segment is 7488 bytes. Which ONE of the following choices has the CORRECT total number of fragments transmitted and the size of the last fragment including IPv4 header?
- 291.** Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection. Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?
- 292.** Assume that you have made a request for a web page through your web browser to a web server. Initially the browser cache is empty. Further, the browser is configured to send HTTP requests in non-persistent mode. The web page contains text and five very small images. The minimum number of TCP connections required to display the web page completely in your browser is _____.
- 293.** Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been

restarted.

294. Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2 and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?

295. TCP client P successfully establishes a connection to TCP server Q. Let denote the sequence number in the SYN sent from P to Q. Let denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

296. Suppose two hosts are connected by a point-to-point link and they are configured to use Stop-end-Wait protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest.

297. Consider the three-way handshake mechanism followed during TCP connection establishment between hosts P and Q. Let X and Y be two random 32-bit starting sequence numbers chosen by P and Q respectively. Suppose P sends a TCP connection request message to Q with a TCP segment having SYN bit = 1, SEQ number = X, and ACK bit = 0. Suppose Q accepts the connection request. Which one of the following choices represents the information present in the TCP segment header that is sent by Q to P?

298. Suppose that everyone in a group of N people wants to communicate secretly with the N - 1 others using symmetric key cryptographic system. The communication between any two persons should not be decodable by the others in the group. The number of keys required in the system as a whole to satisfy the confidentiality requirement is

299. Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10⁸ bits second) over a 1 km(kilometer) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

300. Consider the resolution of the domain name www.gate.org.in by a DNS resolver. Assume that no resource records are cached anywhere across the DNS servers and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is _____.

Operating System

1. Consider a process P running on a CPU. Which one or more of the following events will always trigger a context switch by the OS that results in process P moving to a non-running state (e.g., ready, blocked)?
2. Which of the following process state transitions is/are NOT possible?
3. Dispatch latency is defined as
4. The operating system and the other processes are protected from being modified by an already running process because
5. Consider the following statements about process state transitions for a system using preemptive scheduling. I. A running process can move to ready state. II. A ready process can move to running state. III. A blocked process can move to running state. IV. A blocked process can move to ready state. Which of the above statements are TRUE?
6. Working Set (t, k) at an instant of time t is
7. Suppose a system contains n processes and system uses the round-robin algorithm for CPU scheduling then which data structure is best suited ready queue of the process
8. The maximum number of processes that can be in Ready state for a computer system with n CPUs is
9. The state of a process after it encounters an I/O instruction is?
10. There are three processes in the ready queue. When the currently running process requests for I/O how many process switches take place?
11. Special software to create a job queue is called a
12. Which is the correct definition of a valid process transition in an operating system?
13. In the following process state transition diagram for a uniprocessor system, assume that there are always some processes in the ready state: Now consider the following statements: I. If a process makes a transition D, it would result in another process making transition A immediately. II. A process P2 in blocked state can make transition E while another process P1 is in running state. III. The OS uses preemptive scheduling. IV. The OS uses non-preemptive scheduling. Which of the above statements are TRUE?
14. Which of the following need not necessarily be saved on a Context Switch between processes?
15. Checkpointing a job
16. Which of the following need not necessarily be saved on a context switch between processes?
17. A task in a blocked state
18. What is the name of the technique in which the operating system of a computer executes several programs concurrently by switching back and forth between them?

- 19.** The process state transition diagram of an operating system is as given below. Which of the following must be FALSE about the above operating system?
- 20.** What is the swap space in the disk used for?
- 21.** Which of the following need not necessarily be saved on a context switch between processes?
- 22.** Which of the following actions is/are typically not performed by the operating system when switching context from process A to process B?
- 23.** The process state transition diagram in the below figure is representative of
- 24.** Which of the following does not interrupt a running process ?
- 25.** Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time. A (0, 10), B (2, 6), C (4, 3), and D (6, 7). Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF) CPU scheduling algorithms are applied to the processes?
- 26.** Which one or more of the following CPU scheduling algorithms can potentially cause starvation?
- 27.** Consider four processes P, Q, R, and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?
- 28.** Three processes arrive at time zero with CPU bursts of 16, 20 and 10 milliseconds. If the scheduler has prior knowledge about the length of the CPU bursts, the minimum achievable average waiting time for these three processes in a non-preemptive scheduler (rounded to nearest integer) is _____ milliseconds.
- 29.** Three CPU-bound tasks, with execution times of 15, 12 and 5 time units respectively arrive at times 0, t and 8, respectively. If the operating system implements a shortest remaining time first scheduling algorithm, what should be the value of t to have 4 context switches? Ignore the context switches at time 0 and at the end.
- 30.** Which of the following algorithms defines time quantum?
- 31.** Consider the following set of processes, assumed to have arrived at time 0. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order P1■, P2■, P3■, P4■. If the time quantum for RR is 4 ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to 2 decimal places) is _____
- 32.** Consider the following four processes with arrival times (in milliseconds) and their length of CPU burst (in milliseconds) as shown below: These processes are run on a single processor using preemptive

Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is _____.

33. Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds), and priority (0 is the highest priority) shown below. None of the processes have I/O burst time. The average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm is _____

34. In which one of the following page replacement policies, Belady's anomaly may occur?

35. Consider the following CPU processes with arrival times (in milliseconds) and length of CPU burst (in milliseconds) as given below: If the pre-emptive shortest remaining time first scheduling algorithm is used to schedule the processes, then the average waiting time across all processes is _____ milliseconds.

36. For the real time operating system, which of the following is the most suitable scheduling scheme?

37. Consider the following processes, with the arrival time and the length of the CPU burst given in milliseconds. The scheduling algorithm used is preemptive shortest remaining-time first. The average turn around time of these processes is milliseconds.

38. Consider an arbitrary set of CPU-bound processes with unequal CPU burst lengths submitted at the same time to a computer system. Which one of the following process scheduling algorithms would minimize the average waiting time in the ready queue?

39. Suppose two jobs, each of which needs 10 minutes of CPU time, start simultaneously. Assume 50% I/O wait time. How long will it take for both to complete, if they run sequentially?

40. In a lottery scheduler with 40 tickets, how we will distribute the tickets among 4 processes P1, P2, P3 and P4 such that each process gets 10%, 5%, 60% and 25% respectively?

41. For the processes listed in the following table, which of the following scheduling schemes will give the lowest average turnaround time?

42. Consider a uniprocessor system executing three tasks T1, T2 and T3, each of which is composed of an infinite sequence of jobs (or instances) which arrive periodically at intervals of 3, 7 and 20 milliseconds, respectively. The priority of each task is the inverse of its period, and the available tasks are scheduled in order of priority, with the highest priority task scheduled first. Each instance of T1, T2 and T3 requires an execution time of 1, 2 and 4 milliseconds, respectively. Given that all tasks initially arrive at the beginning of the 1st millisecond and task preemptions are allowed, the first instance of T3 completes its execution at the end of _____ milliseconds.

43. An operating system uses shortest remaining time first scheduling algorithm for pre-emptive scheduling of processes. Consider the following set of processes with their arrival times and CPU burst times (in milliseconds): The average waiting time (in milliseconds) of the processes is _____.

44. Three processes A, B and C each execute a loop of 100 iterations. In each iteration of the loop, a process performs a single computation that requires t_c CPU milliseconds and then initiates a single I/O operation that lasts for t_{io} milliseconds. It is assumed that the computer where the processes execute

has sufficient number of I/O devices and the OS of the computer assigns different I/O devices to each process. Also, the scheduling overhead of the OS is negligible. The processes have the following characteristics: The processes A, B, and C are started at times 0, 5 and 10 milliseconds respectively, in a pure time sharing system (round robin scheduling) that uses a time slice of 50 milliseconds. The time in milliseconds at which process C would complete its first I/O operation is _____.

45. Which of the following strategy is employed for overcoming the priority inversion problem?

46. Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds Using the shortest remaining time first scheduling algorithm, the average process turnaround time (in msec) is _____.

47. A particular parallel program computation requires 100 seconds when executed on a single CPU. If 20% of this computation is strictly sequential, then theoretically the best possible elapsed times for this program running on 2 CPUs and 4 CPUs respectively are

48. A starvation free job scheduling policy guarantees that no job indefinitely waits for a service. Which of the following job scheduling policies is starvation free?

49. A CPU scheduling algorithm determines an order for the execution of its scheduled processes. Given 'n' processes to be scheduled on one processor, how many possible different schedules are there?

50. Consider the following set of processes, with arrival times and the required CPU-burst times given in milliseconds. What is the sequence in which the processes are completed? Assume round robin scheduling with a time quantum of 2 milliseconds?

51. A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

52. Consider the 3 processes, P1, P2 and P3 shown in the table. The completion order of the 3 processes under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units) are

53. In a system using single processor, a new process arrives at the rate of six processes per minute and each such process requires seven seconds of service time. What is the CPU utilization?

54. Below is the precedence graph for a set of tasks to be executed on a parallel processing system S. What is the efficiency of this precedence graph on S if each of the tasks T1,...,T8 takes the same time and the system S has five processors?

55. Which of the following statements are true? I. Shortest remaining time first scheduling may cause starvation. II. Preemptive scheduling may cause starvation. III. Round robin is better than FCFS in terms of response time.

56. Consider three CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at

the end.

57. Four jobs to be executed on a single processor system arrive at time 0 in the order A,B,C,D. Their burst CPU time requirements are 4,1,8,1 time units respectively. The completion time of A under round robin scheduling with time slice of one time unit is

58. If the time-slice used in the round-robin scheduling policy is more than the maximum time required to execute any process, then the policy will

59. On a system using non-preemptive scheduling, processes with expected run times of 5, 18, 9 and 12 are in the ready queue. In what order should they be run to minimize wait time?

60. Round Robin schedule is essentially the pre-emptive version of

61. Consider a set of n tasks with known runtimes $r_1, r_2, \dots, r_n$ to be run on a uniprocessor machine. Which of the following processor scheduling algorithms will result in the maximum throughput?

62. An operating system uses Shortest Remaining Time first (SRT) process scheduling algorithm. Consider the arrival times and execution times for the following processes: What is the total waiting time for process P2?

63. The arrival time, priority, and duration of the CPU and I/O bursts for each of three processes P1, P2 and P3 are given in the table below. Each process has a CPU burst followed by an I/O burst followed by another CPU burst. Assume that each process has its own I/O resource. The multi-programmed operating system uses preemptive priority scheduling. What are the finish times of the processes P1, P2 and P3?

64. Consider three processes, all arriving at time zero, with total execution time of 10, 20 and 30 units, respectively. Each process spends the first 20% of execution time doing I/O, the next 70% of time doing computation, and the last 10% of time doing I/O again. The operating system uses a shortest remaining compute time first scheduling algorithm and schedules a new process either when the running process get blocked on I/O or when the running process finishes its compute burst. Assume that all I/O operations can be overlapped as much as possible. For what percentage of time does the CPU remain idle?

65. Consider three processes (process id 0,1,2, respectively) with compute time bursts 2,4, and 8 time units. All processes arrive at time zero. Consider the longest remaining time first (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turn around time is

66. Consider three CPU-intensive processes, which require 10,20 and 30 time units and arrive at times 0,2, and 6, respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end

67. Which combination of the following features will suffice to characterize an OS as a multi-programmed OS ? (A) More than one program may be loaded into main memory at the same time for execution. (B) If a program waits for certain events such as I/O, another program is immediately scheduled for execution. (C) If the execution of a program terminates, another program is immediately scheduled for execution.

68. An aid to determine the deadlock occurrence is

69. Raymonds tree based algorithm ensures

70. Consider the following snapshot of a system running n concurrent processes. Process i is holding X_i instances of a resource R , $1 \leq i \leq n$. Assume that all instances of R are currently in use. Further, for all i , process i can place a request for at most Y_i additional instances of R while holding the X_i instances it already has. Of the n processes, there are exactly two processes p and q such that $Y_p = Y_q = 0$. Which one of the following conditions guarantees that no other process apart from p and q can complete execution?

71. Consider a system having m resources of the same type. These resources are shared by 3 processes A,B,C, which have peak time demands of 3,4,6 respectively. The minimum value of m that ensures that deadlock will never occur is:

72. Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is ____.

73. What problem is solved by Dijkstra banker' algorithm?

74. A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for three processes are shown below: Which of the following best describes current state of the system ?

75. A multithreaded program P executes with x number of threads and uses y number of locks for ensuring mutual exclusion while operating on shared memory locations. All locks in the program are non-reentrant, i.e., if a thread holds a lock l , then it cannot re-acquire lock l without releasing it. If a thread is unable to acquire a lock, it blocks until the lock becomes available. The minimum value of x and the minimum value of y together for which execution of P can result in a deadlock are:

76. A system has 3 processes sharing 4 resources. If each process needs a maximum of 2 units, then

77. With single resource, deadlock occurs

78. Consider the following policies for preventing deadlock in a system with mutually exclusive resources. I. Processes should acquire all their resources at the beginning of execution. If any resource is not available, all resources acquired so far are released. II. The resources are numbered uniquely, and processes are allowed to request for resources only in increasing resource numbers. III. The resources are numbered uniquely, and processes are allowed to request for resources only in decreasing resource numbers. IV. The resources are numbered uniquely. A process is allowed to request only for a resource with resource number larger than its currently held resources. Which of the above policies can be used for preventing deadlock?

79. A system has 6 identical resources and N processes competing for them. Each process can request atmost 2 resources. Which one of the following values of N could lead to a deadlock?

80. What is the minimum number of resources required to ensure that deadlock will never occur, if there are currently three processes P_1, P_2 and P_3 running in a system whose maximum demand for the

resources of same type are 3, 4, and 5 respectively.

81. A system contains three programs and each requires three tape units for its operation. The minimum number of tape units which the system must have such that deadlocks never arise is _____.

82. An operating system uses the Banker's algorithm for deadlock avoidance when managing the allocation of three resource types X, Y, and Z to three processes P0, P1, and P2. The table given below presents the current system state. Here, the Allocation matrix shows the current number of resources of each type allocated to each process and the Max matrix shows the maximum number of resources of each type required by each process during its execution. There are 3 units of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in a safe state. Consider the following independent requests for additional resources in the current state: REQ1: P0 requests 0 units of X, 0 units of Y and 2 units of Z REQ2: P1 requests 2 units of x, 0 units of Y and 0 units of Z. Which one of the following is TRUE?

83. Which of the following is not a necessary condition for deadlock?

84. A system has n resources $R_0, \dots, R_{n-1}$, and k processes $P_0, \dots, P_{k-1}$. The implementation of the resource request logic of each process P_i is as follows: In which one of the following situations is a deadlock possible?

85. When a process is rolled back as a result of deadlock the difficulty which arises is

86. Consider a system having " n " resources of same type. These resources are shared by 3 processes, A, B, C. These have peak demands of 3, 4, and 6 respectively. For what value of " n " deadlock won't occur

87. Consider a system with 4 types of resources R1 (3 units), R2 (2 units), R3 (3 units), R4 (2 units). A non-preemptive resource allocation policy is used. At any given instance, a request is not entertained if it cannot be completely satisfied. Three processes P1, P2, P3 request the sources as follows if executed independently. Which one of the following statements is TRUE if all three processes run concurrently starting at time $t=0$?

88. An operating system implements a policy that requires a process to release all resources before making a request for another resource. Select the TRUE statement from the following:

89. Which of the following is NOT true of deadlock prevention and deadlock avoidance schemes?

90. Consider a system having ' m ' resources of the same type. The resources are shared by 3 processes A, B, C, which have peak time demands of 3, 4, 6 respectively. The minimum value of ' m ' that ensures that deadlock will never occur is

91. Consider a multi-threaded program with two threads T1 and T2. The threads share two semaphores: s1 (initialized to 1) and s2 (initialized to 0). The threads also share a global variable x (initialized to 0). The threads execute the code shown below. Which of the following outcomes is/are possible when threads T1 and T2 execute concurrently?

92. Consider the following two threads T1 and T2 that update two shared variables a and b. Assume that initially $a=b=1$. Though context switching between threads can happen at any time, each statement of T1

or T2 is executed atomically without interruption. Which one of the following options lists all the possible combinations of values of a and b after both T1 and T2 finish execution?

93. Consider the two functions incr and decr shown below. There are 5 threads each invoking incr once, and 3 threads each invoking decr once, on the same shared variable X. The initial value of X is 10. Suppose there are two implementations of the semaphore s, as follows: I-1: s is a binary semaphore initialized to 1. I-2: s is a counting semaphore initialized to 2. Let V1, V2 be the values of X at the end of execution of all the threads with implementations I-1, I-2, respectively. Which one of the following choices corresponds to the minimum possible values of V1, V2, respectively?

94. Consider a computer system with multiple shared resource types, with one instance per resource type. Each instance can be owned by only one process at a time. Owning and freeing of resources are done by holding a global lock (L). The following scheme is used to own a resource instance: Which of the following choice(s) about the above scheme is/are correct?

95. Consider the following pseudocode, where S is a semaphore initialized to 5 in line #2 and counter is a shared variable initialized to 0 in line #1. Assume that the increment operation in line #7 is not atomic. If five threads execute the function parop concurrently, which of the following program behavior(s) is/are possible?

96. Remote Procedure Calls are used for

97. The hardware implementation which provides mutual exclusion is

98. In multi-programmed systems, it is advantageous if some programs such as editors and compilers can be shared by several users. Which of the following must be true of multi-programmed systems in order that a single copy of a program can be shared by several users? I. The program is a macro. II. The program is recursive. III. The program is reentrant

99. Procedures P1 and P2 have a producer-consumer relationship, communicating by the use of a set of shared buffers. P1 : repeat Obtain an empty buffer Fill it Return a full buffer forever, P2: repeat Obtain a full buffer Empty it Return an empty buffer forever. Increasing the number of buffers is likely to do which of the following? I. Increase the rate at which requests are satisfied (throughput) II. Decrease the likelihood of deadlock. III. Increase the ease of achieving a correct implementation

100. Consider the following solution to the producer-consumer synchronization problem. The shared buffer size is N. Three semaphores empty, full and mutex are defined with respective initial values of 0, N and 1. Semaphore empty denotes the number of available slots in the buffer, for the consumer to read from. Semaphore full denotes the number of available slots in the buffer, for the producer to write to. The placeholder variables, denoted by P, Q, R, and S, in the code below can be assigned either empty or full. The valid semaphore operations are: wait() and signal(). Which one of the following assignments to P, Q, R and S will yield the correct solution?

101. Mutual exclusion problem occurs

102. At a particular time of computation the value of a counting semaphore is 7. Then 20 P operations and x V operations were completed on this semaphore. If the new value of semaphore is 5, x will be

103. Semaphores are used to solve the problem of I. Race Condition. II. Process Synchronization. III. Mutual Exclusion. IV. None of the above

104. A shared variable x , initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to memory, and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?

105. At a particular time the value of counting semaphore is 10. It will become 7 after:

106. The `enter_CS()` and `leave_CS()` functions to implement critical section of a process are realized using test-and-set instruction as follows: In the above solution, x is a memory location associated with the CS and is initialized to 0. Now consider the following statements: I. The above solution to CS problem is deadlock-free. II. The solution is starvation free. III. The processes enter CS in FIFO order. IV. More than one process can enter CS at the same time. Which of the above statements is TRUE?

107. Which of the following tasks is/are the responsibility/responsibilities of the memory management unit (MMU) in a system with paging-based memory management?

108. Consider a memory management system that uses a page size of 2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0,1,2, and 3 are stored in the page frames 1,3,2, and 0, respectively. The physical address (in decimal format) corresponding to the virtual address 2500 (in decimal format) is ____

109. Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with L levels for virtual to physical address translation. The page size is 4 KB (1 KB = 1024 B) and a page table entry at any of the levels occupies 8 bytes. The value of L is ____.

110. Consider the following two-dimensional array D in the C programming language, which is stored in row-major order: `int D[128][128];` Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array D. The Least Recently Used (LRU) page-replacement policy is used by the operating system. A total of 30 physical page frames are allocated to a process which executes the following code snippet: The number of page faults generated during the execution of this code snippet is ____.

111. Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string 7, 2, 7, 3, 2, 5, 3, 4, 6, 7, 7, 1, 5, 6, 1 the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place) is

112. In the context of operating systems, which of the following statements is/are correct with respect to paging?

113. Consider the following page reference string. 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6 What are the minimum number of frames required to get a single page fault for the above sequence assuming LRU

replacement strategy?

114. Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is _____

115. Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statement is TRUE?

116. Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressable. The page size is 8kB and the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?

117. The Operating System of a computer may periodically collect all the free memory space to form contiguous block of free space. This is called:

118. A computer has 1000K of main memory. The jobs arrive and finish in the following sequence. Job 1 requiring 200 K arrives Job 2 requiring 350 K arrives Job 3 requiring 300 K arrives Job 1 finishes Job 4 requiring 120 K arrives Job 5 requiring 150 K arrives Job 6 requiring 80 K arrives. Among best fit and first fit, which performs better for this sequence?

119. Determine the number of page faults when references to pages occur in the order 1,2,4,5,2,1,2,4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page 1 brought earlier than page 2. (assume LRU i.e., Least-Recently-Used algorithm is applied)

120. Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory, and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units. Which one of the following is the correct expression for the page fault rate experienced by the process?

121. Given reference to the following pages by a program 0,9,0,1,8,1,8,7,8,7,1,2,8,2,7,8,2,3,8,3 How many page faults will occur if the program has three page frames available to it and uses an optimal replacement?

122. Recall that Belady's anomaly is that the pages-fault rate may increase as the number of allocated frames increases. Now consider the following statements: S1: Random page replacement algorithm (where a page chosen at random is replaced) suffers from Belady's anomaly S2: LRU page replacement algorithm suffers from Belady's anomaly. Which of the following is CORRECT ?

123. A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The

minimum size of the TLB tag is:

124. Determine the number of page faults when references to pages occur in the following order: 1, 2, 4, 5, 2, 1, 2, 4. Assume that the main memory can accommodate 3 pages and the main memory already has the pages 1 and 2, with page one having brought earlier than page 2. (LRU page replacement algorithm is used)

125. Let the page fault service time be 10 ms in a computer with average memory access time being 20 ns. If one page fault is generated for every 106 memory accesses, what is the effective access time for the memory?

126. In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frames increases?

127. Consider a computer system with ten physical page frames. The system is provided with an access sequence $(a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20})$, where each a_i is a distinct virtual page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is _____.

128. A processor can support a maximum memory of 4GB, where the memory is word-addressable (a word consists of two bytes). The size of the address bus of the processor is at least bits _____.

129. Dirty bit for a page in a page table

130. Increasing the RAM of a computer typically improves performance because:

131. A computer system implements 8 kilobyte pages and a 32-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the virtual address supported by the system is _____ bits.

132. A computer system implements a 40-bit virtual address, page size of 8 kilobytes, and a 128-entry translation look-aside buffer (TLB) organized into 32 sets each having four ways. Assume that the TLB tag does not store any process id. The minimum length of the TLB tag in bits is _____.

133. The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointer. The disk block size is 4 kB and the disk block addresses 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) _____ GB.

134. In a file allocation system, which of the following allocation schemes(s) can be used if no external fragmentation is allowed? I. Contiguous II. Linked III. Indexed

135. A FAT (file allocation table) based file system is being used and the total overhead of each entry in the FAT is 4 bytes in size. Given a 100x106 bytes disk on which the file system is stored and data block size is 103 bytes, the maximum size of a file that can be stored on this disk in units of 106 bytes is _____.

136. A file system with 300 GByte disk uses a file descriptor with 8 direct block addresses, 1 indirect block address and 1 doubly indirect block address. The size of each disk block is 128 Bytes and the size of each

disk block address is 8 Bytes. The maximum possible file size in this file system is

137. Using a larger block size in a fixed block size file system leads to

138. The data blocks of a very large file in the Unix file system are allocated using

139. In a particular Unix OS, each data block is of size 1024 bytes, each node has 10 direct data block addresses and three additional addresses: one for single indirect block, one for double indirect block and one for triple indirect block. Also, each block can contain addresses for 128 blocks. Which one of the following is approximately the maximum size of a file in the file system?

140. A unix-style I-node has 10 direct pointers and one single, one double and one triple indirect pointers. Disk block size is 1 Kbyte, disk block address is 32 bits, and 48-bit integers are used. What is the maximum possible file size?

141. The root directory of a disk should be placed

142. Which of the following requires a device driver ?

143. Disk requests come to a disk driver for cylinders in the order 10, 22, 20, 2, 40, 6 and 38 at a time when the disk drive is reading from cylinder 20. The seek time is 6 ms/cylinder. The total seek time, if the disk arm scheduling algorithms is first-come-first-served is

144. Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ... , 199), and 256 sectors per track (numbered as 0, 1, ... , 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time: [120, 72, 2] , [180, 134, 1] , [60, 20, 0] , [212, 86, 3] , [56, 116, 2] , [118, 16, 1] Currently the head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible. The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____.

145. Consider the disk system with 100 cylinders. The request to access the cylinders occur in the following sequence. 4, 37, 10, 7, 19, 73, 2, 15, 6, 20. Assuming the head is currently at cylinder 50 what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and shortest seek time first algorithm is used.

146. Consider a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is _____.

147. Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, 25, 70. Assume that the initial position of the R/W head is on track 50. The additional distance that will be traversed by the R/W head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves

towards 100 when it starts execution) is _____ tracks.

148. Suppose a disk has 201 cylinders, numbered from 0 to 200. At some time the disk arm is at cylinder 100, and there is a queue of disk access requests for cylinders 30, 85, 90, 100, 105, 110, 135 and 145. If Shortest-Seek Time First (SSTF) is being used for scheduling the disk access, the request for cylinder 90 is serviced after servicing _____ number of requests.

149. Consider a disk system with 100 cylinders. The requests to access the cylinders occur in following sequence: 4, 34, 10, 7, 19, 73, 2, 15, 6, 20. Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1ms to move from one cylinder to adjacent one and shortest seek time first policy is used?

150. There are 200 tracks on a disc platter and the pending requests have come in the order - 36, 69, 167, 76, 42, 51, 126, 12 and 199. Assume the arm is located at the 100th track and moving towards track 200. If sequence of disc access is 126, 167, 199, 12, 36, 42, 51, 69 and 76 then which disc access scheduling policy is used?

151. Disk requests are received by a disk drive for cylinder 5, 25, 18, 3, 39, 8 and 35 in that order. A seek takes 5 msec per cylinder moved. How much seek time is needed to serve these requests for a Shortest Seek First (SSF) algorithm? Assume that the arm is at cylinder 20 when the last of these requests is made with none of the requests yet served

152. The head of a hard disk serves requests following the shortest seek time first (SSTF) policy. What is the maximum cardinality of the request set, so that the head changes its direction after servicing every request if the total number of tracks are 2048 and the head can start from any track?

153. The head of a hard disk serves requests following the shortest seek time first (SSTF) policy. The head is initially positioned at track number 180. Which of the request sets will cause the head to change its direction after servicing every request assuming that the head does not change direction if there is a tie in SSTF and all the requests arrive before the servicing starts?

154. A disk has 200 tracks (numbered 0 through 199). At a given time, it was servicing the request of reading data from track 120, and at the previous request, service was for track 90. The pending requests (in order of their arrival) are for track numbers. 30 70 115 130 110 80 20 25. How many times will the head change its direction for the disk scheduling policies SSTF(Shortest Seek Time First) and FCFS (First Come First Serve)?

155. Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is First Come First Served (FCFS). If FCFS is replaced by Shortest Seek Time First (SSTF), claimed by the vendor to give 50% better benchmark results, what is the expected improvement in the I/O performance of user programs?

156. Which of the following disk scheduling strategies is likely to give the best throughput?

157. Consider the following code snippet using the fork() and wait() system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors. The total number of times the printf statement is executed is ____

158. Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mode?

159. The following C program is executed on a Unix / Linux system: The total number of child process created is _____ .

160. Which of the following standard C library functions will always invoke a system call when executed from a single-threaded process in a UNIX/Linux operating system?

161. The difference between a named pipe and a regular file in Unix is that

162. The following C program: { fork(); fork(); printf("yes"); } If we execute this code segment, how many times the string yes will be printed?

163. The Linux command mknod myfifo b 4 16

164. Identify the correct order in which a server process must invoke the function calls accept, bind, listen, and recv according to UNIX socket API.

165. A process executes the code fork(); fork(); fork(); The total number of child processes created is

166. Which of the following UNIX command allows scheduling a program to be executed at the specified time?

167. Consider the execution of the following commands in a shell on a Linux operating system. bash\$ cat alpha Mathematics bash\$ ln alpha beta bash\$ rm alpha bash\$ cat >> beta << SAME Information Technology SAME bash\$ cat beta The output of the last command will be:

168. A process executes the following code for (i = 0; i < n; i++) fork(); The total number of child processes created is

169. A client process P needs to make a TCP connection to a server process S. Consider the following situation: the server process S executes a socket(), a bind() and a listen() system call in that order, following which it is preempted. Subsequently, the client process P executes a socket() system call followed by connect() system call to connect to the server process S. The server process has not executed any accept() system call. Which one of the following events could take place?

170. Which of the following system calls results in the sending of SYN packets?

171. The contents of the text file t1.txt containing four lines are as follows : a1 b1 a2 b2 a3 b2 a4 b1 The contents of the text file t2.txt containing five lines are as follows : a1 c1 a2 c2 a3 c3 a4 c3 a5 c4 Consider the following Bourne shell script : awk -F ' ' '{Print \$1, \$2}' t1.txt | while read a b ; do awk -v aV = \$a -v bV = \$b -F ' ' aV == \$1 (print aV, bV, \$2) ' t2.txt done Which one of the following strings will NOT be present in the output generated when the above script is run? (Note that the given strings may be substrings of a printed line.)

172. A user level process in Unix traps the signal sent on a Ctrl-C input, and has a signal handling routine that saves appropriate files before terminating the process. When a Ctrl-C input is given to this process, what is the mode in which the signal handling routine executes?

173. A student wishes to create symbolic links in a computer system running Unix. Three text files named "file 1", "file 2" and "file 3" exist in her current working directory, and the student has read and write permissions for all three files. Assume that file 1 contains information about her hobbies, file 2 contains information about her friends and file 3 contains information about her courses. The student executes the following sequence of commands from her current working directory. `ln -s file 1 file 2` `ln -s file 2 file 3` Which of the following types of information would be lost from her file system? I. Hobbies II. Friends III. Courses

174. The shell command `find -name passwd -print` is executed in `/etc` directory of a computer system running Unix. Which of the following shell commands will give the same information as the above command when executed in the same directory?

175. Consider the following code fragment: `if (fork() == 0) { a = a + 5; printf("%d,%dn", a, &a); } else { a = a - 5; printf("%d, %dn", a, &a); }` Let `u`, `v` be the values printed by the parent process, and `x`, `y` be the values printed by the child process. Which one of the following is TRUE?

176. A process executes the following segment of code : `for(i = 1; i <= n; i++) fork ();` The number of new processes created is

177. Which of the following commands or sequences of commands will rename a file `x` to file `y` in a Unix system ? I. `mv y, x` II. `mv x, y` III. `cp y, x (rm x)` IV. `cp x, y (rm x)`

178. System calls are usually invoked by using

179. When an interrupt occurs, an operating system

180. Which one or more of the following need to be saved on a context switch from one thread (`T1`) of a process to another thread (`T2`) of the same process?

181. Which of the following is/are shared by all the threads in a process ? I. Program counter II. Stack III. Address space IV. Registers

182. Consider the following statements with respect to user-level threads and kernel-supported threads (i) context switch is faster with kernel-supported threads (ii) for user-level threads, a system call can block the entire process (iii) Kernel supported threads can be scheduled independently (iv) User level threads are transparent to the kernel. Which of the above statements are true?

183. Let the time taken to switch between user and kernel modes of execution be `t1` while the time taken to switch between two processes be `t2`. Which of the following is TRUE?

184. A thread is usually defined as a 'light weight process' because an operating system (OS) maintains smaller data structures for a thread than for a process. In relation to this, which of the followings is TRUE?

185. Consider the following statements about user level threads and kernel level threads. Which one of the following statements is FALSE?

186. Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time. A (0, 10), B (2, 6), C (4, 3), and D (6, 7). Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF)

CPU scheduling algorithms are applied to the processes?

187. Consider four processes P, Q, R and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

188. A computer has two processors, M1 and M2. Four processes P1, P2, P3, P4 with CPU bursts of 20, 16, 25, and 10 milliseconds, respectively, arrive at the same time and these are the only processes in the system. The scheduler uses non-preemptive priority scheduling, with priorities decided as follows: M1 uses priority of execution for the processes as, $P1 > P3 > P2 > P4$, i.e., P1 and P4 have highest and lowest priorities, respectively. M2 uses priority of execution for the processes as, $P2 > P3 > P4 > P1$, i.e., P2 and P1 have highest and lowest priorities, respectively. A process P_i is scheduled to a processor M_k , if the processor is free and no other process P_j is waiting with higher priority. At any given point of time, a process can be allocated to any one of the free processors without violating the execution priority rules. Ignore the context switch time. What will be the average waiting time of the processes in milliseconds?

189. Consider four processes P, Q, R and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time $t = 0$. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

190. Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time. A (0, 10), B (2, 6), C (4, 3), and D (6, 7). Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF) CPU scheduling algorithms are applied to the processes?

191. A shared variable x , initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to memory, and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?

192. Three concurrent processes X, Y, and Z execute three different code segments that access and update certain shared variables. Process X executes the P operation (i.e., wait) on semaphores a, b and c; process Y executes the P operation on semaphores b, c and d; process Z executes the P operation on semaphores c, d, and a before entering the respective code segments. After completing the execution of its code segment, each process invokes the V operation (i.e., signal) on its three semaphores. All semaphores are binary semaphores initialized to one. Which one of the following represents a

deadlock-free order of invoking the P operations by the processes?

193. At a particular time of computation the value of a counting semaphore is 7. Then 20 P operations and 15 V operations were completed on this semaphore. The resulting value of the semaphore is:

194. Consider a non-negative counting semaphore S. The operation P(S) decrements S, and V(S) increments S. During an execution, 20 P(S) operations and 12 V(S) operations are issued in some order. The largest initial value of S for which at least one P(S) operation will remain blocked is _____.

195. Consider a computer system with multiple shared resource types, with one instance per resource type. Each instance can be owned by only one process at a time. Owning and freeing of resources are done by holding a global lock (L). The following scheme is used to own a resource instance: function OWNRESOURCES(Resource R) Acquire lock L // a global lock if R is available then Acquire R Release lock L else if R is owned by another process P then Terminate P, after releasing all resources owned by P Acquire R Restart P Release lock L end if end if end function. Which of the following choice(s) about the above scheme is/are correct?

196. A shared variable x, initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to memory, and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?

197. A certain computation generates two arrays a and b such that $a[i]=f(i)$ for $0 \leq i < n$ and $b[i] = g(a[i])$ for $0 \leq i < n$. Suppose this computation is decomposed into two concurrent processes X and Y such that X computes the array a and Y computes the array b. The processes employ two binary semaphores R and S, both initialized to zero. The array a is shared by the two processes.

198. Fetch_And_Add (X, i) is an atomic Read-Modify-Write instruction that reads the value of memory location X, increments it by the value i, and returns the old value of X. It is used in the pseudocode shown below to implement a busy-wait lock. L is an unsigned integer shared variable initialized to 0. The value of 0 corresponds to lock being available, while any non-zero value corresponds to the lock being not available. AcquireLock(L){ While (Fetch_And_Add(L,1)) L = 1; } Release Lock(L) { L = 0; } This implementation

199. The atomic fetch-and-set x, y instruction unconditionally sets the memory location x to 1 and fetches the old value of x n y without allowing any intervening access to the memory location x. consider the following implementation of P and V functions on a binary semaphore S.

200. Barrier is a synchronization construct where a set of processes synchronizes globally i.e. each process in the set arrives at the barrier and waits for all others to arrive and then all processes leave the barrier. Let the number of processes in the set be three and S be a binary semaphore with the usual P and V functions. Consider the following C implementation of a barrier with line numbers shown on left. void barrier (void) { 1: P(S); 2: process_arrived++; 3: V(S); 4: while (process_arrived !=3); 5: P(S); 6: process_left++; 7: if (process_left==3) { 8: process_arrived = 0; 9: process_left = 0; 10: } 11: V(S); } The

variables `process_arrived` and `process_left` are shared among all processes and are initialized to zero. In a concurrent program all the three processes call the barrier function when they need to synchronize globally. Which one of the following rectifies the problem in the implementation?

201. The concurrent programming constructs `fork` and `join` are as below: `Fork (label)` which creates a new process executing from the specified label `join (variable)` which decrements the specified synchronization variable (by 1) and terminates the process if the new value is not 0. Show the precedence graph for S1, S2, S3, S4 and S5 of the concurrent program below. $N = 2$ $M = 2$ `fork L3 fork L4 S1 L1 : join N S3 L2: join M S5 L3:S2 goto L1 L4:S4 goto L2 next:`

202. A solution to the Dining Philosophers Problem which avoids deadlock is

203. At a particular time of computation the value of a counting semaphore is 7. Then 20 P operations and 15 V operations were completed on this semaphore. The resulting value of the semaphore is:

204. Semaphore operations are atomic because they are implemented within the OS

205. Consider three processes (process id 0, 1, 2, respectively) with compute time bursts 2, 4, and 8 time units. All processes arrive at time zero. Consider the longest remaining time first (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turn around time is

206. A uni-processor computer system only has two processes, both of which alternate 10 ms CPU bursts with 90 ms I/O bursts. Both the processes were created at nearly the same time. The I/O of both processes can proceed in parallel. Which of the following scheduling strategies will result in the least CPU utilization (over a long period of time) for this system?

207. Which of the following actions is/are typically not performed by the operating system when switching context from process A to process B?

208. A multi-user, multi-processing operating system cannot be implemented on hardware that does not support:

209. Four jobs are waiting to be run. Their expected run times are 6, 3, 5 and x. In what order should they be run to minimize the average response time?

210. Four jobs to be executed on a single processor system arrive at time 0+ in the order A, B, C, D. their burst CPU time requirements are 4, 1, 8, 1 time units respectively. The completion time of A under round robin scheduling with time slice of one time units.

211. State an undesirable characteristic of each of the following criteria for measuring performance of an operating system: (a) Turn around time. (b) Waiting time.

212. A computer has six tape drives, with n processes competing for them. Each process may need two drives. What is the maximum value of n for the system to be deadlock free?

213. $P = \{P1, P2, P3, P4\}$ consists of all active processes in an operating system. $R = \{R1, R2, R3, R4\}$ consists of single instances of distinct types of resources in the system. The resource allocation graph has the following assignment and claim edges. Assignment edges: $R1 \rightarrow P1$, $R2 \rightarrow P2$, $R3 \rightarrow P3$, $R4 \rightarrow P4$

(the assignment edge $R1 \rightarrow P1$ means resource $R1$ is assigned to process $P1$, and so on for others)
Claim edges: $P1 \rightarrow R2$, $P2 \rightarrow R3$, $P3 \rightarrow R1$, $P2 \rightarrow R4$, $P4 \rightarrow R2$ (the claim edge $P1 \rightarrow R2$ means process $P1$ is waiting for resource $R2$, and so on for others) Which of the following statement(s) is/are CORRECT?

214. Consider the following snapshot of a system running n concurrent processes. Process i is holding X_i instances of a resource R , $1 \leq i \leq n$. Assume that all instances of R are currently in use. Further, for all i , process i can place a request for at most Y_i additional instances of R while holding the X_i instances it already has. Of the n processes, there are exactly two processes p and q such that $Y_p = Y_q = 0$. Which one of the following conditions guarantees that no other process apart from p and q can complete execution?

215. In a system, there are three types of resources: E , F and G . Four processes $P0$, $P1$, $P2$ and $P3$ execute concurrently. At the outset, the processes have declared their maximum resource requirements using a matrix named Max as given below. For example, $Max[P2, F]$ is the maximum number of instances of F that $P2$, would require. The number of instances of the resources allocated to the various processes at any given state is given by a matrix named $Allocation$. Consider a state of the system with the $Allocation$ matrix as shown below, and in which 3 instances of E and 3 instances of F are the only resources available.

216. Consider a non-negative counting semaphore S . The operation $P(S)$ decrements S , and $V(S)$ increments S . During an execution, 20 $P(S)$ operations and 12 $V(S)$ operations are issued in some order. The largest initial value of S for which at least one $P(S)$ operation will remain blocked is _____.

217. Consider the following policies for preventing deadlock in a system with mutually exclusive resources. 1. Processes should acquire all their resources at the beginning of execution. If any resource is not available, all resources acquired so far are released. 2. The resources are numbered uniquely, and processes are allowed to request for resources only in increasing resource numbers. 3. The resources are numbered uniquely, and processes are allowed to request for resources only in decreasing resource numbers. 4. The resources are numbered uniquely. A process is allowed to request only for a resource with resource number larger than its currently held resources. Which of the above policies can be used for preventing deadlock?

218. An operating system uses the Banker's algorithm for deadlock avoidance when managing the allocation of three resource types X , Y , and Z to three processes $P0$, $P1$, and $P2$. The table given below presents the current system state. Here, the $Allocation$ matrix shows the current number of resources of each type allocated to each process and the Max matrix shows the maximum number of resources of each type required by each process during its execution. There are 3 units of type X , 2 units of type Y and 2 units of type Z still available. The system is currently in a safe state. Consider the following independent requests for additional resources in the current state: REQ1: $P0$ requests 0 units of X , 0 units of Y and 2 units of Z , REQ2: $P1$ requests 2 units of X , 0 units of Y and 0 units of Z . Which one of the following is TRUE?

219. A single processor system has three resource types X , Y and Z , which are shared by three processes. There are 5 units of each resource type. Consider the following scenario, where the column $alloc$ denotes the number of units of each resource type allocated to each process, and the column $request$ denotes the number of units of each resource type requested by a process in order to complete

execution. Which of these processes will finish LAST?

220. Consider the following snapshot of a system running n processes. Process i is holding x_i instances of a resource R , for $1 \leq i \leq n$. Currently, all instances of R are occupied. Further, for all i , process i has placed a request for an additional y_i instances while holding the x_i instances it already has. There are exactly two processes p and q such that $y_p = y_q = 0$. Which one of the following can serve as a necessary condition to guarantee that the system is not approaching a deadlock?

221. Suppose n processes, $P_1, \dots, P_n$ share m identical resource units, which can be reserved and released one at a time. The maximum resource requirement of process P_i is S_i , where $S_i > 0$. Which one of the following is a sufficient condition for ensuring that deadlock does not occur?

222. An operating system contains 3 user processes each requiring 2 units of resource R . The minimum number of units of R such that no deadlocks will ever arise is

223. Consider a system having m resources of the same type. These resources are shared by 3 processes A , B and C , which have peak demands of 3, 4 and 6 respectively. For what value of m deadlock will not occur?

224. A computer system has 6 tape drives, with n process competing for them. Each process may need 3 tape drives. The maximum value of n for which the system is guaranteed to be deadlock free is:

225. Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?

226. Which of the following tasks is/are the responsibility/responsibilities of the memory management unit (MMU) in a system with paging-based memory management?

227. Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mode?

228. In the context operating systems, which of the following statements is/are correct with respect to paging?

229. Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statements is TRUE?

230. In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frames increases?

231. A computer system implements a 40-bit virtual address, page size of 8 kilobytes, and a 128-entry translation look-aside buffer (TLB) organized into 32 sets each having four ways. Assume that the TLB tag does not store any process id. The minimum length of the TLB tag in bits is _____.

232. Consider a system with byte-addressable memory, 32-bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. The size of the page table in the system in megabytes is _____.

233. A system uses 3 page frames for storing process pages in main memory. It uses the Least Recently Used (LRU) page replacement policy. Assume that all the page frames are initially empty. What is the total number of page faults that will occur while processing the page reference string given below? 4, 7, 6, 1, 7, 6, 1, 2, 7, 2

234. A system uses FIFO policy for page replacement. It has 4 pages frames with no pages loaded to begin with. The system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order. How many page faults will occur?

235. How many 32 K * 1 RAM chips are needed to provide a memory capacity of 256 K-bytes?

236. In which one of the following page replacement policies, Belady's anomaly may occur?

237. What is the swap space in the disk used for

238. The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by.

239. In a system with 32 bit virtual addresses and 1 KB page size, use of one-level page tables for virtual to physical address translation is not practical because of

240. The optimal page replacement algorithm will select the page that.

241. The process of assigning load addresses to the various parts of the program and adjusting the code and data in the program to reflect the assigned addresses is called

242. Consider a virtual memory system with FIFO page replacement policy. For an arbitrary page access pattern, increasing the number of page frames in main memory will

243. In a resident -OS computer, which of the following systems must reside in the main memory under all situations?

244. Locality of reference implies that the page reference being made by a process

245. A ROM is used to store the table for multiplication of two 8-bit unsigned integers. The size of ROM required is

246. In a paged segmented scheme of memory management, the segment table itself must have a page table because:

247. A linker is given object modules for a set of programs that were compiled separately. What information need to be included in an object module?

248. Which page replacement policy sometimes leads to more page faults when size of memory is increased?

249. A computer system supports a logical address space of 232 bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size. The value of b is

_____. (Answer in integer)

250. In optimal page replacement algorithm, information about all future page references is available to the operating system (OS). A modification of the optimal page replacement algorithm is as follows: The OS correctly predicts only up to next 4 page references (including the current page) at the time of allocating a frame to a page. A process accesses the pages in the following order of page numbers: 1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4. If the system has three memory frames that are initially empty, the number of page faults that will occur during execution of the process is _____ (Answer in integer)

251. Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume $1 \text{ KB} = 2^{10}$ bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry. An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process. The value of $X + Y$ is _____.

252. Consider a memory management system that uses a page size of 2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0, 1, 2, and 3 are stored in the page frames 1, 3, 2, and 0, respectively. The physical address (in decimal format) corresponding to the virtual address 2500 (in decimal format) is _____

253. Consider the following two-dimensional array D in the C programming language, which is stored in row-major order: `int D[128][128]`; Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array D. The Least Recently Used (LRU) (page-replacement policy) is used by the operating system. A total of 30 physical page frames are allocated to a process which executes the following code snippet: `for (int i = 0; i < 128; i++) for (int j = 0; j < 128; j++) D[i][j] *= 10;` The number of page faults generated during the execution of this code snippet is _____

254. Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with L levels for virtual to physical address translation. The page size is 4 KB ($1 \text{ KB} = 1024 \text{ B}$) and a page table entry at any of the levels occupies 8 bytes. The value of L is _____

255. Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string 7, 2, 7, 3, 2, 5, 3, 4, 6, 7, 7, 1, 5, 6, 1 the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place) is _____

256. Consider a paging system that uses a 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read in from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is _____.

257. Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressable. The page size is 8 kB and the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?

258. Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory, and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units. Which one of the following is the correct expression for the page fault rate experienced by the process?

259. Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then the size of the per-process page table _____ is megabytes.

260. Consider a computer system with ten physical page frames. The system is provided with an access sequence $(a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20})$ where each a_i is a distinct virtual page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is

261. A computer system implements 8 kilobyte pages and a 32-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the virtual address supported by the system is _____ bits.

262. Consider six memory partitions of sizes 200KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB where KB refers to kilobyte. These partitions need to be allotted to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB and in that order. If the best fit algorithm is used, which partitions are NOT allotted to any process?

263. A computer system implements 8 kilobyte pages and a 32-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the virtual address supported by the system is _____ bits.

264. Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is _____.

265. A computer has twenty physical page frames which contain pages numbered 101 through 120. Now a program accesses the pages numbered 1, 2, ..., 100 in that order, and repeats the access sequence THRICE. Which one of the following page replacement policies experiences the same number of page faults as the optimal page replacement policy for this program?

266. Assume that there are 3 page frames which are initially empty. If the page reference string is 1, 2, 3, 4, 2, 1, 5, 3, 2, 4, 6 the number of page faults using the optimal replacement policy is _____.

267. Consider the virtual page reference string 1, 2, 3, 2, 4, 1, 3, 2, 4, 1. On a demand paged virtual memory system running on a computer system that has main memory size of 3 page frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacement policy. Then

268. A multilevel page table is preferred in comparison to a single level page table for translating virtual address to physical address because.

269. A processor uses 36 bit physical addresses and 32 bit virtual addresses, with a page frame size of 4 Kbytes. Each page table entry is of size 4 bytes. A three level page table is used for virtual to physical address translation, where the virtual address is used as follows: * Bits 30-31 are used to index into the first level page table. * Bits 21-29 are used to index into the second level page table. * Bits 12-20 are used to index into the third level page table, and. * Bits 0-11 are used as offset within the page. The number of bits required for addressing the next level page table (or page frame) in the page table entry of the first, second and third level page tables are respectively

270. A process has been allocated 3 page frames. Assume that none of the pages of the process are available in the memory initially. The process makes the following sequence of page references (reference string): 1, 2, 1, 3, 7, 4, 5, 6, 3, 1. If optimal page replacement policy is used, how many page faults occur for the above reference string?

271. A process has been allocated 3 page frames. Assume that none of the pages of the process are available in the memory initially. The process makes the following sequence of page references (reference string): 1, 2, 1, 3, 7, 4, 5, 6, 3, 1. Least Recently Used (LRU) page replacement policy is a practical approximation to optimal page replacement. For the above reference string, how many more page faults occur with LRU than with the optimal page replacement policy?

272. A virtual memory system uses First In First Out (FIFO) page replacement policy and allocates a fixed number of frames to a process. Consider the following statements: P: Increasing the number of page frames allocated to a process sometimes increases the page fault rate. Q: Some programs do not exhibit locality of reference.

273. A Computer system supports 32-bit virtual addresses as well as 32-bit physical addresses. Since the virtual address space is of the same size as the physical address space, the operating system designers decide to get rid of the virtual memory entirely. Which one of the following is true?

274. Consider a System with a two-level paging scheme in which a regular memory access takes 150 nanoseconds, and servicing a page fault takes 8 milliseconds. An average instruction takes 100 nanoseconds of CPU time, and two memory accesses. The TLB hit ratio is 90% and the page fault rate is one in every 10,000 instructions. What is the effective average instruction execution time?

275. A processor uses 2-level page tables for virtual to physical address translation. Page tables for both levels are stored in the main memory. Virtual and physical addresses are both 32 bits wide. The memory is byte addressable. For virtual to physical address translation, the 10 most significant bits of the virtual address are used as index into the first level page table while the next 10 bits are used as index into the second level page table. The 12 least significant bits of the virtual address are used as offset within the page. Assume that the page table entries in both levels of page tables are 4 bytes wide. Further, the

processor has a translation look-aside buffer (TLB), with a hit rate of 96%. The TLB caches recently used virtual page numbers and the corresponding physical page numbers. The processor also has a physically addressed cache with a hit rate of 90%. Main memory access time is 10ns, cache access time is 1ns, and TLB access time is also 1ns. Suppose a process has only the following pages in its virtual address space: two contiguous code pages starting at virtual address 0×00000000 , two contiguous data pages starting at virtual address 0×00400000 and a stack page starting at virtual address $0 \times \text{FFFFF000}$. The amount of memory required for storing the page tables of this process is

276. A processor uses 2-level page tables for virtual to physical address translation. Page tables for both levels are stored in the main memory. Virtual and physical addresses are both 32 bits wide. The memory is byte addressable. For virtual to physical address translation, the 10 most significant bits of the virtual address are used as index into the first level page table while the next 10 bits are used as index into the second level page table. The 12 least significant bits of the virtual address are used as offset within the page. Assume that the page table entries in both levels of page tables are 4 bytes wide. Further, the processor has a translation look-aside buffer (TLB), with a hit rate of 96%. The TLB caches recently used virtual page numbers and the corresponding physical page numbers. The processor also has a physically addressed cache with a hit rate of 90%. Main memory access time is 10 ns, cache access time is 1 ns, and TLB access time is also 1 ns. Assuming that no page faults occur, the average time taken to access a virtual address is approximately (to the nearest 0.5 ns)

277. Which of the following is NOT an advantage of using shared, dynamically linked libraries as opposed to using statically linked libraries?

278. Consider a machine with 64 MB physical memory and a 32-bit virtual address space. If the page size is 4KB, what is the approximate size of the page table?

279. Suppose the time to service a page fault is on the average 10 milliseconds, while a memory access takes 1 microsecond. Then a 99.99% hit ratio results in average memory access time of.

280. If an instruction takes microseconds and a page fault takes an additional microseconds, the effective instruction time if on the average a page fault occurs every instruction is:

281. A 1000 Kbyte memory is managed using variable partitions but to compaction. It currently has two partitions of sizes 200 Kbytes and 260 Kbytes respectively. The smallest allocation request in Kbytes that could be denied is for

282. A memory page containing a heavily used variable that was initialized very early and is in constant use is removed when

283. Under paged memory management scheme simple lock and key memory protection arrangement may still be required if the.....processors do not have address mapping hardware.

284. A computer system uses 32-bit virtual address, and 32-bit physical address. The physical memory is byte addressable, and the page size is 4 kbytes. It is decided to use two level page tables to translate from virtual address to physical address. Equal number of bits should be used for indexing first level and second level page table, and the size of each page table entry is 4 bytes. (a) Give a diagram showing how a virtual address would be translated to a physical address. (b) What is the number of page table entries

that can be contained in each page? (c) How many bits are available for storing protection and other information in each page table entry?

285. A certain computer system has the segmented paging architecture for virtual memory. The memory is byte addressable. Both virtual and physical address spaces contain 2^{16} bytes each. The virtual address space is divided into 8 non-overlapping equal size segments. The memory management unit (MMU) has a hardware segment table, each entry of which contains the physical address of the page table for the segment. Page tables are stored in the main memory and consist of 2 byte page table entries. (a) What is the minimum page size in bytes so that the page table for a segment requires at most one page to store it? Assume that the page size can only be a power of 2. (b) Now suppose that the page size is 512 bytes. It is proposed to provide a TLB (Translation look-aside buffer) for speeding up address translation. The proposed TLB will be capable of storing page table entries for 16 recently referenced virtual pages, in a fast cache that will use the direct mapping scheme. What is the number of tag bits that will need to be associated with each cache entry. (c) Assume that each page table entry contains (besides other information) 1 valid bit, 3 bits for page protection and 1 dirty bit. How many bits are available in page table entry for storing the aging information for the page? Assume that the page size is 512 bytes.

286. Let the page reference and the working set window be c c d b c e c e a d and 4, respectively. The initial working set at time $t = 0$ contains the pages {a, d, e}, where a was referenced at time $t = 0$, d was referenced at time $t = -1$ and e was referenced at time $t = -2$. Determine the total number of page faults and the average number of page frames used by computing the working set at each reference.

287. A FAT (file allocation table) based file system is being used and the total overhead of each entry in the FAT is 4 bytes in size. Given a 100×10^6 bytes disk on which the file system is stored and data block size is 10^3 bytes, the maximum size of a file that can be stored on this disk in units of 10^6 bytes is _____.

288. Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively:

289. Normally user programs are prevented from handling I/O directly by I/O instructions in the for CPU having explicit I/O instructions, such I/O protection is ensured by having the I/O instructions privilege in CPU with memory mapped I/O there is no explicit I/O instruction. Which one of the following is true for a CPU with memory mapped I/O?

290. Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is First Come First Served (FCFS). If FCFS is replaced by Shortest Seek Time First (SSTF) claimed by the vendor to give 50% better benchmark results, what is the expected improvement in the I/O performance of user programs?

291. Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512-byte sectors. A file has content stored in 3000 sectors located randomly on the disk. Assuming average rotational latency, the total time (in seconds, rounded off to 2 decimal places) to read the entire file from the disk is _____

292. Consider the following five disk access requests of the form (request id, cylinder number) that are present in the disk scheduler queue at a given time. (P, 155), (Q, 85), (R, 110), (S, 30), (T, 115). Assume the head is positioned at cylinder 100. The scheduler follows Shortest Seek Time First scheduling to service the requests. Which one of the following statements is FALSE?

293. The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointers. The disk block size is 4 kB, and the disk block address is 32 -bits long. The maximum possible file size is (rounded off to 1 decimal place) _____ GB.

294. Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ... , 199) and 256 sectors per track (numbered as 0, 1, ... , 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time: [120, 72, 2], [180, 134, 1], [60, 20, 0], [212, 86, 3], [56, 116, 2], [118, 16, 1] Currently the head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible. The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____.

295. Consider a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is _____.

296. Consider a main memory with five page frames and the following sequence of page references: 3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3. Which one of the following is true with respect to page replacement policies First In First Out (FIFO) and Least Recently Used (LRU)?

297. Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, 25, 70. Assume that the initial position of the R/W head is on track 50. The additional distance that will be traversed by the R/W head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves towards 100 when it starts execution) is _____ tracks.

298. A system contains three programs and each requires three tape units for its operation. The Minimum number of tape units which the system must have such that deadlocks never arise is _____.

299. A file system with 300 G Byte disk uses a file descriptor with 8 direct block addresses, 1 indirect block address and 1 doubly indirect block address. The size of each disk block is 128 Bytes and the size of each disk block address is 8 Bytes. The maximum possible file size in this file system is

300. An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to random location is given as 10 ms. Rotational speed of disk is 6000 rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected).

Mathematics

1. Let p and q be the following propositions: p : Fail grade can be given. q : Student scores more than 50% marks. Consider the statement: "Fail grade cannot be given when student scores more than 50% marks." Which one of the following is the CORRECT representation of the above statement in propositional logic?
2. Geetha has a conjecture about integers, which is of the form $\forall x[P(x) \Rightarrow \exists y Q(x,y)]$ where P is a statement about integers, and Q is a statement about pairs of integers. Which of the following (one or more) option(s) would imply Geetha's conjecture?
3. Choose the correct choice(s) regarding the following propositional logic assertion S :
 $S: ((P \wedge Q) \rightarrow R) \rightarrow ((P \wedge Q) \rightarrow (Q \rightarrow R))$
4. Let p and q be two propositions. Consider the following two formulae in propositional logic.
 $S1: (\neg p \wedge (p \vee q)) \rightarrow q$ $S2: q \rightarrow (\neg p \wedge (p \vee q))$. Which one of the following choices is correct?
5. Given that $B(a)$ means "a is a bear" $F(a)$ means "a is a fish" and $E(a,b)$ means "a eats b" Then what is the best meaning of $\forall x[F(x) \rightarrow \forall y(E(y,x) \rightarrow b(y))]$
6. Which one of the following predicate formulae is NOT logically valid? Note that W is a predicate formula without any free occurrence of x .
7. Consider the first-order logic sentence $\phi \equiv \exists s \exists t \exists u \forall v \forall w \forall x \forall y \phi(s,t,u,v,w,x,y)$ where $\phi(s,t,u,v,w,x,y)$ is a quantifier-free first-order logic formula using only predicate symbols, and possibly equality, but no function symbols. Suppose ϕ has a model with a universe containing 7 elements. Which one of the following statements is necessarily true?
8. Which one of the following Boolean expressions is NOT a tautology?
9. Let p , q , r denote the statements "It is raining", "It is cold", and "It is pleasant," respectively. Then the statement "It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold" is represented by
10. Let p , q , and r be propositions and the expression $(p \rightarrow q) \rightarrow r$ be a contradiction. Then, the expression $(r \rightarrow p) \rightarrow q$ is
11. Consider the first-order logic sentence $F: \forall x(\exists y R(x,y))$. Assuming non-empty logical domains, which of the sentences below are implied by F ? I. $\exists y(\exists x R(x,y))$ II. $\exists y(\forall x R(x,y))$ III. $\forall y(\exists x R(x,y))$ IV. $\neg \exists x(\forall y \neg R(x,y))$
12. The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below? I. $p \Rightarrow q$ II. $q \Rightarrow p$ III. $(\neg q) \vee p$ IV. $(\neg p) \vee q$
13. Consider the following expressions: (i) false (ii) Q (iii) true (iv) $P \vee Q$ (v) $\neg Q \vee P$. The number of expressions given above that are logically implied by $P \wedge (P \Rightarrow Q)$ is _____.
14. Let p, q, r, s represent the following propositions. $p: x \in \{8,9,10,11,12\}$, $q: x$ is a composite number, $r: x$ is a perfect square, $s: x$ is a prime number. The integer $x \geq 2$ which satisfies $\neg((p \Rightarrow q) \wedge (\neg r \vee \neg s))$ is _____.

- 15.** In a room there are only two types of people, namely Type 1 and Type 2. Type 1 people always tell the truth and Type 2 people always lie. You give a fair coin to a person in that room, without knowing which type he is from and tell him to toss it and hide the result from you till you ask for it. Upon asking, the person replies the following "The result of the toss is head if and only if I am telling the truth." Which of the following options is correct?
- 16.** Consider the following two statements.S1: If a candidate is known to be corrupt, then he will not be elected. S2: If a candidate is kind, he will be elected. Which one of the following statements follows from S1 and S2 as per sound inference rules of logic?
- 17.** Which one of the following is NOT equivalent to $p \leftrightarrow q$?
- 18.** The CORRECT formula for the sentence, "not all rainy days are cold" is
- 19.** Consider the following statements:P: Good mobile phones are not cheap, Q: Cheap mobile phones are not good, L: P implies Q, M: Q implies P, N: P is equivalent to Q. Which one of the following about L, M, and N is CORRECT?
- 20.** Consider the statement "Not all that glitters is gold". Predicate glitters(x) is true if x glitters and predicate gold(x) is true if x is gold. Which one of the following logical formulae represents the above statement?
- 21.** Which one of the following is NOT logically equivalent to $\neg \exists x(\forall y(\alpha) \wedge \forall z(\beta))$?
- 22.** What is the logical translation of the following statement?"None of my friends are perfect."
- 23.** What is the correct translation of the following statement into mathematical logic?"Some real numbers are rational"
- 24.** Consider the following logical inferences. I1: If it rains then the cricket match will not be played. The cricket match was played. Inference: There was no rain. I2: If it rains then the cricket match will not be played. It did not rain. Inference: The cricket match was played. Which of the following is TRUE?
- 25.** Which one of the following options is CORRECT given three positive integers x, y and z, and a predicate $P(x) = \neg(x=1) \wedge \forall y(\exists z(x=y*z)) \Rightarrow (y=x) \vee (y=1)$
- 26.** Suppose the predicate $F(x,y,t)$ is used to represent the statement that person x can fool person y at time t. which one of the statements below expresses best the meaning of the formula $\forall x \exists y \exists t (\neg F(x,y,t))$?
- 27.** Consider the following well-formed formulae:I. $\neg \forall x(P(x))$. II. $\neg \exists x(P(x))$. III. $\neg \exists x(\neg P(x))$. IV. $\exists x(\neg P(x))$. Which of the above are equivalent?
- 28.** Which one of the following is the most appropriate logical formula to represent the statement? "Gold and silver ornaments are precious". The following notations are used: $G(x)$: x is a gold ornament, $S(x)$: x is a silver ornament, $P(x)$: x is precious
- 29.** Which of the following is the negation of $[\forall x, \alpha \rightarrow (\exists y, \beta \rightarrow (\forall u, \exists v, y))]$

30. Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first order formula with x as a free variable, and β is a first order formula with no free variable.

31. Let P be the partial order defined on the set $\{1,2,3,4\}$ as follows

$P = \{(x,x) \mid x \in \{1,2,3,4\}\} \cup \{(1,2), (3,2), (3,4)\}$ The number of total orders on $\{1,2,3,4\}$ that contain P is _____

32. Let X be a set and 2^X denote the powerset of X . Define a binary operation Δ on 2^X as follows:

$A \Delta B = (A - B) \cup (B - A)$. Let $H = (2^X, \Delta)$. Which of the following statements about H is/are correct?

33. Let S be a set of consisting of 10 elements. The number of tuples of the form (A,B) such that A and B are subsets of S , and $A \subseteq B$ is _____

34. If $A = \{x,y,z\}$ and $B = \{u,v,w,x\}$, and the universe is $\{s,t,u,v,w,x,y,z\}$. Then $(A \cup B^c) \cap (A \cap B)$ is equal to

35. Consider the first order predicate formula: $\forall x [\forall z (x \Rightarrow ((z=x) \vee (z=1))) \Rightarrow$

$\exists w (w > x) \wedge (\forall z (w \Rightarrow ((w=z) \vee (z=1))))]$ Here ' $a|b$ ' denotes that ' a divides b ', where a and b are integers.

Consider the following sets: $S_1: \{1, 2, 3, \dots, 100\}$ S_2 : Set of all positive integers S_3 : Set of all integers.

Which of the above sets satisfy ϕ ?

36. Let $U = \{1,2,\dots,n\}$. Let $A = \{(x,X) \mid x \in X, X \subseteq U\}$. Consider the following two statements on $|A|$. I. $|A| = n2^{n-1}$. II. $|A| = \sum_{k=1}^n k \binom{n}{k}$. Which of the above statements is/are TRUE?

37. The following paradigm can be used to find the solution of the problem in minimum time: Given a set of non-negative integer and a value K , determine if there is a subset of the given set with sum equal to K :

38. Let N be the set of natural numbers. Consider the following sets. P : Set of Rational numbers (positive and negative), Q : Set of functions from $\{0, 1\}$ to N , R : Set of functions from N to $\{0, 1\}$, S : Set of finite subsets of N . Which of the sets above are countable?

39. The symmetric difference of sets $A = \{1,2,3,4,5,6,7,8\}$ and $B = \{1,3,5,6,7,8,9\}$ is:

40. Consider a set U of 23 different compounds in a Chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U . Consider the following statements: I. Each compound in $U \setminus S$ reacts with an odd number of compounds. II. At least one compound in $U \setminus S$ reacts with an odd number of compounds. III. Each compound in $U \cap S$ reacts with an even number of compounds. Which one of the above statements is ALWAYS TRUE?

41. Suppose U is the power set of the set $S = \{1,2,3,4,5,6\}$. For any $T \in U$, let $|T|$ denote the number of elements in T and T' denote the complement of T . For any $T, R \in U$, let $T \setminus R$ be the set of all elements in T which are not in R . Which one of the following is true?

42. Let X and Y denote the sets containing 2 and 20 distinct objects respectively and F denote the set of all possible functions defined from X to Y . Let f be randomly chosen from F . The probability of f being one-to-one is

43. The number of onto functions (surjective functions) from set $X = \{1,2,3,4\}$ to set $Y = \{a,b,c\}$ is

44. The cardinality of the power set of $\{0, 1, 2, \dots, 10\}$ is

45. Let x and Y be finite sets and $f: x \rightarrow Y$ be a function. Which one of the following statements is TRUE?
46. The number of elements in the power set of the set $\{A, B, C\}$ is
47. A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is TRUE about $\oplus$?
48. What is the possible number of reflexive relations on a set of 5 elements?
49. If P, Q, R are subsets of the universal set U , then $(P \cap Q \cap R) \cup (P \cap Q \cap R^c) \cup (P \cap Q^c \cap R) \cup (P \cap Q^c \cap R^c) \cup (P^c \cap Q \cap R) \cup (P^c \cap Q \cap R^c) \cup (P^c \cap Q^c \cap R) \cup (P^c \cap Q^c \cap R^c)$ is
50. The set of all Equivalence Classes of a set A of Cardinality C
51. Let S be a set of n elements. The number of ordered pairs in the largest and the smallest equivalence relations on S are:
52. What is the cardinality of the set of integers X defined below? $X = \{n \mid 1 \leq n \leq 123, n \text{ is not divisible by either } 2, 3 \text{ or } 5\}$
53. Given a set of elements $N = \{1, 2, \dots, n\}$ and two arbitrary subsets $A \subseteq N$ and $B \subseteq N$, how many of the $n!$ permutations p from N to N satisfy $\min[p(A)] = \min[p(B)]$, where $\min(S)$ is the smallest integer in the set of integers S and $p(S)$ is the set of integers obtained by applying permutation p to each element of S ?
54. We are given a set $X = \{x_1, \dots, x_n\}$ where $x_i = 2^i$. A sample $S \subseteq X$ is drawn by selecting each x_i independently with probability $p_i = \frac{1}{2^i}$. The expected value of the smallest number in sample S is:
55. Let X, Y, Z be sets of sizes x, y and z respectively. Let $W = X \times Y$ and E be the set of all subsets of W . The number of functions from Z to E is
56. Let $n = p^2q$, where p and q are distinct prime numbers. How many numbers m satisfy $1 \leq m \leq n$ and $\gcd(m, n) = 1$?
57. Let A be a set with n elements. Let C be a collection of distinct subsets of A such that for any two subsets S_1 and S_2 in C , either $S_1 \subset S_2$ or $S_2 \subset S_1$. What is the maximum cardinality of C ?
58. In a class of 200 students, 125 students have taken Programming Language course, 85 students have taken Data Structures course, 65 students have taken Computer Organization course; 50 students have taken both Programming Language and Data Structures, 35 students have taken both Programming Language and Computer Organization; 30 students have taken both Data Structures and Computer Organization, 15 students have taken all the three courses. How many students have not taken any of the three courses?
59. Let $P(S)$ denotes the power set of set S . Which of the following is always true?
60. A partial order $\leq$ is defined on the set $S = \{x, a_1, a_2, \dots, a_n, y\}$ as $x \leq a_i$ for all i and $a_i \leq y$ for all i , where $n \geq 1$. The number of total orders on the set S which contain the partial order $\leq$ is
61. Given $\Sigma = \{a, b\}$, which one of the following sets is not countable?

- 62.** Let $X=\{2,3,6,12,24\}$, Let $\leq$ be the partial order defined by $X\leq Y$ if x divides y . Number of edges in the Hasse diagram of $(X,\leq)$ is
- 63.** Let A and B be sets and let A^c and B^c denote the complements of the sets A and B . The set $(A-B)\cup(B-A)\cup(A\cap B)$ is equal to
- 64.** The number of elements in the power set $P(S)$ of the set $S=\{\{\emptyset\},1,\{2,3\}\}$ is:
- 65.** Let A be a finite set of size n . The number of elements in the power set of $A\times A$ is:
- 66.** Let S be an infinite set and $S_1, \dots, S_n$ be sets such that $S_1 \cup S_2 \cup \dots \cup S_n = S$. Then
- 67.** Consider the set of integers $\{1,2,3,4,6,8,12,24\}$ together with the two binary operations LCM (lowest common multiple) and GCD (greatest common divisor). Which of the following algebraic structures does this represent?
- 68.** The number of arrangements of six identical balls in three identical bins is
- 69.** The number of integers between 1 and 500 (both inclusive) that are divisible by 3 or 5 or 7 is
- 70.** The number of 4 digit numbers having their digits in non-decreasing order (from left to right) constructed by using the digits belonging to the set $\{1, 2, 3\}$ is
- 71.** The number of divisors of 2100 is
- 72.** The number of bit strings of length 8 that will either start with 1 or end with 00 is?
- 73.** The number of distinct positive integral factors of 2014 is
- 74.** A pennant is a sequence of numbers, each number being 1 or 2. An n -pennant is a sequence of numbers with sum equal to n . For example, $(1,1,2)$ is a 4-pennant. The set of all possible 1- pennants is $\{(1)\}$, the set of all possible 2-pennants is $\{(2), (1,1)\}$ and the set of all 3-pennants is $\{(2,1), (1,1,1), (1,2)\}$. Note that the pennant $(1,2)$ is not the same as the pennant $(2,1)$. The number of 10- pennants is
- 75.** There are 5 bags labelled 1 to 5. All the coins in a given bag have the same weight. Some bags have coins of weight 10 gm, others have coins of weight 11 gm. I pick 1, 2, 4, 8, 16 coins respectively from bags 1 to 5. Their total weight comes out to 323 gm. Then the product of the labels of the bags having 11 gm coins is
- 76.** In how many ways can b blue balls and r red balls be distributed in n distinct boxes?
- 77.** A line L in a circuit is said to have a stuck-at-0 fault if the line permanently has a logic value 0. Similarly a line L in a circuit is said to have a stuck-at-1 fault if the line permanently has a logic value 1. A circuit is said to have a multiple stuck-at fault if one or more lines have stuck at faults. The total number of distinct multiple stuck-at faults possible in a circuit with N lines is
- 78.** In how many ways can we distribute 5 distinct balls, $B_1, B_2, \dots, B_5$ in 5 distinct cells, $C_1, C_2, \dots, C_5$ such that Ball B_i is not in cell $C_i, \forall i=1,2,\dots,5$ and each cell contains exactly one ball?

79. Mala has a colouring book in which each English letter is drawn two times. She wants to paint each of these 52 prints with one of k colours, such that the colour pairs used to colour any two letters are different. Both prints of a letter can also be coloured with the same colour. What is the minimum value of k that satisfies this requirement?

80. n couples are invited to a party with the condition that every husband should be accompanied by his wife. However, a wife need not be accompanied by her husband. The number of different gatherings possible at the party is

81. The minimum number of cards to be dealt from an arbitrarily shuffled deck of 52 cards to guarantee that three cards are from same suit is

82. The number of binary strings of n zeros and k ones in which no two ones are adjacent is

83. How many sub strings of different lengths (non-zero) can be formed from a character string of length n ?

84. The number of substrings (of all lengths inclusive) that can be formed from a character string of length n is

85. How many 4-digit even numbers have all 4 digits distinct?

86. The number of ways in which 5 A's, 5 B's and 5 C's can be arranged in a row is:

87. When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (i.e., 1, 2, 3, 4, 5, and 6) is

88. A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is

89. Let A and B be two events in a probability space with $P(A)=0.3$, $P(B)=0.5$, and $P(A \cap B)=0.1$. Which of the following statements is/are TRUE?

90. Consider a permutation sampled uniformly at random from the set of all permutations of $\{1, 2, 3, \dots, n\}$ for some $n \geq 4$. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4. Which one of the following statements is TRUE?

91. Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on the first throw, and C be the event that denotes HEAD on the second throw. Which of the following statements is/are TRUE?

92. A bag has r red balls and b black balls. All balls are identical except for their colours. In a trial, a ball is randomly drawn from the bag, its colour is noted and the ball is placed back into the bag along with another ball of the same colour. Note that the number of balls in the bag will increase by one, after the trial. A sequence of four such trials is conducted. Which one of the following choices gives the probability of drawing a red ball in the fourth trial?

- 93.** For a given biased coin, the probability that the outcome of a toss is a head is 0.4. This coin is tossed 1,000 times. Let X denote the random variable whose value is the number of times that head appeared in these 1,000 tosses. The standard deviation of X (rounded to 2 decimal place) is
- 94.** The lifetime of a component of a certain type is a random variable whose probability density function is exponentially distributed with parameter 2. For a randomly picked component of this type, the probability that its lifetime exceeds the expected lifetime (rounded to 2 decimal places) is
- 95.** For $n > 2$, let $a \in \{0,1\}^n$ be a non-zero vector. Suppose that x is chosen uniformly at random from $\{0,1\}^n$. Then, the probability that $\sum_{i=1}^n a_i x_i$ is an odd number is
- 96.** Suppose Y is distributed uniformly in the open interval $(1,6)$. The probability that the polynomial $3x^2 + 6xY + 3Y + 6$ has only real roots is (rounded off to 1 decimal place)
- 97.** Two numbers are chosen independently and uniformly at random from the set $\{1, 2, \dots, 13\}$. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is
- 98.** A class of 30 students occupy a classroom containing 5 rows of seats, with 8 seats in each row. If the students seat themselves at random, the probability that sixth seat in the fifth row will be empty is:
- 99.** Two people, P and Q , decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q . Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is
- 100.** P and Q are considering to apply for a job. The probability that P applies for the job is $1/4$. The probability that P applies for the job given that Q applies for the job is $1/2$, and the probability that Q applies for the job given that P applies for the job $1/3$. Then the probability that P does not apply for the job given that Q does not apply for the job is
- 101.** Let X be a Gaussian random variable mean 0 and variance σ^2 . Let $Y = \max(X, 0)$ where $\max(a, b)$ is the maximum of a and b . The median of Y is
- 102.** Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is
- 103.** A probability density function on the interval $[a, 1]$ is given by $1/x^2$ and outside this interval the value of the function is zero. The value of a is
- 104.** The probability that two friends are born in the same month is
- 105.** Let S be a sample space and two mutually exclusive events A and B be such that $A \cup B = S$. If $P(\cdot)$ denotes the probability of the event, the maximum value of $P(A)P(B)$ is
- 106.** The probability that a given positive integer lying between 1 and 100 (both inclusive) is NOT divisible by 2, 3 or 5 is

107. Each of the nine words in the sentence "The quick brown fox jumps over the lazy dog" is written on a separate piece of paper. These nine pieces of paper are kept in a box. One of the pieces is drawn at random from the box. The expected length of the word drawn is

108. The security system at an IT office is composed of 10 computers of which exactly four are working. To check whether the system is functional, the officials inspect four of the computers picked at random (without replacement). The system is deemed functional if at least three of the four computers inspected are working. Let the probability that the system is deemed functional be denoted by p . Then $100p =$

109. Suppose a fair six-sided die is rolled once. If the value on the die is 1, 2, or 3, the die is rolled a second time. What is the probability that the sum total of values that turn up is at least 6?

110. Suppose p is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM, and p has a Poisson distribution with mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

111. Let $P(E)$ denote the probability of the occurrence of event E . If $P(A) = 0.5$ and $P(B) = 1$ then the values of $P(A|B)$ and $P(B|A)$ respectively are

112. Suppose you break a stick of unit length at a point chosen uniformly at random. Then the expected length of the shorter stick is

113. For fair six-sided dice are rolled. The probability that the sum of the results being 22 is $\frac{1296X}{\text{■}}$. The value of X is

114. If the difference between the expectation of the square of random variable ($E[X^2]$) and the square of the expectation of the random variable ($(E[X])^2$) is denoted by R then

115. Consider a finite sequence of random values $X = [x_1, x_2, \dots, x_n]$. Let μ_x be the mean and σ_x be the standard deviation of X . Let another finite sequence Y of equal length be derived from this as $y_i = a \cdot x_i + b$, where a and b are positive constants. Let μ_y be the mean and σ_y be the standard deviation of this sequence. Which one of the following statements is INCORRECT?

116. A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at a time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?

117. Three coins are tossed simultaneously. The probability that they will fall two heads and one tail is

118. Consider a random variable X that takes values $+1$ and -1 with probability 0.5 each. The values of the cumulative distribution function $F(x)$ at $x = -1$ and $+1$ are

119. Let $f(x)$ be the continuous probability density function of a random variable x , the probability that $a < x \leq b$, is :

120. Consider a company that assembles computers. The probability of a faulty assembly of any computer is p . The company therefore subjects each computer to a testing process. This testing process gives the correct result for any computer with a probability of q . What is the probability of a computer being declared faulty?

121. What is the probability that divisor of 1099 is a multiple of 1096?

122. If two fair coins are flipped and at least one of the outcomes is known to be a head, what is the probability that both outcomes are heads?

123. A program consists of two modules executed sequentially. Let $f_1(t)$ and $f_2(t)$ respectively denote the probability density functions of time taken to execute the two modules. The probability density function of the overall time taken to execute the program is given by

124. Aishwarya studies either computer science or mathematics everyday. If she studies computer science on a day, then the probability that she studies mathematics the next day is 0.6. If she studies mathematics on a day, then the probability that she studies computer science the next day is 0.4. Given that Aishwarya studies computer science on Monday, what is the probability that she studies computer science on Wednesday?

125. What is the probability that in a randomly chosen group of r people at least three people have the same birthday?

126. When a coin is tossed, the probability of getting a Head is p , $0 < p < 1$. Let N be the random variable denoting the number of tosses till the first Head appears, including the toss where the Head appears. Assuming that successive tosses are independent, the expected value of N is

127. Suppose we uniformly and randomly select a permutation from the $20!$ Permutations of $1, 2, 3, \dots, 20$. What is the probability that 2 appears at an earlier position than any other even number in the selected permutation?

128. Suppose there are two coins. The first coin gives heads with probability $\frac{85}{100}$ when tossed, while the second coin gives heads with probability $\frac{41}{100}$. One of the two coins is picked up at random with equal probability and tossed. What is the probability of obtaining heads

129. In a multi-user operating system on an average, 20 requests are made to use a particular resource per hour. The arrival of requests follows a Poisson distribution. The probability that either one, three or five requests are made in 45 minutes is given by

130. Company X shipped 5 computer chips, 1 of which was defective. and company Y shipped 4 computer chips, 2 of which were defective. One computer chip is to be chosen uniformly at a random from the 9 chips shipped by the companies. If the chosen chip is found to be defective, what is the probability that the chip came from the company Y?

131. A bag contains 10 blue marbles, 20 green marbles and 30 red marbles. A marble is drawn from the bag, its colour recorded and it is put back in the bag. This process is repeated 3 times. The probability that no two of the marbles drawn have the same colour is

132. An unbiased coin is tossed repeatedly until the outcome of two successive tosses is the same. Assuming that the trials are independent, the expected number of tosses is

133. In a communication network, a packet of length L bits takes link L_1 with a probability of p_1 or link L_2 with a probability of p_2 . Link L_1 and L_2 have bit error probability of b_1 and b_2 respectively.

The probability that the packet will be received without error via either L1 or L2 is

134. For each element in set of size $2n$, an unbiased coin is tossed. The $2n$ coin tossed are independent. An element is chosen if the corresponding coin toss were head. The probability that exactly n elements are chosen is

135. In a certain town, the probability that it will rain in the afternoon is known to be 0.6. Moreover, meteorological data indicates that if the temperature at noon is less than or equal to 25°C , the probability that it will rain in the afternoon is 0.4. The temperature at noon is equally likely to be above 25°C , or at/below 25°C . What is the probability that it will rain in the afternoon on a day when the temperature at noon is above 25°C ?

136. In a population of N families, 50% of the families have three children, 30% of the families have two children and the remaining families have one child. What is the probability that a randomly picked child belongs to a family with two children?

137. Let X and Y be two exponentially distributed and independent random variables with mean α and β , respectively. If $Z = \min(X, Y)$, then the mean of Z is given by

138. Let $f(x)$ be the continuous probability density function of a random variable X . The probability that $a < X \leq b$, is:

139. Box P has 2 red balls and 3 blue balls and box Q has 3 balls and 1 blue ball. A ball is selected as follows: (i) select a box (ii) choose a ball from the selected box such that each ball in the box is equally likely to be chosen. The probabilities of selecting boxes P and Q are $1/3$ and $2/3$ respectively. Given that a ball selected in the above process is a red ball, the probability that it came from the box P is:

140. A random bit string of length n is constructed by tossing a fair coin n times and setting a bit to 0 or 1 depending on outcomes head and tail, respectively. The probability that two such randomly generated strings are not identical is:

141. Let $P(E)$ denote the probability of the event E . Given $P(A) = 1$, $P(B) = 1/2$, the values of $P(A|B)$ and $P(B|A)$ respectively are

142. A program consists of two modules executed sequentially. Let $f_1(t)$ and $f_2(t)$ respectively denote the probability density functions of time taken to execute the two modules. The probability density function of the overall time taken to execute the program is given by

143. If a fair coin is tossed four times. What is the probability that two heads and two tails will result?

144. An examination paper has 150 multiple choice questions of one mark each, with each question having four choices. Each incorrect answer fetches -0.25 marks. Suppose 1000 students choose all their answers randomly with uniform probability. The sum total of the expected marks obtained by all these students is

145. Two n bit binary strings, S_1 and S_2 , are chosen randomly with uniform probability. The probability that the Hamming distance between these strings (the number of bit positions where the two strings differ) is equal to d is

- 146.** A die is rolled three times. The probability that exactly one odd number turns up among the three outcomes is
- 147.** Suppose that the expectation of a random variable X is 5. Which of the following statements is true?
- 148.** Consider two events E_1 and E_2 such that probability of E_1 , $\Pr[E_1] = \frac{1}{2}$, probability of E_2 , $\Pr[E_2] = \frac{1}{3}$, and probability of E_1 and E_2 , $\Pr[E_1 \text{ and } E_2] = \frac{1}{6}$. Which of the following statements is/are true?
- 149.** Four fair coins are tossed simultaneously. The probability that at least one head and one tail turn up is
- 150.** The probability that a number selected at random between 100 and 999 (both inclusive) will not contain the digit 7 is:
- 151.** A bag contains 10 white balls and 15 black balls. Two balls are drawn in succession. The probability that one of them is black and the other is white is:
- 152.** Two dice are thrown simultaneously. The probability that at least one of them will have 6 facing up is
- 153.** The probability that top and bottom cards of a randomly shuffled deck are both aces is
- 154.** The probability that it will rain today is 0.5. The probability that it will rain tomorrow is 0.6. The probability that it will rain either today or tomorrow is 0.7. What is the probability that it will rain today and tomorrow?
- 155.** Seven (distinct) car accidents occurred in a week. What is the probability that they all occurred on the same day?
- 156.** Let a_n represent the number of bit strings of length n containing two consecutive 1s. What is the recurrence relation for a_n ?
- 157.** Let $H_1, H_2, H_3, \dots$ be harmonic numbers. Then, for $n \in \mathbb{Z}^+$, $\sum_{j=1}^n H_j$ can be expressed as
- 158.** Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from $A \times A$ to $A \cup B$. The number of possible values of $|A|$ is
- 159.** Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \max\{x, x^3\}$, $x \in \mathbb{R}$, where $\mathbb{R}$ is the set of all real numbers. The set of all points where $f(x)$ is NOT differentiable is
- 160.** If the ordinary generating function of a sequence $\{a_n\}_{n=0}^{\infty}$ is $(1-z)^{-3}$ then $a_3 - a_0$ is equal to
- 161.** Let A be a finite set having x elements and let B be a finite set having y elements. What is the number of distinct functions mapping B into A .
- 162.** Let S denote the set of all functions $f: \{0,1\}^4 \rightarrow \{0,1\}$. Denote by N the number of functions from S to the set $\{0,1\}$. The value of $\log_2 \log_2 N$ is
- 163.** How many onto (or surjective) functions are there from an n -element ($n \geq 2$) set to a 2-element set?

- 164.** Let f be a function from a set A to a set B , g a function from B to C , and h a function from A to C , such that $h(a)=g(f(a))$ for all $a \in A$. Which of the following statements is always true for all such functions f and g ?
- 165.** Let $f: B \rightarrow C$ and $g: A \rightarrow B$ be two functions and let $h = f \circ g$. Given that h is an onto function which one of the following is TRUE?
- 166.** The number of functions from an m element set to an n element set is
- 167.** Consider the function $y=|x|$ in the interval $[-1, 1]$. In this interval, the function is
- 168.** Let R denote the set of real numbers. Let $f: R \times R \rightarrow R \times R$ be a bijective function defined by $f(x,y)=(x+y,x-y)$. The inverse function of f is given by
- 169.** Suppose X and Y are sets and $|X|$ and $|Y|$ are their respective cardinality. It is given that there are exactly 97 functions from X to Y . From this one can conclude that
- 170.** Let A and B be sets with cardinalities m and n respectively. The number of one-one mappings from A to B , when $m < n$, is
- 171.** If $f(x_i) \cdot f(x_{i+1}) < 0$ then
- 172.** Let R be the set of all binary relations on the set $\{1,2,3\}$. Suppose a relation is chosen from R at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is
- 173.** The time complexity of computing the transitive closure of binary relation on a set of n elements is known to be
- 174.** The time complexity of computing the transitive closure of a binary relation on a set of n elements is known to be
- 175.** Let R be a relation on the set of ordered pairs of positive integers such that $((p,q),(r,s)) \in R$ if and only if $p-s=q-r$. Which one of the following is true about R ?
- 176.** Let R be the relation on the set of positive integers such that aRb if and only if a and b are distinct and have a common divisor other than 1. Which one of the following statements about R is true?
- 177.** Consider the binary relation $R = \{(x,y), (x,z), (z,x), (z,y)\}$ on the set $\{x,y,z\}$. Which one of the following is TRUE?
- 178.** Let R and S be any two equivalence relations on a non-empty set A . Which one of the following statements is TRUE?
- 179.** The binary relation $S = \emptyset$ (empty set) on set $A = \{1,2,3\}$ is
- 180.** A relation R is defined on the set of integers as xRy iff $(x + y)$ is even. Which of the following statements is true?
- 181.** Let L be a set with a relation R which is transitive, anti-symmetric and reflexive and for any two elements $a,b \in L$, let the least upper bound $\text{lub}(a,b)$ and the greatest lower bound $\text{glb}(a,b)$ exist. Which of the following is/are true?

- 182.** The number of binary relations on a set with n elements is:
- 183.** The binary relation $R=\{(1,1),(2,1),(2,2),(2,3),(2,4),(3,1),(3,2),(3,3),(3,4)\}$ on the set $A=\{1,2,3,4\}$ is
- 184.** Suppose A is a finite set with n elements. The number of elements in the largest equivalence relation of A is
- 185.** The number of equivalence relations of the set $\{1,2,3,4\}$ is
- 186.** Let R be a non-empty relation on a collection of sets defined by $A \mathrel{R} B$ if and only if $A \cap B = \emptyset$. Then, (pick the true statement)
- 187.** Let R be a symmetric and transitive relation on a set A . Then
- 188.** The less-than relation, $<$, on reals is
- 189.** Let $\text{Graph}(x)$ be a predicate which denotes that x is a graph. Let $\text{Connected}(x)$ be a predicate which denotes that x is connected. Which of the following first order logic sentences DOES NOT represent the statement: "Not every graph is connected"?
- 190.** The following propositional statement is $(P \rightarrow (Q \vee R)) \rightarrow ((P \vee Q) \rightarrow R)$
- 191.** Let P , Q and R be three atomic propositional assertions. Let X denote $(P \vee Q) \rightarrow R$ and Y denote $(P \rightarrow R) \vee (Q \rightarrow R)$. Which one of the following is a tautology?
- 192.** "If X then Y unless Z " is represented by which of the following formulas in propositional logic? (" $\neg$ ", is negation, " $\wedge$ " is conjunction, and " $\rightarrow$ " is implication)
- 193.** Let a , b , c , d be propositions. Assume that the equivalence $a \Leftrightarrow (b \vee \neg b)$ and $b \Leftrightarrow c$ hold. Then the truth-value of the formula $(a \wedge b) \rightarrow (a \wedge c) \vee d$ is always
- 194.** Which of the following propositions is a tautology?
- 195.** Which of the following is false? Read $\wedge$ as AND, $\vee$ as OR, $\neg$ as NOT, $\rightarrow$ as one way implication and $\leftrightarrow$ as two way implication
- 196.** If the proposition $\neg p \rightarrow q$ is true, then the truth value of the proposition $\neg p \vee (p \rightarrow q)$, where $\neg$ is negation, $\vee$ is inclusive OR and $\rightarrow$ is implication, is
- 197.** The proposition $p \wedge (\neg p \vee q)$ is:
- 198.** Which of the following predicate calculus statements is/are valid?
- 199.** Which of the following well-formed formulas are equivalent?
- 200.** Let Z_n be the group of integers $\{0, 1, 2, \dots, n-1\}$ with addition modulo n as the group operation. The number of elements in the group $Z_2 \times Z_3 \times Z_4$ that are their own inverses is
- 201.** Consider the operators $\oplus$ and $\otimes$ defined by $a \oplus b = a + 2b$, $a \otimes b = ab$, for positive integers. Which of the following statements is/are TRUE?

- 202.** Let G be a group of order 6, and H be a subgroup of G such that $1 < |H| < 6$. Which one of the following options is correct?
- 203.** Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is
- 204.** $(G, *)$ is an abelian group. Then
- 205.** Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is
- 206.** $(G, \cdot)$ is a group such that $(ab)^{-1} = a^{-1}b^{-1}, \forall a, b \in G$, then G is a/an
- 207.** Let G be a group with 15 elements. Let L be a subgroup of G . It is known that $L \neq G$ and that the size of L is at least 4. The size of L is
- 208.** The arithmetic mean of attendance of 49 students of class A is 40% and that of 53 students of class B is 35%. Then the percentage of arithmetic mean of attendance of class A and B is
- 209.** Consider the set $S = \{1, \omega, \omega^2\}$, where ω and ω^2 are cube roots of unity. If $*$ denotes the multiplication operation, the structure $(S, *)$ forms
- 210.** Which one of the following is NOT necessarily a property of a Group?
- 211.** How many different non-isomorphic Abelian groups of order 4 are there?
- 212.** The set $\{1, 2, 3, 5, 7, 8, 9\}$ under multiplication modulo 10 is not a group. Given below are four plausible reasons. Which one of them is false?
- 213.** The set $\{1, 2, 4, 7, 8, 11, 13, 14\}$ is a group under multiplication modulo 15. The inverses of 4 and 7 are respectively:
- 214.** Consider the set Σ^* of all strings over the alphabet $\Sigma = \{0, 1\}$. Σ^* with the concatenation operator for strings
- 215.** Two girls have picked 10 roses, 15 sunflowers and 15 daffodils. What is the number of ways they can divide the flowers among themselves?
- 216.** In a room containing 28 people, there are 18 people who speak English, 15 people who speak Hindi and 22 people who speak Kannada. 9 persons speak both English and Hindi, 11 persons speak both Hindi and Kannada whereas 13 persons speak both Kannada and English. How many speak all three languages?
- 217.** Let $(\mathbb{Z}, *)$ be an algebraic structure where $\mathbb{Z}$ is the set of integers and the operation $*$ is defined by $n * m = \max(n, m)$. Which of the following statements is true for $(\mathbb{Z}, *)$?
- 218.** Let A be the set of all non-singular matrices over real number and let $*$ be the matrix multiplication operation. Then
- 219.** Some group (G, o) is known to be abelian. Then, which one of the following is true for G ?

- 220.** Let A be the adjacency matrix of a simple undirected graph G . Suppose A is its own inverse. Which one of the following statements is always TRUE?
- 221.** The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40. The number of connected components in G is
- 222.** The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. Let G be any graph with n vertices and chromatic number k . Which of the following statements is/are always TRUE?
- 223.** Which of the properties hold for the adjacency matrix A of a simple undirected unweighted graph having n vertices?
- 224.** Consider a simple undirected unweighted graph with at least three vertices. If A is the adjacency matrix of the graph, then the number of 3-cycles in the graph is given by the trace of
- 225.** Consider a simple undirected graph of 10 vertices. If the graph is disconnected, then the maximum number of edges it can have is .
- 226.** G is an undirected graph with vertex set $\{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ and edge set $\{v_1v_2, v_1v_3, v_1v_4, v_2v_4, v_2v_5, v_3v_4, v_4v_5, v_4v_6, v_5v_6, v_6v_7\}$. A breadth first search of the graph is performed with v_1 as the root node. Which of the following is a tree edge?
- 227.** Graph G is obtained by adding vertex s to $K_{3,4}$ and making s adjacent to every vertex of $K_{3,4}$. The minimum number of colours required to edge-colour G is
- 228.** Let G be an undirected complete graph on n vertices, where $n > 2$. Then, the number of different Hamiltonian cycles in G is equal to
- 229.** The number of edges in a regular graph of degree: d and n vertices is
- 230.** G is undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is
- 231.** $g(\cdot)$ is a function from A to B , $f(\cdot)$ is a function from B to C , and their composition defined as $f(\cdot)$ is a mapping from A to C . If $f(\cdot)$ and $g(\cdot)$ are onto (surjective) functions, which ONE of the following is TRUE about the function ?
- 232.** Let P be the partial order defined on the set $\{1,2,3,4\}$ as follows: $P = \{(x,x) \mid x \in \{1, 2, 3, 4\}\} \cup \{(1,2),(3,2),(3,4)\}$. The number of total orders on $\{1,2,3,4\}$ that contain P is _____.
- 233.** Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from $A * A$ to $A \cup B$. The number of possible values of $|A|$ is _____
- 234.** Consider the following sets, where $n > 2$: S_1 : Set of all $n \times n$ matrices with entries from the set $\{a, b, c\}$ S_2 : Set of all functions from the set $\{0,1, 2, \dots, n-1\}$ to the set $\{0, 1, 2\}$. Which of the following choice(s) is/are correct?

235. Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is _____.

236. Let R be the set of all binary relations on the set $\{1,2,3\}$. Suppose a relation is chosen from R at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is _____.

237. Let G be an arbitrary group. Consider the following relations on G : $R_1: \forall a,b \in G, aR_1b$ if and only if $\exists g \in G$ such that $a = g^{-1}bg$. $R_2: \forall a,b \in G, aR_2b$ if and only if $a = b^{-1}$. Which of the above is/are equivalence relation/relation(s)?

238. Suppose U is the power set of the set $S = \{1, 2, 3, 4, 5, 6\}$. For any $T \in U$, let $|T|$ denote the number of elements in T and T' denote the complement of T . For any $T, R \in U$, let $T \setminus R$ be the set of all elements in T which are not in R . Which one of the following is true?

239. The cardinality of the power set of $\{0, 1, 2, \dots, 10\}$ is _____.

240. Let R be the relation on the set of positive integers such that aRb if only if a and b are distinct and have a common divisor other than 1. Which one of the following statements about R is true?

241. Let X and Y be finite sets and $f: X \rightarrow Y$ be a function. Which one of the following statements is TRUE?

242. Which one of the following functions is continuous at $x = 3$?

243. What is the possible number of reflexive relations on a set of 5 elements?

244. Let S be a set of n elements. The number of ordered pairs in the largest and the smallest equivalence relations on S are

245. What is the maximum number of different Boolean functions involving n Boolean variables?

246. Let $A = \{x \in \mathbb{Z} \mid 1 \leq x \leq 20\}$, $B = \{x \in \mathbb{Z} \mid x \text{ is a multiple of } 3\}$, and $C = \{x \in \mathbb{Z} \mid x \text{ is a prime number}\}$. Find $(A \cap B) \cup (A \cap C)$ and determine its cardinality. Justify each step.

247. Define a relation R on the set of integers such that $aRb \iff a-b$ is divisible by 4. Show that R is an equivalence relation and determine its equivalence classes for the set $\{0,1,2,\dots,9\}$.

248. Given the sets A, B, C such that $A \subseteq B \cup C$, under what conditions can $A \subseteq B$ and $A \subseteq C$? Provide counterexamples or proofs.

249. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^2 + 1$. Determine whether f is injective, surjective, or bijective. Provide detailed reasoning.

250. Consider a finite group G with identity element e and binary operation $*$. Prove that for every $a \in G$, the inverse of a is unique and $a^{-1} * a = e$.

251. Prove or disprove: The Cartesian product of two countable infinite sets is countable. Provide detailed justification.

- 252.** Let $f:A \rightarrow B$ and $g:B \rightarrow C$ be functions such that $g \circ f$ is bijective. What can you say about the injectivity or surjectivity of f and g ? Justify with examples.
- 253.** Given a poset $(P, \leq)$, define the concepts of maximal and minimal elements. Provide a Hasse diagram of a partially ordered set with multiple maximal and minimal elements and explain it.
- 254.** Find the number of onto functions from a set with 4 elements to a set with 3 elements. Use inclusion-exclusion and explain all steps.
- 255.** Let G be a cyclic group of order 12. List all subgroups of G , specify their orders, and explain how they are derived.
- 256.** Let A be a 3×3 matrix with real entries such that $A^3 = I$ (identity matrix). What are the possible eigenvalues of A ? Explain with justification.
- 257.** Given two linearly independent vectors u and v in $\mathbb{R}^3$, construct a matrix whose column space spans the plane defined by u and v . Describe how to find the orthogonal projection of any vector $w \in \mathbb{R}^3$ onto this plane.
- 258.** Let A be a symmetric matrix. Show that all eigenvalues of A are real and provide a practical example.
- 259.** If a square matrix A has determinant zero, what does it imply about the invertibility of A ? Use the concepts of linear dependence and rank to justify your answer.
- 260.** Prove that a connected undirected graph with n vertices and $n-1$ edges is a tree. Give an example to support your proof.
- 261.** Given a directed graph, explain how you would detect whether it contains a cycle using DFS. Apply your method to a sample graph.
- 262.** For a connected undirected graph, define and find a minimum spanning tree using Prim's algorithm on the graph: Vertices = $\{A, B, C, D, E\}$; Edges = $\{(A,B,2), (A,C,3), (B,C,1), (B,D,4), (C,E,5), (D,E,2)\}$.
- 263.** In a simple graph with n vertices, what is the maximum number of edges possible without forming a triangle? Justify using Turán's theorem.
- 264.** Prove that in every undirected graph, the number of vertices with odd degree is even. Give examples to support your answer.
- 265.** Define chromatic number. For the following graph, determine the chromatic number and give the coloring scheme: A triangle with one additional vertex connected to each of the triangle's vertices.
- 266.** A graph is said to be bipartite if it does not contain odd-length cycles. Prove this property and apply it to a sample graph.
- 267.** Given an adjacency matrix, write a programmatic approach to count the number of paths of length 2 between all pairs of nodes.
- 268.** Define Eulerian and Hamiltonian paths. Construct a graph that has a Hamiltonian cycle but no Eulerian circuit and explain why.

- 269.** Determine whether the given graph is planar. If not, provide a planar embedding by edge rearrangement (Kuratowski's theorem may be referenced).
- 270.** In how many ways can 5 boys and 5 girls be seated in a row such that no two girls sit together? Solve using combinatorial reasoning.
- 271.** Find the number of 5-digit numbers formed using digits 1 to 9 such that no digit repeats and the number is divisible by 5.
- 272.** Using inclusion-exclusion, find the number of integers between 1 and 1000 that are divisible by 3 or 5 or 7.
- 273.** A committee of 5 members is to be formed from a group of 6 men and 4 women. In how many ways can this be done if at least 2 women must be included?
- 274.** How many different circular permutations are there of 6 distinct objects? Also, how would the count change if 2 of the objects are indistinguishable?
- 275.** Evaluate $\sum_{k=0}^n \binom{n}{k} 2^k$ and interpret the result combinatorially.
- 276.** A password of 8 characters is to be formed using uppercase letters and digits such that at least one digit is included. How many such passwords are possible?
- 277.** Translate the statement "If it is raining, then the ground is wet" into logical symbols and determine its contrapositive and converse.
- 278.** Construct a truth table for $(P \wedge Q) \rightarrow (\neg R \vee Q)$ and identify whether the statement is a tautology, contradiction, or contingency.
- 279.** Use resolution to determine the satisfiability of the formula: $\neg P \vee Q, \neg Q \vee R, P \vee \neg R$
- 280.** Using natural deduction, prove: $(P \rightarrow Q) \wedge (Q \rightarrow R) \Rightarrow (P \rightarrow R)$
- 281.** Define a well-formed formula (wff). How many distinct wffs can be formed with 3 atomic propositions and 2 binary connectives?
- 282.** What is the principal conjunctive normal form (PCNF) of $\neg(P \rightarrow (Q \vee \neg R))$?
- 283.** Show the equivalence between $P \leftrightarrow Q$ and $(P \wedge Q) \vee (\neg P \wedge \neg Q)$ using a truth table.
- 284.** Define satisfiability and validity. Give an example of a formula that is satisfiable but not valid.
- 285.** A bag contains 4 red balls, 5 green balls, and 3 blue balls. Two balls are drawn at random without replacement. Find the probability that both balls are of different colors.
- 286.** A fair die is rolled twice. Find the probability that the sum is divisible by 4 or at least one die shows a 6.
- 287.** In a class of 100 students, 70 like Math, 60 like Physics, and 50 like both. A student is selected at random. What is the probability the student likes at least one of the two?

- 288.** A random variable X has the following probability mass function: $P(X=x) = 1/2^x$, $x=1,2,3,\dots$. Find the expected value and variance of X .
- 289.** A coin is biased such that the probability of heads is 0.6. It is tossed 5 times. What is the probability of getting exactly 3 heads?
- 290.** A continuous random variable X is uniformly distributed in $[2, 10]$. Find the probability that X is greater than 6 and the expected value of X .
- 291.** Define conditional probability and independence. If $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$, determine if A and B are independent.
- 292.** Find the local maxima, minima, and points of inflection of the function $f(x)=x^3-6x^2+9x+15$ using first and second derivatives.
- 293.** Evaluate the limit $\lim_{x \rightarrow 0} x^3 \sin(x) - x$. Justify using L'Hôpital's Rule.
- 294.** Let $f(x) = \int_1^x t \, dt$. Show that $f'(x) = x$ and compute its second derivative.
- 295.** Determine whether the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges or diverges using appropriate convergence tests.
- 296.** Compute the area under the curve $y=x^2$ between $x = 0$ and $x = 3$ using definite integration.
- 297.** Let A be an $n \times n$ matrix such that $A^2=A$ (i.e., A is idempotent). Prove that the rank of A is equal to its trace. Justify with a numerical example for $n=3$.
- 298.** Define Bayes' Theorem. A factory has three machines A , B , and C producing 30%, 45%, and 25% of the total output respectively. Machine A produces 1% defective, B produces 2%, and C produces 3% defective items. If an item is found to be defective, what is the probability that it was produced by Machine B ?
- 299.** Prove that if G is a graph with n vertices and more than $2(n-1)(n-2)$ edges, then G is connected. Use graph theoretic definitions and justify via contradiction.
- 300.** For a propositional formula, define the concept of a dual. Show that the dual of $(P \vee (Q \wedge R))$ is logically equivalent to $(\neg P \wedge (\neg Q \vee \neg R))$ under De Morgan's laws and provide a truth table for verification.

General Aptitude

1. A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of
2. In the sequence 6, 9, 14, x, 30, 41, a possible value of x is
3. In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is $\frac{1}{4}$ th of the number of students who like other branches. The number of students who like both their core and other branches is 500.
4. Two wizards try to create a spell using all the four elements, water, air, fire, and earth. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work. How many attempts does each wizard make before coming to this conclusion, independently?
5. If ' $\rightarrow$ ' denotes increasing order of intensity, then the meaning of the words [walk $\rightarrow$ jog $\rightarrow$ sprint] is analogous to [bothered $\rightarrow$ _____ $\rightarrow$ daunted].
6. The least number of squares to be added in the figure to make AB a line of symmetry is
7. A rectangular paper of 20cm x 8cm is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is
8. A rectangular paper sheet of dimensions 54cm x 4cm is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken. Then, the ratio of the volume of the cylindrical tube to the volume of the cube is
9. The number of coins of Rs. 1, Rs. 5, and Rs. 10 denominations that a person has are in the ratio 5:3:13. Of the total amount, the percentage of money in Rs. 5 coins is
10. If ' $\rightarrow$ ' denotes increasing order of intensity, then the meaning of the words [dry $\rightarrow$ arid $\rightarrow$ parched] is analogous to [diet $\rightarrow$ fast $\rightarrow$ _____].
11. $f(x)$ and $g(y)$ are functions of x and y , respectively, and $f(x)=g(y)$ for all real values of x and y . Which one of the following options is necessarily TRUE for all x and y ?
12. The country of Zombieland is in distress since more than 75% of its working population is suffering from serious health issues. Studies conducted by competent health experts concluded that a complete lack of physical exercise among its working population was one of the leading causes of their health issues. As one of the measures to address the problem, the Government of Zombieland has decided to provide monetary incentives to those who ride bicycles to work. Based only on the information provided above, which one of the following statements can be logically inferred with certainty?
13. Looking at the surface of a smooth 3-dimensional object from the outside, which one of the following options is TRUE?
14. A survey for a certain year found that 90% of pregnant women received medical care at least once before giving birth. Of these women, 60% received medical care from doctors, while 40% received

medical care from other healthcare providers.

15. Kind : _____ : : Often : Frequently

16. We reached the station late, and _____ missed the train.

17. A plot of land must be divided between four families. They want their individual plots to be similar in shape, not necessarily equal in area. The land has equally spaced poles, marked as dots in the below figure. Two ropes, R1 and R2, are already present and cannot be moved. What is the least number of additional straight ropes needed to create the desired plots? A single rope can pass through three poles that are aligned in a straight line.

18. A box contains five balls of same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball. First ball is drawn. What is the probability of getting an orange ball in the next draw?

19. The _____ is too high for it to be considered _____

20. The number of students in three classes is in the ratio 3:13:6. If 18 students are added to each class, the ratio changes to 15:35:21. The total number of students in all the three classes in the beginning was:

21. Pen : Write :: Knife : _____

22. If $(x-21)^2 - (x-23)^2 = x+2$, then the value of x is:

23. Gauri said that she can play the keyboard _____ her sister.

24. Some people suggest anti-obesity measures (AOM) such as displaying calorie information in restaurant menus. Such measures sidestep addressing the core problems that cause obesity: poverty and income inequality.

25. There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag. The probability that at least two chocolates are identical is

26. We have 2 rectangular sheets of paper, M and N, of dimensions 6 cm x 1 cm each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is

27. _____ is to surgery as writer is to _____

28. The ratio of boys to girls in a class is 7 to 3. Among the options below, an acceptable value for the total number of students in the class is:

29. The total revenue of a company during 2014 - 2018 is shown in the bar graph. If the total expenditure of the company in each year is 500 million rupees, then the aggregate profit or loss (in percentage) on the total expenditure of the company during 2014 - 2018 is

30. Two straight lines are drawn perpendicular to each other in X-Y plane. If α and β are the acute angles the straight lines make with the X-axis, then $\alpha + \beta$ is

31. If $P=3$, $R=27$, $T=243$, then $Q + S$

32. Cook : Cook :: Fly :

33. His knowledge of the subject was excellent but his classroom performance was

34. Raman is confident of speaking English _____ six months as he has been practising regularly _____ the last three weeks.

35. In a college, there are three student clubs, 60 students are only in the Drama club, 80 students are only in the Dance club, 30 students are only in Maths club, 40 students are in both Drama and Dance clubs, 12 students are in both Dance and Maths clubs, 7 students are in both Drama and Maths clubs, and 2 students are in all clubs. If 75% of the students in the college are not in any of these clubs, then the total number of students in the college is

36. A court is to a judge as _____ is to a teacher.

37. Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs. 150 more. The cost of the gift was Rs

38. Two cars at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h. The number of hours it takes for the distance between the two cars to be 20 km is

39. The search engine's business model _____ around the fulcrum of trust.

40. The expenditure on the project _____ as follows: equipment Rs.20 lakhs, salaries Rs.12 lakhs, and contingency Rs.3 lakhs.

41. A six sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?

42. In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees in the party?

43. In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift Rs. 750 to each male senior citizen in the town and Rs. 1000 to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only 98th of the eligible men and 32rd of the eligible women claimed the gift. How much money (in Rupees) did the philanthropist give away in total?

44. What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

45. The area of a square is d . What is the area of the circle which has the diagonal of the square as its diameter?
46. A _____ investigation can sometimes yield new facts, but typically organized ones are more successful.
47. "From where are they bringing their books? _____ bringing _____ books from _____"
48. The number of roots of $ex+0.5x^2-2=0$ in the range $[-5,5]$ is
49. X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have
50. There are three boxes. One contains apples, another contains oranges and the last one contains both apples and oranges. All three are known to be incorrectly labeled. If you are permitted to open just one box and then pull out and inspect only one fruit, which box would you open to determine the contents of all three boxes?
51. There are 3 red socks, 4 green socks and 3 blue socks. You choose 2 socks. The probability that they are of the same colour is
52. A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have?
53. There are five buildings called V, W, X, Y and Z in a row (not necessarily in that order). V is to the West of W. Z is to the East of X and the West of V. W is to the West of Y. Which is the building in the middle?
54. Saturn is _____ to be seen on a clear night with the naked eye.
55. Choose the option with words that are not synonyms.
56. Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?
57. The expression $2(x+y) - [x-y]$ is equal to :
58. Six people are seated around a circular table. There are at least two men and two women. There are at least three right-handed persons. Every woman has a left-handed person to her immediate right. None of the women are right-handed. The number of women at the table is
59. The probability that a k -digit number does NOT contain the digits 0, 5, or 9 is
60. Find the smallest number y such that $y \times 162$ is a perfect cube.
61. Rahul, Murali, Srinivas and Arul are seated around a square table. Rahul is sitting to the left of Murali. Srinivas is sitting to the right of Arul. Which of the following pairs are seated opposite each other?

62. Research in the workplace reveals that people work for many reasons _____.
63. After Rajendra Chola returned from his voyage to Indonesia, he _____ to visit the temple in Thanjavur.
64. Among 150 faculty members in an institute, 55 are connected with each other through Facebook and 85 are connected through Whatsapp. 30 faculty members do not have Facebook or Whatsapp accounts. The numbers of faculty members connected only through Facebook accounts is _____.
65. Pick the odd one from the following options.
66. The man who is now Municipal Commissioner worked as _____
67. In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is _____.
68. If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?
69. A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.
70. If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?
71. Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world." The sentence above is an example of a _____ statement.
72. rewording of something written or spoken is a _____.
73. Out of the following four sentences, select the most suitable sentence with respect to grammar and usage.
74. Four branches of a company are located at M, N, O and P. M is north of N at a distance of 4km; P is south of O at a distance of 2 km; N is southeast of O by 1km. What is the distance between M and P in km?
75. A generic term that includes various items of clothing such as a skirt, a pair of trousers and a shirt is _____.
76. Consider a function $f(x) = 1 - x^2$ on $-1 \leq x \leq 1$. The value of x at which the function attains a maximum, and the maximum value of the function are: _____.
77. Choose the statement where underlined word is used correctly.
78. Given Set A = {2, 3, 4, 5} and Set B = {11, 12, 13, 14, 15}, two numbers are randomly selected, one from each set. What is the probability that the sum of the two numbers equals 16?
79. Didn't you buy _____ when you went shopping?
80. Consider the equation: $(7526)_8 - (Y)_8 = (4364)_8$, where $(X)_N$ stands for X to the base N. Find Y _____.

81. The Gross Domestic Product (GDP) in Rupees grew at 7% during 2012-2013. For international comparison, the GDP is compared in US Dollars (USD) after conversion based on the market exchange rate. During the period 2012-2013 the exchange rate for the USD increased from Rs. 50/USD to Rs.60/USD. India's GDP in USD during the period 2012-2013
82. A dance programme is scheduled for 10.00 a.m. Some students are participating in the programme and they need to come an hour earlier than the start of the event. These students should be accompanied by a parent. Other students and parents should come in time for the programme. The instruction you think that is appropriate for this is
83. Which number does not belong in the series below?
84. Choose the word that is opposite in meaning to the word "coherent".
85. If she _____ how to calibrate the instrument, she _____ done the experiment.
86. At what time between 6 a. m. and 7 a. m. will the minute hand and hour hand of a clock make an angle closest to 60° ?
87. If x is real and $x^2 - 2x + 3 = 11$, then possible values of $-x^3 + x^2 - x$ include
88. The old city of Koenigsberg, which had a German majority population before World War 2, is now called Kaliningrad. After the events of the war, Kaliningrad is now a Russian territory and has a predominantly Russian population. It is bordered by the Baltic Sea on the north and the countries of Poland to the south and west and Lithuania to the east respectively. Which of the statements below can be inferred from this passage?
89. The value of $12 + 12 + 12 + \dots$ is
90. What is the average of all multiples of 10 from 2 to 198?
91. Who _____ was coming to see us this evening?
92. When a point inside of a tetrahedron (a solid with four triangular surfaces) is connected by straight lines to its corners, how many (new) internal planes are created with these lines?
93. Round-trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is Rs 100, a group of 5 tourists purchasing round-trip tickets will be charged Rs _____
94. The roots of $ax^2 + bx + c = 0$ are real and positive. a, b and c are real. Then $ax^2 + b^2x + c^2 = 0$ has
95. In a press meet on the recent scam, the minister said, "The buck stops here". What did the minister convey by the statement?
96. The current erection cost of a structure is Rs. 13,200. If the labour wages per day increase by $\frac{1}{5}$ of the current wages and the working hours decrease by $\frac{1}{24}$ of the current period, then the new cost of erection in Rs. Is

97. A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

98. After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again, the spider failed but that did not deter it to refrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

99. Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7 ?

100. Were you a bird, you _____ in the sky.

101. What will be the maximum sum of 44, 42, 40, ... ?

102. Universalism is to particularism as diffuseness is to _____.

103. Which one of the following options is the closest in meaning to the word given below?

104. Given the sequence of terms, AD CG FK JP, the next term is

105. A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y=2x-0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

106. Choose the grammatically INCORRECT sentence:

107. The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of Rs. 50 per unit. The number of units to be produced by the firm such that the profit is maximized is

108. A container originally contains 10 litres of pure spirit. From this container, 1 litre of spirit replaced with 1 litre of water. Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?

109. A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders. Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10th day. What is the minimum number of trucks required so that there will be no pending order at the end of 5th day?

110. The variable cost (V) of manufacturing a product varies according to the equation $V=4q$, where q is the quantity produced. The fixed cost (F) of production of same product reduces with q according to the equation $F=q100$. How many units should be produced to minimize the total cost ($V+F$)?

111. Few school curricula include a unit on how to deal with bereavement and grief, and yet all students at some point in their lives suffer from losses through death and parting. Based on the above passage which

topic would not be included in a unit on bereavement?

112. If $\log(P) = (1/2)\log(Q) = (1/3)\log(R)$, then which of the following options is TRUE?

113. Given digits 2, 2, 3, 3, 3, 4, 4, 4, 4 how many distinct 4 digit numbers greater than 3000 can be formed?

114. 5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long it will take to build the wall?

115. If $137 + 276 = 435$ how much is $731 + 672$?

116. The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.

117. 25 persons are in a room. 15 of them play hockey, 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is:

118. The exponent of 11 in the prime factorization of $300!$ is

119. The Fibonacci sequence is the sequence of integers

120. Four equilateral triangles are used to form a regular closed three-dimensional object by joining along the edges. The angle between any two faces is

121. How many combinations of non-null sets A,B,C are possible from the subsets of $\{2,3,5\}$ satisfying the conditions : (i) A is a subset of B, and (ii) B is a subset of C ?

122. Find the odd one out in the set : $\{19,37,21,17,23,29,31,11\}$

123. How many pairs of sets (S,T) are possible among the subsets of $\{1, 2, 3, 4, 5, 6\}$ that satisfy the condition that S is a subset of T?

124. In a recently held parent-teacher meeting, the teachers had very few complaints about Ravi. After all, Ravi was a hardworking and kind student. Incidentally, almost all of Ravi's friends at school were hardworking and kind too. But the teachers drew attention to Ravi's complete lack of interest in sports. The teachers believed that, along with some of his friends who showed similar disinterest in sports, Ravi needed to engage in some sports for his overall development. Based only on the information provided above, which one of the following statements can be logically inferred with certainty?

125. The minute-hand and second-hand of a clock cross each other _____ times between 09:15:00 AM and 09:45:00 AM on a day.

126. For the past m days, the average daily production at a company was 100 units per day. If today's production of 180 units changes the average to 110 units per day, what is the value of m ?

127. If $f(x) = 2\ln x$ what is the area bounded by $f(x)$ for the interval $[0,2]$ on the x -axis?

128. Writing too many things on the _____ while teaching could make the students get _____.

129. In a 12-hour clock that runs correctly, how many times do the second, minute, and hour hands of the clock coincide, in a 12-hour duration from 3 PM in a day to 3 AM the next day?

130. A person travelled 80 km in 6 hours. If the person travelled the first part with a uniform speed of 10 kmph and the remaining part with a uniform speed of 18 kmph.

131. The average of the monthly salaries of M, N and S is Rs. 4000. The average of the monthly salaries of N, S and P is Rs. 5000. The monthly salary of P is Rs. 6000. What is the monthly salary of M as a percentage of the monthly salary of P?

132. The world is going through the worst pandemic in the past hundred years. The air travel industry is facing a crisis, as the resulting quarantine requirement for travelers led to weak demand. In relation to the first sentence above, what does the second sentence do?

133. Consider a square sheet of side 1 unit. The sheet is first folded along the main diagonal. This is followed by a fold along its line of symmetry. The resulting folded shape is again folded along its line of symmetry. The area of each face of the final folded shape, in square units, equal to

134. A box contains 15 blue balls and 45 black balls. If 2 balls are selected randomly, without replacement, the probability of an outcome in which the first selected is a blue ball and the second selected is a black ball, is _____

135. The front door of Mr. X's house faces East. Mr. X leaves the house, walking 50 m straight from the back door that is situated directly opposite to the front door. He then turns to his right, walks for another 50 m and stops. The direction of the point Mr. X is now located at with respect to the starting point is

136. A digital watch X beeps every 30 seconds while watch Y beeps every 32 seconds. They beeped together at 10 AM. The immediate next time that they will beep together is

137. Five persons P, Q, R, S and T are to be seated in a row, all facing the same direction, but not necessarily in the same order. P and T cannot be seated at either end of the row. P should not be seated adjacent to S. R is to be seated at the second position from the left end of the row. The number of distinct seating arrangements possible is:

138. Five persons P, Q, R, S and T are sitting in a row not necessarily in the same order. Q and R are separated by one person, and S should not be seated adjacent to Q. The number of distinct seating arrangements possible is:

139. The number of hens, ducks and goats in farm P are 65, 91 and 169, respectively. The total number of hens, ducks and goats in a nearby farm Q is 416. The ratio of hens:ducks:goats in farm Q is 5:14:13. All the hens, ducks and goats are sent from farm Q to farm P. The new ratio of hens:ducks:goats in farm P is

140. Ms. X came out of a building through its front door to find her shadow due to the morning sun falling to her right side with the building to her back. From this, it can be inferred that building is facing _____

141. The two pie-charts given below show the data of total students and only girls registered in different streams in a university. If the total number of students registered in the university is 5000, and the total number of the registered girls is 1500; then, the ratio of boys enrolled in Arts to the girls enrolled in

Management is _____

142. IT was estimated that 52 men can complete a strip in a newly constructed highway connecting cities P and Q in 10 days, Due to an emergency, 12 men were sent to another project. How many number of days, more than the original estimate, will be required to complete the strip?

143. The recent measures to improve the output would _____ the level of production to our satisfaction.

144. The bar graph shows the data of the students who appeared and passed in an examination for four schools P, Q, R, and S. The average of success rates (in percentage) of these four schools is _____.

145. The sum of the first n terms in the sequence 8, 88, 888, 8888, ... is _____

146. P, Q, R and S are to be uniquely coded using α and β . If P is coded as $\alpha\alpha$ and Q as $\alpha\beta$, then R and S, respectively, can be coded as _____.

147. Jofra Archer, the England fast bowler, is _____ than accurate.

148. Define $[x]$ as the greatest integer less than or equal to x , for each $x \in (-\infty, \infty)$. If $y = [x]$, then area under y for $x \in [1, 4]$ is _____.

149. Two pipes P and Q can fill a tank in 6 hours and 9 hours respectively, while a third pipe R can empty the tank in 12 hours. Initially, P and R are open for 4 hours, Then P is closed and Q is opened. After 6 more hours R is closed. The total time taken to fill the tank (in hours) is _____

150. Fiscal deficit was 4% of the GDP in 2015 and that increased to 5% in 2016. If the GDP increased by 10% from 2015 to 2016, the percentage increase in the actual fiscal deficit is _____

151. The product of three integers X, Y and Z is 192. Z is equal to 4 and P is equal to the average of X and Y. What is the minimum possible value of P?

152. If IMHO=JNIP; IDK=JEL; and SO=TP, then IDC=

153. M and N had four children P, Q, R and S. Of them, only P and R were married. They had children X and Y respectively. If Y is a legitimate child of W, which of the following statements is necessarily FALSE?

154. A firm hires employees at five different skill levels P, Q, R, S, T. The shares of employment at these skills levels of total employment in 2010 is given in the pie chart as shown. There were a total of 600 employees in 2010 and the total employment increased by 15% from 2010 to 2016. The total employment at skill levels P, Q and R remained unchanged during this period. If the employment at skill level S increased by 40% from 2010 to 2016, how many employees were there at skill level T in 2016?

155. A person divided an amount of Rs. 100,000 into two parts and invested in two different schemes. In one he got 10% profit and in the other he got 12%. If the profit percentages are interchanged with these investments he would have got Rs. 120 less. Find the ratio between his investments in the two schemes.

156. Under a certain legal system, prisoners are allowed to make one statement. If their statements turns out to be true then they are hanged. If the statements turns to be false then they are shot. One prisoner made a statement and the judge had no option but to set him free. Which one of the following could be

that statement?

157. The minister avoided any mention of the issue of women's reservation in the private sector. He was accused of _____ the issue.

158. The sum and product of two integers are 26 and 165 respectively. The difference between these two integers is _____

159. A worker noticed that the hour hand on the factory clock had moved by 225 degrees during her stay at the factory. For how long did she stay in the factory?

160. _____ I permitted him to leave, I wouldn't have had any problem with him being absent, _____ I?

161. A contract is to be completed in 52 days and 125 identical robots were employed, each operational for 7 hours a day. After 39 days, five-seventh of the work was completed. How many additional robots would be required to complete the work on time, if each robot is now operational for 8 hours a day?

162. A wire would enclose an area of 1936m², if it is bent to a square. The wire is cut into two pieces. The longer piece is thrice as long as the shorter piece. The long and the short pieces are bent into a square and a circle, respectively. Which of the following choices is closest to the sum of the areas enclosed by the two pieces in square meters?

163. Forty students watched films A, B and C over a week. Each student watched either only one film or all three. Thirteen students watched film A, sixteen students watched film B and nineteen students watched film C. How many students watched all three films?

164. The perimeters of a circle, a square, and an equilateral triangle are equal. Which one of the following statements is true?

165. Find the missing group of letters in the following series:

166. From the time the front of a train enters a platform, it takes 25 seconds for the back of the train to leave the platform, while traveling at a constant speed of 54 km/h. At the same speed, it takes 14 seconds to pass a man running at 9 km/h in the same direction as the train. What is the length of the train and that of the platform in meters, respectively?

167. Given that a and b are integers and $a+a^2b^3$ is odd, which of the following statements is correct?

168. For integers, a,b and c, what would be the minimum and maximum values respectively of $a+b+c$ if $\log a + \log b + \log c = 0$?

169. A number consists of two digits. The sum of the digits is 9. If 45 is subtracted from the number, its digits are interchanged. What is the number?

170. A rectangle becomes a square when its length and breadth are reduced by 10m and 5m, respectively. During this process, the rectangle loses 650m² of area. What is the area of the original rectangle in square meters?

171. Seven machines take 7 minutes to make 7 identical toys. At the same rate, how many minutes would it take for 100 machine to make 100 toys?

172. There are 4 women P, Q, R, S and 5 men V, W, X, Y, Z in a group. We are required to form pairs each consisting of one woman and one man. P is not to be paired with Z, and Y must necessarily be paired with someone. In how many ways can 4 such pairs be formed?

173. X bullocks and Y tractors take 8 days to plough a field. If we have the number of bullocks and double the number of tractors, it takes 5 days to plough the same field. How many days will it take X bullocks alone to plough the field?

174. P looks at Q while Q looks at R. P is married, R is not. The number of pairs of people in which a married person is looking at an unmarried person is

175. If a and b are integers and $a-b$ is even, which of the following must always be even?

176. If you choose plan P, you will have to _____ plan Q, as these two are mutually _____.

177. Two very famous sportsmen Mark and Steve happened to be brothers and played for country K. Mark teased James, an opponent from country E, "There is no way you are good enough to play for your country." James replied, "Maybe not, but at least I am the best player in my own family." Which one of the following can be inferred from this conversation?

178. Let S_1 be the plane figure consisting of the points (x,y) given by the inequalities $|x-1| \leq 2$ and $|y+3| \leq 3$. Let S_2 be the plane figure given by the inequalities $x-y \geq -2, y \geq 1$ and $x \leq 3$. Let S be the union of S_1 and S_2 . The area of S is.

179. P, Q and R talk about S's car collection. P states that S has at least 3 cars. Q believes that S has less than 3 cars. R indicates that to its knowledge, S has at least one car. Only one of P, Q and R is right. The number of cars owned by S is.

180. In a company with 100 employees, 45 earn Rs.20,000 per month, 25 earn Rs.30000, 20 earn Rs.40000, 8 earn Rs.60000, and 2 earn Rs.150,000. The median of the salaries is

181. As the two speakers become increasingly agitated, the debate became _____

182. A wire of length 340 mm is to be cut into two parts. One of the parts is to be made into a square and the other into a rectangle where sides are in the ratio of 1:2. What is the length of the side of the square (in mm) such that the combined area of the square and the rectangle is a MINIMUM?

183. M and N starts from same location. M travels 10 km East then 10 km North-East. N travels 5 km South then 4 km South-East. What is the shortest distance (in km) between M and N at the end of their travel?

184. Two and a quarter hours back, when seen in the mirror, the reflection of the wall clock without number marking seemed to show 1:30. What is the actual current time shown by the clock?

185. S, M, E and F are working in shifts in a team to finish a project. M works with twice the efficiency of others but for half as many days as E worked. S and M have 6 hour shifts in a day, whereas E and F have

12 hours shifts. What is the ratio of contribution of M to contribution of E in the project?

186. Given $(9 \text{ inches})^{1/2} = (0.25 \text{ yards})^{1/2}$, which one of the following statements is TRUE?

187. After India's cricket world cup victory in 1985, Shrotria who was playing both tennis and cricket till then, decided to concentrate only on cricket. And the rest is history What does the phrase in bold letters mean in this context?

188. The binary operation $\blacksquare$ is defined as $a \blacksquare b = ab + (a+b)$, where a and b are any two real numbers. The value of the identity element of this operation, defined as the number x such that $a \blacksquare x = a$, for any a , is

189. Students taking an exam are divided into two groups, P and Q such that each group has the same number of students. The performance of each of the students in a test was evaluated out of 200 marks. It was observed that the mean of group P was 105, while that of group Q was 85. The standard deviation of group P was 25, while that of group Q was 5. Assuming that the marks were distributed on a normal distribution, which of the following statements will have the highest probability of being TRUE?

190. A window is made up of a square portion and an equilateral triangle portion above it. The base of the triangular portion coincides with the upper side of the square. If the perimeter of the window is 6 m, the area of the window in m^2 is ____

191. The volume of a sphere of diameter 1 unit is _____ than the volume of a cube of side 1 unit.

192. P, Q, R and S are working on a project. Q can finish the task in 25 days, working alone for 12 hours a day. R can finish the task in 50 days, working alone for 12 hours per day. Q worked 12 hours a day but took sick leave in the beginning for two days. R worked 18 hours a day on all days. What is the ratio of work done by Q and R after 7 days from the start of the project?

193. A person moving through a tuberculosis prone zone has a 50% probability of becoming infected. However, only 30% of infected people develop the disease. What percentage of people moving through a tuberculosis prone zone remains infected but does not show symptoms of disease?

194. Michael lives 10 km away from where I live. Ahmed lives 5 km away and Susan lives 7 km away from where I live. Arun is farther away than Ahmed but closer than Susan from where I live. From the information provided here, what is one possible distance (in km) at which I live from Arun's place?

195. In a huge pile of apples and oranges, both ripe and unripe mixed together, 15% are unripe fruits. Of the unripe fruits, 45% are apples. Of the ripe ones, 66% are oranges. If the pile contains a total of 5692000 fruits, how many of them are apples?

196. A coin is tossed thrice. Let C be the event that head occurs in each of first two tosses. Let Y be the event that tails occurs in third toss. Let Z be the event that two tails occur in three tosses. Based on the above information, which one of the following statement is TRUE?

197. Right triangle PQR is to be constructed in xy-plane so that the right angle is at P and line PR is parallel to x-axis. The x and y coordinates of P, Q and R are to be integers that satisfies the inequalities: $-4 \leq x \leq 5$ and $6 \leq y \leq 16$. How many different triangles could be constructed with these properties?

198. Five teams have to compete in a league, with every team playing every other team exactly once, before going to the next round. How many matches will have to be held to complete the league round of matches?

199. From a circular sheet of paper of radius 30 cm, a sector of 10% area is removed. If the remaining part is used to make a conical surface, then the ratio of radius and height of cone is

200. Ram and Shyam shared a secret and promised to each other that it would remain between them. Ram expressed himself in one of the following ways as given in the choices below. Identify the correct way as per standard English.

201. A tiger is 50 leaps of its own behind a deer. The tiger takes 5 leaps per minute to the deer's 4. If the tiger and the deer covers 8 meter and 5 meter per leap respectively, what distance in meters will the tiger have to run before it catches the deer?

202. A five digit number is formed using the digits 1, 3, 5, 7 and 9 without repeating any of them. What is the sum of all such possible five digit numbers?

203. Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the most appropriate option to complete the above sentence.

204. Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select the option with the correct sequence of articles to fill in the blanks.

205. The CEO's decision to downsize the workforce was considered myopic because it sacrificed long-term stability to accommodate short-term gains. Select the most appropriate option that can replace the word "myopic" without changing the meaning of the sentence.

206. If '→' denotes increasing order of intensity, then the meaning of the words [walk → jog → sprint] is analogous to [bothered → _____ → daunted]. Which one of the given options is appropriate to fill the blank?

207. If '→' denotes increasing order of intensity, then the meaning of the words [dry → arid → parched] is analogous to [diet → fast → _____]. Which one of the given options is appropriate to fill the blank?

208. We reached the station late, and _____ missed the train.

209. Kind : _____ :: Often : Frequently (By word meaning)

210. A survey for a certain year found that 90% of pregnant women received medical care at least once before giving birth. Of these women, 60% received medical care from doctors, while 40% received medical care from other healthcare providers. Given this information, which one of the following statements can be inferred with certainty?

211. Looking at the surface of a smooth 3-dimensional object from the outside, which one of the following options is TRUE?

212. The _____ is too high for it to be considered _____.

213. Given below are four statements : Statement 1 : All students are inquisitive. Statement 2 : Some students are inquisitive. Statement 3 : No student is inquisitive. Statement 4 : Some students are not inquisitive. From the given four statements, find the two statements that CANNOT BE TRUE simultaneously, assuming that there is at least one student in the class.

214. A palindrome is a word that reads the same forwards and backwards. In a game of words, a player has the following two plates painted with letters. From the additional plates given in the options, which one of the combinations of additional plates would allow the player to construct a five-letter palindrome. The player should use all the five plates exactly once. The plates can be rotated in their plane.

215. Gauri said that she can play keyboard _____ her sister.

216. A circular sheet of paper is folded along the lines in the directions shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like _____.

217. Consider the following sentences : (i) Everybody in the class is prepared for the exam. (ii) Babu invited Danish to his home because he enjoys playing chess, Which of the following is the CORRECT observation about the above two sentences?

218. _____ is to surgery as writer is to _____. Which one of the following options maintains a similar logical relation in the above sentence?

219. A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon. Which one of the following is NOT a convex polygon?

220. The dawn of the 21st century witnessed the melting glaciers oscillating between giving too much and too little to billions of people who depend on them for fresh water. The UN climate report estimates that without deep cuts to man-made emissions, at least 30% of the northern hemisphere's surface permafrost could melt by the end of the century. Given this situation of imminent global exodus of billions of people displaced by rising seas, nation-states need to rethink their carbon footprint for political concerns, if not for environmental ones. Which one of the following statements can be inferred from the given passage?

221. Select the word that fits the analogy: Cook : Cook :: Fly : _____

222. Raman is confident of speaking English _____ six months as he has been practising regularly _____ the last three weeks.

223. His knowledge of the subject was excellent but his classroom performance was _____.

224. The expenditure on the project _____ as follows; equipment Rs.20 lakhs, salaries Rs.12 lakhs, and contingency Rs.3 lakhs.

225. A court is to a judge as _____ is to a teacher.

226. The search engine's business model _____ around the fulcrum of trust.

227. "From where are they bringing their books? _____ bringing _____ books from _____." The words that best fill the blanks in the above sentence are

228. "A _____ investigation can sometimes yield new facts, but typically organized ones are more successful." The word that best fills the blank in the above sentence is

229. Find the odd one in the following group of words mock, deride, praise, jeer

230. Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required. Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?(i). The author believes that computers are not good for us. (ii). Mobile computers and the internet are both intended inventions

231. All hill-stations have a lake. Ooty has two lakes. Which of the statement(s) below is/are logically valid and can be inferred from the above sentences? (i). Ooty is not a hill-station.(ii). No hill-station can have more than one lake.

232. Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia. Choose the option which is closest in meaning to the underlined phrase in the above sentence.

233. The man who is now Municipal Commissioner worked as _____.

234. If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare' ?

235. Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world." The sentence above is an example of a _____ statement.

236. A rewording of something written or spoken is a _____.

237. Which of the following options is the closest in meaning to the sentence below? She enjoyed herself immensely at the party.

238. "I put the brown paper in my pocket along with the chinks, and possibly other things. I suppose every one must have reflected how primeval and how poetical are the things that one carries in one's pocket: the pocket-knife, for instance the type of all human tools, the infant of the sword. Once I planned to write a book of poems entirely about the things in my pocket. But I found it would be too long: and the age of the great epics is past." (From G.K. Chesterton's "A Piece of Chalk") Based only on the information provided in the above passage, which one of the following statements is true?

239. Didn't you buy _____ when you went shopping?

240. Sequence the following sentences in a coherent passage.P: This fortuitous geological event generated a colossal amount of energy and heat that resulted in the rocks rising to an average height of 4 km across the contact zone. Q: Thus, the geophysicists tend to think of the Himalayas as an active geological event rather than as a static geological feature. R: The natural process of the cooling of this massive edifice absorbed large quantities of atmospheric carbon dioxide, altering the earth's atmosphere

and making it better suited for life.S: Many millennia ago, a breakaway chunk of bedrock from the Antarctic Plate collided with the massive Eurasian Plate.

241. In the given text, the blanks are numbered (i)–(iv). Select the best match for all the blanks.Steve was advised to keep his head _____ (i) _____ before heading _____ (ii) _____ to bat; for, while he had a head _____ (iii) _____ batting, he could only do so with a cool head _____ (iv) _____ his shoulders.

242. The country of Zombieland is in distress since more than 75% of its working population is suffering from serious health issues. Studies conducted by competent health experts concluded that a complete lack of physical exercise among its working population was one of the leading causes of their health issues. As one of the measures to address the problem, the Government of Zombieland has decided to provide monetary incentives to those who ride bicycles to work. Based only on the information provided above, which one of the following statements can be logically inferred with certainty?

243. Which one of the following sentence sequences creates a coherent narrative?(i) Once on the terrace, on her way to her small room in the corner, she notices the man right away. (ii) She begins to pant by the time she has climbed all the stairs.(iii) Mina has bought vegetables and rice at the market, so her bags are heavy. (iv) He was leaning against the parapet, watching the traffic below.

244. Some people believe that "what gets measured, improves". Some other believe that "what gets measured, gets gamed". One possible reason for the difference in the beliefs is the work culture in organizations. In organizations with good work culture, metrics help improve outcomes. However, the same metrics are counterproductive in organizations with poor work culture.Which one of the following is the CORRECT logical inference based on the information in the above passage?

245. Listening to music during exercise improves exercise performance and reduces discomfort. Scientists researched whether listening to music while studying can help students learn better and the results were inconclusive. Students who needed external stimulation for studying fared worse while students who did not need any external stimulation benefited from music. Which one of the following statements is the CORRECT inference of the above passage?

246. Given below are two statements I and II and two conclusions I and II : Statement : I. All bacteria are microorganisms. II. All pathogens are microorganisms. Conclusions :I. Some pathogens are bacteria. II. All pathogens are not bacteria. Based on the above statements and conclusions, which one of the following options is logically CORRECT?

247. Some people suggest anti-obesity measures (AOM) such as displaying calorie information in restaurant menus. Such measures sidestep addressing the core problems that cause obesity: poverty and income inequality. Which one of the following statements summarizes the passage?

248. Goods and Services Tax (GST) is an indirect tax introduced in India in 2017 that is imposed on the supply of goods and services, and it subsumes all indirect taxes except few. It is a destination-based tax imposed on goods and services used, and it is not imposed at the point of origin from where goods come. GST also has a few components specific to state governments, central government and Union Territories (UTs). Which one of the following statements can be inferred from the given passage?

- 249.** The police arrested four criminals - P, Q, R and S. The criminals knew each other. They made the following statements: P says "Q committed the crime." Q says "S committed the crime." R says "I did not do it." S says "What Q said about me is false." Assume only one of the arrested four committed the crime and only one of the statements made above is true. Who committed the crime?
- 250.** "A recent High Court judgement has sought to dispel the idea of begging as a disease — which leads to its stigmatization and criminalization — and to regard it as a symptom. The underlying disease is the failure of the state to protect citizens who fall through the social security net." Which of the following statements can be inferred from the given passage?
- 251.** Consider the following statements relating to the level of poker play of four players P, Q, R, and S. (i) P always beats Q (ii) R always beats S. (iii) S loses to P only sometimes. (iv) R always loses to Q. Which of the following can be logically inferred from the above statements? (i) P is likely to beat all the three other players (ii) S is the absolute worst player in the set
- 252.** Indian currency notes show the denomination indicated in at least seventeen languages. If this is not an indication of the nation's diversity, nothing else is. Which of the following can be logically inferred from the above sentences?
- 253.** Select the alternative meaning of the underlined part of the sentence. The chain snatchers took to their heels when the police party arrived.
- 254.** Bird : Nest :: Bee : _____ Select the correct option to complete the analogy.
- 255.** The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube? Note: The figures shown are representative.
- 256.** Consider the relationships among P, Q, R, S, and T: P is the brother of Q. S is the daughter of Q. T is the sister of S. R is the mother of Q. The following statements are made based on the relationships given above. (1) R is the grandmother of S (2) P is the uncle of S and T. (3) R has only one son. (4) Q has only one daughter. Which one of the following options is correct?
- 257.** Pen : Write :: Knife : _____ Which one of the following options maintains a similar logical relation in the above?
- 258.** A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.
- 259.** There are multiple routes to reach from node 1 to node 2, as shown in the network. The cost of travel on an edge between two nodes is given in rupees. Nodes 'a', 'b', 'c', 'd', 'e', and 'f' are toll booths. The toll price at toll booths marked 'a' and 'e' is Rs.200, and is Rs.100 for the other toll booths. Which is the cheapest route from node 1 to node 2?
- 260.** What is the missing number in the following sequence? 2, 12, 60, 240, 720, 1440, __, 0
- 261.** Based on the given statements, select the most appropriate option to solve the given question. If two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from

the first floor to the second floor of the building? Statements: (I) Each step is $\frac{3}{4}$ foot high. (II) Each step is 1 foot wide.

262. Based only on the conversation below, identify the logically correct inference: "Even if I had known that you were in the hospital, I would not have gone there to see you", Ramya told Josephine.

263. If IMAGE and FIELD are coded as FHB NJ and EMFJG respectively then, which one among the given options is the most appropriate code for BEACH ?

264. The diagram below shows a river system consisting of 7 segments, marked P, Q, R, S, T, U, and V . It splits the land into 5 zones, marked Z1, Z2, Z3, Z4, and Z5. We need to connect these zones using the least number of bridges. Out of the following options, which one is correct? Note: The figure shown is representative.

265. A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded? Note: The figures shown are representative.

266. A cube is to be cut into 8 pieces of equal size and shape. Here, each cut should be straight and it should not stop till it reaches the other end of the cube. The minimum number of such cuts required is

267. In the 4*4 array shown below, each cell of the first three rows has either a cross (X) or a number. The number in a cell represents the count of the immediate neighboring cells (left, right, top, bottom, diagonals) NOT having a cross (X). Given that the last row has no crosses (X), the sum of the four numbers to be filled in the last row is

268. In a recently conducted national entrance test, boys constituted 65% of those who appeared for the test. Girls constituted the remaining candidates and they accounted for 60% of the qualified candidates. Which one of the following is the correct logical inference based on the information provided in the above passage?

269. The corners and mid-points of the sides of a triangle are named using the distinct letters P, Q, R, S, T and U, but not necessarily in the same order. Consider the following statements :1. The line joining P and R is parallel to the line joining Q and S. 2. P is placed on the side opposite to the corner T. 3. S and U cannot be placed on the same side. Which one of the following statements is correct based on the above information?

270. A plot of land must be divided between four families. They want their individual plots to be similar in shape, not necessarily equal in area. The land has equally spaced poles, marked as dots in the below figure. Two ropes, R1 and R2, are already present and cannot be moved. What is the least number of additional straight ropes needed to create the desired plots? A single rope can pass through three poles that are aligned in a straight line.

271. A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given option for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

272. Six students P, Q, R, S, T and U, with distinct heights, compare their heights and make the following observations: Observation I : S is taller than R. Observation II : Q is the shortest of all. Observation III : U is taller than only one student. IV : T is taller than S but is not the tallest. The number of students that are taller than R is the same as the number of students shorter than _____.

273. Three of the five students allocated to a hostel put in special requests to the warden. Given the floor plan of the vacant rooms, select the allocation plan that will accommodate all their requests. Request by X: Due to pollen allergy, I want to avoid a wing next to the garden. Request by Y: I want to live as far from the washrooms as possible, since I am very sensitive to smell. Request by Z: I believe in Vaastu and so want to stay in the South-west wing. The shaded rooms are already occupied. WR is washroom.

274. The given statement is followed by some courses of action. Assuming the statement to be true, decide the correct option. Statement: There has been a significant drop in the water level in the lakes supplying water to the city. Course of action: (I) The water supply authority should impose a partial cut in supply to tackle the situation. (II) The government should appeal to all the residents through mass media for minimal use of water. (III) The government should ban the water supply in lower areas.

275. Let p_1 and p_2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of p_1 and p_2 ?

276. The average marks obtained by a class in an examination were calculated as 30.8 . However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42 . After correcting the marks, the average becomes 31.4. How many students does the class have?

277. Two wizards try to create a spell using all the four elements, water, air, fire, and earth. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combinations of elements, they conclude that the spell does not work. How many attempts does each wizard make before coming to this conclusion, independently?

278. In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is $\frac{1}{4}$ th of the number of students who like other branches. The number of students who like both their core and other branches is 500. The number of students who like their core branches is

279. In the sequence 6, 9, 14, x, 30, 41, a possible value of x is

280. Consider the following sample of numbers: 9, 18, 11, 14, 15, 17, 10, 69, 11, 13. The median of the sample is

281. The number of coins of ₹1, ₹5, and ₹10 denominations that a person has are in the ratio 5:3:13. Of the total amount, the percentage of money in ₹5 coins is

282. If two distinct non-zero real variables x and y are such that $(x+y)$ is proportional to $(x-y)$ then the value of x/y

283. Let r be a root of the equation $x^2 + 2x + 6 = 0$. Then the value of the expression $(r + 2)(r + 3)(r + 4)(r + 5)$ is

284. If $(x - 1/2)^2 - (x - 3/2)^2 = x + 2$, then the value of x is :

285. If θ is the angle, in degrees, between the longest diagonal of the cube and any one of the edges of the cube, then $\cos\theta =$ _____

286. The ratio of boys to girls in a class is 7 to 3. Among the options below, an acceptable value for the total number of students in the class is :

287. Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs 150 more. The cost of the gift was Rs. _____.

288. Two cars start at the same time from the same location and go in the same direction. The speed of the first car is 50 Km/h and the speed of the second car is 60 Km/h. The number of hours it takes for the distance between the two cars to be 20 km is _____.

289. What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

290. The area of a square is d . What is the area of the circle which has the diagonal of the square as its diameter?

291. Among 150 faculty members in an institute, 55 are connected with each other through Facebook® and 85 are connected through WhatsApp®. 30 faculty members do not have Facebook® or WhatsApp® accounts. The number of faculty members connected only through Facebook® accounts is _____.

292. In a 2×4 rectangle grid shown below, each cell is a rectangle. How many rectangles can be observed in the grid?

293. A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.

294. Given set $A = \{2, 3, 4, 5\}$ and Set $B = \{11, 12, 13, 14, 15\}$, two numbers are randomly selected, one from each set. What is probability that the sum of the two numbers equals 16?

295. In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR. What is the ratio of the area of the triangle PST to the area of the trapezium SQRT? Note: The figure shown is representative.

296. A fair six-faced dice, with the faces labelled '1', '2', '3', '4', '5', and '6', is rolled thrice. What is the probability of rolling '6' exactly once?

297. A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. The order in which the scoops are purchased is inconsequential. If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?

298. A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of

299. A rectangular paper sheet of dimensions $54 \text{ cm} \times 4 \text{ cm}$ is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken. Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

300. A box contains five balls of same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball. First ball is drawn. What is the probability of getting an orange ball in the next draw?

Programming and Data Structure

1. Consider the string abbccddeee. Each letter in the string must be assigned a binary code satisfying the following properties: For any two letters, the code assigned to one letter must not be a prefix of the code assigned to the other letter. For any two letters of the same frequency, the letter which occurs earlier in the dictionary order is assigned a code whose length is at most the length of the code assigned to the other letter. Among the set of all binary code assignments which satisfy the above two properties, what is the minimum length of the encoded string?
2. Define R_n to be the maximum amount earned by cutting a rod of length n meters into one or more pieces of integer length and selling them. For $i > 0$, let $p[i]$ denote the selling price of a rod whose length is i meters. Consider the array of prices: $p[1]=1, p[2]=5, p[3]=8, p[4]=9, p[5]=10, p[6]=17, p[7]=18$. Which of the following statements is/are correct about R_7 ?
3. Huffman tree is constructed for the following data: $\{A, B, C, D, E\}$ with frequency $\{0.17, 0.11, 0.24, 0.33$ and $0.15\}$ respectively. 100 00 01101 is decoded as
4. Consider the weights and values of items listed below. Note that there is only one unit of each item. The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt} . A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy} . The value of $V_{opt} - V_{greedy}$ is _____.
5. Suppose P, Q, R, S, T are sorted sequences having lengths 20, 24, 30, 35, 50 respectively. They are to be merged into a single sequence by merging together two sequences at a time. The number of comparisons that will be needed in the worst case by the optimal algorithm for doing this is _____.
6. Consider a job scheduling problem with 4 jobs J_1, J_2, J_3 and J_4 with corresponding deadlines: $(d_1, d_2, d_3, d_4) = (4, 2, 4, 2)$. Which of the following is not a feasible schedule without violating any job schedule?
7. Consider n jobs $J_1, J_2, \dots, J_n$ such that job J_i has execution time t_i and a non-negative integer weight w_i . The weighted mean completion time of the jobs is defined to be $\sum_{i=1}^n w_i \sum_{i=1}^n w_i T_i$, where T_i is the completion time of job J_i . Assuming that there is only one processor available, in what order must the jobs be executed in order to minimize the weighted mean completion time of the jobs?
8. Suppose the letters a, b, c, d, e, f have probabilities $1/2, 1/4, 1/8, 1/16, 1/32, 1/32$ respectively. What is the average length of the Huffman code for the letters a, b, c, d, e, f ?
9. The characters a to h have the set of frequencies based on the first 8 Fibonacci numbers as follows: $a:1, b:1, c:2, d:3, e:5, f:8, g:13, h:21$. A Huffman code is used to represent the characters. What is the sequence of characters corresponding to the following code? 110111100111010
10. The number of distinct minimum-weight spanning trees of the following graph is _____
11. Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree in G has even weight. Which of the following statements is/are TRUE for every

such graph G ?

- 12.** The number of spanning trees in a complete graph of 4 vertices labelled A, B, C, and D is _____
- 13.** Consider a simple undirected weighted graph G , all of whose edge weights are distinct. Which of the following statements about the minimum spanning trees of G is/are TRUE?
- 14.** Let G be a connected undirected weighted graph. Consider the following two statements. S1: There exists a minimum weight edge in G which is present in every minimum spanning tree of G . S2: If every edge in G has distinct weight, then G has a unique minimum spanning tree.
- 15.** Let $G=(V,E)$ be a weighted undirected graph and let T be a Minimum Spanning Tree (MST) of G maintained using adjacency lists. Suppose a new weighed edge $(u,v) \in V \times V$ is added to G . The worst case time complexity of determining if T is still an MST of the resultant graph is _____
- 16.** Consider a graph $G=(V,E)$, where $V=\{v_1, v_2, \dots, v_{100}\}$, $E=\{(v_i, v_j) \mid 1 \leq i < j \leq 100\}$, and weight of the edge (v_i, v_j) is $j-i$. The weight of minimum spanning tree of G is _____
- 17.** Let G be any connection, weighted, undirected graph:
I. G has a unique minimum spanning tree if no two edges of G have the same weight.
II. G has a unique minimum spanning tree if, for every cut of G , there is a unique minimum weight edge crossing the cut. Which of the above two statements is/are TRUE?
- 18.** $G = (V,E)$ is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G . Which of the following statements about the minimum spanning trees (MSTs) of G is/are TRUE?
I. If e is the lightest edge of some cycle in G , then every MST of G includes e .
II. If e is the heaviest edge of some cycle in G , then every MST of G excludes e
- 19.** Let G be a complete undirected graph on 4 vertices, having 6 edges with weights being 1,2,3,4,5, and 6. The maximum possible weight that a minimum weight spanning tree of G can have is .
- 20.** Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are TRUE?
P: Minimum spanning tree of G does not change
Q: Shortest path between any pair of vertices does not change
- 21.** The number of spanning trees for a complete graph with seven vertices is
- 22.** Let G be a connected undirected graph of 100 vertices and 300 edges. The weight of a minimum spanning tree of G is 500. When the weight of each edge of G is increased by five, the weight of a minimum spanning tree becomes _____.
- 23.** An undirected graph $G(V, E)$ contains n ($n > 2$) nodes named $v_1, v_2, \dots, v_n$. Two nodes v_i, v_j are connected if and only if $0 < |i-j| \leq 2$. Each edge (v_i, v_j) is assigned a weight $i+j$. A sample graph with $n=4$ is shown below.
- 24.** Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm?
- 25.** For the undirected, weighted graph given below, which of the following sequences of edges represents a correct execution of Prim's algorithm to construct a Minimum Spanning Tree?

- 26.** G is a graph on n vertices and $2n-2$ edges. The edges of G can be partitioned into two edge-disjoint spanning trees. Which of the following is NOT true for G ?
- 27.** What is the largest integer m such that every simple connected graph with n vertices and n edges contains at least m different spanning trees ?
- 28.** Let w be the minimum weight among all edge weights in an undirected connected graph. Let e be a specific edge of weight w . Which of the following is FALSE?
- 29.** Let G be a weighted undirected graph and e be an edge with maximum weight in G . Suppose there is a minimum weight spanning tree in G containing the edge e . Which of the following statements is always TRUE?
- 30.** Let s and t be two vertices in a undirected graph $G=(V, E)$ having distinct positive edge weights. Let $[X, Y]$ be a partition of V such that $s \in X$ and $t \in Y$. Consider the edge e having the minimum weight amongst all those edges that have one vertex in X and one vertex in Y . Let the weight of an edge e denote the congestion on that edge. The congestion on a path is defined to be the maximum of the congestions on the edges of the path. We wish to find the path from s to t having minimum congestion. Which one of the following paths is always such a path of minimum congestion?
- 31.** An undirected graph G has n nodes. Its adjacency matrix is given by an $n \times n$ square matrix whose (i) diagonal elements are 0's and (ii) non-diagonal elements are 1's. which one of the following is TRUE?
- 32.** Using Prim's algorithm to construct a minimum spanning tree starting with node A , which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree?
- 33.** Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are TRUE? P: Minimum spanning tree of G does not change Q: Shortest path between any pair of vertices does not change
- 34.** Let $G = (V, E)$ be a simple undirected graph, and s be a particular vertex in it called the source. For $x \in V$, let $d(x)$ denote the shortest distance in G from s to x . A breadth first search (BFS) is performed starting at s . Let T be the resultant BFS tree. If (u,v) is an edge of G that is not in T , then which one of the following CANNOT be the value of $d(u)-d(v)$?
- 35.** Consider the tree arcs of a BFS traversal from a source node W in an unweighted, connected, undirected graph. The tree T formed by the tree arcs is a data structure for computing
- 36.** What is the time complexity of Bellman-Ford single-source shortest path algorithm on a complete graph of n vertices?
- 37.** Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T . Which one will be reported by Dijkstra's shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered
- 38.** Which of the following statement(s) is/are correct regarding Bellman-Ford shortest path algorithm? P. Always finds a negative weighted cycle, if one exists. Q. Finds whether any negative weighted cycle is

reachable from the source.

- 39.** Dijkstra's single source shortest path algorithm when run from vertex a in the above graph, computes the correct shortest path distance to
- 40.** Consider a weighted, undirected graph with positive edge weights and let uv be an edge in the graph. It is known that the shortest path from the source vertex s to u has weight 53 and the shortest path from s to v has weight 65. Which one of the following statements is always TRUE?
- 41.** In an unweighted, undirected connected graph, the shortest path from a node S to every other node is computed most efficiently, in terms of time complexity, by
- 42.** Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j iff either $j=i+1$ or $j=3i$. The minimum number of edges in a path in G from vertex 1 to vertex 100 is
- 43.** Suppose we run Dijkstra's single source shortest-path algorithm on the following edge-weighted directed graph with vertex P as the source. In what order do the nodes get included into the set of vertices for which the shortest path distances are finalized?
- 44.** Consider the undirected unweighted graph G . Let a breadth-first traversal of G be done starting from a node r . Let $d(r,u)$ and $d(r,v)$ be the lengths of the shortest paths from r to u and v respectively in G . If u is visited before v during the breadth-first traversal, which of the following statements is correct ?
- 45.** Let $U=\{1,2,3\}$. Let 2^U denote the powerset of U . Consider an undirected graph G whose vertex set is 2^U . For any $A,B \in 2^U$, (A,B) is an edge in G if and only if (i) $A \cap B = \emptyset$, and (ii) either $A \subseteq B$ or $B \subseteq A$. For any vertex A in G , the set of all possible orderings in which the vertices of G can be visited in a Breadth First Search (BFS) starting from A is denoted by $B(A)$. If $\emptyset$ denotes the empty set, then the cardinality of $B(\emptyset)$ is _____.
- 46.** An articulation point in a connected graph is a vertex such that removing the vertex and its incident edges disconnects the graph into two or more connected components. Let T be a DFS tree obtained by doing DFS in a connected undirected graph G . Which of the following options is/are correct?
- 47.** G is an undirected graph with vertex set $\{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ and edge set $\{v_1v_2, v_1v_3, v_1v_4, v_2v_4, v_2v_5, v_3v_4, v_4v_5, v_4v_6, v_5v_6, v_6v_7\}$. A breadth first search of the graph is performed with v_1 as the root node. Which of the following is a tree edge?
- 48.** Let G be a simple undirected graph. Let TD be a depth first search tree of G . Let TB be a breadth first search tree of G . Consider the following statements. (I) No edge of G is a cross edge with respect to TD . (A cross edge in G is between two nodes neither of which is an ancestor of the other in TD .) (II) For every edge (u,v) of G , if u is at depth i and v is at depth j in TB , then $|i-j|=1$.
- 49.** The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?
- 50.** Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n -th vertex in this BFS traversal, then the maximum possible

value of n is _____ .

- 51.** Let $G = (V, E)$ be a simple undirected graph, and s be a particular vertex in it called the source. For $x \in V$, let $d(x)$ denote the shortest distance in G from s to x . A breadth first search (BFS) is performed starting at s . Let T be the resultant BFS tree. If (u, v) is an edge of G that is not in T , then which one of the following CANNOT be the value of $d(u) - d(v)$?
- 52.** Suppose depth first search is executed on the graph below starting at some unknown vertex. Assume that a recursive call to visit a vertex is made only after first checking that the vertex has not been visited earlier. Then the maximum possible recursion depth (including the initial call) is _____.
- 53.** Let G be a graph with n vertices and m edges. What is the tightest upper bound on the running time of Depth First Search on G , when G is represented as an adjacency matrix?
- 54.** The most efficient algorithm for finding the number of connected components in an undirected graph on n vertices and m edges has time complexity
- 55.** A depth-first search is performed on a directed acyclic graph. Let $d[u]$ denote the time at which vertex u is visited for the first time and $f[u]$ the time at which the DFS call to the vertex u terminates. Which of the following statements is always TRUE for all edges (u, v) in the graph ?
- 56.** Let T be a depth first search tree in a undirected graph G . Vertices u and v are leaves of this tree T . The degrees of both u and v in G are at least 2.
- 57.** In a depth-first traversal of a graph G with n vertices, k edges are marked as tree edges. The number of connected components in G is
- 58.** Let G be an undirected graph. Consider a depth-first traversal of G , and let T be the resulting depth-first search tree. Let u be a vertex in G and let v be the first new (unvisited) vertex visited after visiting u in the traversal. Which of the following statement is always true?
- 59.** Consider the undirected unweighted graph G . Let a breadth-first traversal of G be done starting from a node r . Let $d(r, u)$ and $d(r, v)$ be the lengths of the shortest paths from r to u and v respectively in G . If u is visited before v during the breadth-first traversal, which of the following statements is correct ?
- 60.** Consider product of three matrices M_1 , M_2 and M_3 having w rows and x columns, x rows and y columns, and y rows and z columns. Under what condition will it take less time to compute the product as $(M_1 M_2) M_3$ than to compute $M_1 (M_2 M_3)$?
- 61.** Assume that multiplying a matrix G_1 of dimension $p \times q$ with another matrix G_2 of dimension $p \times q$ requires pqr scalar multiplications. Computing the product of n matrices $G_1 G_2 G_3 \dots G_n$ can be done by parenthesizing in different ways. Define $G_i G_{i+1}$ as an explicitly computed pair for a given paranthesization if they are directly multiplied. For example, in the matrix multiplication chain $G_1 G_2 G_3 G_4 G_5 G_6$ using parenthesization $(G_1 (G_2 G_3)) (G_4 (G_5 G_6))$, $G_2 G_3$ and $G_5 G_6$ are the only explicitly computed pairs. Consider a matrix multiplication chain $F_1 F_2 F_3 F_4 F_5$, where matrices F_1, F_2, F_3, F_4 and F_5 are of dimensions $2 \times 25, 25 \times 3, 3 \times 16, 16 \times 1$ and 1×1000 , respectively. In the paranthesization of $F_1 F_2 F_3 F_4 F_5$ that minimizes the total number of scalar multiplications, the explicitly computed pairs is/are

- 62.** Let A_1, A_2, A_3 , and A_4 be four matrices of dimensions 10×5 , 5×20 , 20×10 , and 10×5 , respectively. The minimum number of scalar multiplications required to find the product $A_1 A_2 A_3 A_4$ using the basic matrix multiplication method is _____.
- 63.** The Floyd-Warshall algorithm for all-pair shortest paths computation is based on
- 64.** A Young tableau is a 2D array of integers increasing from left to right and from top to bottom. Any unfilled entries are marked with ∞ , and hence there cannot be any entry to the right of, or below a ∞ . The following Young tableau consists of unique entries. When an element is removed from a Young tableau, other elements should be moved into its place so that the resulting table is still a Young tableau (unfilled entries may be filled in with a ∞). The minimum number of entries (other than 1) to be shifted, to remove 1 from the given Young tableau is _____.
- 65.** Consider two strings $A = \text{"qpqrr"}$ and $B = \text{"pqprrqp"}$. Let x be the length of the longest common subsequence (not necessarily contiguous) between A and B and let y be the number of such longest common subsequences between A and B . Then $x + 10y = \underline{\hspace{1cm}}$.
- 66.** Four matrices M_1, M_2, M_3 and M_4 are dimensions $p \times q, q \times r, r \times s$ and $s \times t$ respectively can be multiplied in several ways with different number of total scalar multiplications. For example When multiplied as $((M_1 \blacksquare \times M_2 \blacksquare) \times (M_3 \blacksquare \times M_4 \blacksquare))$ the total number of scalar multiplications is $pqr + rst + prt$. When multiplied as $((M_1 \blacksquare \times M_2 \blacksquare) \times M_3 \blacksquare) \times M_4 \blacksquare$, the total number of scalar multiplications is $pqr + prs + pst$. If $p=10, q=100, r=20, s=5$ and $t=80$, then the minimum number of scalar multiplications needed is
- 67.** A sub-sequence of a given sequence is just the given sequence with some elements (possibly none or all) left out. We are given two sequences $X[m]$ and $Y[n]$ of lengths m and n , respectively, with indexes of X and Y starting from 0. We wish to find the length of the longest common sub-sequence (LCS) of $x[m]$ and $Y[n]$ as $l(m,n)$, where an incomplete recursive definition for the function $l(i,j)$ to compute the length of the LCS of $x[m]$ and $Y[n]$ is given below: The values of $l(i,j)$ could be obtained by dynamic programming based on the correct recursive definition of $l(i,j)$ of the form given above, using an array $L[M,N]$, where $M = m+1$ and $N = n+1$, such that $L[i,j] = l(i,j)$. Which one of the following statements would be TRUE regarding the dynamic programming solution for the recursive definition of $l(i,j)$?
- 68.** A sub-sequence of a given sequence is just the given sequence with some elements (possibly none or all) left out. We are given two sequences $X[m]$ and $Y[n]$ of lengths m and n , respectively, with indexes of X and Y starting from 0. We wish to find the length of the longest common sub-sequence (LCS) of $x[m]$ and $Y[n]$ as $l(m,n)$, where an incomplete recursive definition for the function $l(i,j)$ to compute the length of the LCS of $x[m]$ and $Y[n]$ is given below:
- 69.** Suppose that a robot is placed on the Cartesian plane. At each step it is allowed to move either one unit up or one unit right, i.e., if it is at (i,j) then it can move to either $(i+1,j)$ or $(i,j+1)$. How many distinct paths are there for the robot to reach the point $(10,10)$ starting from the initial position $(0,0)$?
- 70.** If a node has K children in B tree, then the node contains exactly _____ keys.
- 71.** A B-tree of order 4 is built from scratch by 10 successive insertions. What is the maximum number of node splitting operations that may take place?

- 72.** Choose the correct alternatives (more than one may be correct) and write the corresponding letters only: A 2-3 tree is such that a. All internal nodes have either 2 or 3 children b. All paths from root to the leaves have the same length. The number of internal nodes of a 2-3 tree having 9 leaves could be
- 73.** In a B+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases?
- 74.** B+ Trees are considered BALANCED because
- 75.** With reference to the B+ tree index of order 1 shown below, the minimum number of nodes (including the Root node) that must be fetched in order to satisfy the following query: "Get all records with a search key greater than or equal to 7 and less than 15" is _____.
- 76.** Consider a B+-tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?
- 77.** The following key values are inserted into a B+ tree in which order of the internal nodes is 3, and that of the leaf nodes is 2, in the sequence given below. The order of internal nodes is the maximum number of tree pointers in each node, and the order of leaf nodes is the maximum number of data items that can be stored in it. The B+ - tree is initially empty. 10, 3, 6, 8, 4, 2, 1 The maximum number of times leaf nodes would get split up as a result of these insertions is
- 78.** Consider the B+ tree in the adjoining figure, where each node has at most two keys and three links. Keys K15 and then K25 are inserted into this tree in that order. Now the key K50 is deleted from the B+ tree resulting after the two insertions made earlier. Consider the following statements about the B+ tree resulting after this deletion. i. The height of the tree remains the same. ii. The node (disregarding the links) is present in the tree. iii. The root node remains unchanged (disregarding the links) 1 Which one of the following options is true ?
- 79.** Consider a binary min-heap containing 105 distinct elements. Let k be the index (in the underlying array) of the maximum element stored in the heap. The number of possible values of k is
- 80.** An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?
- 81.** Let A be a priority queue for maintaining a set of elements. Suppose A is implemented using a max-heap data structure. The operation Extract-Max(A) extracts and deletes the maximum element from A. The operation Insert(A, key) inserts a new element key in A. The properties of a max-heap are preserved at the end of each of these operations. When A contains n elements, which one of the following statements about the worst case running time of these two operations is TRUE?
- 82.** Let H be a binary min-heap consisting of n elements implemented as an array. What is the worst case time complexity of an optimal algorithm to find the maximum element in H?
- 83.** Consider the array representation of a binary min-heap containing 1023 elements. The minimum number of comparisons required to find the maximum in the heap is _____.

- 84.** Consider the following statements:
 I. The smallest element in a max-heap is always at a leaf node.
 II. The second largest element in a max-heap is always a child of the root node.
 III. A max-heap can be constructed from a binary search tree in $\Theta(n)$ time.
 IV. A binary search tree can be constructed from a max-heap in $\Theta(n)$ time.
 Which of the above statements is/are TRUE?
- 85.** Given a binary-max heap. The elements are stored in an array as 25,14,16,13,10,8,12. What is the content of the array after two delete operations?
- 86.** The number of possible min-heaps containing each value from {1, 2, 3, 4, 5, 6, 7} exactly once is _____.
- 87.** A complete binary min-heap is made by including each integer in [1,1023] exactly once. The depth of a node in the heap is the length of the path from the root of the heap to that node. Thus, the root is at depth 0. The maximum depth at which integer 9 can appear is _____.
- 88.** An operator delete(i) for a binary heap data structure is to be designed to delete the item in the i-th node. Assume that the heap is implemented in an array and i refers to the i-th index of the array. If the heap tree has depth d (number of edges on the path from the root to the farthest leaf), then what is the time complexity to re-fix the heap efficiently after the removal of the element?
- 89.** Consider the following array of elements. (89,19,50,17,12,15,2,5,7,11,6,9,100) The minimum number of interchanges needed to convert it into a max-heap is _____.
- 90.** Consider a complete binary tree where the left and the right subtrees of the root are max-heaps. The lower bound for the number of operations to convert the tree to a heap is _____.
- 91.** A priority queue is implemented as a Max-Heap. Initially, it has 5 elements. The level-order traversal of the heap is: 10, 8, 5, 3, 2. Two new elements 1 and 7 are inserted into the heap in that order. The level-order traversal of the heap after the insertion of the elements is: _____.
- 92.** A data structure is required for storing a set of integers such that each of the following operations can be done in $O(\log n)$ time, where n is the number of elements in the set.
 1. Deletion of the smallest element.
 2. Insertion of an element if it is not already present in the set
 Which of the following data structures can be used for this purpose?
- 93.** A max-heap is a heap where the value of each parent is greater than or equal to the value of its children. Which of the following is a max-heap?
- 94.** We have a binary heap on n elements and wish to insert n more elements (not necessarily one after another) into this heap. The total time required for this is _____.
- 95.** Consider the process of inserting an element into a Max Heap, where the Max Heap is represented by an array. Suppose we perform a binary search on the path from the new leaf to the root to find the position for the newly inserted element, the number of comparisons performed is: _____.
- 96.** A 3-ary max heap is like a binary max heap, but instead of 2 children, nodes have 3 children. A 3-ary heap can be represented by an array as follows: The root is stored in the first location, a[0], nodes in the next level, from left to right, is stored from a[1] to a[3]. The nodes from the second level of the tree from left

to right are stored from $a[4]$ location onward. An item x can be inserted into a 3-ary heap containing n items by placing x in the location $a[n]$ and pushing it up the tree to satisfy the heap property. Suppose the elements 7, 2, 10 and 4 are inserted, in that order, into the valid 3-ary max heap found in the above question, Which one of the following is the sequence of items in the array representing the resultant heap?

97. A Priority-Queue is implemented as a Max-Heap. Initially, it has 5 elements. The level-order traversal of the heap is given below: 10, 8, 5, 3, 2. Two new elements '1' and '7' are inserted in the heap in that order. The level-order traversal of the heap after the insertion of the elements is:

98. An array of integers of size n can be converted into a heap by adjusting the heaps rooted at each internal node of the complete binary tree starting at the node $\lfloor (n-1)/2 \rfloor$, and doing this adjustment up to the root node (root node is at index 0) in the order $\lfloor (n-1)/2 \rfloor, \lfloor (n-3)/2 \rfloor, \dots, 0$. The time required to construct a heap in this manner is

99. A data structure is required for storing a set of integers such that each of the following operations can be done in $O(\log n)$ time, where n is the number of elements in the set. 1. Deletion of the smallest element. 2. Insertion of an element if it is not already present in the set. Which of the following data structures can be used for this purpose?

100. In a heap with n elements with the smallest element at the root, the 7th smallest element can be found in time

101. The minimum number of interchanges needed to convert the array into a max-heap is 89, 19, 40, 17, 12, 10, 2, 5, 7, 11, 6, 9, 70

102. Of the following, which best approximates the ratio of the number of nonterminal nodes in the total number of nodes in a complete K -ary tree of depth N ?

103. Consider any array representation of an n element binary heap where the elements are stored from index 1 to index n of the array. For the element stored at index i of the array ($i \leq n$), the index of the parent is

104. A complete n -ary tree is a tree in which each node has n children or no children. Let I be the number of internal nodes and L be the number of leaves in a complete n -ary tree. If $L = 41$, and $I = 10$, what is the value of n ?

105. In a complete k -ary tree, every internal node has exactly k children. The number of leaves in such a tree with n internal nodes is:

106. A complete n -ary tree is one in which every node has 0 or n sons. If x is the number of internal nodes of a complete n -ary tree, the number of leaves in it is given by

107. The number of leaf nodes in a rooted tree of n nodes, with each node having 0 or 3 children is

108. An algorithm has to store several keys generated by an adversary in a hash table. The adversary is malicious who tries to maximize the number of collisions. Let k be the number of keys, m be the number of slots in the hash table, and $k > m$. Which one of the following is the best hashing strategy to counteract the adversary?

109. Suppose we are given n keys, m hash table slots, and two simple uniform hash functions h_1 and h_2 . Further suppose our hashing scheme uses h_1 for the odd keys and h_2 for the even keys. What is the expected number of keys in a slot?

110. Consider a dynamic hashing approach for 4-bit integer keys: 1. There is a main hash table of size 4. 2. The 2 least significant bits of a key is used to index into the main hash table. 3. Initially, the main hash table entries are empty. 4. Thereafter, when more keys are hashed into it, to resolve collisions, the set of all keys corresponding to a main hash table entry is organized as a binary tree that grows on demand. 5. First, the 3rd least significant bit is used to divide the keys into left and right subtrees. 6. To resolve more collisions, each node of the binary tree is further sub-divided into left and right subtrees based on the 4th least significant bit. 7. A split is done only if it is needed, i.e., only when there is a collision. Consider the following state of the hash table. Which of the following sequences of key insertions can cause the above state of the hash table (assume the keys are in decimal notation)?

111. Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 23$, and the secondary hash function is $h_2(k) = 1 + (k \bmod 19)$. Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value $k=90$ is _____.

112. A hash table with 10 buckets with one slot per bucket is depicted here. The symbols, S1 to S7 are initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is:

113. A Hash Function f defined as $f(\text{key}) = \text{key} \bmod 7$. With linear probing while inserting the keys 37, 38, 72, 48, 98, 11, 56 into a table indexed from 0, in which location key 11 will be stored (Count table index 0 as 0th location)?

114. A hash table with ten buckets with one slot per bucket is shown in the following figure. The symbols S1 to S7 initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is

115. Given a hash table T with 25 slots that stores 2000 elements, the load factor α for T is _____.

116. Consider a 13 element hash table for which $f(\text{key}) = \text{key} \bmod 13$ is used with integer keys. Assuming linear probing is used for collision resolution, at which location would the key 103 be inserted, if the keys 661, 182, 24 and 103 are inserted in that order?

117. Consider a hash table with 9 slots. The hash function is $h(k) = k \bmod 9$. The collisions are resolved by chaining. The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10. The maximum, minimum, and average chain lengths in the hash table, respectively, are

118. The keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function $h(k) = k \bmod 10$ and linear probing. What is the resultant hash table?

119. Consider a hash table of size 11 that uses open addressing with linear probing. Let $h(k) = k \bmod 11$ be the hash function used. A sequence of records with keys 43 36 92 87 11 4 71 13 14 is inserted into an

initially empty hash table, the bins of which are indexed from zero to ten. What is the index of the bin into which the last record is inserted?

120. Consider a hash function that distributes keys uniformly. The hash table size is 20. After hashing of how many keys will the probability that any new key hashed collides with an existing one exceed 0.5.

121. A hash table contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is $\text{key} \% 10$. If the values 43, 165, 62, 123, 142 are inserted in the table, in what location would the key value 142 be inserted?

122. Consider a hash table of size seven, with starting index zero, and a hash function $(3x + 4) \bmod 7$. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Note that '-' denotes an empty location in the table.

123. An advantage of chained hash table (external hashing) over the open addressing scheme is

124. Let A be an array containing integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in non-decreasing order. The distance of the array [2,5,3,1,4,2,6] is _____

125. What is the worst-case number of arithmetic operations performed by recursive binary search on a sorted array of size n?

126. Let P be an array containing n integers. Let t be the lowest upper bound on the number of comparisons of the array elements, required to find the minimum and maximum values in an arbitrary array of n elements. Which one of the following choices is correct?

127. If an array A contains the items 10, 4, 7, 23, 67, 12 and 5 in that order, what will be the resultant array A after third pass of insertion sort?

128. Consider a 2-dimensional array x with 10 rows and 4 columns, with each element storing a value equivalent to the product of row number and column number. The array is stored in row-major format. If the first element $x[0][0]$ occupies the memory location with address 1000 and each element occupies only one memory location, which all locations (in decimal) will be holding a value of 10?

129. Let A be an array of 31 numbers consisting of sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i that $A[i]$ is 1 by probing the minimum numbers of locations in A. The worst case number of probes performed by an optimal algorithm is _____.

130. The average number of key comparisons required for a successful search for sequential search on n items is

131. A frame buffer array is addressed in row major order for a monitor with pixel locations starting from (0,0) and ending with (100,100). What is address of the pixel(6,10)? Assume one bit storage per pixel and starting pixel location is at 0.

132. In an array of $2N$ elements that is both 2-ordered and 3-ordered, what is the maximum number of positions that an element can be from its position if the array were 1-ordered?

133. An array of n numbers is given, where n is an even number. The maximum as well as the minimum of these n numbers needs to be determined. Which of the following is TRUE about the number of comparisons needed?

134. Let `SLLdel` be a function that deletes a node in a singly-linked list given a pointer to the node and a pointer to the head of the list. Similarly, let `DLLdel` be another function that deletes a node in a doubly-linked list given a pointer to the node and a pointer to the head of the list. Let n denote the number of nodes in each of the linked lists. Which one of the following choices is TRUE about the worst-case time complexity of `SLLdel` and `DLLdel`?

135. Consider a singly linked list of the form where F is a pointer to the first element in the linked list and L is the pointer to the last element in the list. The time of which of the following operations depends on the length of the list?

136. In a doubly linked list the number of pointers affected for an insertion operation will be

137. Given two statements -Insertion of an element should be done at the last node of the circular list
,Deletion of an element should be done at the last node of the circular list

138. An unordered list contains n distinct elements. The number of comparisons to find an element in this list that is neither maximum nor minimum is

139. An algorithm performs $(\log N)^{1/2}$ find operations, N insert operations, $(\log N)^{1/2}$ delete operations, and $(\log N)^{1/2}$ decrease-key operations on a set of data items with keys drawn from a linearly ordered set. For a delete operation, a pointer is provided to the record that must be deleted. For the decrease-key operation, a pointer is provided to the record that has its key decreased. Which one of the following data structures is the most suited for the algorithm to use, if the goal is to achieve the best total asymptotic complexity considering all the operations?

140. Consider a single linked list where F and L are pointers to the first and last elements respectively of the linked list. The time for performing which of the given operations depends on the length of the linked list?

141. The minimum number of fields with each node of doubly linked list is

142. What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be maintained in sorted order?

143. A circularly linked list is used to represent a Queue. A single variable p is used to access the Queue. To which node should p point such that both the operations `enQueue` and `deQueue` can be performed in constant time?

144. In the worst case, the number of comparisons needed to search a single linked list of length n for a given element is

145. The concatenation of two lists is to be performed on $O(1)$ time. Which of the following implementations of a list should be used?

- 146.** Which of the following statements is true?
 I. As the number of entries in a hash table increases, the number of collisions increases.
 II. Recursive programs are efficient
 III. The worst case complexity for Quicksort is $O(n^2)$
 IV. Binary search using a linear linked list is efficient
- 147.** For merging two sorted lists of sizes m and n into a sorted list of size $m+n$, we require comparisons of
- 148.** Consider the following sequence of operations on an empty stack.
`push(54); push(52); pop(); push(55); push(62); s=pop();`
 Consider the following sequence of operations on an empty queue.
`enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q=dequeue();`
 The value of $s+q$ is _____.
- 149.** The best data structure to check whether an arithmetic expression has balanced parenthesis is a:
- 150.** If the sequence of operations - push (1), push (2), pop, push (1), push (2), pop, pop, pop, push (2), pop are performed on a stack, the sequence of popped out values
- 151.** The five items: A, B, C, D, and E are pushed in a stack, one after other starting from A. The stack is popped four items and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is
- 152.** Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE with respect to this modified stack?
- 153.** Assume that the operators $+, -, \times$ are left associative and $^$ is right associative. The order of precedence (from highest to lowest) is $^, \times, +, -$. The postfix expression corresponding to the infix expression $a+b \times c-d^e^f$ is
- 154.** Stack A has the entries a, b, c (with a on top). Stack B is empty. An entry popped out of stack A can be printed immediately or pushed to stack B. An entry popped out of the stack B can be only be printed. In this arrangement, which of the following permutations of a, b, c are not possible?
- 155.** A program attempts to generate as many permutations as possible of the string, 'abcd' by pushing the characters a,b,c,d in the same order onto a stack, but it may pop off the top character at any time. Which one of the following strings CANNOT be generated using this program?
- 156.** A function f defined on stacks of integers satisfies the following properties. $f(\emptyset)=0$ and $f(\text{push}(S,i))=\max(f(S),0)+i$ for all stacks S and integers i . If a stack S contains the integers 2, -3, 2, -1, 2 in order from bottom to top, what is $f(S)$?
- 157.** Assume that the operators $+, -, \times$, are left associative and $^$ is right associative. the order of precedence (from highest to lowest) is $^, \times, +, -$. The postfix expression corresponding to the infix expression $a + b \times c-d^e^f$ is
- 158.** The best data structure to check whether an arithmetic expression has balanced parentheses is a
- 159.** A single array $A[1...MAXSIZE]$ is used to implement two stacks. The two stacks grow from opposite ends of the array. Variables $top1$ and $top2$ ($top1 < top2$) point to the location of the topmost element in each of the stacks. If the space is to be used efficiently, the condition for "stack full" is

160. Let S be a stack of size $n \geq 1$. Starting with the empty stack, suppose we push the first n natural numbers in sequence, and then perform n pop operations. Assume that Push and Pop operation take X seconds each, and Y seconds elapse between the end of the one such stack operation and the start of the next operation. For $m \geq 1$, define the stack-life of m as the time elapsed from the end of Push (m) to the start of the pop operation that removes m from S . The average stack-life of an element of this stack is

161. The postfix expression for the infix expression $A+B*(C+D)/F+D*E$ is:

162. A circular queue has been implemented using a single linked list where each node consists of a value and a single pointer pointing to the next node. We maintain exactly two external pointers FRONT and REAR pointing to the front node and the rear node of the queue, respectively. Which of the following statements is/are CORRECT for such a circular queue, so that insertion and deletion operation can be performed in $O(1)$ time ?
I. Next pointer of front node points to the rear node.
II. Next pointer of rear node points to the front node.

163. Consider a standard Circular Queue implementation (which has the same condition for Queue Full and Queue Empty) whose size is 11 and the elements of the queue are $q[0], q[1], \dots, q[10]$. The front and rear pointers are initialized to point at $q[2]$. In which position will the ninth element be added?

164. A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT (n refers to the number of items in the queue)?

165. Suppose a circular queue of capacity $(n-1)$ elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variables, respectively. Initially, $REAR = FRONT = 0$. The conditions to detect queue full and queue empty are

166. Suppose you are given an implementation of a queue of integers. The operations that can be performed on the queue are:
i. $isEmpty(Q)$ returns true if the queue is empty, false otherwise.
ii. $delete(Q)$ deletes the element at the front of the queue and returns its value.
iii. $insert(Q, i)$ inserts the integer i at the rear of the queue.
Consider the following function

167. A priority queue Q is used to implement a stack that stores characters. PUSH (C) is implemented as INSERT (Q, C, K) where K is an appropriate integer key chosen by the implementation. POP is implemented as DELETEMIN (Q). For a sequence of operations, the keys chosen are in

168. Consider the following statements:
(i). First-in-first out types of computations are efficiently supported by STACKS.
(ii). Implementing LISTS on linked lists is more efficient than implementing LISTS on an array for almost all the basic LIST operations.
(iii). Implementing QUEUES on a circular array is more efficient than implementing QUEUES on a linear array with two indices.
(iv). Last-in-first-out type of computations are efficiently supported by QUEUES.

169. Consider a complete binary tree with 7 nodes. Let A denote the set of first 3 elements obtained by performing Breadth-First Search (BFS) starting from the root. Let B denote the set of first 3 elements obtained by performing Depth-First Search (DFS) starting from the root. The value of $|A-B|$ is

170. Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.) Suppose two leaves a and b of T are chosen uniformly and independently at random. The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places) _____.

171. The postorder traversal of a binary tree is 8,9,6,7,4,5,2,3,1. The inorder traversal of the same tree is 8,6,9,4,7,2,5,1,3. The height of a tree is the length of the longest path from the root to any leaf. The height of the binary tree above is _____.

172. Consider the following New-order strategy for traversing a binary tree: Visit the root; Visit the right sub tree using New-order; Visit the left sub tree using New-order; The New-order traversal of the expression tree corresponding to the reverse polish expression $3\ 4\ *\ 5\ -\ 2\ ^\ 6\ 7\ *\ 1\ +\ -$ is given by:

173. The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 5 are

174. Consider the pseudocode given below. The function `DoSomething ()` takes as argument a pointer to the root of an arbitrary tree represented by the leftMostChild-rightSibling representation. Each node of the tree is of type `TreeNode`. When the pointer to the root of a tree is passed as the argument to `DoSomething`, the value returned by the function corresponds to the

175. In a binary tree with n nodes, every node has an odd number of descendants. Every node is considered to be its own descendant. What is the number of nodes in the tree that have exactly one child?

176. The inorder and preorder traversal of a binary tree are $d\ b\ e\ a\ f\ c\ g$ and $a\ b\ d\ e\ c\ f\ g$, respectively. The postorder traversal of the binary tree is:

177. The height of a binary tree is the maximum number of edges in any root to leaf path. The maximum number of nodes in a binary tree of height h is:

178. An array X of n distinct integers is interpreted as a complete binary tree. The index of the first element of the array is 0. If only the root node does not satisfy the heap property, the algorithm to convert the complete binary tree into a heap has the best asymptotic time complexity of

179. Consider the label sequences obtained by the following pairs of traversals on a labeled binary tree. Which of these pairs identify a tree uniquely? (i) preorder and postorder (ii) inorder and postorder (iii) preorder and inorder (iv) level order and postorder

180. Let `LASTPOST`, `LASTIN` and `LASTPRE` denote the last vertex visited in a postorder, inorder and preorder traversal respectively, of a complete binary tree. Which of the following is always true?

181. Suppose a binary search tree with 1000 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with 0, the 3rd largest element of the tree is stored at index _____

182. A binary search tree T contains n distinct elements. What is the time complexity of picking an element in T that is smaller than the maximum element in T ?

- 183.** The number of ways in which the numbers 1,2,3,4,5,6,7 can be inserted in an empty binary search tree, such that the resulting tree has height 6, is _____.
- 184.** While inserting the elements 71, 65, 84, 69, 67, 83 in an empty binary search tree (BST) in the sequence shown, the element in the lowest level is
- 185.** What are the worst-case complexities of insertion and deletion of a key in a binary search tree?
- 186.** We are given a set of n distinct elements and an unlabeled binary tree with n nodes. In how many ways can we populate the tree with the given set so that it becomes a binary search tree?
- 187.** The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)?
- 188.** Suppose the numbers 7,5,1,8,3,6,0,9,4,2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the usual ordering on natural numbers. What is the in-order traversal sequence of the resultant tree?
- 189.** You are given the postorder traversal, P , of a binary search tree on the n elements $1, 2, \dots, n$. You have to determine the unique binary search tree that has P as its postorder traversal. What is the time complexity of the most efficient algorithm for doing this?
- 190.** When searching for the key value 60 in a binary search tree, nodes containing the key values 10, 20, 40, 50, 70, 80, 90 are traversed, not necessarily in the order given. How many different orders are possible in which these key values can occur on the search path from the root to the node containing the value 60?
- 191.** A binary search tree contains the numbers 1,2,3,4,5,6,7,8. When the tree is traversed in pre-order and the values in each node printed out, the sequence of values obtained is 5,3,1,2,4,6,8,7. If the tree is traversed in post-order, the sequence obtained would be
- 192.** The numbers $1, 2, \dots, n$ are inserted in a binary search tree in some order. In the resulting tree, the right subtree of the root contains p nodes. The first number to be inserted in the tree must be
- 193.** The following numbers are inserted into an empty binary search tree in the given order: 10, 1, 3, 5, 15, 12, 16. What is the height of the binary search tree (the height is the maximum distance of a leaf node from the root)?
- 194.** Suppose the numbers 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are inserted in that order into an initially empty binary search tree. The binary search tree uses the usual ordering on natural numbers. What is the inorder transversal sequence of the resultant tree ?
- 195.** A binary search tree contains the value 1,2,3,4,5,6,7,8. The tree is traversed in pre-order and the values are printed out. Which of the following sequences is a valid output?
- 196.** The minimum height of an AVL tree with n nodes is

197. In a balanced binary search tree with n elements, what is the worst case time complexity of reporting all elements in range $[a,b]$? Assume that the number of reported elements is k .

198. What is the worst case time complexity of inserting n^2 elements into an AVL-tree with n elements initially?

199. Access time of the symbolic table will be logarithmic if it is implemented by

200. Suppose we have a balanced binary search tree T holding n numbers. We are given two numbers L and H and wish to sum up all the numbers in T that lie between L and H . Suppose there are m such numbers in T . If the tightest upper bound on the time to compute the sum is $O(n \log b n + m \log d n)$, the value of $a + 10b + 100c + 1000d$ is _____.

201. The number of rotations required to insert a sequence of elements 9, 6, 5, 8, 7, 10 into an empty AVL tree is?

202. The worst case running time to search for an element in a balanced binary search tree with n^2 elements is

203. A program takes as input a balanced binary search tree with n leaf nodes and computes the value of a function $g(x)$ for each node x . If the cost of computing $g(x)$ is $\min\{\text{no. of leaf-nodes in left-subtree of } x, \text{ no. of leaf-nodes in right-subtree of } x\}$ then the worst-case time complexity of the program is

204. A weight-balanced tree is a binary tree in which for each node, the number of nodes in the left subtree is at least half and at most twice the number of nodes in the right subtree. The maximum possible height (number of nodes on the path from the root to the furthest leaf) of such a tree on n nodes is best described by which of the following ?

205. In the balanced binary tree in the below figure, how many nodes will become unbalanced when a node is inserted as a child of the node g ?

206. Given an integer array of size N , we want to check if the array is sorted (in either ascending or descending order). An algorithm solves this problem by making a single pass through the array and comparing each element of the array only with its adjacent elements. The worst-case time complexity of this algorithm is

207. There are n unsorted arrays: $A_1, A_2, \dots, A_n$. Assume that n is odd. Each of $A_1, A_2, \dots, A_n$ contains n distinct elements. There are no common elements between any two arrays. The worst-case Asymptotic Notation of computing the median of the medians of $A_1, A_2, \dots, A_n$ is _____.

208. In an adjacency list representation of an undirected simple graph $G = (V, E)$, each edge (u, v) has two adjacency list entries: $[v]$ in the adjacency list of u , and $[u]$ in the adjacency list of v . These are called twins of each other. A twin pointer is a pointer from an adjacency list entry to its twin. If $|E|=m$ and $|V|=n$, and the memory size is not a constraint, what is the Asymptotic Notation of the most efficient algorithm to set the twin pointer in each entry in each adjacency list?

209. The given diagram shows the flowchart for a recursive function $A(n)$. Assume that all statements, except for the recursive calls, have $O(1)$ Asymptotic Notation. If the worst case Asymptotic Notation of this

function is $O(n^\alpha)$, then the least possible value (accurate up to two decimal positions) of α is .

210. Two alternative packages A and B are available for processing a database having 10k records. Package A requires $0.0001n^2$ time units and package B requires $10n \log_{10} n$ time units to process n records. What is the smallest value of k for which package B will be preferred over A?

211. The time taken by binary search algorithm to search a key in a sorted array of n elements is

212. Exponentiation is a heavily used operation in public key cryptography. Which of the following options is the tightest upper bound on the number of multiplications required to compute $b^n \bmod m$, $0 \leq b, n \leq m$?

213. Suppose we modify the above function `foo()` and store the values of `foo(i)`, $0 \leq i < n$, as and when they are computed. With this modification, the time complexity for function `foo()` is significantly reduced. The space complexity of the modified function would be:

214. The Asymptotic Notation of computing the transitive closure of a binary relation on a set of n elements is known to be:

215. Let $A[1, \dots, n]$ be an array storing a bit (1 or 0) at each location, and $f(m)$ is a function whose Asymptotic Notation is $\theta(m)$. Consider the following program fragment written in a C like language:

216. The cube root of a natural number n is defined as the largest natural number m such that $m^3 \leq n$. The complexity of computing the cube root of n (n is represented in binary notation) is

217. Let S be a sorted array of n integers. Let $T(n)$ denote the time taken for the most efficient algorithm to determine if there are two elements with sum less than 1000 in S . Which of the following statement is true?

218. Consider a sequence of 14 elements: $A = [-5, -10, 6, 3, -1, -2, 13, 4, -9, -1, 4, 12, -3, 0]$. The subsequence sum $S(i, j) = \sum_{k=i}^j A[k]$. Determine the maximum of $S(i, j)$, where $0 \leq i \leq j < 14$. (Divide and conquer approach may be used). Answer: _____

219. The minimum number of comparisons required to find the minimum and the maximum of 100 numbers is _____.

220. If n is a power of 2, then the minimum number of multiplications needed to compute a^n is

221. Of the following sort algorithms, which has execution time that is least dependant on initial ordering of the input?

222. Given two sorted list of size m and n respectively. The number of comparisons needed the worst case by the merge sort algorithm will be:

223. An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is _____

224. The number of swappings needed to sort the numbers 8, 22, 7, 9, 31, 5, 13 in ascending order using bubble sort is

225. Algorithm design technique used in quicksort algorithm is?

226. Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in ascending order, which of the following are TRUE? I. Quicksort runs in $\Theta(n^2)$ time II. Bubblesort runs in $\Theta(n^2)$ time III. Merge sort runs in $\Theta(n)$ time IV. Insertion sort runs in $\Theta(n)$ time

227. The worst case running times of Insertion sort, Mergesort and Quicksort, respectively, are:

228. A machine needs a minimum of 100 sec to sort 1000 names by quick sort. The minimum time needed to sort 100 names will be approximately

229. Assume that a mergesort algorithm in the worst case takes 30 seconds for an input of size 64. Which of the following most closely approximates the maximum input size of a problem that can be solved in 6 minutes?

230. Suppose you are provided with the following function declaration in the C programming language. `int partition(int a[], int n);` The function treats the first element of `a[]` as a pivot, and rearranges the array so that all elements less than or equal to the pivot is in the left part of the array, and all elements greater than the pivot is in the right part. In addition, it moves the pivot so that the pivot is the last element of the left part. The return value is the number of elements in the left part. The following partially given function in the C programming language is used to find the k th smallest element in an array `a[]` of size n using the partition function. We assume $k \leq n$

231. Which one of the following is the recurrence equation for the worst case time complexity of the Quicksort algorithm for sorting n (≥ 2) numbers? In the recurrence equations given in the options below, c is a constant.

232. You have an array of n elements. Suppose you implement quick sort by always choosing the central element of the array as the pivot. Then the tightest upper bound for the worst case performance is

233. Consider the following sorting algorithms. I. Quicksort. II. Heapsort. III. Mergesort. Which of them perform in least time in the worst case?

234. Let P be a quick sort program to sort numbers in ascending order using the first element as the pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs `[1 2 3 4 5]` and `[4 1 5 3 2]` respectively. Which one of the following holds?

235. Which of the following sorting algorithms has the minimum running time complexity in the best and average case?

236. The number of elements that can be sorted in $\Theta(\log n)$ time using heap sort is

237. Which one of the following is the tightest upper bound that represents the number of swaps required to sort n numbers using selection sort?

238. A list of n strings, each of length n , is sorted into lexicographic order using the merge-sort algorithm. The worst case running time of this computation is

- 239.** In quick sort, for sorting n elements, the $(n/4)$ th smallest element is selected as pivot using an $O(n)$ time algorithm. What is the worst case time complexity of the quick sort?
- 240.** How many comparisons are needed to sort an array of length 5 if a straight selection sort is used and array is already in the opposite order?
- 241.** What is the number of swaps required to sort n elements using selection sort, in the worst case?
- 242.** If we use Radix Sort to sort n integers in the range $(nk/2, nk)$, for some $k > 0$ which is independent of n , the time taken would be?
- 243.** Consider the Quicksort algorithm. Suppose there is a procedure for finding a pivot element which splits the list into two sub-lists each of which contains at least one-fifth of the elements. Let $T(n)$ be the number of comparisons required to sort n elements. Then
- 244.** Selection sort algorithm design technique is an example of
- 245.** The median of n elements can be found in $O(n)$ time. Which one of the following is correct about the complexity of quick sort, in which median is selected as pivot?
- 246.** The tightest lower bound on the number of comparisons, in the worst case, for comparison-based sorting is of the order of
- 247.** In a permutation $a_1, \dots, a_n$ of n distinct integers, an inversion is a pair $(a_i > a_j)$ such that $i < j$ and $a_i > a_j$. What would be the worst case time complexity of the insertion Sort algorithm, if the inputs are restricted to permutations of $1, \dots, n$ with at most n inversions?
- 248.** The usual $\Theta(n^2)$ implementation of Insertion Sort to sort an array uses linear search to identify the position where an element is to be inserted into the already sorted part of the array. If, instead, we use binary search to identify the position, the worst case running time will
- 249.** If one uses straight two-way merge sort algorithm to sort the following elements in ascending order: 20, 47, 15, 8, 9, 4, 40, 30, 12, 17 then the order of these elements after second pass of the algorithm is:
- 250.** Algorithm design technique used in quicksort algorithm is?
- 251.** Randomized quicksort is an extension of quicksort where the pivot is chosen randomly. What is the worst case complexity of sorting n numbers using randomized quicksort ?
- 252.** Choose the correct alternatives (more than one may be correct) and write the corresponding letters only: Following algorithm(s) can be used to sort n in the range $[1 \dots n^3]$ in $O(n)$ time
- 253.** The complexity of comparison based sorting algorithms is:
- 254.** Let P be a quicksort program to sort numbers in ascending order. Let t_1 and t_2 be the time taken by the program for the inputs $[1\ 2\ 3\ 4]$ and $[5\ 4\ 3\ 2\ 1]$, respectively. Which of the following holds?
- 255.** The integer value printed by the ANSI-C program given below is _____.

256. A language with string manipulation facilities uses the following operations. `head(s)`- returns the first character of the string `s` `tail(s)`- returns all but the first character of the string `s` `concat(s1,s2)`- concatenates string `s1` with `s2`. The output of `concat(head(s), head(tail(tail(s))))`, where `s` is `acbc` is:

257. Consider the following program written in pseudo-code. Assume that `x` and `y` are integers. The number of times that the print statement is executed by the call `Count(1024,1024)` is _____.

258. Which one of the following is correct about the statements given below?
I. All function calls are resolved at compile time in C lang II. All function calls are resolved at compile time in C++ lang

259. What will be the output of the following C program?

260. Consider the following C program. Which one of the following expressions, when placed in the blank above, will NOT result in a type checking error?

261. What will be the output of the following pseudo-code when parameters are passed by reference and dynamic scoping is assumed?

262. Suppose `c=(c[0],...,c[k-1])` is an array of length `k`, where all the entries are from the set `{0,1}`. For any positive integers `a` and `n`, consider the following pseudocode.

263. What is the return value of `f(p,p)`, if the value of `p` is initialized to 5 before the call? Note that the first parameter is passed by reference, whereas the second parameter is passed by value.

264. If `get(6)` function is being called in `main()` then how many times will the `get()` function be invoked before returning to the `main()`?

265. Consider an array `X` that contains `n` positive integers. A subarray of `X` is defined to be a sequence of array locations with consecutive indices. The C code snippet given below has been written to compute the length of the longest subarray of `X` that contains at most two distinct integers. The code has two missing expressions labelled (P) and (Q).

266. Consider the C program fragment below which is meant to divide `x` by `y` using repeated subtractions. The variables `x`, `y`, `q` and `r` are all unsigned int.

267. The following function computes the maximum value contained in an integer array `p[]` of size `n` (`n>=1`).

268. The following three 'C' language statements is equivalent to which single statement?

269. Suppose `n` and `p` are unsigned int variables in a C program. We wish to set `p` to `nC3`. If `n` is large, which one of the following statements is most likely to set `p` correctly?

270. In Java, after executing the following code what are the values of `x`, `y` and `z`?
`int x, y=10; z=12; x=y++ + z++;`

271. In the following Pascal program segment, what is the value of `X` after the execution of the program segment?

272. The conditional expansion facility of macro processor is provided to

273. Assume that X and Y are non-zero positive integers. What does the following Pascal program segment do?

274. S1: The compiler will generate code to allocate a temporary nameless cell, initialize it to 13, and pass the address of the cell to swap S2: On execution the code will generate a runtime error on line L1 S3: On execution the code will generate a runtime error on line L2 S4: The program will print 13 and 8 S5: The program will print 13 and -2 Exactly the following set of statement(s) is correct:

275. Choose the correct option to fill ? 1 and ? 2 so that the program below prints an input string in reverse order. Assume that the input string is terminated by a newline character.

276. The following program fragment is written in a programming language that allows variables and does not allow nested declarations of functions. If the programming language uses dynamic scoping and call by name parameter passing mechanism, the values printed by the above program are

277. In which of the following cases is it possible to obtain different results for call-by-reference and call-by-name parameter passing methods?

278. What is printed by the print statements in the program P1 assuming call by reference parameter passing ?

279. Assume that the variable y points to a struct (allocated on the heap) containing two fields f1 and f2, and the local variables x, y, z, p, q, and i are allotted registers. Common sub-expression elimination (CSE) optimization is applied on the code. The number of addition and the dereference operations (of the form y -> f1 or y -> f2) in the optimized code, respectively, are:

280. Let P be an array containing n integers. Let t be the lowest upper bound on the number of comparisons of the array elements, required to find the minimum and maximum values in an arbitrary of n elements. Which one of the following choices is correct?

281. A program P reads in 500 integers in the range [0, 100] representing the cores of 500 students. It then print the frequency of each score above 50. What would be the best way for P to store the frequencies?

282. A single array A[1..MAXSIZE] is used to implement two stacks, The two stacks grow from opposite ends of the array. Variables top1 and top2 (top1 < top2) point to the location of the topmost element in each of the stacks, If the space is to be used efficiently, the condition for "stack full" is

283. Two matrices M1 and M2 are to be stored in arrays A and B respectively. Each array can be stored either in row-major or column-major order in contiguous memory locations. The time complexity of an algorithm to compute $M1 * M2$ will be

284. Let A be a two dimensional array declared as follows: A : array [1... 10] [1... 15] of integer; Assuming that each integer takes one memory locations the array is stored in row-major order and the first element of the array is stored at location 100, what is the address of the element A[i] [j]?

285. In a compact single dimensional array representation for lower triangular matrices (i.e all the elements above the diagonal are zero) of size n * n, non-zero elements (i.e., elements of the lower

triangle) of each row are stored one after another, starting from the first row, the index of the (i, j) th element of the lower triangular matrix in this new representation is

286. Consider the following sequence of operations on an empty stack. push(54); push(52); pop(); push(55); push(62); s = pop(); Consider the following sequence of operations on an empty queue. enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q = dequeue(); The value of s + q is _____

287. A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT (refers to the number of items in the queue)?

288. A function f defined on stacks of integers satisfies the following properties. $f(\emptyset) = 0$ and $f(\text{push}(S, i)) = \max(f(S), 0) + i$ for all stacks S and integers i . If a stack S contains the integers 2, -3, 2, -1, 2 in order from bottom to top, what is $f(S)$?

289. The best data structure to check whether an arithmetic expression has balanced parentheses is a

290. Consider the following statements: (i) First-in-first out types of computations are efficiently supported by STACKS. (ii) Implementing LISTS on linked lists is more efficient than implementing LISTS on an array for almost all the basic LIST operations. (iii) Implementing QUEUES on a circular array is more efficient than implementing QUEUES on a linear array with two indices. (iv) Last-in-first-out type of computations are efficiently supported by QUEUES.

291. Consider a stack data structure into which we can PUSH and POP records. Assume that each record pushed in the stack has a positive integer key and that all keys are distinct. We wish to augment the stack data structure with an $O(1)$ time MIN operation that returns a pointer to the record with smallest key present in the stack. 1. without deleting the corresponding record, and 2. without increasing the complexities of the standard stack operations. Which one or more of the following approach(es) can achieve it?

292. Consider a sequence a of elements $a_0 = 1, a_1 = 5, a_2 = 7, a_3 = 8, a_4 = 9$, and $a_5 = 2$. The following operations are performed on a stack S and a queue Q , both of which are initially empty. I: push the elements of a from a_0 to a_5 in that order into S . II: enqueue the elements of a from a_0 to a_5 in that order into Q . III: pop an element from S . IV: dequeue an element from Q . V: pop an element from S . VI: dequeue an element from Q . VII: dequeue an element from Q and push the same VIII: Repeat operation VII three times. IX: pop an element from S . X: pop an element from S . The top element of S after executing the above operations is _____.

293. Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE with respect to this modified stack?

294. Suppose a circular queue of capacity $(n - 1)$ elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variables, respectively. Initially, REAR = FRONT = 0. The conditions to detect queue full and queue empty are

295. The following postfix expression with single digit operands is evaluated using a stack: $8\ 2\ 3\ ^\wedge\ /\ 2\ 3\ *\ +\ 5\ 1\ *\ -$. Note that $^\wedge$ is the exponentiation operator. The top two elements of the stack after the first $*$ is evaluated are:

296. A program attempts to generate as many permutation as possible of the string "abcd" by pushing the character a,b,c,d in the same order onto a stack, but it may pop off the top character at any time. Which one of the following strings CANNOT be generated using this program?

297. Let S be a stack of size $n \geq 1$. Starting with the empty stack, suppose we push the first n natural numbers in sequence, and then perform n pop operations. Assume that Push and Pop operations take X seconds each, and Y seconds elapse between the end of one such stack operation and the start of the next operation. For $m \geq 1$, define the stack-life of m as the time elapsed from the end of Push(m) to the start of the pop operation that removes m from S. The average stack-life of an element of this stack is

298. Assume that the operators $+$, $-$, $\times$, are left associative and $^\wedge$ is right associative. The order of precedence (from highest to lowest) is $^\wedge$, $\times$, $+$, $-$. The postfix expression corresponding to the infix expression $a + b \times c - d^\wedge e^\wedge f$ is

299. A priority queue Q is used to implement a stack that stores characters. PUSH (C) is implemented INSERT (Q, C, K) where K is an appropriate integer key chosen by the implementation. POP is implemented as DELETEMIN(Q). For a sequence of operations, the keys chosen are in

300. Which of the following permutations can be obtained in the output (in the same order) using a stack assuming that the input is the sequence 1, 2, 3, 4, 5 in that order?

Computer Organization and Architecture

1. A processor with 16 general purpose registers uses a 32-bit instruction format. The instruction format consists of an opcode field, an addressing mode field, two register operand fields, and a 16-bit scalar field. If 8 addressing modes are to be supported, the maximum number of unique opcodes possible for every addressing mode is _____.
2. Consider a 32-bit processor which supports 70 instructions. Each instruction is 32 bit long and has 4 fields namely opcode, two register identifiers and an immediate operand of unsigned integer type. Maximum value of the immediate operand that can be supported by the processor is 8191. How many registers the processor has?
3. A non-pipelined CPU has 12 general purpose registers (R0, R1, R2, ..., R12). Following operations are supported ADD Ra, Rb, Rr Add Ra to Rb and store the result in Rr, MUL Ra, Rb, Rr Multiply Ra to Rb and store the result in Rr, MUL operation takes two clock cycles, ADD takes one clock cycle. Calculate minimum number of clock cycles required to compute the value of the expression $XY + XYZ + YZ$. The variable X, Y, Z are initially available in registers R0, R1 and R2 and contents of these registers must not be modified.
4. A processor has 64 registers and uses 16-bit instruction format. It has two types of instructions: I-type and R-type. Each I-type instruction contains an opcode, a register name, and a 4-bit immediate value. Each R-type instruction contains an opcode and two register names. If there are 8 distinct I-type opcodes, then the maximum number of distinct R-type opcodes is _____.
5. A byte addressable computer has a memory capacity of 2mKB (k bytes) and can perform 2^n operations. An instruction involving 3 operands and one operator needs maximum of:
6. A data driven machine is one that executes an instruction if the needed data is available. The physical ordering of the code listing does not dictate the course of execution. Consider the following pseudo-code: (A). Multiply E by 0.5 to get F. (B). Add A and B to get E. (C). Add B with 0.5 to get D. (D). Add E and F to get G. (E). Add A with 10.5 to get C. Assume A, B, C are already assigned values and the desired output is G. Which of the following sequence of execution is valid
7. A processor has 16 integer registers (R0, R1, ..., R15) and 64 floating point registers (F0, F1, ..., F63). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type-4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating point register operand (1F). The maximum value of N is _____.
8. Consider the following processor design characteristics. I. Register-to-register arithmetic operations only. II. Fixed-length instruction format. III. Hardwired control unit. Which of the characteristics above are used in the design of a RISC processor?
9. Consider a processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and a

twelve-bit immediate value. Each instruction must be stored in memory in byte-aligned fashion. If a program has 100 instructions, the amount of memory(in bytes) consumed by the program text is _____ .

10. A processor has 40 distinct instructions and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is _____ .

11. In $X = (M + N \times O) / (P \times Q)$, how many one-address instructions are required to evaluate it?

12. The contents of the flag register after execution of the following program by 8085 microprocessor will be - Program, SUB A, MVI B,(01)H, DCR B, HLT

13. Consider a processor with byte-addressable memory. Assume that all registers, including Program Counter (PC) and Program Status Word (PSW), are of size 2 bytes. A stack in the main memory is implemented from memory location (0100) base 16 and it grows upward. The stack pointer (SP) points to the top element of the stack. The current value of SP is (016E) base 16. The CALL instruction is of two words, the first word is the op-code and the second word is the starting address of the subroutine (one word = 2 bytes). The CALL instruction is implemented as follows: Store the current value of PC in the stack, Store the value of PSW register in the stack, Load the starting address of the subroutine in PC. The content of PC just before the fetch of a CALL instruction is (5FA0) base 16. After execution of the CALL instruction, the value of the stack pointer is

14. Assume that 16-bit CPU is trying to access a double word starting at an odd address. How many memory operations are required to access the data?

15. The use of multiple register windows with overlap causes a reduction in the number of memory accesses for: I. Function locals and parameters. II. Register saves and restores. III. Instruction fetches

16. A CPU has 24-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values in decimal)?

17. The Memory Address Register

18. Which of the following architecture is/are not suitable for realising SIMD?

19. Consider the following Assembly language program
MVI A, 30H
ACI 30H
XRA A
POP H
After the execution of the above program, the contents of the accumulator will be

20. Assume that $EA = (X) +$ is the effective address equal to the contents of location X, with X incremented by one word length after the effective address is calculated; $EA = -(X)$ is the effective address equal to the contents of location X, with X decremented by one word length before the effective address is calculated; $EA = (X) -$ is the effective address equal to the contents of location X, with X decremented by one word length after the effective address is calculated. The format of the instruction is (opcode, source, destination), which means (destination ■ source op destination). Using X as a stack pointer, which of the following instructions can pop the top two elements from the stack, perform the addition operation and push the result back to the stack.

- 21.** A processor that has the carry, overflow and sign flag bits as part of its program status word (PSW) performs addition of the following two 2's complement numbers 01001101 and 11101001. After the execution of this addition operation, the status of the carry, overflow and sign flags, respectively will be:
- 22.** The use of multiple register windows with overlap causes a reduction in the number of memory accesses for: I. Function locals and parameters. II. Register saves and restores. III. Instruction fetches
- 23.** In an instruction execution pipeline, the earliest that the data TLB (Translation Lookaside Buffer) can be accessed is
- 24.** Which of the following must be true for the RFE (Return From Exception) instruction on a general purpose processor? I. It must be a trap instruction. II. It must be a privileged instruction. III. An exception cannot be allowed to occur during execution of an RFE instruction
- 25.** Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal. Assume that the memory is byte addressable and the word size is 32 bits. If an interrupt occurs during the execution of the instruction "INC R3", what return address will be pushed on to the stack?
- 26.** Consider a new instruction named branch-on-bit-set (mnemonic bbs). The instruction "bbs reg, pos, label" jumps to label if bit in position pos of register operand reg is one. a register is 32 bits wide and the bits are numbered 0 to 31, bit in position 0 being the least significant. Consider the following emulation of this instruction on a processor that does not have bbs implemented.
temp → reg & mask
Branch to label if temp is non-zero.
The variable temp is a temporary register. For correct emulation the variable mask must be generated by
- 27.** A CPU has 24-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values in decimal)?
- 28.** If we use internal data forwarding to speed up the performance of a CPU (R1, R2 and R3 are registers and M[100] is a memory reference), then the sequence of operations $R1 \rightarrow M[100] \rightarrow R2 \rightarrow M[100] \rightarrow R3$ can be replaced by
- 29.** Let the clock cycles required for various operations be as follows: Register to/from memory transfer : 3 clock cycles
ADD with both operands in register : 1 clock cycle
Instruction fetch and decode : 2 clock cycles per word.
The total number of clock cycles required to execute the program is
- 30.** Consider that the memory is byte addressable with size 32 bits, and the program has been loaded starting from memory location 1000 (decimal). If an interrupt occurs while the CPU has been halted after executing the HALT instruction, the return address (in decimal) saved in the stack will be
- 31.** What are the states of the Auxiliary Carry (AC) and Carry Flag (CY) after executing the following 8085 program ?
MIV H, 5DH
MIV L, 6BH
MOV A, H
ADD L
- 32.** In 8085 which of the following modifies the program counter ?
- 33.** To put the 8085 microprocessor in the wait state

- 34.** The main difference(s) between a CISC and a RISC processor is/are that a RISC processor typically:
a) has fewer instructions. b) has fewer addressing modes. c) has more registers. d) is easier to implement using hardwired control logic
- 35.** Contents of A register after the execution of the following 8085 microprocessor program is MVI A, 55 H
MVI C, 25 H ADDC DAA
- 36.** Suppose a processor does not have any stack pointer register. Which of the following statements is true ?
- 37.** Number of machine cycles required for RET instruction in 8085 microprocessor is
- 38.** The most relevant addressing mode to write position-independent codes is:
- 39.** Consider two processors P1 and P2 executing the same instruction set. Assume that under identical conditions, for the same input, a program running on P2 takes 25% less time but incurs 20% more CPI (clock cycles per instruction) as compared to the program running on P1. If the clock frequency of P1 is 1GHz, then the clock frequency of P2 (in GHz) is _____.
- 40.** A machine has a 32-bit architecture, with 1-word long instructions. It has 64 registers, each of which is 32 bits long. It needs to support 45 instructions, which have an immediate operand in addition to two register operands. Assuming that the immediate operand is an unsigned integer, the maximum value of the immediate operand is _____.
- 41.** How many number of times the instruction sequence below will loop before coming out of the loop?
MOV AL, 00H A1: INC AL JNZ A1
- 42.** In 8086, the jump condition for the instruction JNBE is?
- 43.** Consider the following sequence of micro-operations. $MBR \leftarrow PC$, $MAR \leftarrow X$, $PC \leftarrow Y$, $Memory \leftarrow MBR$. Which one of the following is a possible operation performed by this sequence?
- 44.** Find the memory address of the next instruction executed by the microprocessor (8086), when operated in real mode for CS=1000 and IP=E000
- 45.** Compared to CISC processors, RISC processors contain
- 46.** A processor that has carry, overflow and sign flag bits as part of its program status word (PSW) performs addition of the following two 2's complement numbers 01001101 and 11101001. After the execution of this addition operation, the status of the carry, overflow and sign flags, respectively will be
- 47.** Consider a RISC machine where each instruction is exactly 4 bytes long. Conditional and unconditional branch instructions use PC- relative addressing mode with Offset specified in bytes to the target location of the branch instruction. Further the Offset is always with respect to the address of the next instruction in the program sequence. If the target of the branch instruction is i, then the decimal value of the Offset is _____
- 48.** A stack organized computer is characterised by instructions with

- 49.** The immediate addressing mode can be used for: 1. Loading internal registers with initial values. 2. Perform arithmetic or logical operation on data contained in instructions. Which of the following is true?
- 50.** Relative mode of addressing is most relevant to writing:
- 51.** Consider a hypothetical processor with an instruction of type `LW R1, 20(R2)`, which during execution reads a 32-bit word from memory and stores it in a 32-bit register R1. The effective address of the memory location is obtained by the addition of constant 20 and the contents of register R2. Which of the following best reflects the Addressing Modes implemented by this instruction for the operand in memory?
- 52.** `MOV [BX], AL` type of data addressing is called ?
- 53.** For computers based on three-address instruction formats, each address field can be used to specify which of the following: (S1) A memory operand. (S2) A processor register. (S3) An implied accumulator register
- 54.** In the Big-Endian system, the computer stores
- 55.** In which addressing mode, the effective address of the operand is generated by adding a constant value to the content of a register?
- 56.** Which of the following is/are true of the auto-increment Addressing Modes? I. It is useful in creating self-relocating code. II. If it is included in an Instruction Set Architecture, then an additional ALU is required for effective address calculation. III. The amount of increment depends on the size of the data item accessed.
- 57.** Which of the following is/are true of the auto-increment addressing mode? I. It is useful in creating self-relocating code. II. If it is included in an Instruction Set Architecture, then an additional ALU is required for effective address calculation. III. The amount of increment depends on the size of the data item accessed.
- 58.** Relative mode of addressing is most relevant to writing:
- 59.** Which of the following addressing modes permits relocation without any change whatsoever in the code?
- 60.** A certain processor supports only the immediate and the direct addressing modes. Which of the following programming language features cannot be implemented on this
- 61.** In the absolute the Addressing Modes :
- 62.** Which of the following Addressing Modes are suitable for program relocation at run time?
- 63.** Consider a digital display system (DDS) shown in the figure that displays the contents of register X. A 16-bit code word is used to load a word in X, either from S or from R. S is a 1024-word memory segment and R is a 32-word register file. Based on the value of mode bit M, T selects an input word to load in X. P and Q interface with the corresponding bits in the code word to choose the addressed word. Which one of the following represents the functionality of P, Q, and T?

64. Consider a main memory system that consists of 8 memory modules attached to the system bus, which is one word wide. When a write request is made, the bus is occupied for 100 nanoseconds (ns) by the data, address, and control signals. During the same 100 ns, and for 500 ns thereafter, the addressed memory module executes one cycle accepting and storing the data. The (internal) operation of different memory modules may overlap in time, but only one request can be on the bus at any time. The maximum number of stores (of one word each) that can be initiated in 1 millisecond is _____

65. Two control signals in microprocessor which are related to Direct Memory Access (DMA) are

66. The microinstructions stored in the control memory of a processor have a width of 26 bits. Each microinstruction is divided into three fields: a micro operation field of 13 bits, a next address field (X), and a MUX select field (Y). There are 8 status bits in the inputs of the MUX. How many bits are there in the X and Y fields, and what is the size of the control memory in number of words?

67. Consider a CPU where all the instructions require 7 clock cycles to complete execution. There are 140 instructions in the instruction set. It is found that 125 control signals are needed to be generated by the control unit. While designing the horizontal microprogrammed control unit, single address field format is used for branch control logic. What is the minimum size of the control word and control address register?

68. Consider the following data path of a CPU. ALU, the bus and all the registers in the data path are of identical size. All operations including incrementation of the PC and the GPRs are to be carried out in the ALU. Two clock cycles are needed for memory read operation - the first one for loading address in the MAR and the next one for loading data from the memory bus into the MDR. The instruction "call Rn, sub" is a two word instruction. Assuming that PC is incremented during the fetch cycle of the first word of the instruction, its register transfer interpretation is

69. An instruction set of a processor has 125 signals which can be divided into 5 groups of mutually exclusive signals as follows: Group 1 : 20 signals, Group 2 : 70 signals, Group 3 : 2 signals, Group 4 : 10 signals, Group 5 : 23 signals.. How many bits of the control words can be saved by using vertical microprogramming over horizontal microprogramming?

70. Consider the following data path of a CPU. ALU, the bus and all the registers in the data path are of identical size. All operations including incrementation of the PC and the GPRs are to be carried out in the ALU. Two clock cycles are needed for memory read operation - the first one for loading address in the MAR and the next one for loading data from the memory bus into the MDR. The instruction "add R0, R1" has the register transfer interpretation $R0 \leftarrow R0 + R1$. The minimum number of clock cycles needed for execution cycle of this instruction is.

71. The first operand (destination) "A[R0]" uses indexed addressing mode with R0 as the index register. The second operand (source) "@B" uses indirect addressing mode. A and B are memory addresses residing at the second and the third words, respectively. The first word of the instruction specifies the opcode, the index register designation and the source and destination addressing modes. During execution of ADD instruction, the two operands are added and stored in the destination (first operand). The number of memory cycles needed during the execution cycle of the instruction is:

72. The microinstructions stored in the control memory of a processor have a width of 26 bits. Each microinstruction is divided into three fields: a micro-operation field of 13 bits, a next address field (X), and

a MUX select field (Y). There are 8 status bits in the inputs of the MUX. How many bits are there in the X and Y fields, and what is the size of the control memory in number of words?

73. Arrange the following configuration for CPU in decreasing order of operating speeds: Hard wired control, Vertical microprogramming, Horizontal microprogramming.

74. How many bits are there in the X and Y fields, and what is the size of the control memory in number of words?

75. ALU, the bus and all the registers in the data path are of identical size. All operations including incrementation of the PC and the GPRs are to be carried out in the ALU. Two clock cycles are needed for memory read operation - the first one for loading address in the MAR and the next one for loading data from the memory bus into the MDR. The instruction "add R0, R1" has the register transfer interpretation $R0 \leftarrow R0 + R1$. The minimum number of clock cycles needed for execution cycle of this instruction is.

76. A micro instruction is to be designed to specify: A). none or one of the three micro operations of one kind and. B). none or upto six micro operations of another kind. The minimum number of bits in the micro-instruction is:

77. A micro program control unit is required to generate a total of 25 control signals. Assume that during any micro instruction, at most two control signals are active. Minimum number of bits required in the control word to generate the required control signals will be:

78. Consider the following data path of a simple non-pipeline CPU. The registers A, B, A1, A2, MDR, the bus and the ALU are 8-bit wide. SP and MAR are 16-bit registers. The MUX is of size $8 \times (2:1)$ and the DEMUX is of size $8 \times (1:2)$. Each memory operation takes 2 CPU clock cycles and uses MAR (Memory Address Register) and MDR (Memory Data Register). SP can be decremented locally. The CPU instruction "push r", where $r = A \text{ or } B$, has the specification $M[SP] \rightarrow r$ $SP \rightarrow SP - 1$. How many CPU clock cycles are needed to execute the "push r" instruction ?

79. A microprogrammed control unit

80. Consider a computer system with DMA support. The DMA module is transferring one 8-bit character in one CPU cycle from a device to memory through cycle stealing at regular intervals. Consider a 2 MHz processor. If 0.5% processor cycles are used for DMA, the data transfer rate of the device is _____ bits per second.

81. Which one of the following facilitates transfer of bulk data from hard disk to main memory with the highest throughput?

82. Consider a computer with a 4 MHz processor. Its DMA controller can transfer 8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following is the data transfer rate (in bits per second) of the DMA controller if 1% of the processor cycles are used for DMA?

83. A hard disk with a transfer rate of 10 Mbytes/second is constantly transferring data to memory using DMA. The processor runs at 600 MHz, and takes 300 and 900 clock cycles to initiate and complete DMA transfer respectively. If the size of the transfer is 20 Kbytes, what is the percentage of processor time consumed for the transfer operation?

- 84.** What is the bit rate of a video terminal unit with 80 characters/line, 8 bits/character and horizontal sweep time of 100 μ s (including 20 μ s of retrace time)?
- 85.** Normally user programs are prevented from handling I/O directly by I/O instructions in them. For CPUs having explicit I/O instructions, such I/O protection is ensured by having the I/O instructions privileged. In a CPU with memory mapped I/O, there is no explicit I/O instruction. Which one of the following is true for a CPU with memory mapped I/O?
- 86.** A device with data transfer rate 10 KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 4 μ sec. The byte transfer time between the device interfaces register and CPU or memory is negligible. What is the minimum performance gain of operating the device under interrupt mode over operating it under program-controlled mode?
- 87.** Consider a disk drive with the following specifications: 16 surfaces, 512 tracks/surface, 512 sectors/track, 1 KB/sector, rotation speed 3000 rpm. The disk is operated in cycle stealing mode whereby whenever one 4 byte word is ready it is sent to memory; similarly, for writing, the disk interface reads a 4 byte word from the memory in each DMA cycle. Memory cycle time is 40 nsec. The maximum percentage of time that the CPU gets blocked during DMA operation is:
- 88.** The ability to temporarily halt the CPU and use this time to send information on buses is called
- 89.** Which of the following DMA transfer modes and interrupt handling mechanisms will enable the highest I/O band-width?
- 90.** The device which is used to connect a peripheral to bus is known as
- 91.** On a non-pipelined sequential processor, a program segment, which is a part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory. Initialize the address register, Initialize the count to 500, LOOP: Load a byte from device, Store in memory at address given by address register, Increment the address register, Decrement the count. If count $\neq$ 0 go to LOOP. Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute. The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory. What is the approximate speedup when the DMA controller based design is used in place of the interrupt driven program based input-output?
- 92.** In DMA transfer scheme, the transfer scheme other than burst mode is
- 93.** A processor is fetching instructions at the rate of 1 MIPS. A DMA module is used to transfer characters to RAM from a device transmitting at 9600 bps. How much time will the processor be slowed down due to DMA activity?
- 94.** The size of the data count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29,154 kilobytes from disk to main memory. The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is ____.

95. For the daisy chain scheme of connecting I/O devices, which of the following statements is true?

96. A low memory can be connected to 8085 by using

97. A keyboard connected to a computer is used at a rate of 1 keystroke per second. The computer system polls the keyboard every 10 ms (milli seconds) to check for a keystroke and consumes 100 μ s (micro seconds) for each poll. If it is determined after polling that a key has been pressed, the system consumes an additional 200 μ s to process the keystroke. Let T1 denote the fraction of a second spent in polling and processing a keystroke. In an alternative implementation, the system uses interrupts instead of polling. An interrupt is raised for every keystroke. It takes a total of 1 ms for servicing an interrupt and processing a keystroke. Let T2 denote the fraction of a second spent in servicing the interrupt and processing a keystroke. The ratio T1/T2 is _____. (Rounded off to one decimal place)

98. Consider the following statements. I. Daisy chaining is used to assign priorities in attending interrupts. II. When a device raises a vectored interrupt, the CPU does polling to identify the source of interrupt. III. In polling, the CPU periodically checks the status bits to know if any device needs its attention. IV. During DMA, both the CPU and DMA controller can be bus masters at the same time. Which of the above statements is/are TRUE?

99. The following are some events that occur after a device controller issues an interrupt while process L is under execution. (P) The processor pushes the process status of L onto the control stack. (Q) The processor finishes the execution of the current instruction. (R) The processor executes the interrupt service routine. (S) The processor pops the process status of L from the control stack. (T) The processor loads the new PC value based on the interrupt. Which one of the following is the correct order in which the events above occur?

100. Which interrupt in 8085 Microprocessor is unmaskable?

101. In 8085 microprocessor, the ISR for handling trap interrupt is at which location?

102. A computer handles several interrupt sources of which the following are relevant for this question. Interrupt from CPU temperature sensor (raises interrupt if CPU temperature is too high). Interrupt from Mouse (raises Interrupt if the mouse is moved or a button is pressed). Interrupt from Keyboard (raises Interrupt if a key is pressed or released). Interrupt from Hard Disk (raises Interrupt when a disk read is completed). Which one of these will be handled at the HIGHEST priority?

103. On receiving an interrupt from an I/O device, the CPU

104. A certain microprocessor requires 4.5 microseconds to respond to an interrupt. Assuming that the three interrupts I1, I2 and I3 require the following execution time after the interrupt is recognized: I. I1 requires 25 microseconds. II. I2 requires 35 microseconds III. I3 requires 20 microseconds I1 has the highest priority and I3 has the lowest. What is the possible range of time for I3 to be executed assuming that it may or may not occur simultaneously with other interrupts?

105. A CPU generally handles an interrupt by executing an interrupt service routine

106. The TRAP is one of the interrupts available in INTEL 8085. Which one of the following statements is true of TRAP?

107. An interrupt in which the external device supplies its address as well as the interrupt requests is known as

108. Which one of the following is true for a CPU having a single interrupt request line and a single interrupt grant line?

109. A device employing INTR line for device interrupt puts the CALL instruction on the data bus while

110. The 8085 microprocessor responds to the presence of an interrupt

111. A CPU has two modes-privileged and non-privileged. In order to change the mode from privileged to non-privileged

112. RST 7.5 interrupt in 8085 microprocessor executes the interrupt service routing from interrupt vector location

113. A processor needs software interrupt to

114. Choose the correct alternatives (more than one may be correct) and write the corresponding letters only: The TRAP interrupts mechanism of the 8085 microprocessor:

115. On receiving an interrupt from a I/O device the CPU:

116. One instruction tries to write an operand before it is written by previous instruction. This may lead to a dependency called

117. A five-stage pipeline has stage delays of 150, 120, 150, 160 and 140 nanoseconds. The registers that are used between the pipeline stages have a delay of 5 nanoseconds each. The total time to execute 100 independent instructions on this pipeline, assuming there are no pipeline stalls, is _____ nanoseconds.

118. Consider a pipelined processor with 5 stages, Instruction Fetch(IF), Instruction Decode(ID), Execute (EX), Memory Access (MEM), and Write Back (WB). Each stage of the pipeline, except the EX stage, takes one cycle. Assume that the ID stage merely decodes the instruction and the register read is performed in the EX stage. The EX stage takes one cycle for ADD instruction and the register read is performed in the EX stage, The EX stage takes one cycle for ADD instruction and two cycles for MUL instruction. Ignore pipeline register latencies. Consider the following sequence of 8 instructions: ADD, MUL, ADD, MUL, ADD, MUL, ADD, MUL. Assume that every MUL instruction is data-dependent on the ADD instruction just before it and every ADD instruction (except the first ADD) is data-dependent on the MUL instruction just before it. The speedup defined as follows. $\text{Speedup} = \frac{\text{Execution time without operand forwarding}}{\text{Execution time with operand forwarding}}$. The Speedup achieved in executing the given instruction sequence on the pipelined processor (rounded to 2 decimal places) is _____

119. A processor X1 operating at 2 GHz has a standard 5-stage RISC instruction pipeline having a base CPI (cycles per instruction) of one without any pipeline hazards. For a given program P that has 30% branch instructions, control hazards incur 2 cycles stall for every branch. A new version of the processor X2 operating at same clock frequency has an additional branch predictor unit (BPU) that completely eliminates stalls for correctly predicted branches. There is neither any savings nor any additional stalls for

wrong predictions. There are no structural hazards and data hazards for X1 and X2. If the BPU has a prediction accuracy of 80%, the speed up (rounded off to two decimal places) obtained by X2 over X1 in executing P is

120. Consider a 3-stage pipelined processor having a delay of 10 ns (nanoseconds), 20 ns, and 14 ns, for the first, second, and the third stages, respectively. Assume that there is no other delay and the processor does not suffer from any pipeline hazards. Also assume that one instruction is fetched every cycle. The total execution time for executing 100 instructions on this processor is _____ ns

121. Consider a 5-stage pipelined processor with Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Register Writeback (WB) stages. Which of the following statements about forwarding is/are CORRECT?

122. The baseline execution time of a program on a 2 GHz single core machine is 100 nanoseconds (ns). The code corresponding to 90% of the execution time can be fully parallelized. The overhead for using an additional core is 10 ns when running on a multicore system. Assume that all cores in the multicore system run their share of the parallelized code for an equal amount of time. The number of cores that minimize the execution time of the program is _____

123. An instruction format has the following structure: Instruction Number:

Opcode destination reg, source reg-1, source reg-2. Consider the following sequence of instructions to be executed in a pipelined processor: I1: DIV R3, R1, R2, I2: SUB R5, R3, R4, I3: ADD R3, R5, R6, I4: MUL R7, R3, R8. Which of the following statements is/are TRUE

124. A non-pipelined instruction execution unit operating at 2 GHz takes an average of 6 cycles to execute an instruction of a program P. The unit is then redesigned to operate on a 5-stage pipeline at 2 GHz. Assume that the ideal throughput of the pipelined unit is 1 instruction per cycle. In the execution of program P, 20% instructions incur an average of 2 cycles stall due to data hazards and 20% instructions incur an average of 3 cycles stall due to control hazards. The speedup (rounded off to one decimal place) obtained by the pipelined design over the non-pipelined design is _____

125. Consider a 5- segment pipeline with a clock cycle time 20 ns in each sub operation. Find out the approximate speed-up ratio between pipelined and non-pipelined system to execute 100 instructions. (if an average, every five cycles, a bubble due to data hazard has to be introduced in the pipeline)

126. Consider a non-pipelined processor operating at 2.5 GHz. It takes 5 clock cycles to complete an instruction. You are going to make a 5-stage pipeline out of this processor. Overheads associated with pipelining force you to operate the pipelined processor at 2 GHz. In a given program, assume that 30% are memory instructions, 60% are ALU instructions and the rest are branch instructions. 5% of the memory instructions cause stalls of 50 clock cycles each due to cache misses and 50% of the branch instructions cause stalls of 2 cycles each. Assume that there are no stalls associated with the execution of ALU instructions. For this program, the speedup achieved by the pipelined processor over the non-pipelined processor (round off to 2 decimal places) is _____.

127. A particular parallel program computation requires 100 sec when executed on a single processor, if 40% of this computation is inherently sequential (i.e. will not benefit from additional processors), then theoretically best possible elapsed times of this program running with 2 and 4 processors, respectively,

are:

128. Instructions execution in a processor is divided into 5 stages. Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Execute (EX), and Write Back (WB). These stages take 5, 4, 20, 10 and 3 nanoseconds (ns) respectively. A pipelined implementation of the processor requires buffering between each pair of consecutive stages with a delay of 2ns. Two pipelined implementations of the processor are contemplated. (i) a naive pipeline implementation (NP) with 5 stages and (ii) an efficient pipeline (EP) where the OF stage is divided into stages OF1 and OF2 with execution times of 12 ns and 8 ns respectively. The speedup (correct to two decimal places) achieved by EP over NP in executing 20 independent instructions with no hazards is _____.

129. The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB). The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the PO stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards. The number of clock cycles required for completion of execution of the sequence of instructions is _____.

130. Register renaming is done in pipelined processors:

131. Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and average cycles per instruction of four. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume that there are no stalls in the pipeline. The speedup achieved in this pipelined processor is _____.

132. The dynamic hazard problem occurs in _____.

133. Consider a 3GHz (gigahertz) processor with a three-stage pipeline and stage latencies T_1, T_2 and T_3 such that $T_1 = 3T_2/4 = 2T_3$. If the longest pipeline stage is split into two pipeline stages of equal latency, the new frequency is _____ GHz, ignoring delays in the pipeline registers.

134. Suppose the functions F and G can be computed in 5 and 3 nano seconds by functional units UF and UG, respectively. Given two instances of UF and two instances of UG, it is required to implement the computation $F(G(X_i))$ for $1 \leq i \leq 10$. Ignoring all other delays, the minimum time required to complete this computation is _____ nanoseconds.

135. The stage delays in a 4-stage pipeline are 800, 500, 400 and 300 picoseconds. The first stage (with delay 800 picoseconds) is replaced with a functionally equivalent design involving two stages with respective delays 600 and 350 picoseconds. The throughput increase of the pipeline is _____ percent.

136. Consider the following code sequence having five instructions I1 to I5. Each of these instructions has the following format. OP Ri, Rj, Rk where operation OP is performed on contents of registers Rj and Rk and the result is stored in register Ri. I1: ADD R1, R2, R3. I2: MUL R7, R1, R3. I3: SUB R4, R1, R5. I4: ADD R3, R2, R4. I5: MUL R7, R8, R9. Consider the following three statements. S1: There is an anti-dependence between instructions I2 and I5. S2: There is an anti-dependence between instructions I2 and I4. S3: Within an instruction pipeline an anti-dependence always creates one or more stalls.

137. Consider the sequence of machine instructions given below: MUL R5, R0, R1 DIV R6, R2, R3 ADD R7, R5, R6 SUB R8, R7, R4. In the above sequence, R0 to R8 are general purpose registers. In the instructions shown, the first register stores the result of the operation performed on the second and the third registers. This sequence of instructions is to be executed in a pipelined instruction processor with the following 4 stages: (1) Instruction Fetch and Decode (IF), (2) Operand Fetch (OF), (3) Perform Operation (PO) and (4) Write back the result (WB). The IF, OF and WB stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD or SUB instruction, 3 clock cycles for MUL instruction and 5 clock cycles for DIV instruction. The pipelined processor uses operand forwarding from the PO stage to the OF stage. The number of clock cycles taken for the execution of the above sequence of instructions is _____.

138. Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and average cycles per instruction of four. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume that there are no stalls in the pipeline. The speed up achieved in this pipelined processor is _____.

139. An instruction pipeline has five stages, namely, instruction fetch (IF), instruction decode and register fetch (ID/RF), instruction execution (Ex), memory access (MEM), and register write back (WB) with stage latencies 1 ns, 2.2 ns, 2 ns, 1 ns, and 0.75 ns, respectively (ns stands for nanoseconds). To gain in terms of frequency, the designers have decided to split the ID/RF stage into three stages (ID, RF1, RF2) each of latency 2.2/3 ns. Also, the Ex stage is split into two stages (Ex1, Ex2) each of latency 1 ns. The new design has a total of eight pipeline stages. A program has 20% branch instructions which execute in the Ex stage and produce the next instruction pointer at the end of the Ex stage in the old design and at the end of the Ex2 stage in the new design. The IF stage stalls after fetching a branch instruction until the next instruction pointer is computed. All instructions other than the branch instruction have an average CPI of one in both the designs. The execution times of this program on the old and the new design are P and Q nanoseconds, respectively. The value of P/Q is _____.

140. Consider the following processors (ns stands for nanoseconds). Assume that the pipeline registers have zero latency. P1: Four-stage pipeline with stage latencies 1 ns, 2 ns, 2 ns, 1 ns. P2: Four-stage pipeline with stage latencies 1 ns, 1.5 ns, 1.5 ns, 1.5 ns. P3: Five-stage pipeline with stage latencies 0.5 ns, 1 ns, 1 ns, 0.6 ns, 1 ns. P4: Five-stage pipeline with stage latencies 0.5 ns, 0.5 ns, 1 ns, 1 ns, 1.1 ns. Which processor has the highest peak clock frequency?

141. Consider a 6-stage instruction pipeline, where all stages are perfectly balanced. Assume that there is no cycle-time overhead of pipelining. When an application is executing on this 6-stage pipeline, the speedup achieved with respect to non-pipelined execution if 25% of the instructions incur 2 pipeline stall cycles is _____.

142. A pipeline P operating at 400 MHz has a speedup factor of 6 and operating at 70% efficiency. How many stages are there in the pipeline?

143. Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI and WO are 5 ns, 7 ns, 10 ns, 8 ns and 6 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 1 ns. A program consisting of

12 instructions I1, I2, I3,..., I12 is executed in this pipelined processor. Instruction I4 is the only branch instruction and its branch target is I9. If the branch is taken during the execution of this program, the time (in ns) needed to complete the program is

144. Register renaming is done in pipelined processors:

145. If a microcomputer operates at 5 MHz with an 8-bit bus and a newer version operates at 20 MHz with a 32-bit bus, the maximum speed-up possible approximately will be

146. A processor takes 12 cycles to complete an instruction I. The corresponding pipelined processor uses 6 stages with the execution times of 3,2,5,4,6 and 2 cycles respectively. What is the asymptotic speedup assuming that a very large number of instructions are to be executed?

147. A 5-stage pipelined processor has Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Write Operand (WO) stages. The IF, ID, OF and WO stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD and SUB instructions, 3 clock cycles for MUL instruction, and 6 clock cycles for DIV instruction respectively. Operand forwarding is used in the pipeline. What is the number of clock cycles needed to execute the following sequence of instructions?

148. Consider a pipelined processor with the following four stages: IF: Instruction Fetch ID: Instruction Decode and Operand Fetch EX: Execute WB: Write Back The IF, ID and WB stages take one clock cycle each to complete the operation. The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle and the MUL instruction needs 3 clock cycles in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions?

149. The performance of a pipelined processor suffers if:

150. A non pipelined single cycle processor operating at 100 MHz is converted into a synchronous pipelined processor with five stages requiring 2.5 nsec, 1.5 nsec, 2 nsec, 1.5 nsec and 2.5 nsec, respectively. The delay of the latches is 0.5 nsec. The speedup of the pipeline processor for a large number of instructions is:

151. Delayed branching can help in the handling of control hazards The following code is to run on a pipelined processor with one branch delay slot: I1: ADD $\leftarrow R2$ R7 +R8, I2 : SUB R4 $\leftarrow R5$ - R6, I3: ADD R1 $\leftarrow R2$ + R3. I4 : STORE Memory [R4] $\leftarrow R1$ BRANCH to Label if R1==0. Which of the instructions I1, I2, I3 or I4 can legitimately occupy the delay slot without any other program modification?

152. Which of the following are NOT true in a pipelined processor? I. Bypassing can handle all RAW hazards. II. Register renaming can eliminate all register carried WAR hazards. III. Control hazard penalties can be eliminated by dynamic branch prediction

153. The floating point unit of a processor using a design D takes 2t cycles compared to t cycles taken by the fixed point unit. There are two more design suggestions D1 and D2. D1 uses 30% more cycles for fixed point unit but 30% less cycles for floating point unit as compared to design D. D2 uses 40% less cycles for fixed point unit but 10% more cycles for floating point unit as compared to design D. For a given program which has 80% fixed point operations and 20% floating point operations, which of the following

ordering reflects the relative performances of three designs? ($D_i > D_j$ denotes that D_i is faster than D_j).

154. A processor takes 12 cycles to complete an instruction I . The corresponding pipelined processor uses 6 stages with the execution times of 3, 2, 5, 4, 6 and 2 cycles respectively. What is the asymptotic speedup assuming that a very large number of instructions are to be executed?

155. Consider a pipelined processor with the following four stages: IF: Instruction Fetch, ID: Instruction Decode and Operand Fetch, EX: Execute, WB: Write Back. The IF, ID and WB stages take one clock cycle each to complete the operation. The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle and the MUL instruction needs 3 clock cycles in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions? ADD R2, R1, R0 ($R2 \leftarrow R1 + R0$)
MUL R4, R3, R2 ($R4 \leftarrow R3 * R2$) SUB R6, R5, R4 ($R6 \leftarrow R5 - R4$)

156. A CPU has five-stages pipeline and runs at 1GHz frequency. Instruction fetch happens in the first stage of the pipeline. A conditional branch instruction computes the target address and evaluates the condition in the third stage of the pipeline. The processor stops fetching new instructions following a conditional branch until the branch outcome is known. A program executes 10^9 instructions out of which 20% are conditional branches. If each instruction takes one cycle to complete on average, then total execution time of the program is

157. We have two designs D_1 and D_2 for a synchronous pipeline processor. D_1 has 5 pipeline stages with execution times of 3 nsec, 2 nsec, 4 nsec, 2 nsec and 3 nsec while the design D_2 has 8 pipeline stages each with 2 nsec execution time. How much time can be saved using design D_2 over design D_1 for executing 100 instructions?

158. A pipelined processor uses a 4-stage instruction pipeline with the following stages: Instruction fetch (IF), Instruction decode (ID), Execute (EX) and Writeback (WB). The arithmetic operations as well as the load and store operations are carried out in the EX stage. The sequence of instructions corresponding to the statement $X = (S - R * (P + Q)) / T$ is given below. The values of variables P, Q, R, S and T are available in the registers R0, R1, R2, R3 and R4 respectively, before the execution of the instruction sequence.

159. In an enhancement of a design of a CPU, the speed of a floating point unit has been increased by 20% and the speed of a fixed point unit has been increased by 10%. What is the overall speedup achieved if the ratio of the number of floating point operations to the number of fixed point operations is 2:3 and the floating point operation used to take twice the time taken by the fixed point operation in the original design?

160. A 4-stage pipeline has the stage delays as 150, 120, 160 and 140 nanoseconds respectively. Registers that are used between the stages have a delay of 5 nanoseconds each. Assuming constant clocking rate, the total time taken to process 1000 data items on this pipeline will be

161. For a pipelined CPU with a single ALU, consider the following situations 1. The $(j + 1)$ -th instruction uses the result of j -th instruction as an operand. 2. The execution of a conditional jump instruction. 3. The j -th and $j + 1$ -st instructions require the ALU at the same time. Which of the above can cause a hazard?

162. The performance of a pipelined processor suffers if:

163. Comparing the time T_1 taken for a single instruction on a pipelined CPU with time T_2 taken on a non-pipelined but identical CPU, we can say that

164. A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stalls is 2. The program exhibits 2% miss rate on instruction cache and 8% miss rate on data cache. The miss penalty is 100 cycles. The speedup (rounded off to two decimal places) achieved with a perfect cache (i.e., with NO data or instruction cache misses) is _____

165. Consider two set-associative cache memory architectures: WBC, which uses the write back policy, and WTC, which uses the write through policy. Both of them use the LRU (Least Recently Used) block replacement policy. The cache memory is connected to the main memory. Which of the following statements is/are TRUE?

166. An 8-way set associative cache of size 64 KB (1 KB = 1024 bytes) is used in a system with 32-bit address. The address is sub-divided into TAG, INDEX, and BLOCK OFFSET. The number of bits in the TAG is _____.

167. Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (i.e., PQRSPQRS...). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

168. A cache memory that has a hit rate of 0.8 has an access latency 10 ns and miss penalty 100 ns. An optimization is done on the cache to reduce the miss rate. However, the optimization results in an increase of cache access latency to 15 ns, whereas the miss penalty is not affected. The minimum hit rate (rounded off to two decimal places) needed after the optimization such that it should not increase the average memory access time is _____.

169. Let WB and WT be two set associative cache organizations that use LRU algorithm for cache block replacement. WB is a write back cache and WT is a write through cache. Which of the following statements is/are FALSE?

170. Assume a two-level inclusive cache hierarchy, L1 and L2, where L2 is the larger of the two. Consider the following statements. S1: Read misses in a write through L1 cache do not result in writebacks of dirty lines to the L2. S2: Write allocate policy must be used in conjunction with write through caches and no-write allocate policy is used with writeback caches. Which of the following statements is correct?

171. Consider a set-associative cache of size 2KB ($1\text{KB}=2^{10}$ bytes) with cache block size of 64 bytes. Assume that the cache is byte-addressable and a 32-bit address is used for accessing the cache. If the width of the tag field is 22 bits, the associativity of the cache is _____

172. Consider a computer system with a byte-addressable primary memory of size 2^{32} bytes. Assume the computer system has a direct-mapped cache of size 32KB ($1\text{KB}=2^{10}$ bytes), and each cache block is of size 64 bytes. The size of the tag field is _____ bits.

173. How many total bits are required for a direct-mapped cache with 128 KB of data and 1 word block size, assuming a 32-bit address and 1 word size of 4 bytes?

174. A certain processor deploys a single-level cache. The cache block size is 8 words and the word size is 4 bytes. The memory system uses a 60-MHz clock. To service a cache-miss, the memory controller first takes 1 cycle to accept the starting address of the block, it then takes 3 cycles to fetch all the eight words of the block, and finally transmits the words of the requested block at the rate of 1 word per cycle. The maximum bandwidth for the memory system when the program running on the processor issues a series of read operations is _____ $\times 10^6$ bytes/sec.

175. A direct mapped cache memory of 1 MB has a block size of 256 bytes. The cache has an access time of 3 ns and a hit rate of 94%. During a cache miss, it takes 20 ns to bring the first word of a block from the main memory, while each subsequent word takes 5 ns. The word size is 64 bits. The average memory access time in ns (round off to 1 decimal place) is _____.

176. A certain processor uses a fully associative cache of size 16 kB, The cache block size is 16 bytes. Assume that the main memory is byte addressable and uses a 32-bit address. How many bits are required for the Tag and the Index fields respectively in the addresses generated by the processor?

177. A computer system with a word length of 32 bits has a 16 MB byte- addressable main memory and a 64 KB, 4-way set associative cache memory with a block size of 256 bytes. Consider the following four physical addresses represented in hexadecimal notation. $A1=0x42C8A4$, $A2=0x546888$, $A3=0x6A289C$, $A4=0x5E4880$

178. The read access time of main memory is 90 nanoseconds. Assume that the caches use the referred word-first read policy and the write back policy. Assume that all the caches are direct mapped caches. Assume that the dirty bit is always 0 for all the blocks in the caches. In execution of a program, 60% of memory reads are for instruction fetch and 40% are for memory operand fetch. The average read access time in nanoseconds (up to 2 decimal places) is _____.

179. Consider a machine with a byte addressable main memory of 2^{32} bytes divided into blocks of size 32 bytes. Assume that a direct mapped cache having 512 cache lines is used with this machine. The size of the tag field in bits is _____.

180. A cache memory needs an access time of 30 ns and main memory 150 ns, what is average access time of CPU (assume hit ratio = 80%)?

181. The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N Bytes. The size of each cache block is 2^M words. For a K-way set-associative cache memory, the length (in number of bits) of the tag field is

182. For a multi-processor architecture, in which protocol a write transaction is forwarded to only those processors that are known to possess a copy of newly altered cache line?

183. Consider a system with 2 level cache. Access times of Level 1 cache, Level 2 cache and main memory are 1 ns, 10 ns, and 500 ns respectively. The hit rates of Level 1 and Level 2 caches are 0.8 and 0.9, respectively. What is the average access time of the system ignoring the search time within the cache?

184. Consider a 2-way set associative cache with 256 blocks and uses LRU replacement, Initially the cache is empty. Conflict misses are those misses which occur due the contention of multiple blocks for the same cache set. Compulsory misses occur due to first time access to the block. The following sequence of accesses to memory blocks. (0,128,256,128,0,128,256,128,1,129,257,129,1,129,257,129) is repeated 10 times. The number of conflict misses experienced by the cache is _____.

185. Consider a two-level cache hierarchy with L1 and L2 caches. An application incurs 1.4 memory accesses per instruction on average. For this application, the miss rate of L1 cache 0.1, the L2 cache experiences, on average, 7 misses per 1000 instructions. The miss rate of L2 expressed correct to two decimal places is _____.

186. A cache memory unit with capacity of N words and block size of B words is to be designed. If it is designed as a direct mapped cache, the length of the TAG field is 10 bits. If the cache unit is now designed as a 16-way set-associative cache, the length of the TAG field is _____bits.

187. In a two-level cache system, the access times of L1 and L2 caches are 1 and 8 clock cycles, respectively. The miss penalty from L2 cache to main memory is 18 clock cycles . The miss rate of L1 cache is twice that of L2. The average memory access time (AMAT) of this cache system is 2 cycles. This miss rates of L1 and L2 respectively are :

188. The width of the physical address on a machine is 40 bits. The width of the tag field in a 512KB 8-way set associative cache is _____ bits.

189. How much speed do we gain by using the cache, when cache is used 80% of the time? Assume cache is faster than main memory.

190. An access sequence of cache block addresses is of length N and contains n unique block addresses. The number of unique block addresses between two consecutive accesses to the same block address is bounded above K. What is the miss ratio if the access sequence is passed through a cache of associativity $A \geq k$ exercising least-recently-used replacement policy?

191. A 4-way set-associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. The number of bits for the TAG field is _____

192. In designing a computer's cache system, the cache block (or cache line) size is an important Parameter. Which one of the following statements is correct in this context?

193. If the associativity of a processor cache is doubled while keeping the capacity and block size unchanged, which one of the following is guaranteed to be NOT affected?

194. The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 2 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations, 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. The average memory access time (in nanoseconds) in executing the sequence of instructions is _____.

195. Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____.

196. Consider a machine with a byte addressable main memory of 2^{20} bytes, block size of 16 bytes and a direct mapped cache having 2^{12} cache lines. Let the addresses of two consecutive bytes in main memory be (E201F) base 16 and (E2020) base 16. What are the tag and cache line address (in hex) for main memory address (E201F) base 16 ?

197. Consider a direct mapped cache with 64 blocks and a block size of 16 bytes. To what block number does the byte address 1206 map to

198. The search concept used in associative memory is

199. A computer has a 256 KByte, 4-way set associative, write back data cache with block size of 32 Bytes. The processor sends 32 bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit. The number of bits in the tag field of an address is

200. A computer has a 256 KByte, 4-way set associative, write back data cache with block size of 32 Bytes. The processor sends 32 bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit. The size of the cache tag directory is

201. In a k-way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set are placed in sequence one after another. The lines in set s are sequenced before the lines in set (s+1). The main memory blocks are numbered 0 onwards. The main memory block numbered j must be mapped to any one of the cache lines from

202. A 32-bit wide main memory unit with a capacity of 1 GB is built using 256M x 4-bit DRAM chips. The number of rows of memory cells in the DRAM chip is 2^{14} . The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closest integer) of the time available for performing the memory read/write operations in the main memory unit is _____.

203. How many 128x8 bit RAMs are required to design 32Kx32 bit RAM?

204. Suppose you want to build a memory with 4 byte words and a capacity of 2^{21} bits. What is type of decoder required if the memory is built using 2Kx8 RAM chips?

205. How many 32K x 1 RAM chips are needed to provide a memory capacity of 256K-bytes?

206. If each address space represents one byte of storage space, how many address lines are needed to access RAM chips arranged in a 4x6 array, where each chip is 8Kx4 bits ?

207. A RAM chip has a capacity of 1024 words of 8 bits each (1K x 8). The number of 2 x 4 decoders with enable line needed to construct a 16K x 16 RAM from 1K x 8 RAM is

- 208.** The amount of ROM needed to implement a 4 bit multiplier is
- 209.** Number of chips (128×8 RAM) needed to provide a memory capacity of 2048 bytes
- 210.** A main memory unit with a capacity of 4 megabytes is built using 1Mx1-bit DRAM chips. Each DRAM chip has 1K rows of cells with 1K cells in each row. The time taken for a single refresh operation is 100 nanoseconds. The time required to perform one refresh operation on all the cells in the memory unit is
- 211.** The process of organizing the memory into two banks to allow 8-and 16-bit data operation is called
- 212.** How many 32K x 1 RAM chips are needed to provide a memory capacity of 256Kbytes?
- 213.** In comparison with static RAM memory, the dynamic Ram memory has
- 214.** A dynamic RAM has a memory cycle time of 64 nsec. It has to be refreshed 100 times per msec and each refresh takes 100 nsec. What percentage of the memory cycle time is used for refreshing?
- 215.** What is the minimum size of ROM required to store the complete truth table of an 8-bit × 8-bit multiplier?
- 216.** A particular disk unit uses a bit string to record the occupancy or vacancy of its tracks, with 0 denoting vacant and 1 for occupied. A 32-bit segment of this string has hexadecimal value D4FE2003. The percentage of occupied tracks for the corresponding part of the disk, to the nearest percentage is:
- 217.** Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each sector holds 1024 bytes of data. The number of cylinders in the hard disk is ____
- 218.** A magnetic disk has 100 cylinders, each with 10 tracks of 10 sectors. If each sector contains 128 bytes, what is the maximum capacity of the disk in kilobytes?
- 219.** Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512-byte sectors. A file has content stored in 3000 sectors located randomly on the disk. Assuming average rotational latency, the total time (in seconds, rounded off to 2 decimal places) to read the entire file from the disk is _____
- 220.** The capacity of a memory unit is defined by the number of words multiplied by the number of bits/word. How many separate address and data lines are needed for a memory of 4K×16?
- 221.** The refreshing rate of dynamic RAMs is in the range of
- 222.** What is the minimum size of ROM required to store the complete truth table of an 8-bit × 8-bit multiplier?
- 223.** A ROM is used to store the table for multiplication of two 8-bit unsigned integers. The size of ROM required is
- 224.** Raid configurations of the disks are used to provide
- 225.** The total time to prepare a disk drive mechanism for a block of data to be read from it is

226. A hard disk has 63 sectors per track, 10 platters each with 2 recording surfaces and 1000 cylinders. The address of a sector is given as a triple (c,h,s), where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addressed as (0,0,0), the 1st sector as (0,0,1), and so on. The address (400, 16, 29) corresponds to sector number:

227. Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stores in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively

228. An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10ms. Rotational speed of disk is 6000rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected)

229. A fast wide SCSI-II disk drive spins at 7200 RPM, has a sector size of 512 bytes, and holds 160 sectors per track. Estimate the sustained transfer rate of this drive

230. Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each cylinder contains 64 sectors (0-63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is (cylinder no., surface no., sector no.). A file of size 42797 KB is stored in the disk and the starting disk location of the file is (1200, 9, 40). What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

231. In case of a DVD, the speed of data transfer is mentioned in multiples of?

232. Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors. Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete rotation. The total time (in milliseconds) needed to read the entire file is _____.

233. Consider a typical disk that rotates at 15000 rotations per minute (RPM) and has a transfer rate of 50×10^6 bytes/sec. If the average seek time of the disk is twice the average rotational delay and the controller's transfer time is 10 times the disk transfer time, the average time (in milliseconds) to read or write a 512-byte sector of the disk is _____.

234. Six files F1, F2, F3, F4, F5 and F6 have 100, 200, 50, 80, 120, 150 records respectively. In what order should they be stored so as to optimize act. Assume each file is accessed with the same frequency

235. A hard disk system has the following parameters : Number of tracks =500, Number of sectors/track =100, Number of bytes /sector =500, Time taken by the head to move from one track to adjacent track =1 ms, Rotation speed =600 rpm. What is the average time taken for transferring 250 bytes from the disk ?

236. Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively:

237. For a magnetic disk with concentric circular tracks, the seek latency is not linearly proportional to the seek distance due to

238. Which of the following RAID level provides the highest Data Transfer Rate (Read/Write)

239. A hard disk system has the following parameters :Number of tracks = 500, Number of sectors/track = 100, Number of bytes /sector = 500, Time taken by the head to move from one track to adjacent track = 1 ms, Rotation speed = 600 rpm.What is the average time taken for transferring 250 bytes from the disk ?

240. A disk has 8 equidistant tracks. The diameters of the innermost and outermost tracks are 1 cm and 8 cm respectively. The innermost track has a storage capacity of 10 MB.What is the total amount of data that can be stored on the disk if it is used with a drive that rotates it with I.Constant Linear Velocity.
II.Constant Angular Velocity?

241. The storage area of a disk has the innermost diameter of 10 cm and outermost diameter of 20 cm. The maximum storage density of the disk is 1400 bits/cm. The disk rotates at a speed of 4200 RPM. The main memory of a computer has 64-bit word length and $1\mu\text{s}$ cycle time. If cycle stealing is used for data transfer from the disk, the percentage of memory cycles stolen for transferring one word is

242. Using a larger block size in a fixed block size file system leads to

243. Raid configurations of the disks are used to provide

244. Formatting for a floppy disk refers to

245. The data transfer rate of a double-density floppy disk system is about

246. A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stalls is 2. The program exhibits 2% miss rate on instruction cache and 8% miss rate on data cache. The miss penalty is 100 cycles. The speedup (rounded off to two decimal places) achieved with a perfect cache (i.e., with NO data or instruction cache misses) is _____

247. Consider two set-associative cache memory architectures: WBC, which uses the write back policy, and WTC, which uses the write through policy. Both of them use the LRU (Least Recently Used) block replacement policy. The cache memory is connected to the main memory. Which of the following statements is/are TRUE?

248. An 8-way set associative cache of size 64 KB (1 KB = 1024 bytes) is used in a system with 32-bit address. The address is sub-divided into TAG, INDEX, and BLOCK OFFSET. The number of bits in the TAG is _____.

249. Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (i.e., PQRSPQRS...). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

250. A cache memory that has a hit rate of 0.8 has an access latency 10 ns and miss penalty 100 ns. An optimization is done on the cache to reduce the miss rate. However, the optimization results in an increase of cache access latency to 15 ns, whereas the miss penalty is not affected. The minimum hit rate (rounded off to two decimal places) needed after the optimization such that it should not increase the average memory access time is _____.

251. Let WB and WT be two set associative cache organizations that use LRU algorithm for cache block replacement. WB is a write back cache and WT is a write through cache. Which of the following statements is/are FALSE?

252. Assume a two-level inclusive cache hierarchy, L1 and L2, where L2 is the larger of the two. Consider the following statements. S1: Read misses in a write through L1 cache do not result in writebacks of dirty lines to the L2. S2: Write allocate policy must be used in conjunction with write through caches and no-write allocate policy is used with writeback caches. Which of the following statements is correct?

253. Consider a set-associative cache of size 2KB ($1\text{KB}=2^{10}$ bytes) with cache block size of 64 bytes. Assume that the cache is byte-addressable and a 32-bit address is used for accessing the cache. If the width of the tag field is 22 bits, the associativity of the cache is _____

254. Consider a computer system with a byte-addressable primary memory of size 2^{32} bytes. Assume the computer system has a direct-mapped cache of size 32KB ($1\text{KB}=2^{10}$ bytes), and each cache block is of size 64 bytes. The size of the tag field is _____ bits.

255. How many total bits are required for a direct-mapped cache with 128 KB of data and 1 word block size, assuming a 32-bit address and 1 word size of 4 bytes?

256. A computer system with a word length of 32 bits has a 16 MB byte-addressable main memory and a 64 KB, 4-way set associative cache memory with a block size of 256 bytes. Consider the following four physical addresses represented in hexadecimal notation. $A1=0x42C8A4$, $A2=0x546888$, $A3=0x6A289C$, $A4=0x5E4880$

257. A direct mapped cache memory of 1 MB has a block size of 256 bytes. The cache has an access time of 3 ns and a hit rate of 94%. During a cache miss, it takes 20 ns to bring the first word of a block from the main memory, while each subsequent word takes 5 ns. The word size is 64 bits. The average memory access time in ns (round off to 1 decimal place) is _____.

258. A certain processor deploys a single-level cache. The cache block size is 8 words and the word size is 4 bytes. The memory system uses a 60-MHz clock. To service a cache-miss, the memory controller first takes 1 cycle to accept the starting address of the block, it then takes 3 cycles to fetch all the eight words of the block, and finally transmits the words of the requested block at the rate of 1 word per cycle. The maximum bandwidth for the memory system when the program running on the processor issues a series of read operations is _____ $\times 10^6$ bytes/sec.

259. A certain processor uses a fully associative cache of size 16 kB, The cache block size is 16 bytes. Assume that the main memory is byte addressable and uses a 32-bit address. How many bits are required for the Tag and the Index fields respectively in the addresses generated by the processor?

260. The read access time of main memory is 90 nanoseconds. Assume that the caches use the referred word-first read policy and the write back policy. Assume that all the caches are direct mapped caches. Assume that the dirty bit is always 0 for all the blocks in the caches. In execution of a program, 60% of memory reads are for instruction fetch and 40% are for memory operand fetch. The average read access time in nanoseconds (up to 2 decimal places) is _____.

261. Consider a machine with a byte addressable main memory of 2^{32} bytes divided into blocks of size 32 bytes. Assume that a direct mapped cache having 512 cache lines is used with this machine. The size of the tag field in bits is _____.

262. A cache memory needs an access time of 30 ns and main memory 150 ns, what is average access time of CPU (assume hit ratio = 80%)?

263. The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N Bytes. The size of each cache block is 2^M words. For a K-way set-associative cache memory, the length (in number of bits) of the tag field is

264. For a multi-processor architecture, in which protocol a write transaction is forwarded to only those processors that are known to possess a copy of newly altered cache line?

265. In a two-level cache system, the access times of L1 and L2 caches are 1 and 8 clock cycles, respectively. The miss penalty from L2 cache to main memory is 18 clock cycles. The miss rate of L1 cache is twice that of L2. The average memory access time (AMAT) of this cache system is 2 cycles. This miss rates of L1 and L2 respectively are :

266. A cache memory unit with capacity of N words and block size of B words is to be designed. If it is designed as a direct mapped cache, the length of the TAG field is 10 bits. If the cache unit is now designed as a 16-way set-associative cache, the length of the TAG field is _____bits.

267. Consider a 2-way set associative cache with 256 blocks and uses LRU replacement, Initially the cache is empty. Conflict misses are those misses which occur due the contention of multiple blocks for the same cache set. Compulsory misses occur due to first time access to the block. The following sequence of accesses to memory blocks.(0,128,256,128,0,128,256,128,1,129,257,129,1,129,257,129) is repeated 10 times. The number of conflict misses experienced by the cache is _____.

268. Consider a two-level cache hierarchy with L1 and L2 caches. An application incurs 1.4 memory accesses per instruction on average. For this application, the miss rate of L1 cache 0.1, the L2 cache experiences, on average, 7 misses per 1000 instructions. The miss rate of L2 expressed correct to two decimal places is _____.

269. Consider a system with 2 level cache. Access times of Level 1 cache, Level 2 cache and main memory are 1 ns, 10 ns, and 500 ns respectively. The hit rates of Level 1 and Level 2 caches are 0.8 and 0.9, respectively. What is the average access time of the system ignoring the search time within the cache?

270. How much speed do we gain by using the cache, when cache is used 80% of the time? Assume cache is faster than main memory.

271. An access sequence of cache block addresses is of length N and contains n unique block addresses. The number of unique block addresses between two consecutive accesses to the same block address is bounded above K . What is the miss ratio if the access sequence is passed through a cache of associativity $A \geq k$ exercising least-recently-used replacement policy?

272. A 4-way set-associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. The number of bits for the TAG field is _____

273. In designing a computer's cache system, the cache block (or cache line) size is an important Parameter. Which one of the following statements is correct in this context?

274. If the associativity of a processor cache is doubled while keeping the capacity and block size unchanged, which one of the following is guaranteed to be NOT affected?

275. The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 2 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations, 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. The average memory access time (in nanoseconds) in executing the sequence of instructions is _____.

276. Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____.

277. Consider a machine with a byte addressable main memory of 2^{20} bytes, block size of 16 bytes and a direct mapped cache having 2^{12} cache lines. Let the addresses of two consecutive bytes in main memory be (E201F) base 16 and (E2020) base 16. What are the tag and cache line address (in hex) for main memory address (E201F) base 16 ?

278. The width of the physical address on a machine is 40 bits. The width of the tag field in a 512KB 8-way set associative cache is _____ bits.

279. An 8KB direct mapped write-back cache is organized as multiple blocks, each of size 32-bytes. The processor generates 32-bit addresses. The cache controller maintains the tag information for each cache block comprising of the following. 1 Valid bit 1 Modified bit As many bits as the minimum needed to identify the memory block mapped in the cache. What is the total size of memory needed at the cache controller to store metadata (tags) for the cache?

280. Consider a direct mapped cache with 64 blocks and a block size of 16 bytes. To what block number does the byte address 1206 map to

281. The main memory of a computer has 2^m blocks while the cache has 2^c blocks. If the cache uses the set associative mapping scheme with 2 blocks per set, then block k of the main memory maps to the set:

282. A computer system has a 4 K word cache organized in block-set-associative manner with 4 blocks per set, 64 words per block. The number of bits in the SET and WORD fields of the main memory address format is:

283. Consider a 2-way set associative cache memory with 4 sets and total 8 cache blocks (0-7) and a main memory with 128 blocks (0-127). What memory blocks will be present in the cache after the following sequence of memory block references if LRU policy is used for cache block replacement? 0 5 3 9 7 0 16 55. Assuming that initially the cache did not have any memory block from the current job?

284. Consider a direct mapped cache of size 32 KB with block size 32 bytes. The CPU generates 32 bit addresses. The number of bits needed for cache indexing and the number of tag bits are respectively.

285. Consider a fully associative cache with 8 cache blocks (numbered 0-7) and the following sequence of memory block requests: 4,3,25,8,19,6,25,8,16,35,45,22,8,3,16,25,7. If LRU replacement policy is used, which cache block will have memory block 7?

286. Consider a system with 2 level cache. Access times of Level 1 cache, Level 2 cache and main memory are 1 ns, 10 ns, and 500 ns respectively. The hit rates of Level 1 and Level 2 caches are 0.8 and 0.9, respectively. What is the average access time of the system ignoring the search time within the cache?

287. Consider a small two-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently used (LRU) scheme. The number of cache misses for the following sequence of block addresses is 8, 12, 0, 12,8

288. Consider two cache organizations: The first one is 32 KB 2-way set associative with 32-byte block size. The second one is of the same size but direct mapped. The size of an address is 32 bits in both cases. A 2-to-1 multiplexer has a latency of 0.6 ns while a kbit comparator has a latency of $k/10$ ns. The hit latency of the set associative organization is h_1 while that of the direct mapped one is h_2 . The value of h_2 is:

289. A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is

290. A CPU has a cache with block size 64 bytes. The main memory has k banks, each bank being c bytes wide. Consecutive c -byte chunks are mapped on consecutive banks with warp-around. All the k banks can be accessed in parallel, but two accesses to the same bank must be serialized. A cache block access may involve multiple iterations of parallel bank accesses depending on the amount of data obtained by accessing all the k banks in parallel. Each iteration requires decoding the bank numbers to be accessed in parallel and this takes $k/2$ ns. The latency of one bank access is 80 ns. If $c = 2$ and $k=24$, then latency of retrieving a cache block starting at address zero from main memory is

291. Consider a machine with a byte addressable main memory of 2^{16} bytes. Assume that a direct mapped data cache consisting of 32 lines of 64 bytes each is used in the system. A 50×50 two-dimensional array of bytes is stored in the main memory starting from memory location 1100H. Assume that the data cache is initially empty. The complete array is accessed twice. Assume that the

contents of the data cache do not change in between the two accesses.

292. Consider a small 2-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently (LRU) scheme. The number of cache misses for the following sequence of block addresses is 8, 12, 0, 12, 8

293. Consider a Direct Mapped Cache with 8 cache blocks (numbered 0-7). If the memory block requests are in the following order 3, 5, 2, 8, 0, 63, 9,16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82,17, 24.Which of the following memory blocks will not be in the cache at the end of the sequence ?

294. An 8KB direct mapped write-back cache is organized as multiple blocks, each of size 32-bytes. The processor generates 32-bit addresses. The cache controller maintains the tag information for each cache block comprising of the following. 1 Valid bit 1 Modified bit As many bits as the minimum needed to identify the memory block mapped in the cache. What is the total size of memory needed at the cache controller to store metadata (tags) for the cache?

295. In a k-way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set are placed in sequence one after another. The lines in set s are sequenced before the lines in set (s+1). The main memory blocks are numbered 0 onwards. The main memory block numbered j must be mapped to any one of the cache lines from

296. A computer has a 256 KByte, 4-way set associative, write back data cache with block size of 32 Bytes. The processor sends 32 bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit.

297. Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____.

298. The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 2 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations, 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. The average memory access time (in nanoseconds) in executing the sequence of instructions is _____.

299. A cache memory unit with capacity of N words and block size of B words is to be designed. If it is designed as a direct mapped cache, the length of the TAG field is 10 bits. If the cache unit is now designed as a 16-way set-associative cache, the length of the TAG field is _____bits.

300. The value of a float type variable is represented using the single-precision -bit floating point format of standard that uses bit for sign, bits for biased exponent and bits for mantissa. float type variable is assigned the decimal value of The representation of in hexadecimal notation is

Digital Logic

1. Which of the following is/are EQUAL to 224 in radix-5 (i.e., base-5) notation?
2. A particular number is written as 132 in radix-4 representation. The same number in radix-5 representation is _____.
3. Consider the IEEE-754 single precision floating point numbers $P=0xC1800000$ and $Q=0x3F5C2EF4$. Which one of the following corresponds to the product of these numbers (i.e., $P \times Q$), represented in the IEEE-754 single precision format?
4. Consider a system that uses 5 bits for representing signed integers in 2's complement format. In this system, two integers A and B are represented as $A=01010$ and $B=11010$. Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow?
5. Consider three floating point numbers A, B and C stored in registers RA, RB and RC, respectively as per IEEE-754 single precision floating point format. The 32-bit content stored in these registers (in hexadecimal form) are as follows. $RA = 0xC1400000$, $RB = 0x42100000$, $RC = 0x41400000$. Which one of the following is FALSE?
6. Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation $R1+R2$, which one of the following values of R1 and R2 gives an arithmetic overflow?
7. If the numerical value of a 2-byte unsigned integer on a little endian computer is 255 more than that on a big endian computer, which of the following choices represent(s) the unsigned integer on a little endian computer?
8. If x and y are two decimal digits and $(0.1101)_{\text{base } 2} = (0.8xy5)_{\text{base } 10}$, the decimal value of $x+y$ is _____.
9. Consider the following representation of a number in IEEE 754 single-precision floating point format with a bias of 127. S:1, E:10000001, F:11110000000000000000000. Here, S,E and F denote the sign, exponent, and fraction components of the floating point representation. The decimal value corresponding to the above representation (rounded to 2 decimal places) is _____.
10. Let the representation of a number in base 3 be 210. What is the hexadecimal representation of the number?
11. Consider three registers R1, R2, and R3 that store numbers in IEEE-754 single precision floating point format. Assume that R1 and R2 contain the values (in hexadecimal notation) $0x42200000$ and $0xC1200000$, respectively. If $R3 = R1/R2$, what is the value stored in R3?
12. Consider $Z = X - Y$ where X, Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of:
13. A computer uses ternary system instead of the traditional system, An n bit string in the binary system will occupy

14. If a variable can take only integral values from 0 to n , where n is an integer, then the variable can be represented as a bit-field whose width is (the log in the answer are to the base 2, and $\lfloor \log_2 n \rfloor$ means the floor of $\log_2 n$)
15. The representation of the value of a 16-bit unsigned integer X in hexadecimal number system is BCA9. The representation of the value of X in octal number system is
16. The n -bit fixed-point representation of an unsigned real number real X uses f bits for the fraction part. Let $i=n-f$. The range of decimal values for X in this representation is
17. Let X be the number of distinct 16-bit integers in 2's complement representation. Let Y be the number of distinct 16 bit integers in sign magnitude representation. Then $X-Y$ is _____.
18. The 16 bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is _____.
19. The decimal number has 64 digits. The number of bits needed for its equivalent binary representation is?
20. The number of 1's in the binary representation of $(3*4096+15*256+5*16+3)$ are:
21. The code which uses 7 bits to represent a character is :
22. The range of integers that can be represented by an n bit 2's complement number system is:
23. Which of the given number has its IEEE-754 32-bit floating point representation as (0 10000000 110 0000 0000 0000 0000 0000)
24. Consider the equation $(43)_{\text{base } x} = (y3)_{\text{base } 8}$ where x and y are unknown. The number of possible solutions is _____
25. How many different BCD numbers can be stored in 12 switches ? (Assume two position or on-off switches).
26. Consider the equation $(123)_5 = (x8)y$ with x and y as unknown. The number of possible solutions is _____.
27. The value of a float type variable is represented using the single-precision 32-bit floating point format of IEEE-754 standard that uses 1 bit for sign, 8 bits for biased exponent and 23 bits for mantissa. A float type variable x is assigned the decimal value of -14.25. The representation of x in hexadecimal notation is
28. In the standard IEEE 754 single precision floating point representation, there is 1 bit for sign, 23 bits for fraction and 8 bits for exponent. What is the precision in terms of the number of decimal digits?
29. The base (or radix) of the number system such that the following equation holds is _____. $312/20 = 13.1$
30. The binary equivalent of the decimal number 42.75 is

31. Two eight bit bytes 11000011 and 01001100 are added. What are the values of the overflow, carry and zero flags respectively, if the arithmetic unit of the CPU uses 2's complement form?
32. The number 1102 in base 3 is equivalent to 123 in which base system?
33. When two BCD numbers 0×14 and 0×08 are added what is the binary representation of the resultant number ?
34. What is the decimal value of the floating-point number C1D00000 (hexadecimal notation)? (Assume 32-bit, single precision floating point IEEE representation)
35. P is a 16-bit signed integer. The 2's complement representation of P is (F87B)₁₆. The 2's complement representation of $8 \times P$ is
36. The addition of 4-bit, two's complement, binary numbers 1101 and 0100 results in
37. The two numbers given below are multiplied using the Booth's algorithm Multiplicand: 0101 1010 1110 1110 Multiplier: 0111 0111 1011 1101 .How many additions/subtractions are required for the multiplication of the above two numbers?
38. The range of integers that can be represented by an n bit 2's complement number system is:
39. Let r denote number system radix. The only value(s) of r that satisfy the equation $(121r)^{1/2} = 11r$ is / are
40. The following bit pattern represents a floating point number in IEEE 754 single precision format 1 10000011 10100000000000000000000. The value of the number in decimal form is
41. A computer uses 8 digit mantissa and 2 digit exponent. If $a=0.052$ and $b=28E+11$ then $b+a-b$ will :
42. Consider a computer system that stores a floating-point numbers with 16-bit mantissa and an 8-bit exponent, each in two's complement. The smallest and largest positive values which can be stored are :
43. One approach to handling fuzzy logic data might be to design a computer using ternary (base-3) logic so that data could be stored as "true," "false," and "unknown." If each ternary logic element is called a flit, how many flits are required to represent at least 256 different values?
44. Consider number represented in 4-bit gray code. Let $h_3h_2h_1h_0$ be the gray code representation of a number n and let $g_3g_2g_1g_0$ be the gray code of (n+1) (modulo 16) value of the number. Which one of the following functions is correct?
45. Using Booth's Algorithm for multiplication, the multiplier -57 will be recoded as
46. Using a 4-bit 2's complement arithmetic, which of the following additions will result in an overflow?
(I).1100 + 1100 (II). 0011 + 0111 (III). 1111 + 0111
47. Let A = 1111 1010 and B = 0000 1010 be two 8-bit 2's complement numbers. Their product in 2's complement is

48. What is the result of evaluating the following two expressions using three-digit floating point arithmetic with rounding? $(113. + - 111.) + 7.51$ $113. + (- 111. + 7.51)$
49. If $73x_{\blacksquare}$ (in base-x number system) is equal to $54y_{\blacksquare}$ (in base y-number system), the possible values of x and y are
50. The exponent of a floating-point number is represented in excess-N code so that:
51. The number of 1's in the binary representation of $(3 \times 4096 + 15 \times 256 + 5 \times 16 + 3)$ are:
52. Booth's algorithm for integer multiplication gives worst performance when the multiplier pattern is
53. The octal representation of an integer is $(342)_{\text{base } 8}$. If this were to be treated as an eight-bit integer in an 8085 based computer, its decimal equivalent is
54. Consider an eight-bit ripple-carry Combinational Circuit for computing the sum of A and B, where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is _____.
55. For a binary half-subtractor having two inputs A and B, the correct set of logical outputs $D(=A \text{ minus } B)$ and $X(=\text{borrow})$ are
56. When two 8-bit numbers $A7_{\blacksquare} \dots A0_{\blacksquare}$ and $B7_{\blacksquare} \dots B0_{\blacksquare}$ in 2's complement representation (with $A0_{\blacksquare}$ and $B0_{\blacksquare}$ as the least significant bits) are added using a ripple-carry Combinational Circuit, the sum bits obtained are $S7_{\blacksquare} \dots S0_{\blacksquare}$ and the carry bits are $C7_{\blacksquare} \dots C0_{\blacksquare}$. An overflow is said to have occurred if
57. A multiplexer is placed between a group of 32 registers and an accumulator to regulate data movement such that at any given point in time the content of only one register will move to the accumulator. The number of select lines needed for the multiplexer is _____.
58. If there are m input lines and n output lines for a decoder that is used to uniquely address a byte addressable 1 KB RAM, then the minimum value of $m+n$ is _____.
59. In a 8-bit ripple carry adder using identical full adders, each full adder takes 34 ns for computing sum. If the time taken for 8-bit addition is 90 ns, find time taken by each full adder to find carry.
60. Minimum number of 2×1 multiplexers required to realize the following function, $f = A \text{ bar } B \text{ bar } C + A \text{ bar } B \text{ bar } C \text{ bar}$. Assume that inputs are available only in true form and Boolean a constant 1 and 0 are available.
61. A half Combinational Circuit is implemented with XOR and AND gates. A full Combinational Circuit is implemented with two half Combinational Circuits and one OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds. A 4-bit ripple-carry binary Combinational Circuit is implemented by using four full Combinational Circuits. The total propagation time of this 4-bit binary Combinational Circuit in microseconds is _____.
62. If half adders and full adders are implements using gates, then for the addition of two 17 bit numbers (using minimum gates) the number of half adders and full adders required will be

- 63.** A circuit outputs a digit in the form of 4 bits. 0 is represented by 0000, 1 by 0001, ..., 9 by 1001. A combinational circuit is to be designed which takes these 4 bits as input and outputs 1 if the digit ≥ 5 , and 0 otherwise. If only AND, OR and NOT gates may be used, what is the minimum number of gates required?
- 64.** Consider a multiplexer with X and Y as data inputs and Z as control input. Z = 0 selects input X, and Z = 1 selects input Y. What are the connections required to realize the 2-variable Boolean function $f = T + R$, without using any additional hardware?
- 65.** A 4-bit carry look ahead Combinational Circuit, which adds two 4-bit numbers, is designed using AND, OR, NOT, NAND, NOR gates only. Assuming that all the inputs are available in both complemented and uncomplemented forms and the delay of each gate is one time unit, what is the overall propagation delay of the Combinational Circuit? Assume that the carry network has been implemented using two-level AND-OR logic.
- 66.** How many 3-to-8 line decoders with an enable input are needed to construct a 6- to-64 line decoder without using any other logic gates?
- 67.** Suppose only one multiplexer and one inverter are allowed to be used to implement any Boolean function of n variables. What is the minimum size of the multiplexer needed?
- 68.** How many 2-input multiplexers are required to construct a 2^{10} -input multiplexer?
- 69.** How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?
- 70.** How many programmable fuses are required in a PLA which takes 16 inputs and gives 8 outputs? It has to use 8 OR gates and 32 AND gates.
- 71.** A half Combinational Circuit is implemented with XOR and AND gates. A full Combinational Circuit is implemented with two half Combinational Circuits and one OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds. A 4-bit ripple-carry binary Combinational Circuit is implemented by using four full Combinational Circuits. The total propagation time of this 4-bit binary Combinational Circuit in microseconds is _____.
- 72.** In a look-ahead carry generator, the carry generate function G and the carry propagate function P_i for inputs A_i and B_i are given by: $P_i = A_i \oplus B_i$ and $G_i = A_i B_i$. The expressions for the sum bit S_i and the carry bit C_{i+1} of the look-ahead carry Combinational Circuit are given by: $S_i = P_i \oplus C_i$ and $C_{i+1} = G_i + P_i C_i$. Consider a two-level logic implementation of the look-ahead carry generator. Assume that all P_i and G_i are available for the carry generator circuit and that the AND and OR gates can have any number of inputs. The number of AND gates and OR gates needed to implement the look-ahead carry generator for a 4-bit Combinational Circuit with S_3, S_2, S_1, S_0 as C_4 its outputs are respectively:
- 73.** Suppose only one multiplexer and one inverter are allowed to be used to implement any Boolean function of n variables. What is the minimum size of the multiplexer needed?
- 74.** How many 3-to-8 line decoders with an enable input are needed to construct a 6- to-64 line decoder without using any other logic gates?

75. A 4-bit carry look ahead Combinational Circuit, which adds two 4-bit numbers, is designed using AND, OR, NOT, NAND, NOR gates only. Assuming that all the inputs are available in both complemented and uncomplemented forms and the delay of each gate is one time unit, what is the overall propagation delay of the Combinational Circuit? Assume that the carry network has been implemented using two-level AND-OR logic.

76. Consider a multiplexer with X and Y as data inputs and Z as control input. Z = 0 selects input X, and Z = 1 selects input Y. What are the connections required to realize the 2-variable Boolean function $f = T + R$, without using any additional hardware?

77. A circuit outputs a digit in the form of 4 bits. 0 is represented by 0000, 1 by 0001, ..., 9 by 1001. A combinational circuit is to be designed which takes these 4 bits as input and outputs 1 if the digit ≥ 5 , and 0 otherwise. If only AND, OR and NOT gates may be used, what is the minimum number of gates required?

78. Consider the ALU shown below: If the operands are in 2's complement representation, which of the following operations can be performed by suitably setting the control lines K and C0 only (+ and - denote addition and subtraction respectively)?

79. An N-bit carry lookahead adder, where N is a multiple of 4, employs ICs 74181 (4 bit ALU) and 74182 (4 bit carry lookahead generator). The minimum addition time using the best architecture for this adder is

80. Advantage of synchronous sequential circuits over asynchronous one is :

81. The functional difference between SR flip-flop and J-K flip-flop is that :

82. We want to design a synchronous counter that counts these quence 0-1-0-2-0-3 and then repeats. The minimum number of J-K flip flop srequired to implement this counteris _____.

83. The minimum number of JK flip-flops required to construct a synchronous counter with the count sequence (0,0,1,1,2,2,3,3,0,0,...) is _____.

84. A modulus -12 ring counter requires a minimum of

85. A positive edge-triggered D flip-flop is connected to a positive edge-triggered JK flip-flop as follows. The Q output of the D flip-flop is connected to both the J and K inputs of the JK flip-flop, while the Q output of the JK flip-flop is connected to the input of the D flip-flop. Initially, the output of the D flip-flop is set to logic one and the output of the JK flip-flop is cleared. Which one of the following is the bit sequence (including the initial state) generated at the Q output of the JK flip-flop when the flip-flops are connected to a free-running common clock? Assume that J=K=1 is the toggle mode and J=K=0 is the state-holding mode of the JK flip-flop. Both the flip-flops have non-zero propagation delays.

86. Consider a 4-bit Johnson counter with an initial value of 0000. The counting sequence of this counter is

87. The above synchronous sequential circuit built using JK flip-flops is initialized with $Q_2 \blacksquare Q_1 \blacksquare Q_0 \blacksquare = 000$. The state sequence for this circuit for the next 3 clock cycles is

88. Let $k=2n$. A circuit is built by giving the output of an n-bit binary counter as input to an n-to- $2n$ bit decoder. This circuit is equivalent to a

- 89.** In a three stage counter, using RS flip flops what will be the value of the counter after giving 9 pulses to its input ? Assume that the value of counter before giving any pulses is 1 :
- 90.** In an RS flip-flop, if the S line (Set line) is set high (1) and the R line (Reset line) is set low (0), then the state of the flip-flop is :
- 91.** The minimum number of D flip-flops needed to design a mod-258 counter is
- 92.** In an SR latch made by cross-coupling two NAND gates, if both S and R inputs are set to 0, then it will result in
- 93.** The characteristic equation of an SR flip-flop is given by :
- 94.** Which of the following input sequences for a cross-coupled R-S flip-flop realized with two NAND gates may lead to an oscillation?
- 95.** How many pulses are needed to change the contents of a 8-bit up counter from 10101100 to 00100111 (rightmost bit is the LSB)?
- 96.** In an SR latch made by cross-coupling two NAND gates, if both S and R inputs are set to 0, then it will result in
- 97.** A 1-input, 2-output synchronous sequential circuit behaves as follows. Let z_k, n_k denote the number of 0's and 1's respectively in initial k bits of the input ($z_k + n_k = k$). The circuit outputs 00 until one of the following conditions holds. (1). $z_k - n_k = 2$. In this case, the output at the k -th and all subsequent clock ticks is 10. 2. $n_k - z_k = 2$. In this case, the output at the k -th and all subsequent clock ticks is 01. What is the minimum number of states required in the state transition graph of the above circuit?
- 98.** Choose the correct alternatives (more than one may be correct) and write the corresponding letters only: Advantage of synchronous sequential circuits over asynchronous ones is:
- 99.** Minimum number of NAND gates required to implement the following binary equation $Y = (\bar{A} + \bar{B})(C + D)$
- 100.** What is the minimum number of 2-input NOR gates required to implement 4-variable function expressed in sum-of-minterms form as $f = \sum(0,2,5,7,8,10,13,15)$? Assume that all the inputs and their complements are available.
- 101.** Any set of Boolean operation that is sufficient to represent all Boolean expression is said to be complete. Which of the following is not complete ?
- 102.** Consider the minterm list form of a Boolean function F given below.
 $F(P,Q,R,S) = \sum m(0,2,5,7,9,11) + d(3,8,10,12,14)$. Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is _____.
- 103.** What is the minimum number of two-input NAND gates used to perform the function of two-input OR gate?

- 104.** Given $f(w,x,y,z) = \sum m(0,1,2,3,7,8,10) + \sum d(5,6,11,15)$, where d represents the don't care condition in Karnaugh maps. Which of the following is a minimum product-of-sums (POS) form of $f(w,x,y,z)$?
- 105.** The simplified SOP (Sum of Product) from the Boolean expression $(P+Q\text{ bar} + R\text{ bar}).(P+Q+R).(P+Q+R\text{ bar})$ is
- 106.** The minimum number of NAND gates required to implement the Boolean function $A+AB\text{ bar}+AB\text{ bar}C$ is equal to
- 107.** Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator. Which one of the following must always be TRUE?
- 108.** The boolean expression $AB+AB'+A'C+AC$ is independent of the boolean variable
- 109.** The complement of the Boolean expression $AB(B+C+AC)$ is
- 110.** The total number of prime implicants of the function $f(w,x,y,z)=\sum(0,2,4,5,6,10)$ is _____.
- 111.** The number of min-terms after minimizing the following Boolean Algebra is _____. $[D\text{ bar} + AB\text{ bar} + A\text{ bar}C + AC\text{ bar}D + A\text{ bar}C\text{ bar}D]\text{ bar}$
- 112.** The output of a tristate buffer when the enable input is 0 is
- 113.** Let $\oplus$ denote the Exclusive OR (XOR) operation. Let '1' and '0' denote the binary constants. Consider the following Boolean Algebra for F over two variables P and Q.
 $F(P,Q)=((1\oplus P)\oplus(P\oplus Q))\oplus((P\oplus Q)\oplus(Q\oplus 0))$. The equivalent expression for F is
- 114.** Consider the following minterm expression for F. $F(P,Q,R,S) = \sum 0, 2, 5, 7, 8, 10, 13, 15$. The minterms 2, 7, 8 and 13 are 'do not care terms'. The minimal sum of-products form for F is
- 115.** The dual of a Boolean function $F(x_1, x_2, \dots, x_n, +, \cdot)$, written as F_D , is the same expression as that of F with + and $\cdot$ swapped. F is said to be self-dual if $F=F_D$. The number of self-dual functions with n Boolean variables is
- 116.** Any set of Boolean operators that is sufficient to represent all Boolean expressions is said to be complete. Which of the following is not complete ?
- 117.** The simplified SOP (Sum of Product) form of the Boolean Algebra $(P+Q+R).(P+Q+R).(P+Q+R)$ is
- 118.** The minterm expansion of $f(P,Q,R)=PQ+QR'+PR'$ is
- 119.** Consider the following boolean function of four variables $f(w,x,y,z)=\sum(1,3,4,6,9,11,12,14)$, the function is
- 120.** The switching expression corresponding to $f(A,B,C,D)=\sum(1,4,5,9,11,12)$ is
- 121.** What is the minimum number of gates required to implement the Boolean function $(AB+C)$ if we have to use only 2-input NOR gates?

- 122.** In the expression $A(A+B)$ by writing the first term A as $A+0$, the expression is best simplified as
- 123.** The Boolean theorem $AB+A'C+BC=AB+A'C$ corresponds to
- 124.** A set of Boolean connectives is functionally complete if all Boolean functions can be synthesized using those. Which of the following sets of connectives is NOT functionally complete?
- 125.** The following expression was to be realized using 2-input AND and OR gates. However, during the fabrication all 2-input AND gates were mistakenly substituted by 2-input NAND gates.
 $(a.b).c+(a'.c).d+(b.c).d+a.d$. What is the function finally realized ?
- 126.** Consider a boolean function $f(w,x,y,z)$. Suppose that exactly one of its inputs is allowed to change at a time. If the function happens to be true for two input vectors $i_1=(w_1,x_1,y_1,z_1)$ and $i_2=(w_2,x_2,y_2,z_2)$, we would like the function to remain true as the input changes from i_1 to i_2 (i_1 and i_2 differ in exactly one bit position), without becoming false momentarily. Let $f(w,x,y,z)=\Sigma(5,7,11,12,13,15)$. Which of the following cube covers of f will ensure that the required property is satisfied?
- 127.** What is the minimum number of NAND gates required to implement a 2-input EXCLUSIVE-OR function without using any other logic gate?
- 128.** The total number of Boolean functions which can be realised with four variables is:
- 129.** The Boolean expression $A \oplus B \oplus A$ is equivalent to
- 130.** What is the equivalent Boolean expression in product-of-sums form for the Karnaugh map
- 131.** Convert the decimal number 9857 to its equivalent in binary, octal, and hexadecimal systems.
- 132.** Perform subtraction: $(1101101)_2 - (101101)_2$ using 2's complement.
- 133.** Add $(1101.101)_2$ and $(100.11)_2$ using binary arithmetic rules.
- 134.** Show how overflow is detected during binary addition using 4-bit signed numbers.
- 135.** Represent the decimal number -127 in 8-bit 2's complement.
- 136.** Represent -45 in 8-bit 2's complement and 1's complement formats.
- 137.** Show the IEEE 754 representation of -18.375 (single precision).
- 138.** Find the base r if $(234)_r = (1010)_{10}$.
- 139.** Define Gray Code and convert $(1011)_2$ to its Gray Code equivalent.
- 140.** Perform multiplication of two binary numbers: $(1011)_2 \times (1101)_2$ and explain each step.
- 141.** Convert the BCD number (1001 0100 0110) to its decimal equivalent.
- 142.** A number in base r is given as $(321)_r$. If its decimal value is 70, find r .
- 143.** Perform BCD addition of 97 and 45. Show intermediate steps and corrections.

144. Convert the hexadecimal number (3F.8) to decimal and binary.
145. What is Excess-3 code? Convert (46) to Excess-3.
146. Perform the operation: (1011.101) + (110.011).
147. Why is base-2 used in digital systems? Justify with circuit-level considerations.
148. Represent 0.8125 in binary and show how floating-point normalization would occur.
149. Convert (10101010101) to its equivalent signed-magnitude format.
150. Convert (777) to binary and then to decimal.
151. Evaluate (A3) – (2F) and show binary intermediate steps.
152. Prove using Boolean laws: $A + A'B = A + B$
153. Convert 0.3 (repeating) in decimal to binary up to 10 fractional places.
154. Simplify the Boolean expression $A'B + AB + AB'$
155. Derive the result of $(732) \times (14)$.
156. Prove DeMorgan's Theorem with the expression: $(A+B)' = A'B'$
157. Convert binary number 11100101 to Gray code and back.
158. Show the canonical SOP form for the function $F(A,B,C) = \Sigma(1,3,5,7)$
159. Show how carry propagation delay affects binary addition.
160. Convert the following SOP to a POS: $A'B + AB' + ABC$
161. How are negative numbers handled in sign-magnitude vs 2's complement?
162. Derive the dual and complement of $AB + C$
163. Explain the method of performing binary division and solve: $(101101) \div (101)$.
164. Express the following in canonical POS: $F(A,B,C) = \Pi(0,2,4,6)$
165. Describe normalization in floating-point representation with an example.
166. Realize $F = A'B + AB + AC$ using only NAND gates.
167. Explain overflow and underflow in floating-point operations with numerical examples.
168. Find the minimal SOP form for $F = A'BC + AB + A'C$
169. Write the dual and inverse of $A + AB' + B$
170. Using Boolean algebra, prove that $(A+B)(A+B') = A$

171. Use Boolean laws to reduce $(A+A')(B+B')$
172. Find complement of the function: $F=(A+B')(C+D)$
173. Convert a logic function from its truth table to both SOP and POS forms.
174. Show that the expression $A+AB=A$ holds using a truth table.
175. Prove using algebra that $A(A+B)=A$
176. Simplify $F=(A+B)(A+C)(B+C)$
177. Simplify $A+AB'C+AB$ to minimal terms.
178. Prove using truth table: $AB+A'B=B$
179. Show using identities that $A+A'B=A+B$
180. Using Boolean algebra, implement the half-adder logic.
181. Derive canonical forms from K-map directly.
182. Show how XOR and XNOR can be derived using basic gates.
183. Implement $F=A \cdot (B+C)+A' \cdot C$ using NOR gates only.
184. Design a 2-to-1 MUX using only Boolean expressions.
185. Minimize $F(A,B,C)=\Sigma(0,2,3,6,7)$ using a 3-variable K-map.
186. Prove $A+AB+ABC=A$ using identities.
187. Draw 4-variable K-map and minimize $F=\Sigma(0,2,3,5,6,7,9,11,13,15)$
188. Implement $F=A \oplus B \oplus C$ using only NAND gates.
189. Differentiate between prime implicants and essential prime implicants.
190. Reduce $(A+B)(A+C)(B+C)$ to minimum terms.
191. For a given K-map, identify don't care conditions and simplify $F=\Sigma(1,3,7)+d(2,6)$
192. Use Boolean algebra to convert a 3-variable SOP to POS.
193. Show the K-map simplification of a full adder's sum output.
194. Demonstrate how any logic function can be implemented using only NAND.
195. Minimize a function with don't cares: $F=\Sigma(1,3,5,7,9)+d(0,2,4,6,8)$
196. Use K-map to implement a 3-variable parity checker.
197. Given a truth table with 4 inputs, construct K-map and minimize to SOP.

- 198.** Describe the steps to simplify a Boolean function using K-map with groups of 8.
- 199.** Explain how overlapping groups help in further minimization in K-map.
- 200.** Reduce $F = \Sigma(1,2,5,6,9,10,13,14)$ using 4-variable K-map.
- 201.** Implement a logic circuit using minimized K-map output: $F = \Sigma(0,4,5,7,12,13)$
- 202.** Explain the role of don't-care conditions in circuit design.
- 203.** How can POS be obtained using K-map grouping of zeros?
- 204.** Derive minimal expression from K-map: $F = \Pi(1,2,4,7)$
- 205.** Show with example how redundant terms affect the K-map grouping.
- 206.** Analyze and minimize a 5-variable Boolean function using two 4-variable K-maps.
- 207.** Minimize a majority function using a 3-variable K-map.
- 208.** Use a 3-variable K-map to verify consensus theorem.
- 209.** Solve a timing issue using K-map minimization of glitch-free logic.
- 210.** Design a 3:8 decoder using 2:4 decoders.
- 211.** Implement a 4-bit binary adder using full adders.
- 212.** Realize a 4-bit comparator and explain with block diagram.
- 213.** Construct a 4x1 multiplexer using basic gates and write its logic equation.
- 214.** Implement a priority encoder using 8 inputs.
- 215.** Design a 3-bit binary to Gray code converter.
- 216.** Realize a 4-bit parallel adder-subtractor and explain mode selection.
- 217.** Build a circuit to detect if a 4-bit input has even parity.
- 218.** Construct a half subtractor and explain its truth table.
- 219.** Create a combinational lock circuit with a fixed 3-bit password.
- 220.** Realize a 2-bit magnitude comparator using combinational logic.
- 221.** Design a 4-bit BCD adder with carry-out detection.
- 222.** Implement a 3-to-8 decoder using logic expressions only.
- 223.** Implement a 2-bit binary multiplier using AND and half adders.
- 224.** Build an ALU block that supports AND, OR, ADD, and SUB.

- 225.** Design a combinational circuit for converting binary to 7-segment display code.
- 226.** Convert a BCD number to Excess-3 code using combinational logic.
- 227.** Show how a 4:1 multiplexer can be used to implement any Boolean function of two variables.
- 228.** Design a combinational circuit that counts the number of 1s in a 4-bit input.
- 229.** Realize a 2's complement circuit for an n-bit binary number.
- 230.** Implement a multiplexer using only NAND gates.
- 231.** Construct a 3-variable function using MUX and give the truth table.
- 232.** Realize a full adder using two half adders.
- 233.** Explain how a logic function can be implemented using a decoder and OR gates.
- 234.** Compare ripple carry adder and carry lookahead adder using logic design.
- 235.** Construct a logic circuit to detect a binary palindrome.
- 236.** Realize a full subtractor using two half subtractors and an OR gate
- 237.** Show how XOR gates can be used to construct a 1-bit parity generator and checker.
- 238.** Implement a 3-variable majority function using basic gates.
- 239.** Design a 3-bit synchronous up counter using JK flip-flops.
- 240.** Draw and explain the working of a T flip-flop with its excitation table.
- 241.** Construct a MOD-6 asynchronous counter using JK flip-flops.
- 242.** Compare and contrast SR, D, T, and JK flip-flops with logic diagrams.
- 243.** Draw the timing diagram for a JK flip-flop with preset and clear inputs.
- 244.** Design a 3-state finite state machine (FSM) with Moore model and transition diagram.
- 245.** Convert a D flip-flop to a T flip-flop and explain the logic.
- 246.** Realize a synchronous decade counter and explain its operation
- 247.** Explain how race-around condition occurs in JK flip-flop and how it is resolved.
- 248.** Draw the excitation table for a T flip-flop and use it in a counter design.
- 249.** Design a sequence detector that detects "101" in a bit stream.
- 250.** Show how master-slave JK flip-flop avoids race condition.
- 251.** Realize a 4-bit shift register with serial input and parallel output.

- 252.** Build a FSM that outputs 1 if the number of 1's in the input stream is even.
- 253.** Draw the state diagram of a MOD-10 synchronous counter.
- 254.** Design a traffic light controller using FSM (Green → Yellow → Red → Green).
- 255.** Compare asynchronous vs synchronous counters with suitable examples.
- 256.** Implement an FSM with overlapping sequence detection "1011".
- 257.** Create a flip-flop-based design that toggles every third clock pulse.
- 258.** Explain the difference between Mealy and Moore machines with diagrams.
- 259.** Implement a state machine that recognizes the pattern "1101".
- 260.** Design a clock divider circuit that produces a 1 Hz output from a 4 Hz clock.
- 261.** Explain how clock skew affects synchronous circuit design.
- 262.** Realize a binary counter using D flip-flops and explain logic.
- 263.** Build a Johnson counter and explain how it differs from ring counter.
- 264.** Build a glitch-free synchronous circuit using edge-triggered flip-flops.
- 265.** Construct a 3-bit bidirectional shift register using D flip-flops.
- 266.** Analyze timing diagram of a shift register fed with 1011 sequence.
- 267.** Explain setup time and hold time using waveform examples.
- 268.** Explain propagation delay and its effect on counter accuracy.
- 269.** Create a sequence generator circuit to produce the binary pattern "1001".
- 270.** Use state reduction to minimize a 5-state FSM.
- 271.** Design a finite state machine for elevator floor logic (ground, 1st, 2nd).
- 272.** Create a pattern recognizer that detects "10010" in serial input.
- 273.** Explain the working of a ring counter and give truth table.
- 274.** Design a self-starting synchronous MOD-5 counter.
- 275.** Analyze a digital circuit combining MUX, decoder, and flip-flops. Determine output behavior.
- 276.** Compare the clocking requirements of ripple and synchronous counters.
- 277.** A circuit uses two JK flip-flops with feedback. Derive the output sequence for 5 clock pulses.
- 278.** Derive and implement the next-state logic for a given FSM table.

- 279.** Show how a combinational circuit can be converted to a sequential circuit using feedback.
- 280.** Build a 2-bit Gray code counter using D flip-flops.
- 281.** Design a logic unit that performs addition and subtraction using control logic.
- 282.** Show how a universal shift register works and explain each mode.
- 283.** Implement a priority encoder and extend it to handle 16 inputs.
- 284.** Implement a 4-bit synchronous down counter using JK flip-flops.
- 285.** Propose a 4-bit ALU that supports AND, OR, NOT, XOR, ADD, SUB, and compare operations.
- 286.** Create a logic diagram for a sequential circuit that remembers last two inputs.
- 287.** Evaluate hazards and glitches in a combinational logic circuit using waveform analysis
- 288.** Use Karnaugh maps to minimize next-state logic for a sequential circuit.
- 289.** Reduce a given sequential machine using state equivalence concepts.
- 290.** Construct a modulo-7 counter and give its truth table.
- 291.** Show how to design a timing-sensitive circuit to detect double keypress.
- 292.** Explain metastability in flip-flops and how to avoid it.
- 293.** Given a flip-flop circuit with feedback loops and clocking, find the output sequence.
- 294.** Build a circuit that detects and corrects single-bit parity errors.
- 295.** Analyze a 4-stage ring counter with asynchronous clear.
- 296.** Given a logic circuit with a combination of AND-OR-INVERT logic, derive its Boolean expression.
- 297.** Implement a BCD to Excess-3 encoder using only NOR gates.
- 298.** Create a finite state machine to control a vending machine that accepts coins in any order.
- 299.** A vending machine accepts only 1¢ and 2¢ coins. It dispenses an item worth 3¢ . Design a Moore finite state machine that detects the input coin sequence and dispenses the item once the total amount becomes 3¢ . Show the state diagram, state table, and implementation using D flip-flops.
- 300.** Design a 4-bit Arithmetic Logic Unit (ALU) that supports the following operations based on 3 control lines: AND, OR, ADD, SUB, and XOR. Include the logic circuit diagram, truth table, control logic, and carry/overflow detection logic in your design. Also, explain how the ALU handles negative numbers in 2's complement format.

Theory of Computation

1. Consider the following two regular expressions over the alphabet $\{0,1\}$: $r=0^*+1^*$ $s=01^*+10^*$. The total number of strings of length less than or equal to 5 , which are neither in r nor in s , is _____
2. Which of the following regular expressions represent(s) the set of all binary numbers that are divisible by three? Assume that the string ϵ is divisible by three.
3. In some programming language, an identifier is permitted to be a letter followed by any number of letters or digits. If L and D denote the sets of letters and digits respectively, which of the following expressions defines an identifier?
4. Let $L=\{w \in (0+1)^* \mid w \text{ has even number of } 1 \text{'s}\}$, i.e. L is the set of all bit strings with even number of 1's. Which one of the regular expression below represents L ?
5. Consider the regular expression $R=(a+b)^*(aa+bb)(a+b)^*$. Which one of the regular expressions given below defines the same language as defined by the regular expression R ?
6. Let S and T be languages over $\Sigma=\{a,b\}$ represented by the regular expressions $(a+b)^*$ and $(a+b)^*$, respectively. Which of the following is true?
7. In some programming language, an identifier is permitted to be a letter followed by any number of letters or digits. If L and D denote the sets of letters and digits respectively, which of the following expressions defines an identifier?
8. If the regular set A is represented by $A=(01+1)^*$ and the regular set B is represented by $B=((01)^*1^*)^*$, which of the following is true?
9. What is the highest type number that can be assigned to the following grammar? $S \rightarrow Aa, A \rightarrow Ba, B \rightarrow abc$
10. Let L_1 be the language represented by the regular expression $b^*ab^*(ab^*ab^*)$ and $L_2=\{w \in (a+b)^* \mid |w| \leq 4\}$, where $|w|$ denotes the length of string w . The number of strings in L_2 which are also in L_1 is _____
11. Consider the following two statements: P : Every regular grammar is $LL(1)$, Q : Every regular set has a $LR(1)$ grammar. Which of the following is TRUE?
12. Consider the regular grammar below $S \rightarrow bS \mid aA \mid \epsilon, A \rightarrow aS \mid bA$. The Myhill-Nerode equivalence classes for the language generated by the grammar are
13. Language L_1 is defined by the grammar: $S_1 \rightarrow aS_1b \mid \epsilon$, Language L_2 is defined by the grammar: $S_2 \rightarrow abS_2 \mid \epsilon$. Consider the following statements: P : L_1 is regular, Q : L_2 is regular
14. For $\Sigma=\{a,b\}$, let us consider the regular language $L=\{x \mid x=a^{2+3k} \text{ or } x=b^{10+12k}, k \geq 0\}$. Which one of the following can be a pumping length (the constant guaranteed by the pumping lemma) for L ?
15. Let $L \subseteq \{0,1\}^*$ be an arbitrary regular language accepted by a minimal DFA with k states. Which one of the following languages must necessarily be accepted by a minimal DFA with k states?

16. Let P be a regular language and Q be a context free language such that $Q \subseteq P$. (For example, let P be the language represented by the regular expression p^*q^* and Q be $\{p^n q^n \mid n \in \mathbb{N}\}$) Then which of the following is ALWAYS regular?

17. If s is a string over $(0+1)^*$, then let $n_0(s)$ denote the number of 0's in s and $n_1(s)$ the number of 1's in s . Which one of the following languages is not regular?

18. A language L satisfies the Pumping Lemma for regular languages, and also the Pumping Lemma for context-free languages. Which of the following statements about L is TRUE?

19. Suppose we want to design a synchronous circuit that processes a string of 0's and 1's. Given a string, it produces another string by replacing the first 1 in any subsequence of consecutive 1's by a 0. Consider the following example. Input sequence: 00100011000011100, Output sequence: 00000001000001100. A Mealy Machine is a state machine where both the next state and the output are functions of the present state and the current input. The above mentioned circuit can be designed as a two state Mealy machine. The states in the Mealy machine can be represented using Boolean values 0 and 1. We denote the current state, the next state, the next incoming bit, and the output bit of the Mealy machine by the variables s , t , b and y respectively. Assume the initial state of the Mealy machine is 0. What are the Boolean expressions corresponding to t and y in terms of s and b ?

20. Let Σ be the set of all bijections from $\{1, \dots, 5\}$ to $\{1, \dots, 5\}$, where id denotes the identity function, i.e. $\text{id}(j)=j, \forall j$. Let $\circ$ denote composition on functions. For a string $x = x_1 x_2 \dots x_n \in \Sigma, n \geq 0$, let $\pi(x) = x_1 \circ x_2 \circ \dots \circ x_n$. Consider the language $L = \{x \in \Sigma^* \mid \pi(x) = \text{id}\}$. The minimum number of states in any DFA accepting L is _____.

21. Consider the language L given by the regular expression $(a+b)^* b (a+b)^*$ over the alphabet $\{a, b\}$. The smallest number of states needed in a deterministic finite-state automaton (DFA) accepting L is _____.

22. The number of states in the minimal deterministic finite automaton corresponding to the regular expression $(0+1)^*(10)$ is _____.

23. How many states are there in a minimum state deterministic finite automaton accepting the language $L = \{w \mid w \in \{0,1\}^*, \text{ number of 0's is divisible by 2 and number of 1's is divisible by 5, respectively}\}$?

24. The number of states required by a Finite State Machine, to simulate the behavior of a computer with a memory capable of storing 'm' words, each of length 'n' bits is?

25. Let w be any string of length n in $\{0, 1\}^*$. Let L be the set of all substrings of w . What is the minimum number of states in a non-deterministic finite automaton that accepts L ?

26. Let $G = (V, \Sigma, S, P)$ be a context-free grammar in Chomsky Normal Form with $\Sigma = \{a, b, c\}$ and V containing 10 variable symbols including the start symbol S . The string $w = a^{30} b^{30} c^{30}$ is derivable from S . The number of steps (application of rules) in the derivation $S \rightarrow^* w$ is _____.

27. Let N_f and N_p denote the classes of languages accepted by non-deterministic finite automata and non-deterministic push-down automata, respectively. Let D_f and D_p denote the classes of languages accepted by deterministic finite automata and deterministic push-down automata respectively. Which one

of the following is TRUE?

28. Consider the regular expression $(0 + 1)^n$... n times. The minimum state finite automaton that recognizes the language represented by this regular expression contains

29. Let L be the set of all binary strings whose last two symbols are the same. The number of states in the minimal state deterministic finite state automaton accepting L is

30. Consider a DFA over $\Sigma = \{a, b\}$ accepting all strings which have number of a's divisible by 6 and number of b's divisible by 8. What is the minimum number of states that the DFA will have ?

31. Given an arbitrary non-deterministic finite automaton (NFA) with N states, the maximum number of states in an equivalent minimized DFA is at least.

32. Choose the correct alternatives (more than one may be correct) and write the corresponding letters only: In which of the cases stated below is the following statement true? "For every non-deterministic machine M1 there exists an equivalent deterministic machine M2 recognizing the same language".

33. For a Turing machine M, $\langle M \rangle$ denotes an encoding of M. Consider the following two languages. $L_1 = \{\langle M \rangle \mid M \text{ takes more than 2021 steps on all inputs}\}$, $L_2 = \{\langle M \rangle \mid M \text{ takes more than 2021 steps on some input}\}$. Which one of the following options is correct?

34. Consider the following problems. $L(G)$ denotes the language generated by a grammar G. $L(M)$ denotes the language accepted by a machine M. (I) For an unrestricted grammar G and a string w, whether $w \in L(G)$. (II) Given a Turing machine M, whether $L(M)$ is regular. (III) Given two grammars G_1 and G_2 , whether $L(G_1) = L(G_2)$. (IV) Given an NFA N, whether there is a deterministic PDA P such that N and P accept the same language. Which one of the following statements is correct?

35. Let A and B be infinite alphabets and let # be a symbol outside both A and B. Let f be a total functional from A^* to B^* . We say f is computable if there exists a Turing machine M which given an input x in A^* , always halts with f(x) on its tape. Let L_f denote the language $\{x\#f(x) \mid x \in A^*\}$. Which of the following statements is true:

36. Consider the following languages. $L_1 = \{\langle M \rangle \mid M \text{ takes at least 2016 steps on some input}\}$, $L_2 = \{\langle M \rangle \mid M \text{ takes at least 2016 steps on all inputs}\}$ and $L_3 = \{\langle M \rangle \mid M \text{ accepts } \epsilon\}$, where for each Turing machine M, $\langle M \rangle$ denotes a specific encoding of M. Which one of the following is TRUE?

37. Consider the following statements. I. The complement of every Turing decidable language is Turing decidable. II. There exists some language which is in NP but is not Turing decidable. III. If L is a language in NP, L is Turing decidable. Which of the above statements is/are true?

38. Let $\langle M \rangle$ be the encoding of a Turing machine as a string over $\Sigma = \{0, 1\}$. Let $L = \{\langle M \rangle \mid M \text{ is a Turing machine that accepts a string of length 2014}\}$. Then, L is

39. Which of the following statements is/are FALSE? 1. For every non-deterministic Turing machine, there exists an equivalent deterministic Turing machine. 2. Turing recognizable languages are closed under union and complementation. 3. Turing decidable languages are closed under intersection and complementation. 4. Turing recognizable languages are closed under union and intersection.

40. Consider the following problem x . Given a Turing machine M over the input alphabet Σ , any state q of M . And a word $w \in \Sigma^*$ does the computation of M on w visit the state q ? Which of the following statements about x is correct?

41. Consider the pushdown automaton (PDA) P below, which runs on the input alphabet $\{a, b\}$, has stack alphabet $\{\perp, A\}$, and has three states $\{s, p, q\}$, with s being the start state. A transition from state u to state v , labelled $c/X/\gamma$, where c is an input symbol or $\perp$, X is a stack symbol, and γ is a string of stack symbols, represents the fact that in state u , the PDA can read c from the input, with X on the top of its stack, pop X from the stack, push in the string γ on the stack, and go to state v . In the initial configuration, the stack has only the symbol $\perp$ in it. The PDA accepts by empty stack.

42. Consider the NPDA $\mathcal{P} = \{Q, \Sigma, \Gamma, \delta, q_0, \perp, F = \{q_2\}\}$, where (as per usual convention) Q is the set of states, Σ is the input alphabet, Γ is the stack alphabet, δ is the state transition function, q_0 is the initial state, $\perp$ is the initial stack symbol, and F is the set of accepting states. The state transition is as follows:

43. Consider the pushdown automaton (PDA) below which runs over the input alphabet $\{a, b, c\}$. It has the stack alphabet $\{Z_0, X\}$ where Z_0 is the bottom-of-stack marker. The set of states of the PDA is $\{s, t, u, f\}$ where s is the start state and f is the final state. The PDA accepts by final state. The transitions of the PDA given below are depicted in a standard manner. For example, the transition $(s, b, X) \rightarrow (t, XZ_0)$ means that if the PDA is in state s and the symbol on the top of the stack is X , then it can read b from the input and move to state t after popping the top of stack and pushing the symbols Z_0 and X (in that order) on the stack. $(s, a, Z_0) \rightarrow (s, XXZ_0)$, $(s, \perp, Z_0) \rightarrow (f, \perp)$, $(s, a, X) \rightarrow (s, XXX)$, $(s, b, X) \rightarrow (t, \perp)$, $(t, b, X) \rightarrow (t, \perp)$, $(t, c, X) \rightarrow (u, \perp)$, $(u, c, X) \rightarrow (u, \perp)$, $(u, \perp, Z_0) \rightarrow (f, \perp)$. The language accepted by the PDA is

44. Let P be a non-deterministic push-down automaton (NPDA) with exactly one state, q , and exactly one symbol, Z , in its stack alphabet. State q is both the starting as well as the accepting state of the PDA. The stack is initialized with one Z before the start of the operation of the PDA. Let the input alphabet of the PDA be Σ . Let $L(P)$ be the language accepted by the PDA by reading a string and reaching its accepting state. Let $N(P)$ be the language accepted by the PDA by reading a string and emptying its stack. Which of the following statements is TRUE?

45. The language accepted by a Pushdown Automaton in which the stack is limited to 10 items is best described as

46. Let L_1 be the set of all languages accepted by a PDA by final state and L_2 the set of all languages accepted by empty stack. Which of the following is true?

47. Given the following statements S_1 : Every context-sensitive language L is recursive, S_2 : There exists a recursive language that is not context-sensitive. Which statements are true?

48. Let X be a recursive language and Y be a recursively enumerable but not recursive language. Let W and Z be two languages such that Y bar reduces to W , and Z reduces to X bar (reduction means the standard many-one-reduction). Which one of the following statements is TRUE?

49. Let L_1 be a recursive language. Let L_2 and L_3 be languages that are recursively enumerable but not recursive. Which of the following statements is not necessarily true?

50. Let L_1 be regular language, L_2 be a deterministic context-free language and L_3 a recursively enumerable, but not recursive, language. Which one of the following statements is false?

51. For $S \in (0+1)^*$, let $d(s)$ denote the decimal value of s (e.g. $d(101)=5$). Let $L = \{s \in (0+1)^* \mid d(s) \bmod 5 = 2 \text{ and } d(s) \bmod 7 \neq 4\}$. Which one of the following statements is true?

52. L_1 is a recursively enumerable language over Σ . An algorithm A effectively enumerates its words as $w_1, w_2, w_3, \dots$ define another language L_2 over $\Sigma \cup \{\#\}$ as $\{w_i \# w_j \mid w_i, w_j \in L_1, i < j\}$. Here $\#$ is a new symbol. Consider the following assertions. S_1 : L_1 is recursive implies L_2 is recursive. S_2 : L_2 is recursive implies L_1 is recursive

53. Consider the following sets: S_1 : Set of all recursively enumerable languages over the alphabet $\{0, 1\}$. S_2 : Set of all syntactically valid C programs. S_3 : Set of all languages over the alphabet $\{0, 1\}$. S_4 : Set of all non-regular languages over the alphabet $\{0, 1\}$. Which of the above sets are uncountable?

54. Let $L(R)$ be the language represented by regular expression R . Let $L(G)$ be the language generated by a context free grammar G . Let $L(M)$ be the language accepted by a Turing machine M . Which of the following decision problems are undecidable ? I. Given a regular expression R and a string w , is $w \in L(R)$? II. Given a context-free grammar G , $L(G) = \emptyset$? III. Given a context-free grammar G , is $L(G) = \Sigma^*$ for some alphabet Σ ? IV. Given a Turing machine M and a string w , is $w \in L(M)$?

55. Which of the following decision problems are undecidable? I. Given NFAs N_1 and N_2 , is $L(N_1) \cap L(N_2) = \emptyset$? II. Given a CFG $G = (N, \Sigma, P, S)$ and a string $x \in \Sigma^*$, does $x \in L(G)$? III. Given CFGs G_1 and G_2 , is $L(G_1) = L(G_2)$? IV. Given a TM M , is $L(M) = \emptyset$?

56. Let $A \leq_m B$ denotes that language A is mapping reducible (also known as many-to-one reducible) to language B . Which one of the following is FALSE?

57. Which of the following is/are undecidable? 1. G is a CFG. Is $L(G) = \emptyset$? 2. G is a CFG. Is $L(G) = \Sigma^*$? 3. M is a Turing machine. Is $L(M)$ regular? 4. A is a DFA and N is an NFA. Is $L(A) = L(N)$?

58. Which of the following problems are decidable? (1) Does a given program ever produce an output? (2) If L is a context-free language, then, is $\bar{L}$ also context-free? (3) If L is a regular language, then, is $\bar{L}$ also regular? (4) If L is a recursive language, then, is $\bar{L}$ also recursive?

59. Which of the following are decidable? (I). Whether the intersection of two regular languages is infinite. (II). Whether a given context-free language is regular. (III). Whether two push-down automata accept the same language. (IV). Whether a given grammar is context-free

60. Consider two languages L_1 and L_2 each on the alphabet Σ . Let $f: \Sigma \rightarrow \Sigma$ be a polynomial time computable bijection such that $(\forall x)[x \in L_1 \text{ iff } f(x) \in L_2]$. Further, let f^{-1} be also polynomial time computable. Which of the following CANNOT be true?

61. The language which is generated by the grammar $S \rightarrow aSa \mid bSb \mid a \mid b$ over the alphabet of $\{a, b\}$ is the set of

62. A CFG (Context Free Grammar) is said to be in Chomsky Normal Form (CNF), if all the productions are of the form $A \rightarrow BC$ or $A \rightarrow a$. Let G be a CFG in CNF. To derive a string of terminals of length x , the

number of products to be used is:

63. Consider the following expression grammar $G: E \rightarrow E-T|T, T \rightarrow T+F|F, F \rightarrow (E)|id$. Which of the following grammars is not left recursive, but is equivalent to G ?
64. Identify the language generated by the following grammar, where S is start variable. $S \rightarrow XY, X \rightarrow aX|a, Y \rightarrow aYb|\epsilon$
65. Which of the following statements about parser is/are CORRECT? I. Canonical LR is more powerful than SLR. II. SLR is more powerful than LALR. III. SLR is more powerful than Canonical LR.
66. If G is grammar with productions $S \rightarrow SaS \mid aSb \mid bSa \mid SS \mid \epsilon$ where S is the start variable, then which one of the following is not generated by G ?
67. Consider the following grammar. $P \rightarrow xQRS, Q \rightarrow yz|z, R \rightarrow w|\epsilon, S \rightarrow y$ What is FOLLOW (Q) ?
68. Consider the following context-free grammar over the alphabet $\Sigma = \{a,b,c\}$ with S as the start symbol. $S \rightarrow abScT|abcT, T \rightarrow bT|b$. Which one of the following represents the language generated by the above grammar ?
69. Consider the following statements about the context free grammar $G = \{S \rightarrow SS, S \rightarrow ab, S \rightarrow ba, S \rightarrow \epsilon\}$. (I). G is ambiguous. (II). G produces all strings with equal number of a 's and b 's. (III). G can be accepted by a deterministic PDA. Which combination below expresses all the true statements about G ?
70. Consider the following context-free grammars: $G_1: S \rightarrow aS|B, B \rightarrow b|bB, G_2: S \rightarrow aA|bB, A \rightarrow aA|B|\epsilon, B \rightarrow bB|\epsilon$. Which one of the following pair of languages is generated by G_1 and G_2 , respectively?
71. A CFG G is given with the following productions where S is the start symbol, A is a non-terminal and a and b are terminals. $S \rightarrow aS \mid A, A \rightarrow aAb \mid bAa \mid \epsilon$. For the string "aabbaab" how many steps are required to derive the string and how many parse trees are there?
72. A CFG G is given with the following productions where S is the start symbol, A is a non-terminal and a and b are terminals. $S \rightarrow aS \mid A, A \rightarrow aAb \mid bAa \mid \epsilon$. Which of the following strings is generated by the grammar above?
73. Consider a CFG with the following productions. $S \rightarrow AA \mid B, A \rightarrow 0A \mid A0 \mid 1, B \rightarrow 0B00 \mid 1$. S is the start symbol, A and B are non-terminals and 0 and 1 are the terminals. The language generated by this grammar is:
74. Which of the following statements are true? (I). Every left-recursive grammar can be converted to a right-recursive grammar and vice-versa. (II). All ϵ -productions can be removed from any context-free grammar by suitable transformations. (III). The language generated by a context-free grammar all of whose productions are of the form $x \rightarrow w$ or $x \rightarrow wY$ (where, w is a string of terminals and Y is a non-terminal), is always regular. (IV). The derivation trees of strings generated by a context-free grammar in Chomsky Normal Form are always binary trees
75. In the context-free grammar below, S is the start symbol, a and b are terminals, and ϵ denotes the empty string. $S \rightarrow aSAb \mid \epsilon, A \rightarrow bA \mid \epsilon$. The grammar generates the language

76. Consider the following statements about the context-free grammar $G = \{S \rightarrow SS, S \rightarrow ab, S \rightarrow ba, S \rightarrow \epsilon\}$ (1). G is ambiguous (2). G produces all strings with equal number of a's and b's (3). G can be accepted by a deterministic PDA. Which combination below expresses all the true statements about G ?

77. Which of the following grammar rules violate the requirements of an operator grammar? P, Q, R are nonterminals, and r, s, t are terminals. (i) $P \rightarrow QR$ (ii) $P \rightarrow QsR$ (iii) $P \rightarrow \epsilon$ (iv) $P \rightarrow QtRr$

78. Consider the grammar shown below. $S \rightarrow C C, C \rightarrow cC \mid d$. The grammar is

79. Consider the following decision problems: (P1): Does a given finite state machine accept a given string? (P2): Does a given context free grammar generate an infinite number of strings? Which of the following statements is true?

80. Consider a grammar with the following productions $S \rightarrow \alpha a b \beta b \alpha c \gamma a B, S \rightarrow \alpha S \beta b, S \rightarrow \alpha b b \gamma ab, S \alpha \rightarrow b d b \gamma b$. The above grammar is:

81. Choose the correct alternatives (more than one may be correct) and write the corresponding letters only: If G is a context free grammar and w is a string of length l in $L(G)$, how long is a derivation of w in G , if G is in Chomsky normal form?

82. Choose the correct alternatives (more than one may be correct) and write the corresponding letters only: Consider the SLR(1) and LALR (1) parsing tables for a context free grammar. Which of the following statement is/are true?

83. Choose the correct alternatives (more than one may be correct) and write the corresponding letters only: For a context free grammar, $FOLLOW(A)$ is the set of terminals that can appear immediately to the right of non-terminal A in some "sentential" form. We define two sets $LFOLLOW(A)$ and $RFOLLOW(A)$ by replacing the word "sentential" by "left sentential" and "right most sentential" respectively in the definition of $FOLLOW(A)$.

84. Let L_1 be a regular language and L_2 be a context-free language. Which of the following languages is/are context-free?

85. For a string w , we define w^R to be the reverse of w . For example, if $w = 01101$ then $w^R = 10110$. Which of the following languages is/are context-free?

86. Consider the language $L = \{a^n \mid n \geq 0\} \cup \{a^n b a^n \mid n \geq 0\}$ and the following statements. I. L is deterministic context-free. II. L is context-free but not deterministic context-free. III. L is not $LL(k)$ for any k . Which of the above statements is/are TRUE?

87. For any two languages L_1 and L_2 such that L_1 is context-free and L_2 is recursively enumerable but not recursive, which of the following is/are necessarily true? (I). L_1^c (complement of L_1) is recursive. (II). L_2^c (complement of L_2) is recursive. (III). L_1^c is context-free. (IV). $L_1^c \cup L_2$ is recursively enumerable

88. Let L be a regular language and M be a context-free language, both over the alphabet Σ . Let L^c and M^c denote the complements of L and M respectively. Which of the following statements about the language $L^c \cup M^c$ is TRUE?

- 89.** If L_1 and L_2 are context free languages and R a regular set, one of the languages below is not necessarily a context free language. Which one?
- 90.** Context-free languages and regular languages are both closed under the operation (s) of :
- 91.** Let $L \subseteq \{0,1\}^*$ be an arbitrary regular language accepted by a minimal DFA with k states. Which one of the following languages must necessarily be accepted by a minimal DFA with k states?
- 92.** Let G_1, G_2 be Context Free Grammars (CFGs) and R be a regular expression. For a grammar G , let $L(G)$ denote the language generated by G . Which ONE among the following questions is decidable?
- 93.** Suppose that L_1 is a regular and L_2 is a context-free language, Which one of the following languages is NOT necessarily context-free?
- 94.** Let Σ denote an encoding of an automaton M . Suppose that $\Sigma = \{0, 1\}$. Which of the following languages is/are NOT recursive?
- 95.** A regular language L is accepted by a non-deterministic finite automaton (NFA) with n states. Which of the following statement(s) is/are FALSE?
- 96.** Let L_1, L_2 be two regular languages and L_3 a language which is not regular. Which of the following statements is/are always TRUE?
- 97.** For $\Sigma = \{a, b\}$, let us consider the regular language $L = \{x \mid x = a^{2+3k} \text{ or } x = b^{10+12k}, k \geq 0\}$. Which one of the following can be a pumping length (the constant guaranteed by the pumping lemma) for L ?
- 98.** The number of states in the minimum sized DFA that accepts the language defined by the regular expression $(0+1)^*(0+1)(0+1)^*$ is _____.
- 99.** Let L_1 recursive language. Let L_2 and L_3 be languages that are recursively enumerable but not recursive. Which of the following statement is not necessarily true?
- 100.** Given an arbitrary non-deterministic finite automaton (NFA) with N states, the maximum number of states in an equivalent minimized DFA is at least
- 101.** Consider the regular expression $(0+1)(0+1) \dots n$ times. The minimum state finite automaton that recognizes the language represented by this regular expression contains:
- 102.** How many substrings of different lengths (non-zero) can be formed from a character string of length n ?
- 103.** If the regular set A is represented by $A = (01 + 1)^*$ and the regular set 'B' is represented by $B = ((01)^*1^*)$, which of the following is true?
- 104.** Let L_1 be the language represented by the regular expression $b^*ab^*(ab^*ab^*)^*$ and $L_2 = \{w \in (a + b)^* \mid |w| \leq 4\}$, where $|w|$ denotes the length of string w . The number of strings in L_2 which are also in L_1 is _____.
- 105.** Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 1 Mbps (= 10⁶ bits per second). The aggregate

number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is

106. Consider the following two statements about regular languages: S1: Every infinite regular language contains an undecidable language as a subset. S2: Every finite language is regular. Which one of the following choices is correct?

107. Consider the following language. $L = \{x \in \{a, b\}^* \mid \text{number of } a\text{'s in } x \text{ is divisible by 2 but not divisible by 3}\}$ The minimum number of states in a DFA that accepts L is _____.

108. Let w be any string of length n in $\{0, 1\}^*$. Let L be the set of all substrings of w . What is the minimum number of states in a non-deterministic finite automaton that accepts L ?

109. A minimum state deterministic finite automaton accepting the language $L = \{w \mid w \in \{0, 1\}^*, \text{ number of } 0\text{'s and } 1\text{'s in } w \text{ are divisible by 3 and 5, respectively}\}$ has

110. If s is a string over $(0 + 1)^*$ then let $n_0(s)$ denote the number of 0's in s and $n_1(s)$ the number of 1's in s . Which one of the following languages is not regular?

111. Consider the regular language $L = (111 + 11111)^*$. The minimum number of states in any DFA accepting this language is

112. Consider a DFA over $\Sigma = \{a, b\}$ accepting all strings which have number of a 's divisible by 6 and number of b 's divisible by 8. What is the minimum number of states that the DFA will have?

113. What can be said about a regular language L over $\{a\}$ whose minimal finite state automaton has two states?

114. Let L be the set of all binary strings whose last two symbols are the same. The number of states in the minimum state deterministic finite-state automaton accepting L is

115. Which one of the following regular expressions over $\{0, 1\}$ denotes the set of all strings not containing 100 as substring?

116. The number of sub-strings (of all lengths inclusive) that can be formed from a character string of length n is

117. If G is a context-free grammar and w is a string of length n in $L(G)$, how long is a derivation of w in G if G is Chomsky normal form?

118. Let $r = 1(1+0)^*$, $s = 11^*0$ and $t = 1^*0$ be three regular expressions. Which one of the following is true?

119. How many substrings (of all lengths inclusive) can be formed from a character string of length n ? Assume all characters to be distinct. Prove your answer.

120. Given that language L_1 is regular and that the language $L_1 \cap L_2$ is regular is the language L_2 is always regular?

- 121.** Give the regular expression over $\{0,1\}$ to denote the set of proper non-null substrings of the string 0110.
- 122.** Give minimal DFA that performs as a Mod-3 1's counter, i.e., outputs a 1 each time the number of 1's in the input sequence is a multiple of 3.
- 123.** Let G_1, G_2 be Context Free Grammars (CFGs) and R be a regular expression. For a grammar G , let $L(G)$ denote the language generated by G . Which ONE among the following questions is decidable?
- 124.** Let L_1 be a regular language and L_2 be a context-free language. Which of the following languages is/are context-free?
- 125.** Suppose that L_1 is a regular and L_2 is a context-free language, Which one of the following languages is NOT necessarily context-free?
- 126.** The lexical analysis for a modern computer language such as java needs the power of which one of the following machine model in a necessary and sufficient sense?
- 127.** The language accepted by a pushdown Automation in which the stack is limited to items is best described as
- 128.** Let L_d be the set of all languages accepted by a PDA by final state and L_e the set of all languages accepted by empty stack. Which of the following is true?
- 129.** Let $G = (V, \Sigma, S, P)$ be a context-free grammar in Chomsky Normal Form with $\Sigma = \{a, b, c\}$ and V containing 10 variable symbols including the start symbol S . The string $w = a^30b^30c^30$ is derivable from S . The number of steps (application of rules) in the derivation $S \Rightarrow^* w$ is _____
- 130.** Consider the following languages: $L_1 = \{a^n w a^n \mid w \in \{a, b\}^*\}$ $L_2 = \{w x w^R \mid w, x \in \{a, b\}^*, |w|, |x| > 0\}$. Note that w^R is the reversal of the string w . Which of the following is/are TRUE?
- 131.** For a string w , we define w^R to be the reverse of w . For example, if $w = 01101$ then $w^R = 10110$. Which of the following languages is/are context-free?
- 132.** Which of the following statements are true? 1. Every left-recursive grammar can be converted to a right-recursive grammar and vice-versa. 2. All ϵ -productions can be removed from any context-free grammar by suitable transformations. 3. The language generated by a context-free grammar all of whose productions are of the form $X \rightarrow w$ or $X \rightarrow wY$ (where, w is a string of terminals and Y is a non terminal), is always regular. 4. The derivation trees of strings generated by a context-free grammar in Chomsky Normal Form are always binary trees
- 133.** Consider the following statements about the context-free grammar $G = \{S \rightarrow SS, S \rightarrow ab, S \rightarrow ba, S \rightarrow \epsilon\}$ 1. G is ambiguous. 2. G produces all strings with equal number of a's and b's. 3. G can be accepted by a deterministic PDA. Which combination below expresses all the true statements about G ?
- 134.** Let L be the set of all binary strings whose last two symbols are the same. The number of states in the minimum state deterministic finite-state automation accepting L is
- 135.** Which of the following languages over $\{a,b,c\}$ is accepted by Deterministic push down automata?

136. If L_1 and L_2 are context free languages and R a regular set, one of the languages below is not necessarily a context free language. Which one?

137. Context free languages and regular languages are both closed under the operation(s) of :

138. Which of the following are decidable? 1. Whether the intersection of two regular languages is infinite. 2. Whether a given context-free language is regular. 3. Whether two push-down automata accept the same language. 4. Whether a given grammar is context-free

139. Language L_1 is polynomial time reducible to language L_2 . Language L_3 is polynomial time reducible to L_2 , which in turn is polynomial time reducible to language L_4 . Which of the following is/are true?

140. Consider three decision problems P_1 , P_2 , and P_3 . It is known that P_1 is decidable and P_2 is un-decidable. Which of the following is TRUE?

141. Consider the following problem X . Given a Turing machine M over the input alphabet Σ , any state q of M And a word $w \in \Sigma^*$, does the computation of M on w visit the state q ". Which of the following statements about X is correct?

142. Consider the following decision problems : P_1 Does a given finite state machine accept a given string. P_2 Does a given context free grammar generate an infinite number of strings. Which of the following statements is true?

143. Which one of the following is the strongest correct statement about a finite language over some finite alphabet Σ ?

144. Consider the language $L = \{a^n | n \geq 0\} \cup \{a^n b^n | n \geq 0\}$ and the following statements. I. L is deterministic context-free. II. L is context-free but not deterministic context-free. III. L is not $LL(k)$ for any k . Which of the above statements is/are TRUE?

145. Consider the following statements. I. The complement of every Turing decidable language is Turing decidable. II. There exists some language which is in NP but is not Turing decidable. III. If L is a language in NP, L is Turing decidable. Which of the above statements is/are true?

146. Which of the following statements is/are FALSE? 1. For every non-deterministic Turing machine, there exists an equivalent deterministic Turing machine. 2. Turing recognizable languages are closed under union and complementation. 3. Turing decidable languages are closed under intersection and complementation. 4. Turing recognizable languages are closed under union and intersection

147. If the strings of a language L can be effectively enumerated in lexicographic (i.e., alphabetic) order, which of the following statements is true?

148. Regarding the power of recognition of languages, which of the following statement is false?

149. Which of the following statements is / are true / false? The language $\{0^n \mid n \text{ is prime}\}$ is not regular

150. Which of the following statements is / are true / false? Union of two recursive languages is recursive

- 151.** For a Turing machine M , $\{M\}$ denotes an encoding of M . Consider the following two languages. $L_1 = \{(M) \mid M \text{ takes more than 2021 steps on all inputs}\}$. $L_2 = \{(M) \mid M \text{ takes more than 2021 steps on some input}\}$. Which one of the following options is correct?
- 152.** Consider the following sets : S1. Set of all recursively enumerable languages over the alphabet $\{0,1\}$. S2. Set of all syntactically valid C programs. S3. Set of all languages over the alphabet $\{0,1\}$. S4. Set of all non-regular languages over the alphabet $\{0,1\}$. Which of the above sets are uncountable?
- 153.** The set of all recursively enumerable languages is
- 154.** Consider the following problems. $L(G)$ denotes the language generated by a grammar G . $L(M)$ denotes the language accepted by a machine M . I. For an unrestricted grammar G and a string w , whether $w \in L(G)$. II. Given a Turing machine M , whether $L(M)$ is regular. III. Given two grammars G_1 and G_2 , whether $L(G_1) = L(G_2)$. IV. Given an NFAN, whether there is a deterministic PDAP such that N , and P accept the same language. Which one of the following statements is correct?
- 155.** Let X be a recursive language and Y be a recursively enumerable but not recursive language. Let W and Z be two languages such that $\bar{Y}$ reduces to W , and Z reduces to $\bar{X}$ (reduction means the standard many-one reduction). Which one of the following statements is TRUE?
- 156.** Let L_1 be a regular language, L_2 be a deterministic context-free language and L_3 a recursively enumerable, but not recursive, language. Which one of the following statement is false?
- 157.** Let L_1 be a recursive language, and Let L_2 be a recursively enumerable but not a recursive language. Which one of the following is TRUE?
- 158.** In which of the cases stated below is the following statement true? "For every non-deterministic machine M_1 there exists an equivalent deterministic machine M_2 recognizing the same language".
- 159.** Design a DFA to accept all binary strings such that the number of 0s is divisible by 3 and the number of 1s is even. Justify the correctness of your design using a transition table and test strings.
- 160.** Prove that for every NFA, there exists an equivalent DFA. Provide the formal construction method using the subset construction algorithm and demonstrate with a concrete example.
- 161.** Explain the concept of DFA minimization. Given a DFA with 7 states, show how you can minimize it using the table-filling algorithm. Provide a detailed walkthrough.
- 162.** Show that the language $L = \{w \mid w \text{ contains equal number of a's and b's}\}$ is not regular using the contradiction method and closure properties of regular languages.
- 163.** Discuss the limitations of finite automata with respect to memory. Give an example of a language that cannot be accepted by any finite automaton and justify your answer.
- 164.** Convert the given ϵ -NFA to its equivalent DFA using epsilon-closure and subset construction method. Use the transition diagram for the ϵ -NFA below and describe each step.
- 165.** Design an NFA that accepts all strings over $\{a, b\}$ that contain at least one 'a' followed somewhere later by a 'b'. Explain your transitions with examples.

- 166.** For the language $L = \{w \in \{0,1\}^* \mid w \text{ ends with } 01 \text{ and has at most one } 0\}$, design a DFA. Explain how each state maintains the required constraints.
- 167.** Define the concept of distinguishable strings with respect to Myhill-Nerode theorem. Using this concept, prove that the language $L = \{a^n b^n \mid n \geq 0\}$ is not regular.
- 168.** Construct a minimized DFA for the language over $\{0,1\}$ which accepts strings that do not contain '101' as a substring. Provide all intermediate steps and explain the logic.
- 169.** Prove that the class of regular languages is closed under intersection and complement. Use DFA constructions to support your proof with diagrams.
- 170.** Let $L = \{w \in \{0,1\}^* \mid \text{the number of 1s mod } 3 \text{ is equal to the number of 0s mod } 2\}$. Design a DFA that accepts L and explain your design choices.
- 171.** Construct a DFA that accepts all binary strings where the second last symbol is a 1. Provide a justification for your state design and test cases.
- 172.** Given two DFAs M_1 and M_2 , describe an algorithm to construct a DFA that accepts the intersection of $L(M_1)$ and $L(M_2)$. Use an example for demonstration.
- 173.** Define state equivalence in DFA minimization. Prove that two non-equivalent states always lead to distinguishable final behaviors using an example.
- 174.** Show that the set of all palindromes over $\{a, b\}$ is not regular using the pumping lemma. Provide a detailed explanation of the contradiction.
- 175.** Construct an NFA that accepts the language $L = \{w \in \{0,1\}^* \mid w \text{ contains '00' or '11' as a substring but not both}\}$. Explain your acceptance criteria clearly.
- 176.** Prove that any regular language can be accepted by a DFA with a number of states exponential in the size of an equivalent regular expression. Provide reasoning.
- 177.** What are the key differences in computational power and efficiency between DFA and NFA? Illustrate your explanation with a real-world use-case example.
- 178.** Design an NFA with minimum number of states for the language $L = \{w \in \{a,b\}^* \mid w \text{ has a length divisible by } 3\}$. Convert it into a DFA and analyze state explosion.
- 179.** Discuss why ϵ -transitions are useful in NFA design. Construct an ϵ -NFA for a language that accepts strings ending with either 'ab' or 'ba' and explain ϵ -closure usage.
- 180.** Construct a DFA that accepts all strings over $\{a, b\}$ such that no two b's occur consecutively. Justify your transitions and final state selections.
- 181.** For the language $L = \{a^n b^m \mid n \geq 2, m \geq 3\}$, determine whether it is regular. If yes, provide a DFA. If not, prove it using pumping lemma or closure properties.
- 182.** Construct a DFA for binary strings such that every '1' is followed by exactly two '0's. Validate your DFA with multiple test cases.

- 183.** Define the extended transition function $\delta^*(q, w)$ in the context of DFA. Explain how it is used in proving the correctness of DFA for a given language.
- 184.** Define regular expressions formally. Construct a regular expression over the alphabet $\{a, b\}$ that denotes the language of all strings that: begin and end with the same symbol, and do not contain the substring "ab" more than once. After constructing the expression, convert it into an equivalent NFA and explain your design choices in detail.
- 185.** Prove that regular expressions and finite automata are equivalent in their expressive power. Use the McNaughton-Yamada-Thompson construction to convert a complex regular expression to an NFA and provide step-by-step transitions and ϵ -moves. Highlight any edge cases that arise during construction.
- 186.** Given the regular expression $R = ((a + b)abb(a + b))$, Construct the equivalent ϵ -NFA using the Thompson construction method. Convert the ϵ -NFA to a DFA, and minimize it. Finally, write the language accepted in set notation and prove whether it is prefix-closed.
- 187.** Demonstrate the limitations of regular expressions. Show that the language $L = \{a^n b^n \mid n \geq 1\}$ cannot be described using any regular expression. Use the pumping lemma to prove non-regularity and explain why no regular expression or finite automaton can accept such a language.
- 188.** Design a regular expression for the following language over $\Sigma = \{0,1\}$: All strings that do not contain two consecutive 1s and contain an odd number of 0s. After constructing the expression, explain how your design ensures both constraints are satisfied.
- 189.** Prove that the language $L = \{w \in \{a, b\}^* \mid \text{number of } a\text{'s is even or number of } b\text{'s is a multiple of } 3\}$ is regular. Construct the regular expression and provide a formal proof using closure properties of regular languages. Convert your expression into a minimized DFA as part of the solution.
- 190.** Construct a regular expression over the alphabet $\{a, b\}$ that generates all strings such that every 'a' is immediately followed by 'b' but 'b' can occur alone as well. Design the equivalent DFA and Justify how it correctly enforces the condition. Also, analyze the state complexity of your design.
- 191.** Given a regular expression R , describe the general algorithm to convert R to a DFA. Apply the algorithm to the expression $R = (a + b)^*abb$ and explain each transition and state formation in detail. Discuss potential ambiguities and how they are resolved during conversion.
- 192.** Let L_1 and L_2 be regular languages over the alphabet $\{a, b\}$. Show that the language $L = \{x \in \Sigma^* \mid x \in L_1 \cap L_2\}$ is also regular. Provide a detailed construction of DFA for L using product automaton. Support your explanation with an example and complete transition tables.
- 193.** Prove that regular languages are closed under the reverse operation. Define the reverse of a language and construct the reverse of a given DFA formally. Use a non-trivial example to demonstrate the reversal of DFA and its conversion to a regular expression.
- 194.** Construct a regular expression for the language of all strings over $\{0,1\}$ such that the total number of 1s is divisible by 4 and total number of 0s is at least 2. Explain how your expression handles both constraints simultaneously, and show its conversion to a minimal DFA.

- 195.** Given a language $L = \{w \mid w \text{ does not contain substring '001'}\}$, prove that L is regular. Construct the DFA for this language and convert it to a regular expression. Then, generalize your method to construct similar regular expressions that avoid arbitrary substrings.
- 196.** Let R be a regular expression such that $R = (ab)^*$. What language does it represent? Convert it to an equivalent DFA and show all steps including state combinations and minimization. Prove whether the language accepted is prefix-closed or suffix-closed.
- 197.** For the regular expression $R = (a+b)(a+b)^*$, Construct a corresponding ϵ -NFA. Describe how backtracking and greedy matching play roles in implementing this expression in lexical analyzers. Discuss the practical limitations of implementing complex REs in compilers.
- 198.** Show that the family of regular languages is closed under intersection, complement, and set difference operations. For each operation, construct a formal proof and illustrate it with an example using transition diagrams.
- 199.** Explain the algebraic laws of regular expressions. Using these laws, simplify the expression $((a + b)a(a + b)) + ((a + b)b(a + b))$ into a more concise form. Prove the correctness of your simplification using automata.
- 200.** Let $L = \{w \mid w \text{ has at most one occurrence of the substring "abc"}\}$. Prove that L is regular by constructing a DFA. Convert this DFA into a regular expression and analyze its complexity in terms of states and transitions.
- 201.** Construct a regular expression that generates strings over $\{a, b\}$ such that the total number of 'a's is a multiple of 2 and the total number of 'b's is a multiple of 3. Demonstrate the construction of a DFA and use modular arithmetic to track constraints.
- 202.** Show that the language $L = \{w \mid w \text{ has equal number of occurrences of substring "ab" and "ba"}\}$ is not regular. Prove this formally using contradiction and the pumping lemma.
- 203.** Design a regular expression for strings over $\{0,1\}$ where every '1' is followed by exactly one '0' and no two 1s appear consecutively. Convert it to a DFA, Minimize the DFA, Prove using Myhill-Nerode theorem that no smaller DFA can accept this language.
- 204.** State the pumping lemma for regular languages. Use it to prove that the language $L = \{a^n b^n \mid n \geq 0\}$ is not regular. Explain all assumptions, the decomposition of the string, and why pumping fails for all possible divisions satisfying the lemma's conditions.
- 205.** Given the language $L = \{a^i b^j c^k \mid i = j \text{ or } j = k\}$, Prove that L is not regular using the pumping lemma. Discuss whether the language can be represented as the union of two regular languages. Explain why the union of two non-regular languages is not necessarily non-regular.
- 206.** Prove using the pumping lemma that the language $L = \{ww \mid w \in \{a, b\}^*\}$ is not regular. Also, explain how this language requires memory beyond the capability of a finite automaton and why a stack-based machine (like PDA) would be more appropriate.

207. Let $L = \{a^n b^m \mid n \neq m\}$. Determine whether this language is regular. If regular, construct a DFA. If not, prove it using contradiction via the pumping lemma. Support your conclusion with clear pumping breakdown and explanation.

208. Use the pumping lemma to prove that $L = \{a^n b^m c^n \mid n, m \geq 0\}$ is not regular. In your proof, discuss how the symmetry between 'a' and 'c' in the string plays a role in violating the pumping lemma.

209. Consider $L = \{w \in \{a, b\}^* \mid \text{number of a's} = \text{number of b's}\}$. Prove that L is not regular using the pumping lemma. Then show a contrast by defining a similar regular language where counting is bounded (e.g., at most two a's), and construct a DFA for it.

210. The language $L = \{a^p \mid p \text{ is a prime number}\}$ is not regular. Use the pumping lemma to prove this, taking care to show that no matter how the string is pumped, the resulting string will eventually violate the prime condition.

211. Is the language $L = \{a^n b^m \mid n \text{ is a perfect square, } m \geq 0\}$ regular? Provide a formal proof using the pumping lemma to show that it is not. Discuss how non-linear counting constraints make a language non-regular.

212. Let $L = \{a^n b^m c^n \mid n \geq 0\}$. Use the pumping lemma to prove that L is not regular. Also discuss why this language is not even context-free, and how that can be shown using the context-free pumping lemma.

213. Prove that the language $L = \{w \mid w \text{ contains more a's than b's}\}$ is not regular. Discuss the contradiction when attempting to pump strings in L . Then contrast with a related regular language: $\{w \mid w \text{ contains more than two a's}\}$, and construct its DFA.

214. Use the pumping lemma to show that the language $L = \{w \in \{0,1\}^* \mid w \text{ is a palindrome}\}$ is not regular. Also explain intuitively why recognizing palindromes requires memory that finite automata cannot provide.

215. Explain why the pumping lemma cannot be used to prove that a language is regular. Give an example of a non-regular language that satisfies the conditions of the pumping lemma for a small number of strings but still fails to be regular.

216. Given $L = \{a^i b^j \mid i \text{ is divisible by } j\}$, prove that L is not regular. Use contradiction and pumping lemma by carefully choosing values of i and j , and explain how the division condition breaks under pumping.

217. Let $L = \{w \mid \text{number of substrings "aa"} = \text{number of substrings "bb"}\}$. Use the pumping lemma to prove that L is not regular. Discuss the challenges in maintaining equal counts of arbitrary overlapping substrings in a finite automaton.

218. Prove that the language $L = \{a^i b^j c^k \mid i = j \text{ or } j = k\}$ is not regular using the pumping lemma. Also determine whether it is context-free or not and justify using appropriate proof technique if possible.

219. Define a Context-Free Grammar (CFG) formally. Convert the following context-free grammar into an equivalent Chomsky Normal Form (CNF): $S \rightarrow aSb \mid \epsilon$. Explain each step of your conversion, highlighting any challenges you faced while maintaining the grammar's language.

220. Prove that the intersection of two context-free languages is not necessarily context-free. Provide a concrete example and show the process of proving this non-closure property. Discuss why this property

affects the design of efficient compilers.

221. Given the grammar $G: S \rightarrow Aa \mid Bb \mid \varepsilon$, $A \rightarrow Ab \mid b$, $B \rightarrow aB \mid b$, Prove whether the grammar G is ambiguous or not. If ambiguous, find the pair of strings that cause the ambiguity and show the two different derivations.

222. Describe the relationship between context-free grammars and pushdown automata (PDA). For the language $L = \{a^n b^n \mid n \geq 0\}$, design an equivalent PDA and show the formal steps of the transition function. Provide a detailed explanation of how the PDA simulates the grammar.

223. Prove that every context-free language can be accepted by a PDA. Construct a PDA for the language $L = \{a^n b^n \mid n \geq 0\}$ and explain its operation. Show how it uses the stack to balance the number of a's and b's.

224. Explain how to convert a given PDA into an equivalent context-free grammar (CFG). Provide a step-by-step method and apply it to a PDA that accepts the language $L = \{a^n b^n c^n \mid n \geq 0\}$. Explain your construction of rules and derivations.

225. Is the language $L = \{ww \mid w \in \{a, b\}^*\}$ context-free? Use the pumping lemma for context-free languages to prove that L is not context-free. Show the steps clearly and discuss why PDA cannot accept this language.

226. Design a context-free grammar for the language $L = \{a^n b^m c^k \mid n, m, k \geq 0\}$. Then, convert this grammar into Greibach Normal Form (GNF) and explain the transformation steps in detail.

227. What is the relationship between determinism and context-free languages? Define deterministic context-free languages (DCFLs). Show whether the language $L = \{a^n b^n c^n \mid n \geq 0\}$ is deterministic context-free and explain the reasoning behind your conclusion.

228. Prove that the language $L = \{a^i b^q c^i \mid i, q \geq 0\}$ is context-free but not regular. Construct a context-free grammar for L and explain why it cannot be recognized by any finite automaton.

229. Consider the language $L = \{w \in \{a, b, c\}^* \mid w \text{ is a palindrome}\}$. Prove that L is not regular using the pumping lemma. Prove that L is context-free by constructing an appropriate CFG and PDA.

230. Is the language $L = \{a^i b^j c^k \mid i \neq j\}$ context-free? Use the pumping lemma for context-free languages to show that L is not context-free. Discuss the failure of the PDA in accepting the language and how it can be rejected.

231. Construct a context-free grammar that generates the language $L = \{a^n b^m \mid n \leq m\}$. Provide a derivation tree for the string "aaabbb" using the grammar and explain each rule step by step.

232. What is the concept of ambiguity in context-free grammars? Given the grammar: $S \rightarrow aSb \mid aS \mid bS \mid ab$, Prove that it is ambiguous by showing two different leftmost derivations for the string "ab". Discuss the significance of ambiguity in parsing and its impact on compiler design.

233. Design a PDA for the language $L = \{w \mid w \text{ has an equal number of a's and b's and has no other character in the string}\}$. Explain how the PDA uses the stack to count and ensure equal occurrences of a's and b's.

- 234.** Prove that the language $L = \{a^n b^m c^n \mid n \geq 0\}$ is not context-free. Use the pumping lemma for context-free languages and provide a clear explanation of why a PDA cannot recognize this language.
- 235.** Design a context-free grammar for the language $L = \{a^n b^m c^n \mid n, m \geq 0, n \neq m\}$. Explain how your grammar ensures that the number of a's is not equal to the number of b's while keeping the structure of the language valid.
- 236.** What is the difference between context-free languages and context-sensitive languages? Give an example of a context-sensitive language that is not context-free. Then, show that a context-free grammar can be converted into an equivalent pushdown automaton (PDA).
- 237.** Use the pumping lemma for context-free languages to prove that the language $L = \{a^n b^m c^n \mid n \geq 0\}$ is not context-free. Describe how each pumpable part of the string violates the context-free nature of the language.
- 238.** Design a context-free grammar for the language $L = \{a^n b^m c^k \mid n + m = k\}$. Provide a derivation tree for the string "aabbcc" and show how the grammar satisfies the condition.
- 239.** Construct a PDA for the language $L = \{a^n b^n \mid n \geq 0\}$. Then, use the PDA to simulate the process of accepting the string "aaabbb". Discuss the steps taken by the PDA in detail.
- 240.** Describe how to minimize a context-free grammar. Explain the steps of minimizing a context-free grammar and apply the process to the grammar: $S \rightarrow aS \mid bS \mid \epsilon$. Provide the minimized form of the grammar and explain each transformation.
- 241.** Prove that context-free languages are closed under union. Let L_1 and L_2 be context-free languages. Show that the language $L = L_1 \cup L_2$ is context-free by constructing a context-free grammar for L .
- 242.** Design a context-free grammar for the language $L = \{a^n b^m \mid n, m \geq 0, n \neq m\}$. Describe why this grammar cannot be minimized further and how it generates the required strings.
- 243.** Prove that the language $L = \{a^n b^m c^n d^m \mid n \geq 0\}$ is context-free but not regular. Show the construction of a context-free grammar for this language. Then, prove using the pumping lemma that it cannot be regular.
- 244.** Define a Turing Machine (TM) formally. Explain its components (tape, head, states, etc.), transition function, and how a TM differs from a finite automaton and a pushdown automaton in terms of computational power.
- 245.** Design a Turing Machine that accepts the language $L = \{w \in \{a, b\}^* \mid w \text{ has an equal number of a's and b's}\}$. Describe the states, transitions, and the algorithm the machine uses to accept strings with an equal number of a's and b's.
- 246.** Prove that the language $L = \{a^n b^n \mid n \geq 0\}$ is decidable. Construct a Turing Machine that accepts this language and explain its working. Then, demonstrate that the language is decidable by discussing how the machine halts for all inputs.
- 247.** Given the language $L = \{w \in \{a, b, c\}^* \mid w \text{ contains an equal number of a's, b's, and c's}\}$, Design a Turing Machine that accepts the language and prove that the language is decidable. Show how the

machine checks the equality of the number of a's, b's, and c's in the string.

248. Design a Turing Machine to decide whether the language $L = \{a^i b^m c^k \mid i = m = k\}$ is decidable or not. Explain the working of the TM and discuss any possible edge cases.

249. What is the Halting Problem? Formally define the Halting Problem and prove that it is undecidable. Provide a proof by contradiction using the concept of reduction to show why there is no Turing Machine that can solve the Halting Problem for all inputs.

250. Design a Turing Machine to decide whether a given string $w \in \{a, b\}^*$ is a palindrome. Describe how the TM verifies if a string is a palindrome by comparing characters at both ends and explains how the tape head moves.

251. Given a Turing Machine M and input w , describe how you would construct a Turing Machine that determines whether the machine M halts on input w . Discuss why this problem is undecidable and refer to the Halting Problem in your explanation.

252. What does it mean for a language to be undecidable? Provide an example of an undecidable language and explain why no Turing Machine can decide it. Use the example to show the connection between undecidability and the limitations of computation.

253. Design a Turing Machine that accepts the language $L = \{w \in \{a, b, c\}^* \mid w \text{ has an odd number of a's and b's and an even number of c's}\}$. Explain the states, transitions, and how the TM distinguishes between odd and even counts.

254. Prove that the language $L = \{w \in \{a, b\}^* \mid w \text{ contains at least one a and at least one b}\}$ is decidable. Construct a Turing Machine to accept this language and describe the machine's steps clearly.

255. Design a Turing Machine that decides whether a string is in the language $L = \{a^n b^m c^n \mid n \geq 0 \text{ and } m \neq n\}$. Prove that this problem is decidable by constructing the Turing Machine and describing its behavior.

256. Prove that the language $L = \{w \in \{a, b\}^* \mid w \text{ contains an odd number of a's and an even number of b's}\}$ is decidable. Construct the Turing Machine that decides this language and explain how the TM halts after reading the input.

257. Given the language $L = \{a^n b^n c^n \mid n \geq 0\}$, prove that this language is not decidable. Use the technique of reduction to show that this language is undecidable and explain why no Turing Machine can decide it.

258. Design a Turing Machine that accepts the language $L = \{w \in \{a, b\}^* \mid w \text{ has the same number of a's as b's, and the string length is even}\}$. Explain the steps and describe how the machine verifies both conditions (equal numbers of a's and b's, even length of the string).

259. Prove that the language $L = \{w \in \{a, b\}^* \mid w \text{ is of the form } a^n b^n \text{ and } n \geq 0\}$ is decidable.* Design a Turing Machine for this language and discuss how it checks the condition of matching numbers of a's and b's. Explain why the machine halts in all cases.

260. What is the connection between the Halting Problem and Turing Machines? Explain how the Halting Problem leads to undecidability by constructing a reduction from the Halting Problem to another

undecidable problem.

261. Design a Turing Machine that decides the language $L = \{w \in \{a, b\}^* \mid w \text{ has an equal number of a's and b's, and the first half of the string contains only a's and the second half contains only b's}\}$. Discuss the working of the machine and how it ensures both conditions are satisfied.

262. Prove that the language $L = \{w \in \{0, 1\}^* \mid w \text{ contains an equal number of 0's and 1's}\}$ is decidable. Design the Turing Machine for this language and explain the steps taken by the machine to ensure equality between 0's and 1's.

263. Discuss the concept of reductions in Turing Machines. Explain how the concept of reductions is used to prove that a problem is undecidable, and show an example by reducing the Halting Problem to another undecidable problem.

264. Explain the Halting Problem. Why is it undecidable?

265. Prove that the Post Correspondence Problem (PCP) is undecidable.

266. Discuss the concept of decidability in relation to Turing machines.

267. Prove that the problem of determining whether a given Turing machine halts on an empty tape is undecidable.

268. Show that the set of all Turing machines is uncountable.

269. Discuss the relationship between decidability and reducibility

270. Define the class of decidable languages and provide an example.

271. Explain the concept of an undecidable language with an example.

272. Prove that the equivalence of two Turing machines is undecidable.

273. What is the significance of Rice's Theorem in the context of decidability?

274. Discuss the undecidability of the Post's Problem.

275. Prove that the problem of determining whether a Turing machine accepts a specific string is undecidable.

276. Explain the concept of a reduction from one undecidable problem to another.

277. What is the difference between a decidable language and a recognizable language?

278. Define and prove the undecidability of the problem of determining whether a Turing machine has a certain number of accept states.

279. Define recursive languages and provide an example.

280. Discuss the difference between recursive and recursively enumerable languages.

- 281.** Prove that the class of recursive languages is closed under complement.
- 282.** Explain how a Turing machine can accept a recursively enumerable language.
- 283.** Show that every recursive language is also recursively enumerable, but not every recursively enumerable language is recursive.
- 284.** Prove that the intersection of two recursively enumerable languages is recursively enumerable.
- 285.** Discuss the closure properties of recursive languages.
- 286.** Describe the process of deciding whether a given language is recursive or recursively enumerable.
- 287.** Prove that the halting problem is recursively enumerable but not recursive.
- 288.** Discuss the concept of undecidable problems within the context of recursively enumerable languages.
- 289.** Design a deterministic finite automaton (DFA) for the language of all strings over the alphabet $\{a, b\}$ that contain an even number of a's and an odd number of b's. Provide the transition diagram and explain the working of the DFA.
- 290.** Consider the grammar $G: S \rightarrow aSb \mid T, T \rightarrow aTa \mid bTb$. Determine if the grammar generates a regular language. If it does not, explain why and prove the non-regularity.
- 291.** For the language $L = \{a^n b^n c^n \mid n \geq 0\}$, prove that L is not a context-free language. Discuss the reasons using the pumping lemma for context-free languages.
- 292.** Consider the Turing machine M that accepts a language L . Define the acceptance condition for M and describe how to determine if M halts on every input. Also, explain the relationship between decidability and the halting problem in this context.
- 293.** Show that the class of recursive languages is closed under intersection. Prove your result by providing a formal construction or argument.
- 294.** For the language $L = \{a^n b^n c^n \mid n \geq 0\}$, show that L is not regular by using the pumping lemma for regular languages. Include all steps of the proof and explain how the lemma applies to the given language.
- 295.** Given the context-free grammar (CFG): $S \rightarrow AB \mid a, A \rightarrow aA \mid \epsilon, B \rightarrow bB \mid \epsilon$. Convert this CFG to an equivalent pushdown automaton (PDA) and explain the steps involved in the conversion.
- 296.** For a Turing machine, define the concepts of a "halting state" and a "non-halting state." Explain how the machine can transition between these states and the significance of each state in terms of the computation. Also, describe how to determine if a Turing machine will halt on a specific input.
- 297.** The relationship between the classes of regular languages, context-free languages, and recursively enumerable languages. Include a detailed discussion on closure properties and the hierarchy of these language classes.

298. Consider a Turing machine M that computes a partial function. Define the concept of "recursively enumerable" (RE) and "recursive" (decidable) languages. Explain the differences with examples and discuss the implications for Turing machines that accept these types of languages.

299. Construct a Turing machine that accepts the language $L = \{w \mid w \text{ contains an equal number of a's, b's, and c's, and the number of a's, b's, and c's is greater than or equal to 1}\}$. Provide a detailed explanation of the working of the Turing machine, the states involved, and how it processes input strings.

300. Given the context-free grammar: $S \rightarrow aSb \mid bSa \mid \epsilon$. Show whether this grammar is ambiguous or not. If it is ambiguous, provide two different derivations for the same string and explain the steps in detail. If it is not ambiguous, justify why.